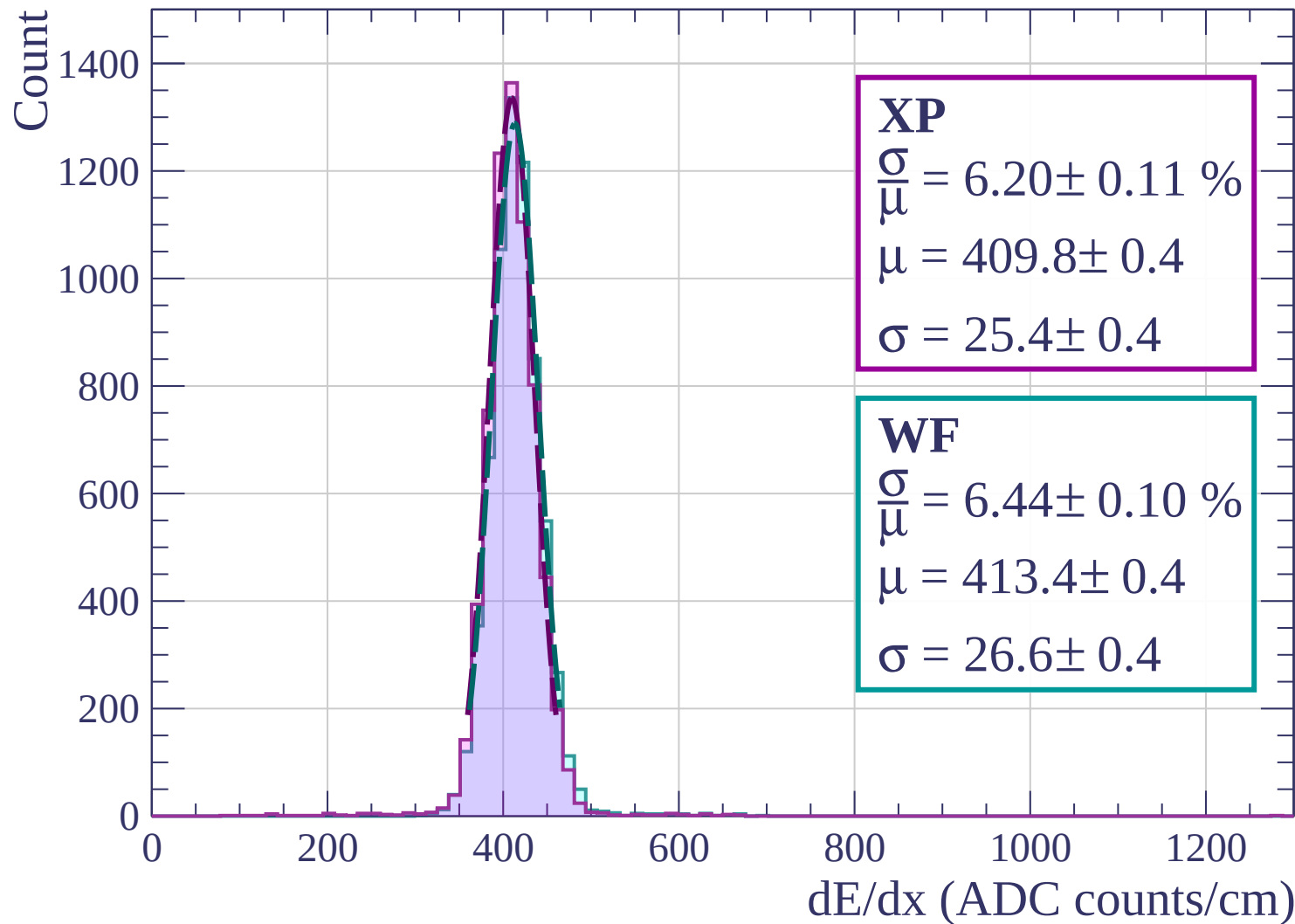
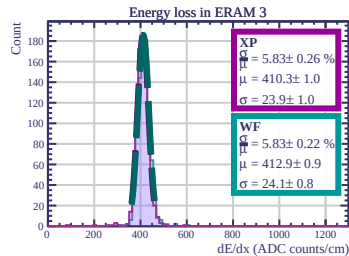
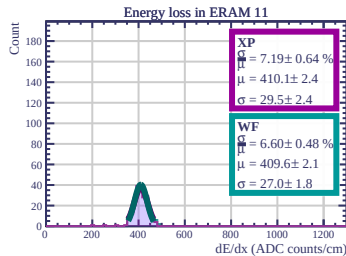
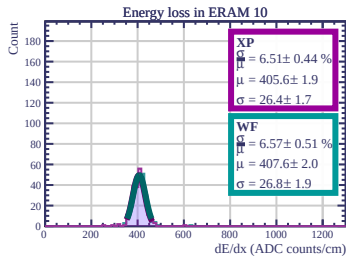
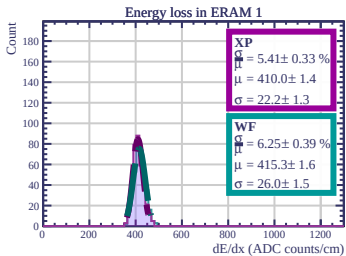
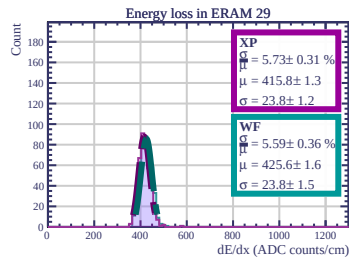
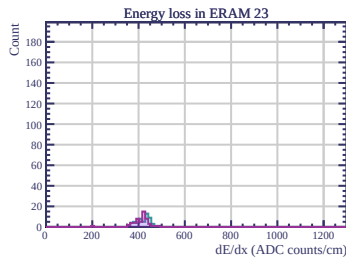
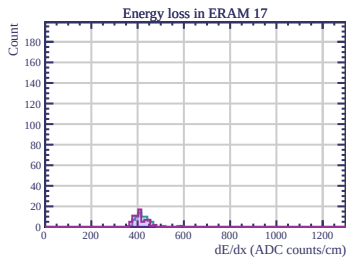
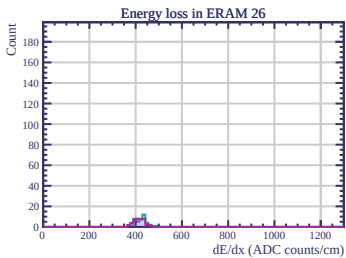
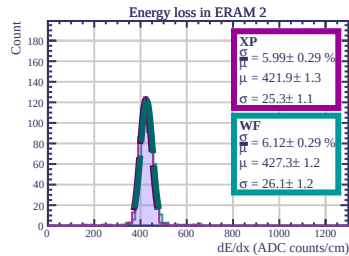
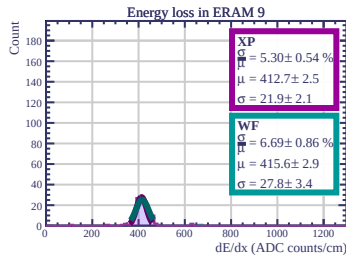
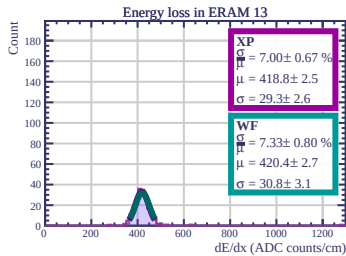
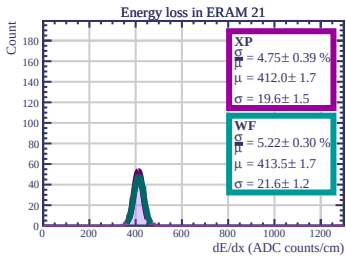
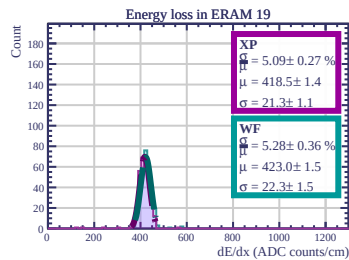
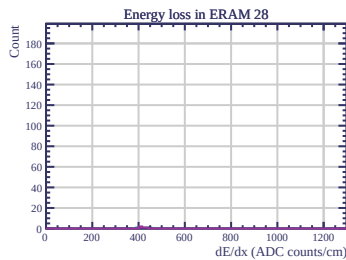
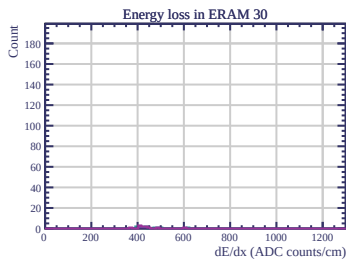
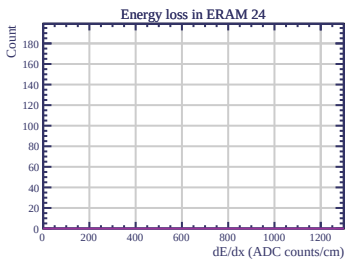
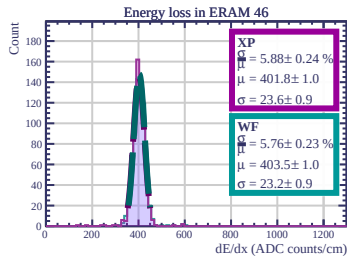
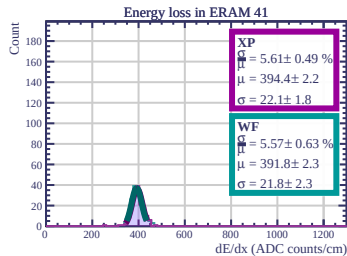
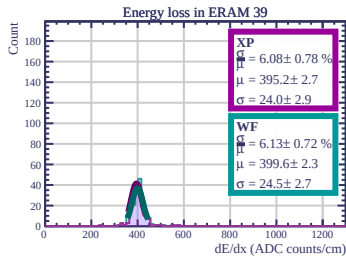
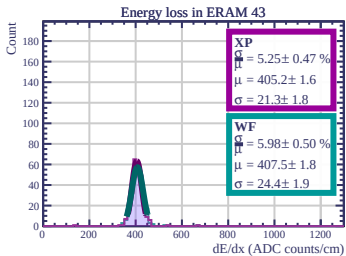
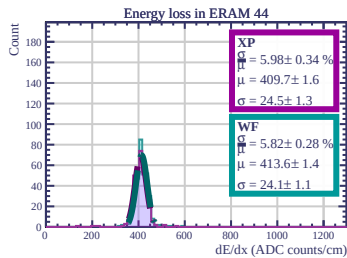
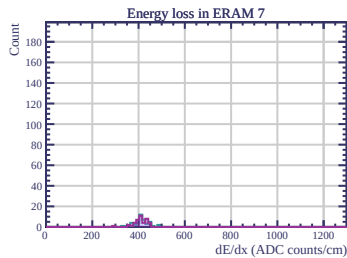
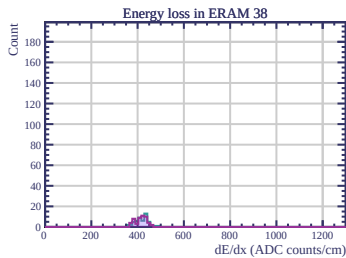
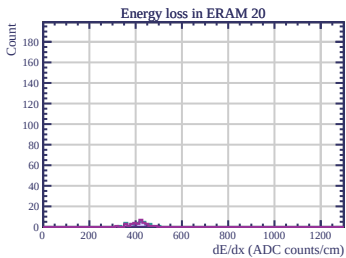
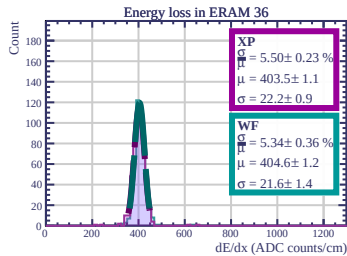
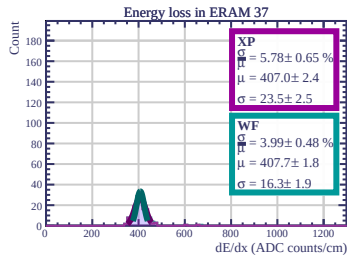
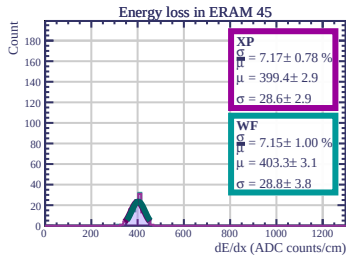
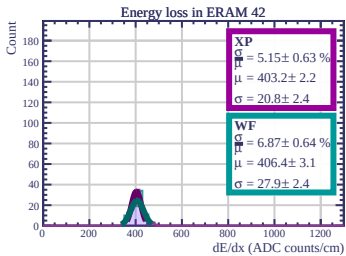
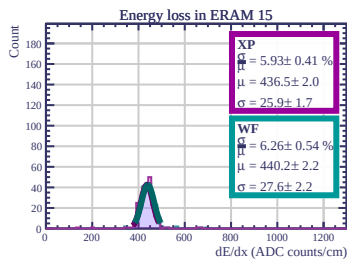
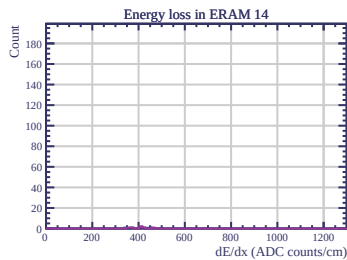
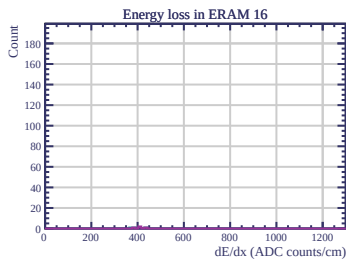
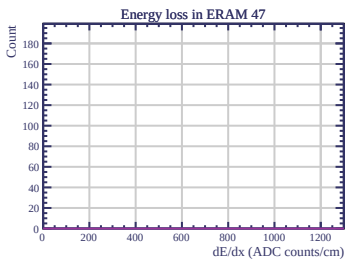
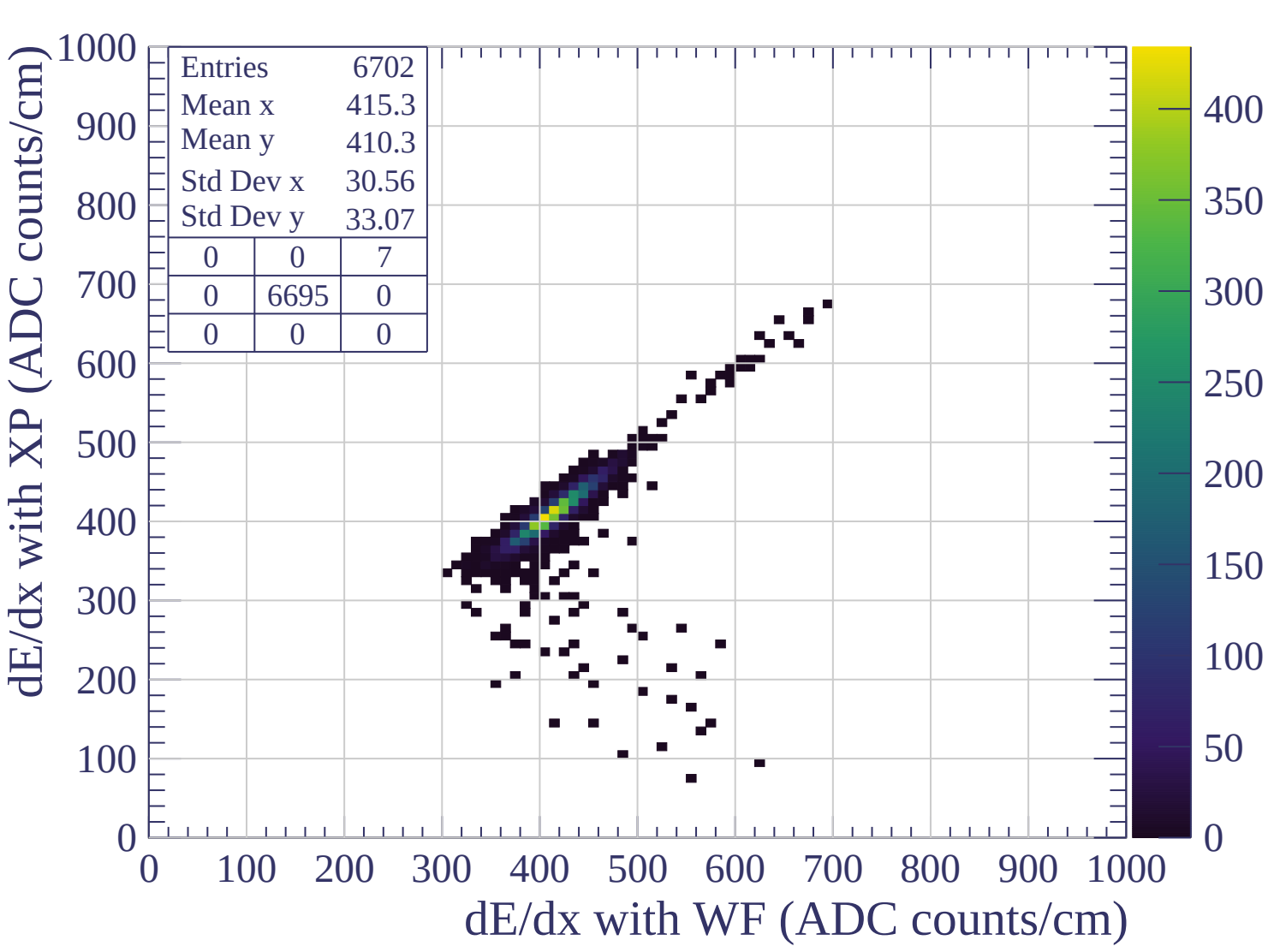
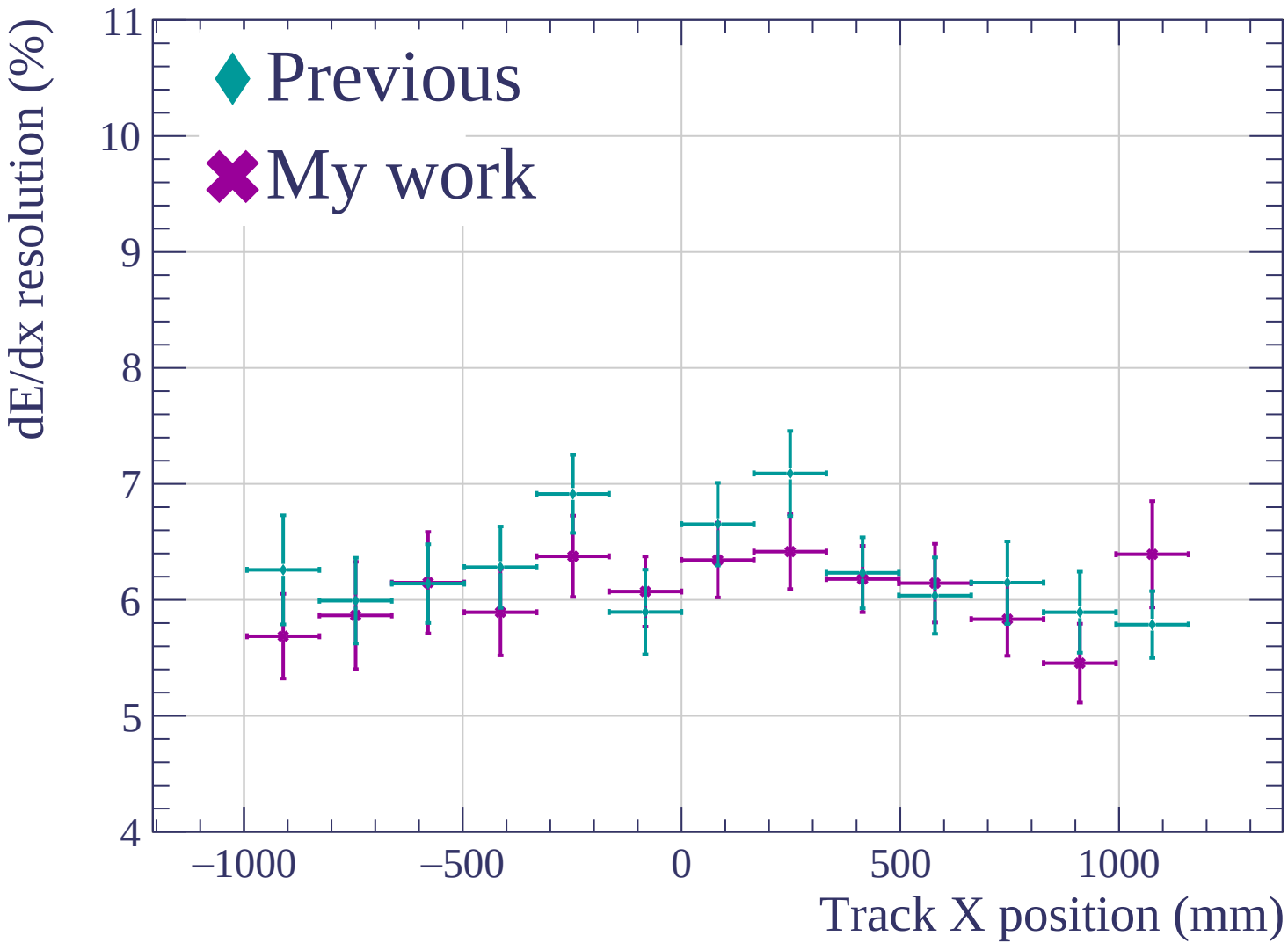


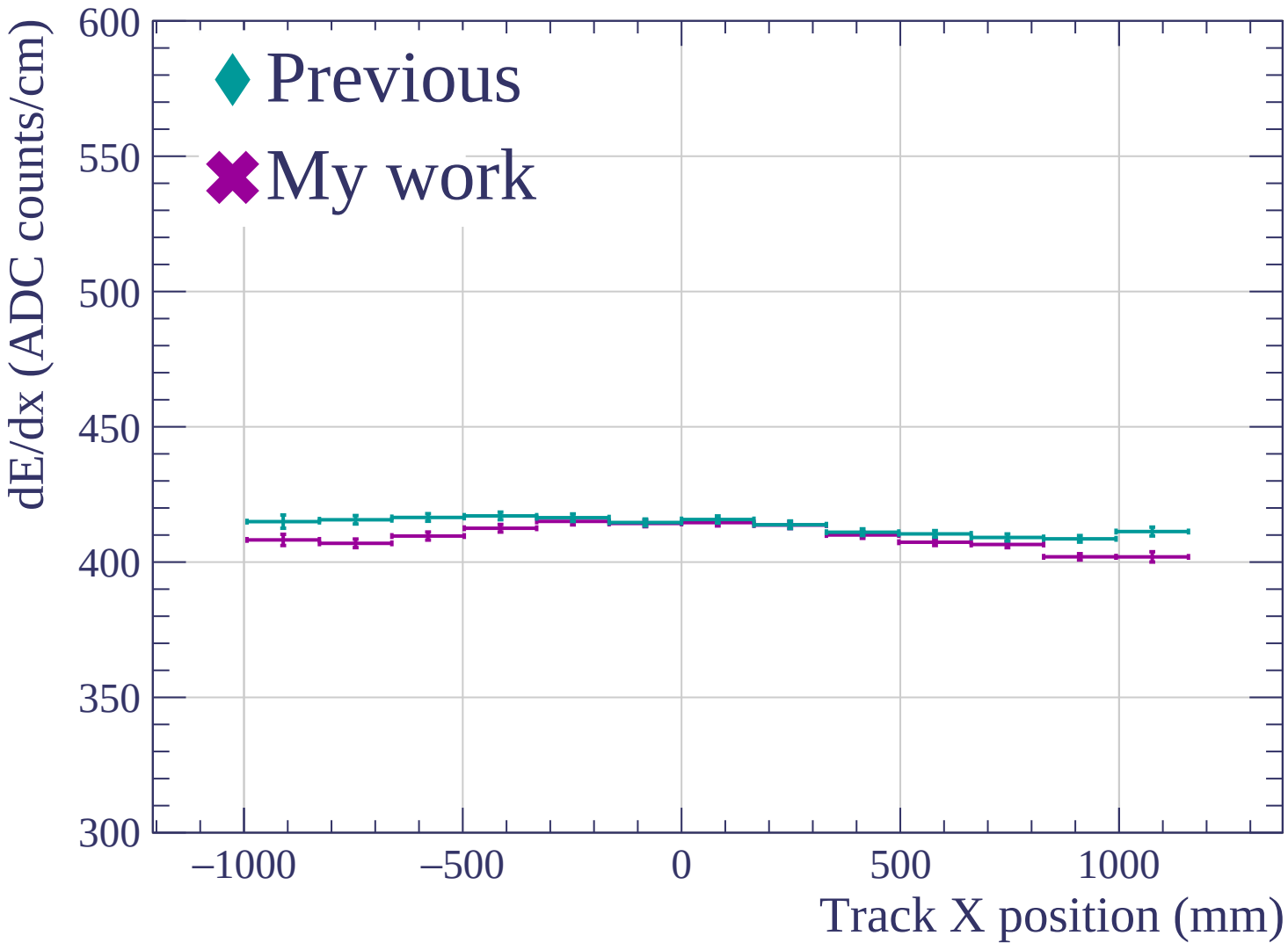


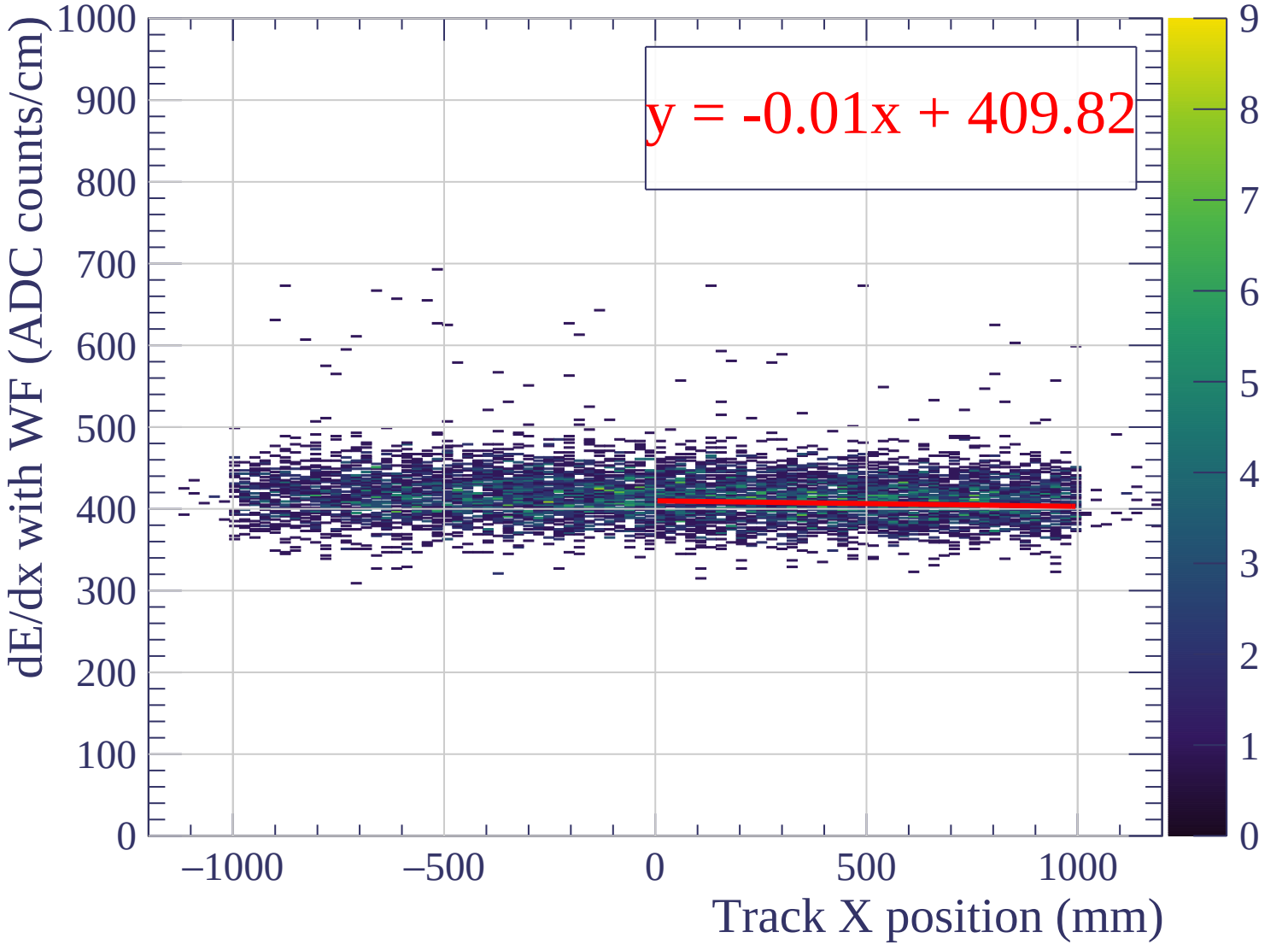


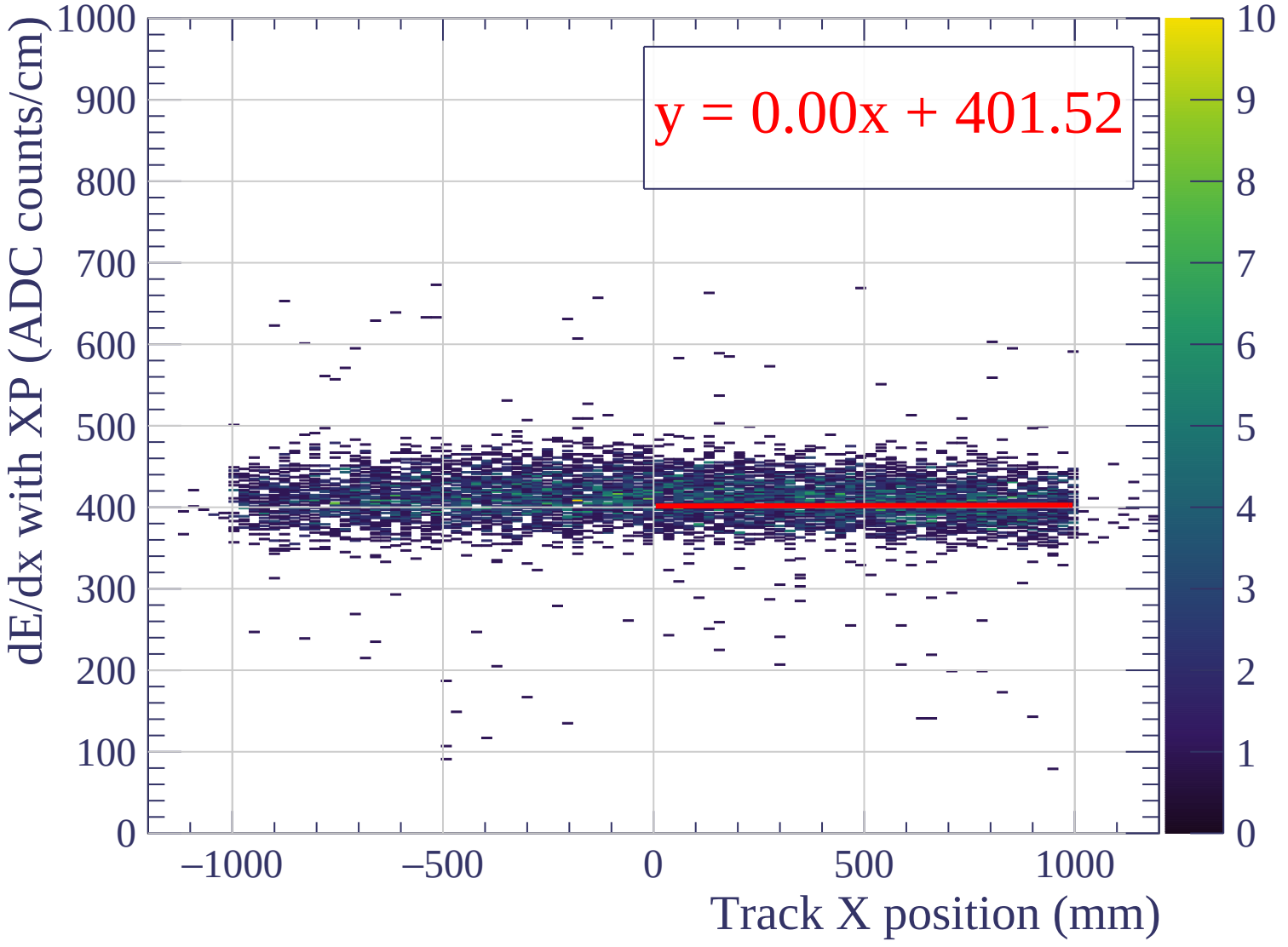


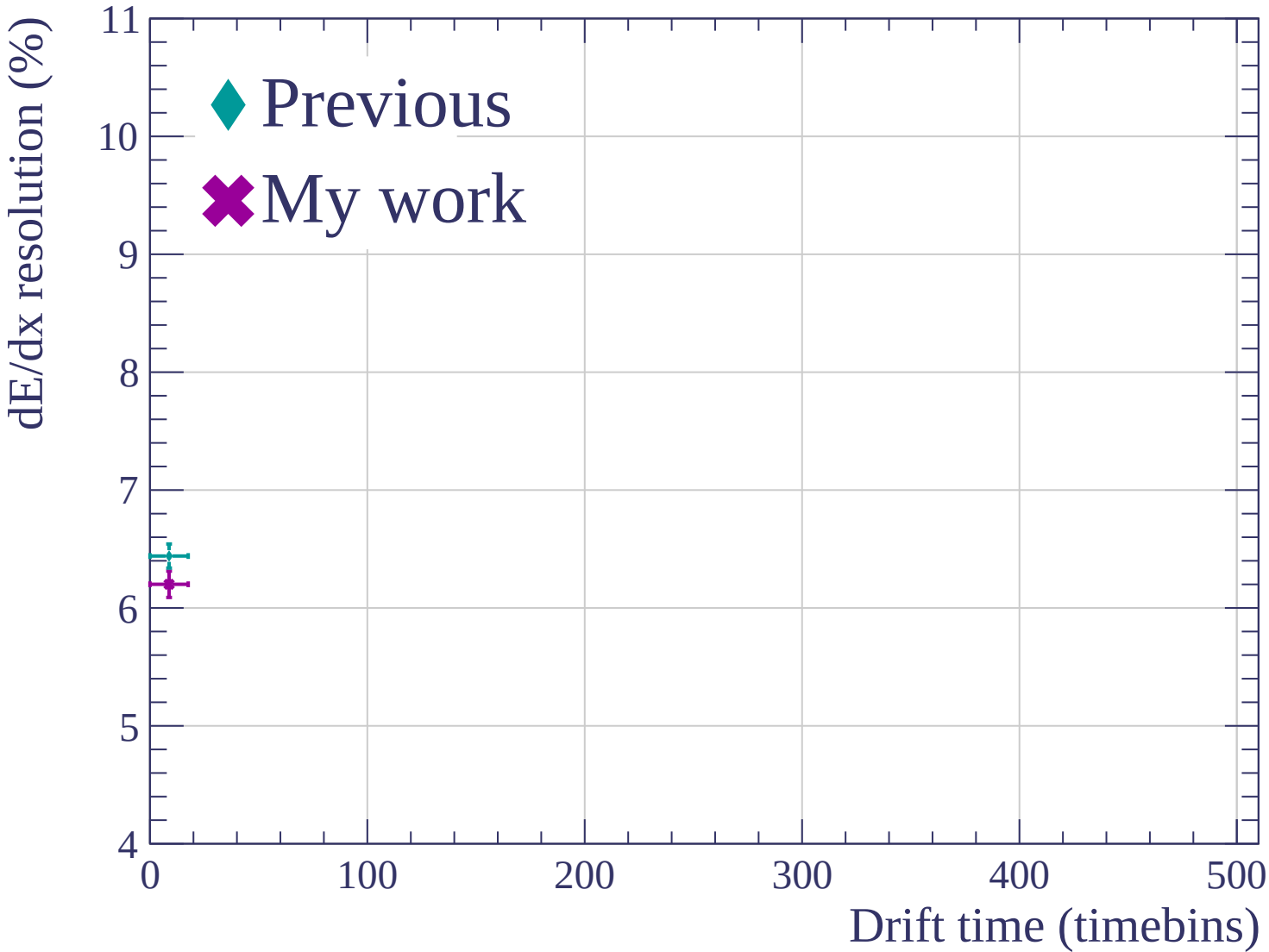


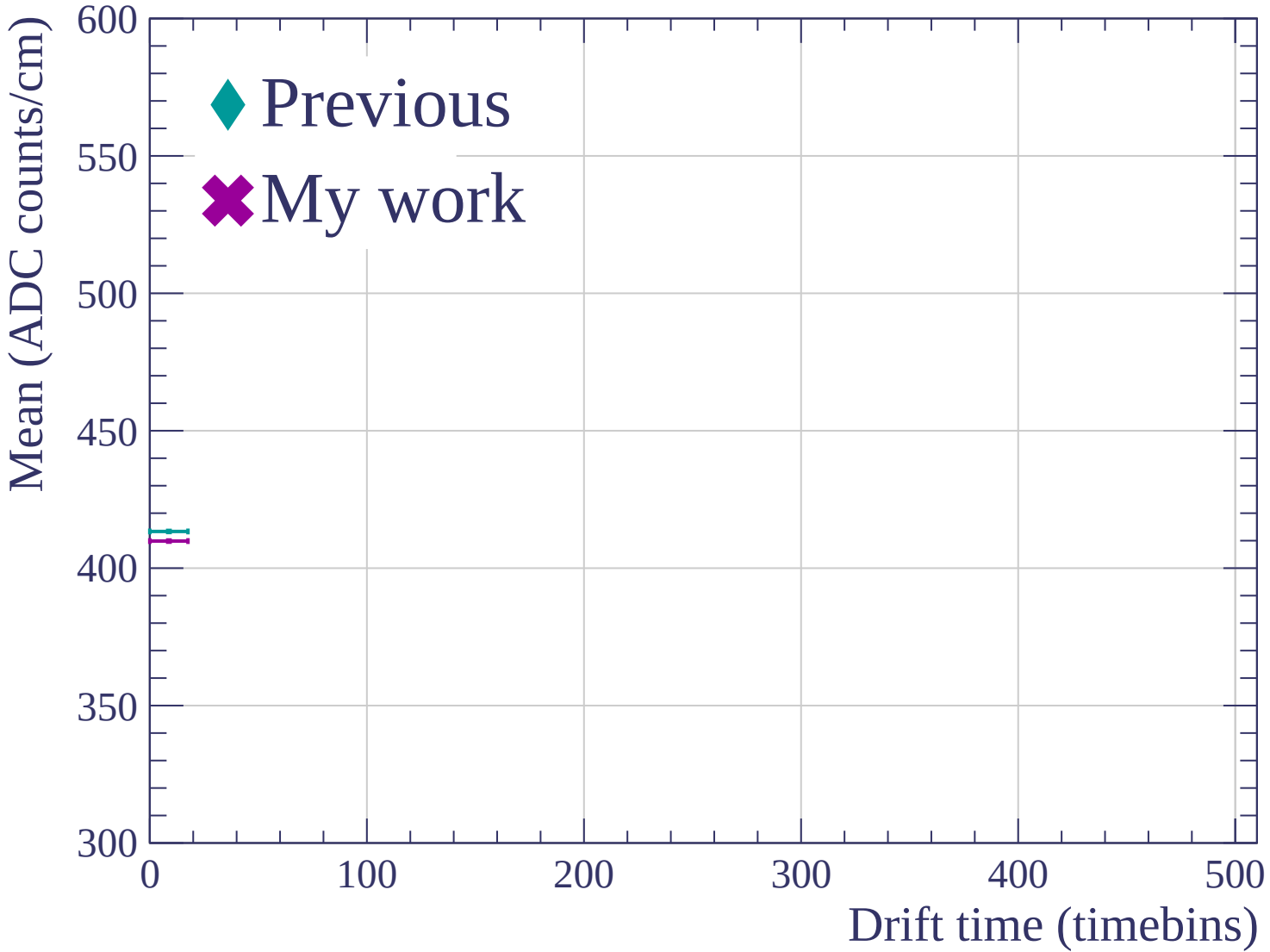


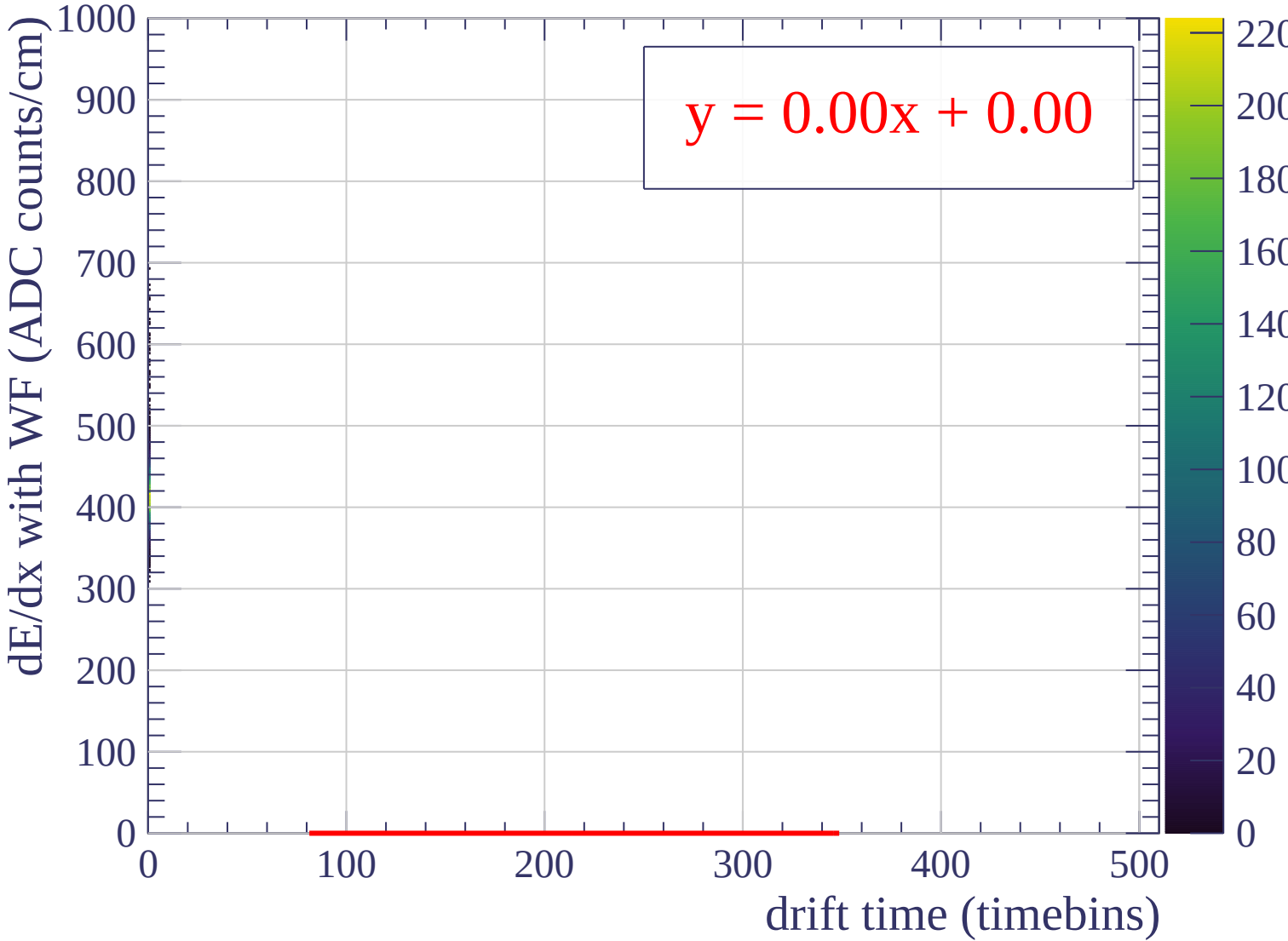


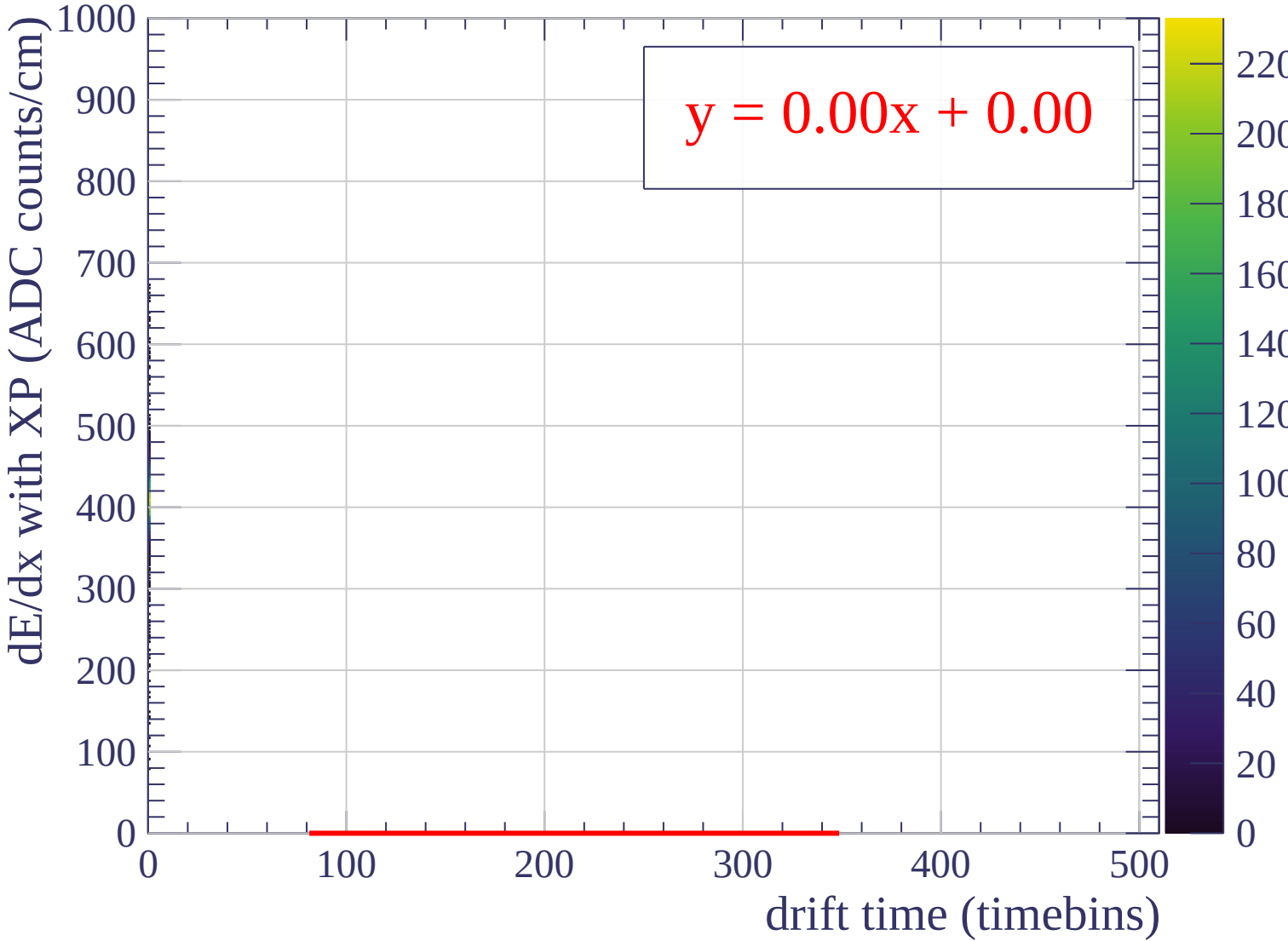


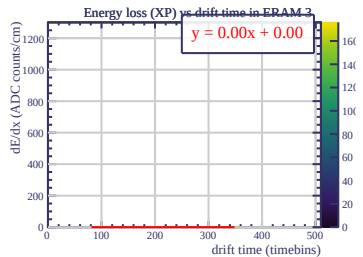
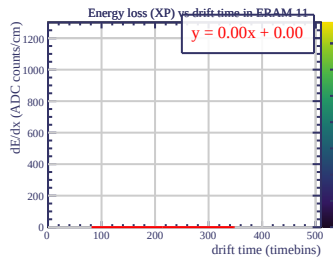
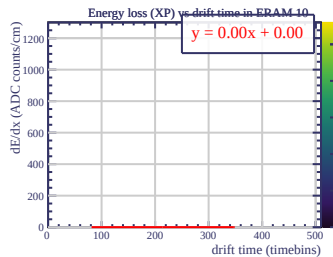
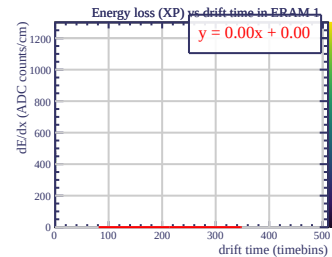
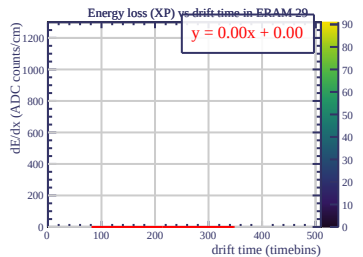
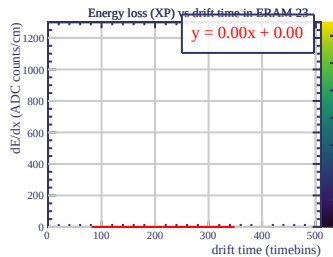
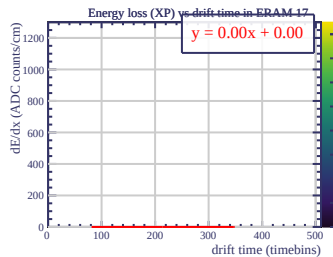
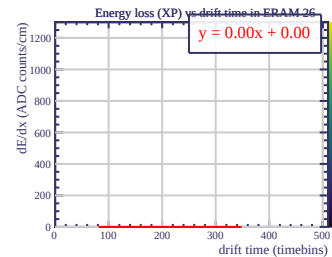
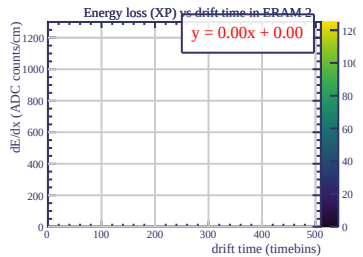
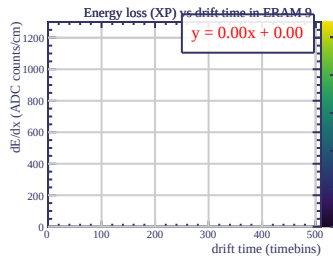
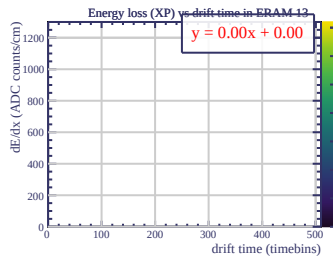
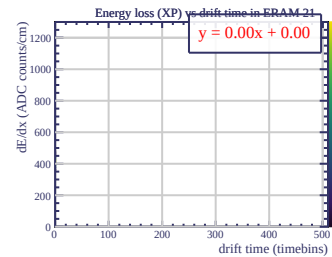
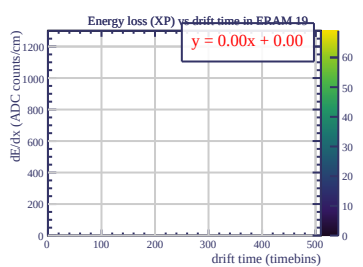
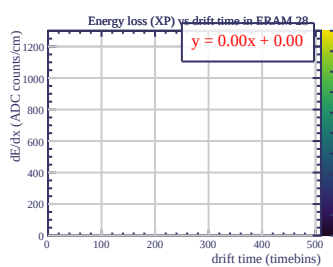
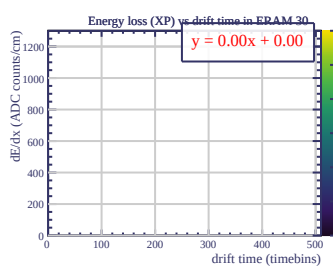
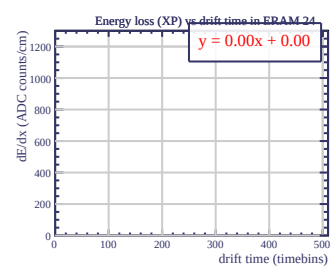


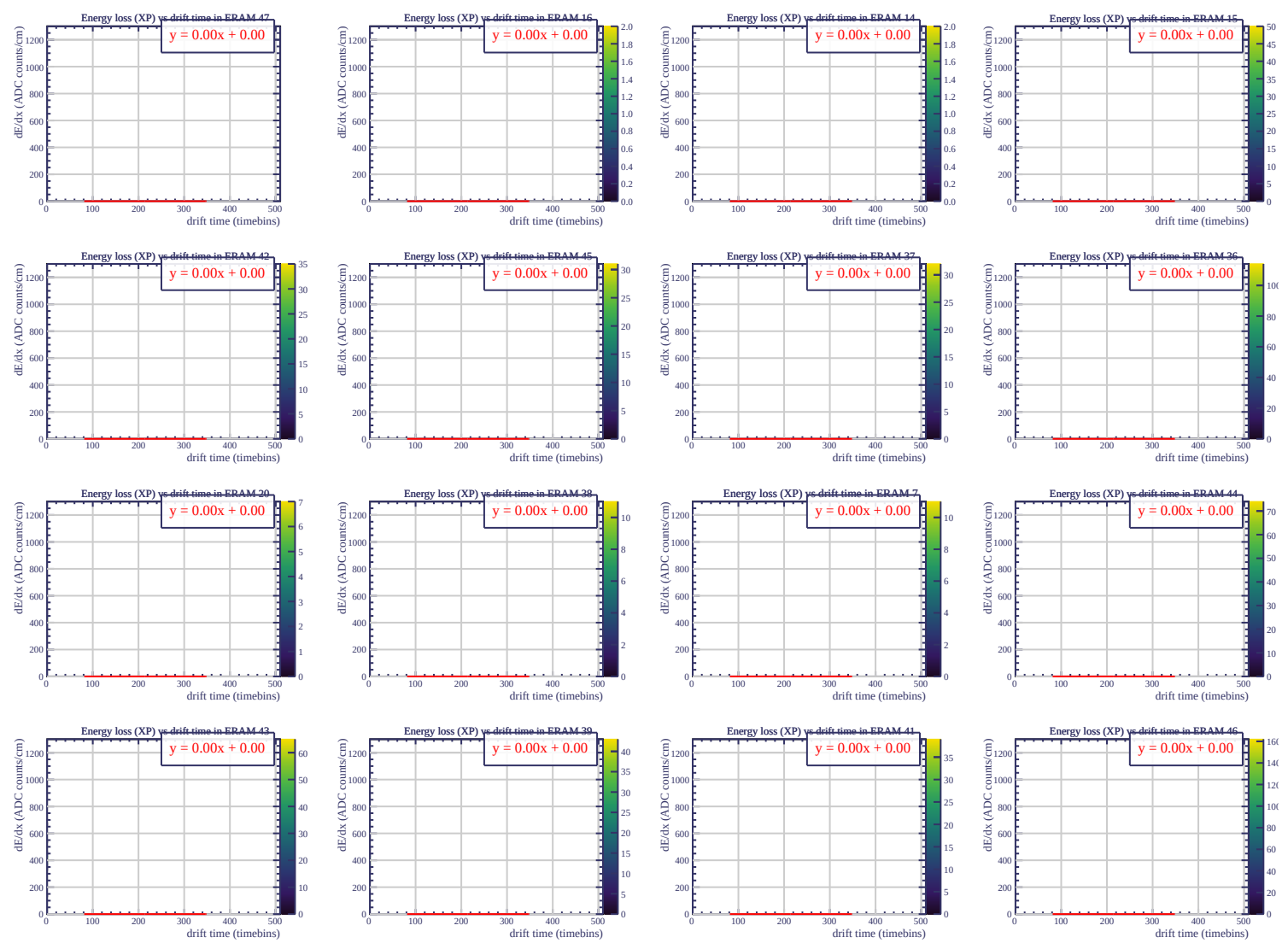


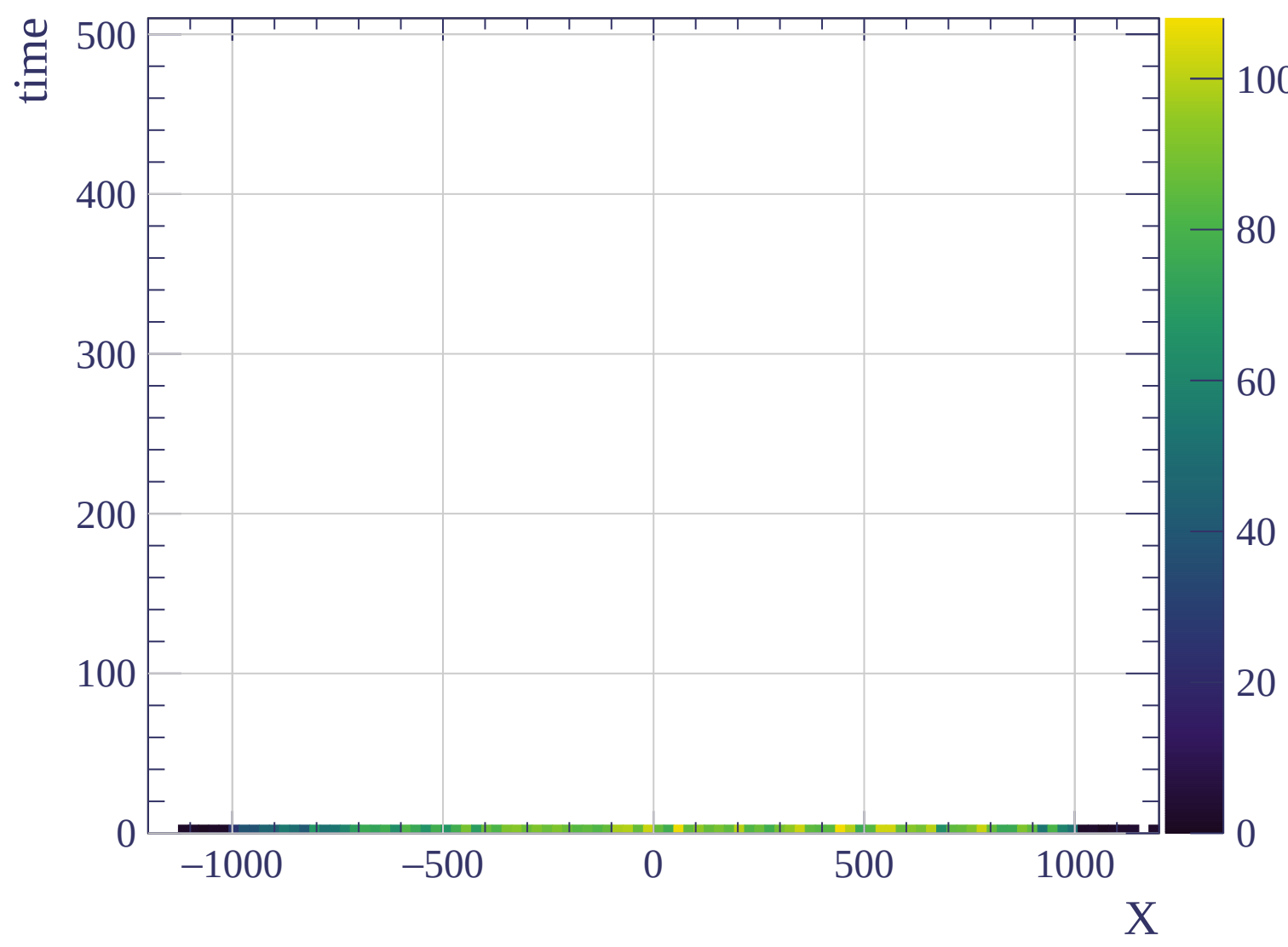


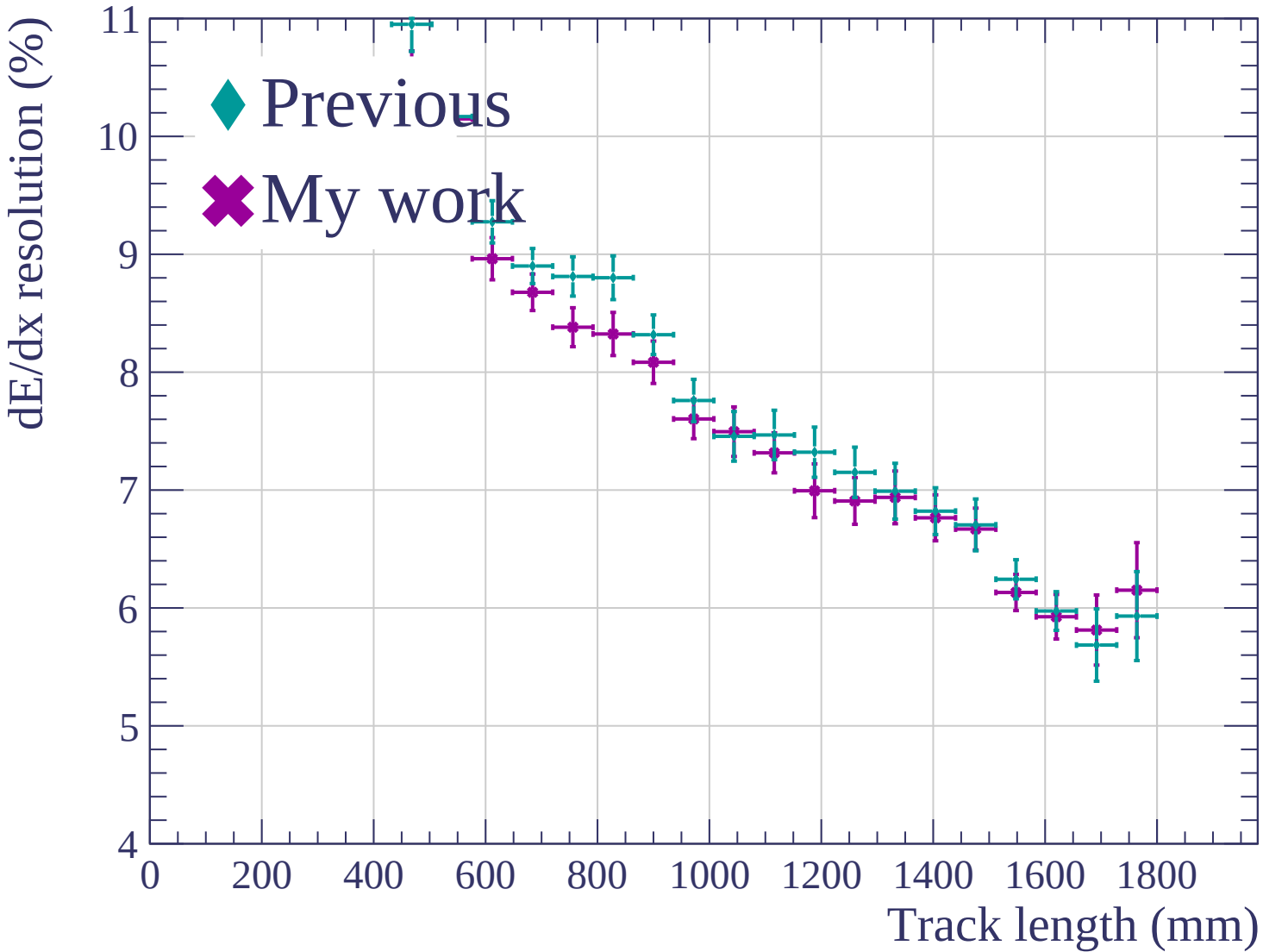


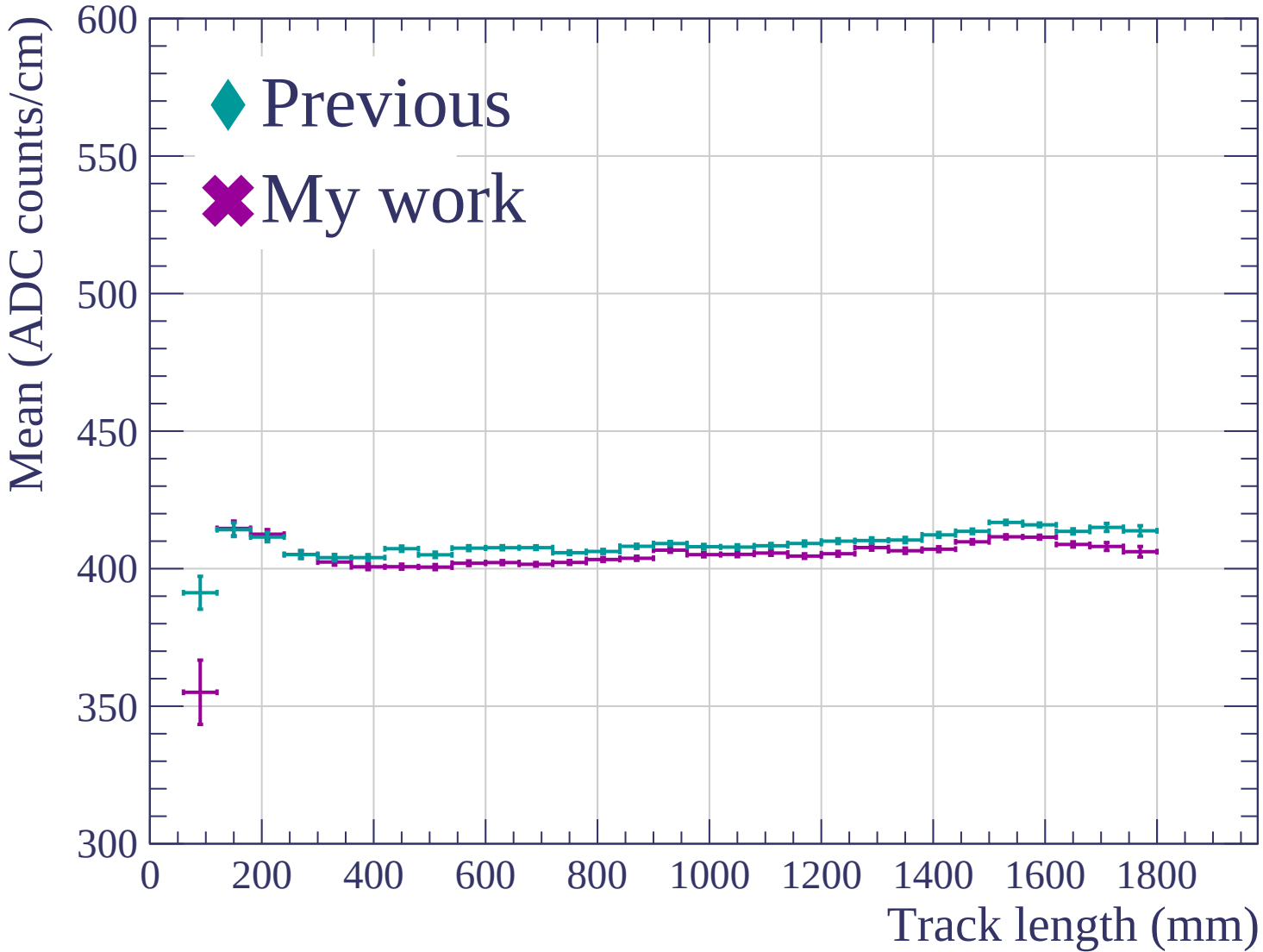


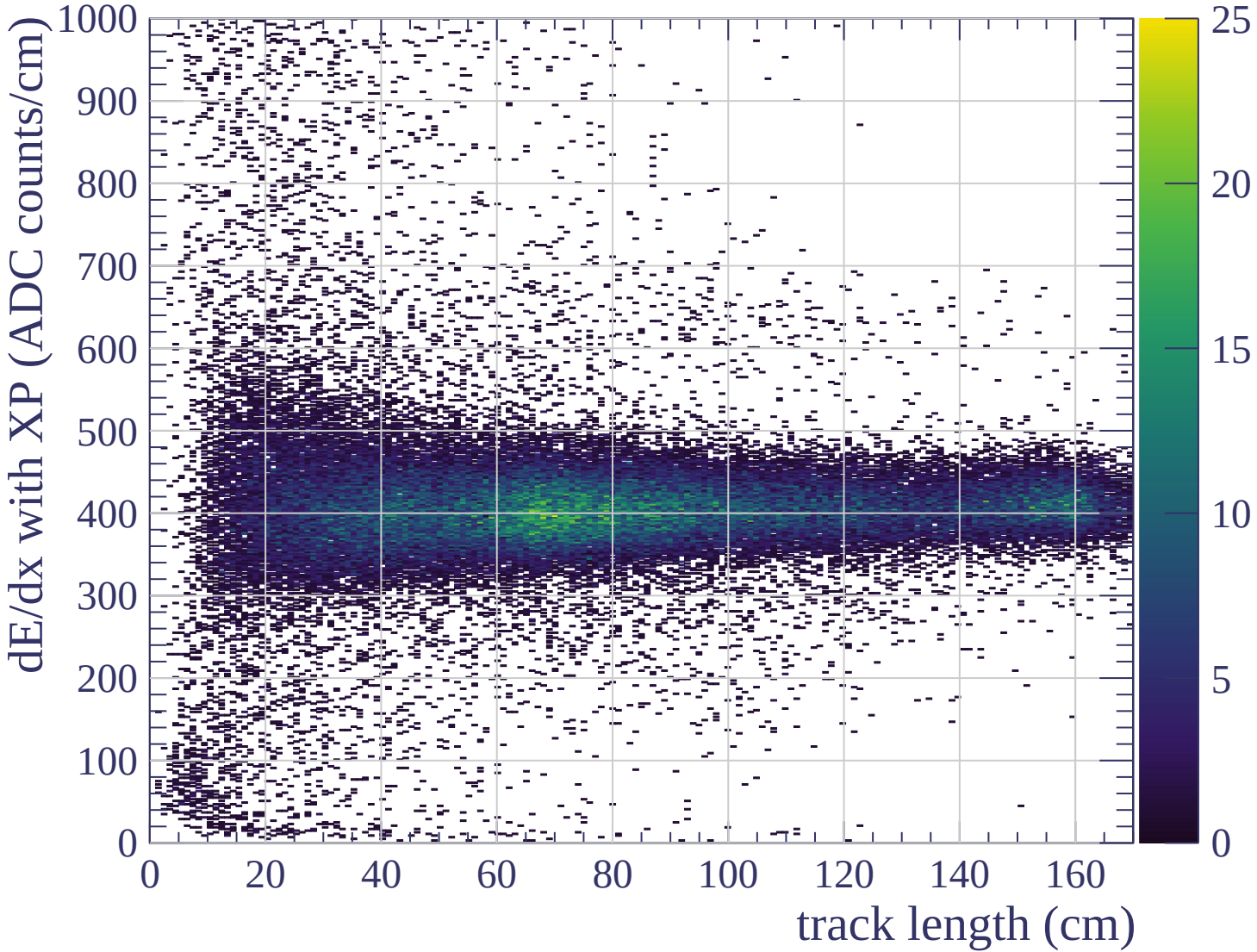


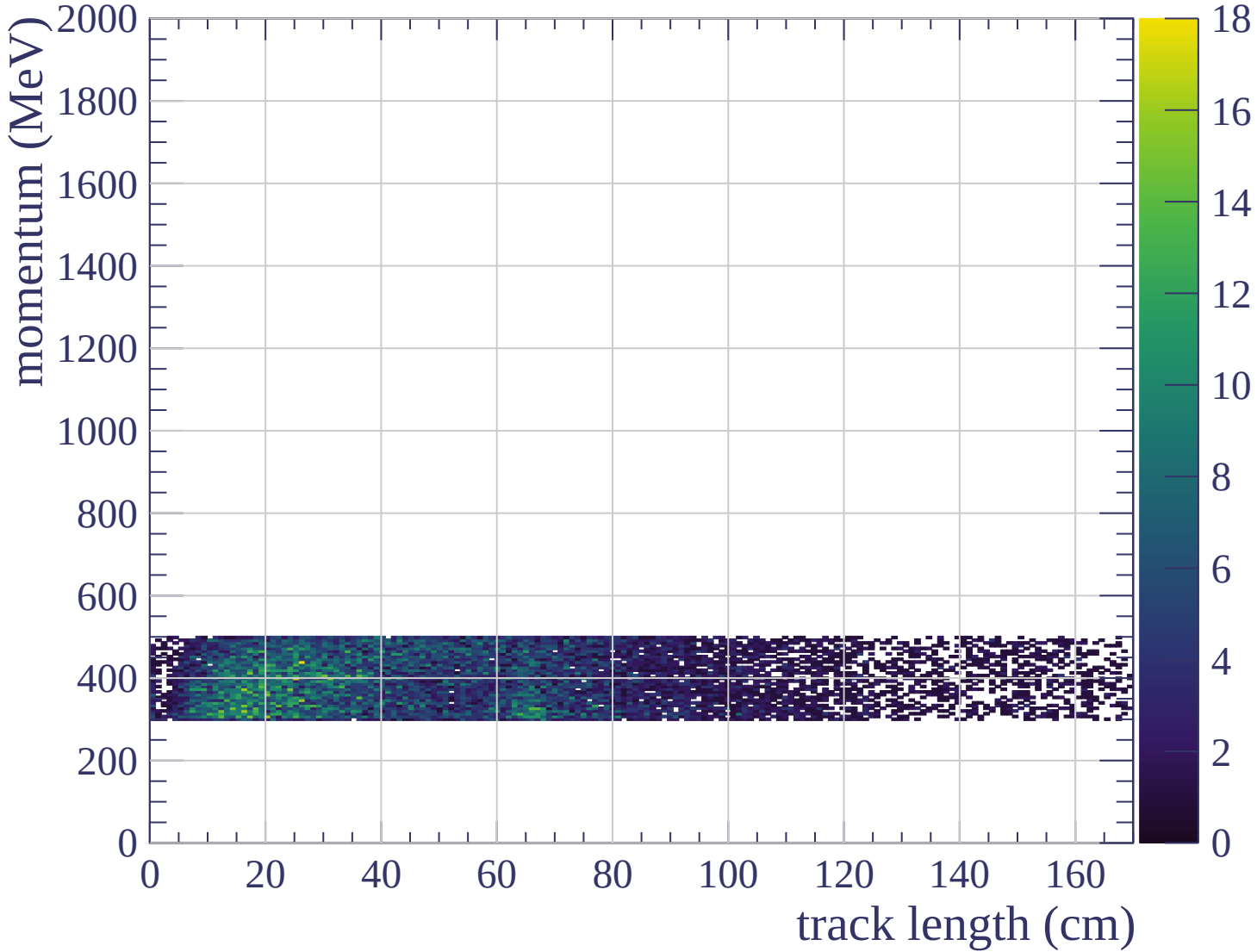


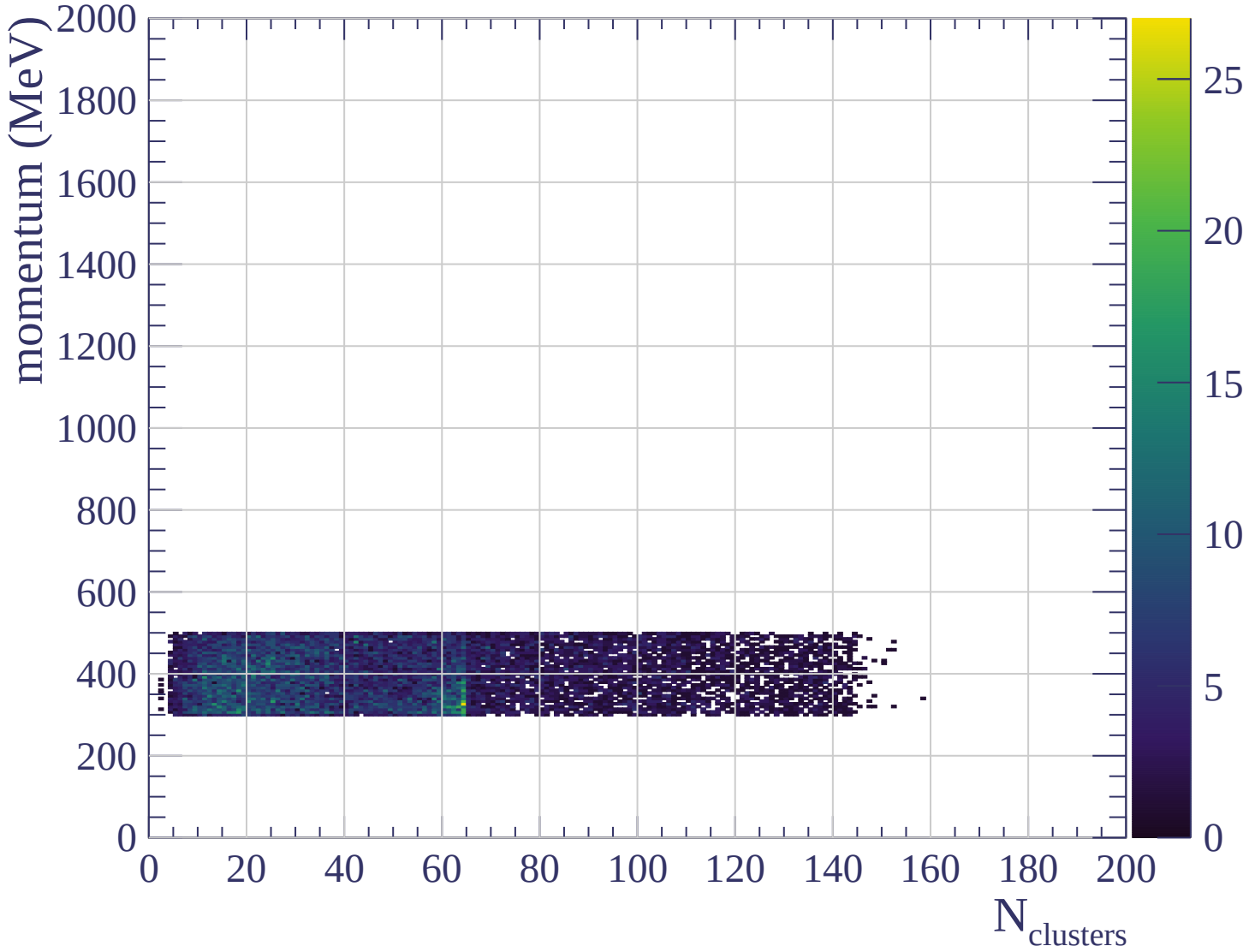


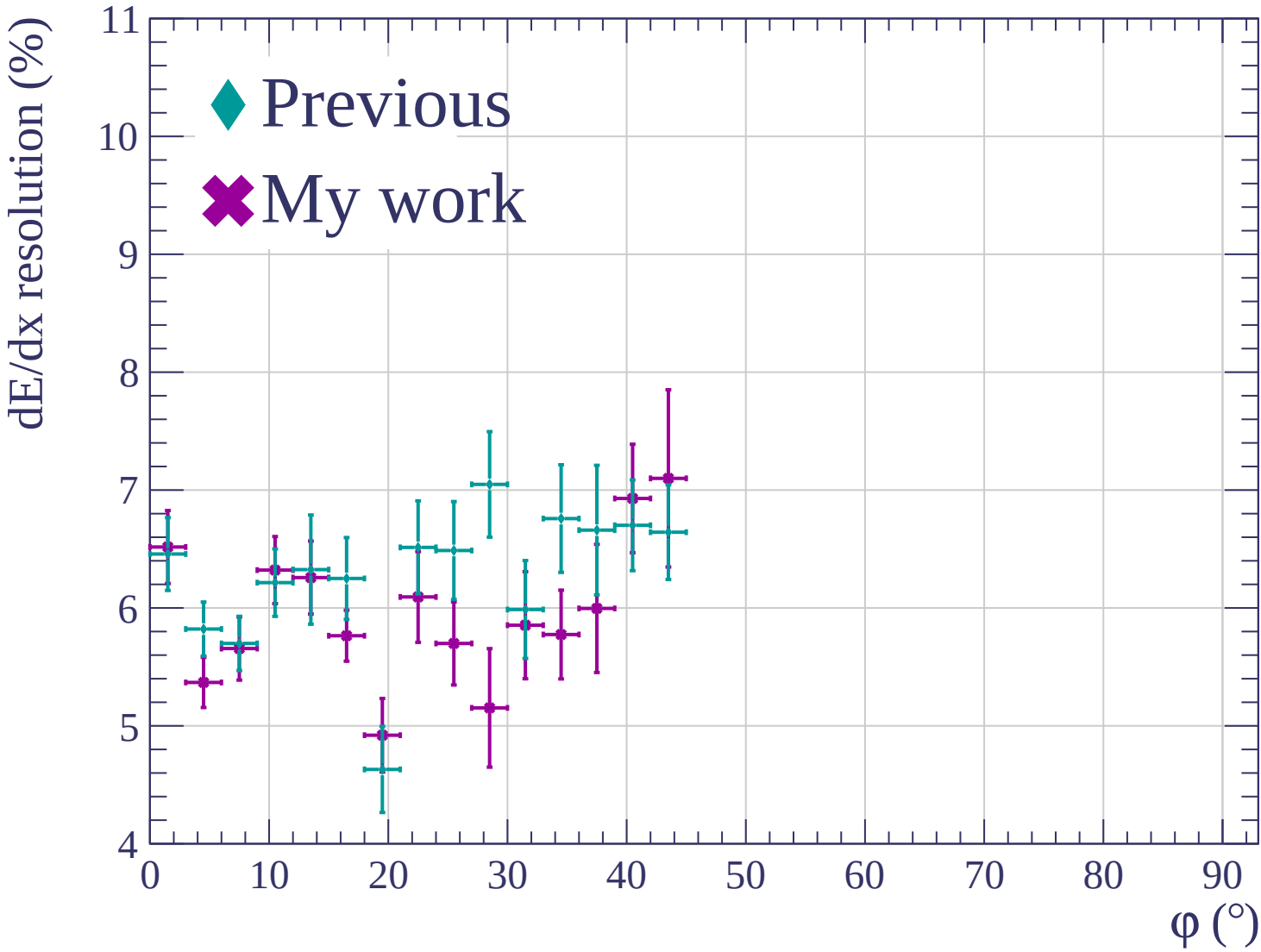


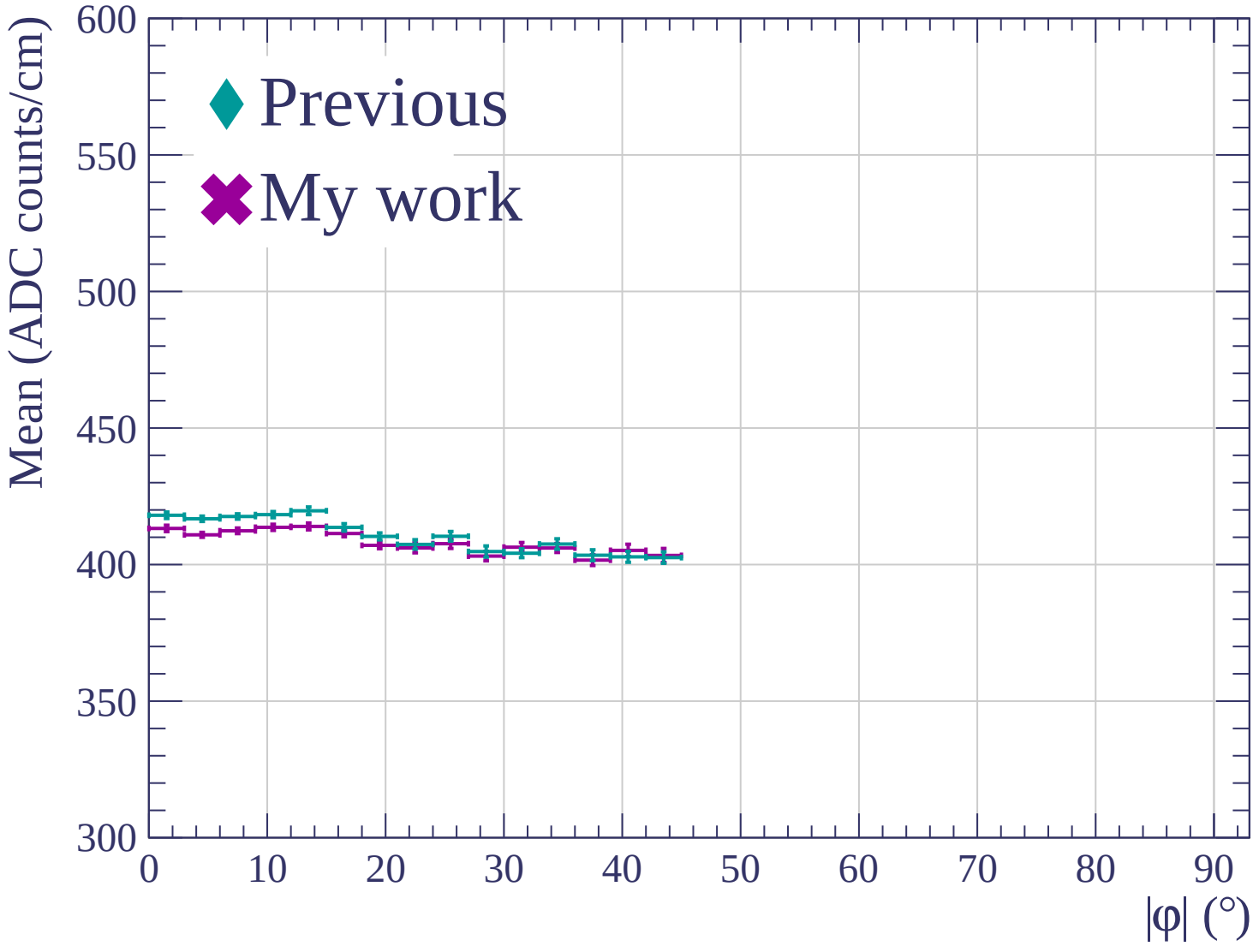


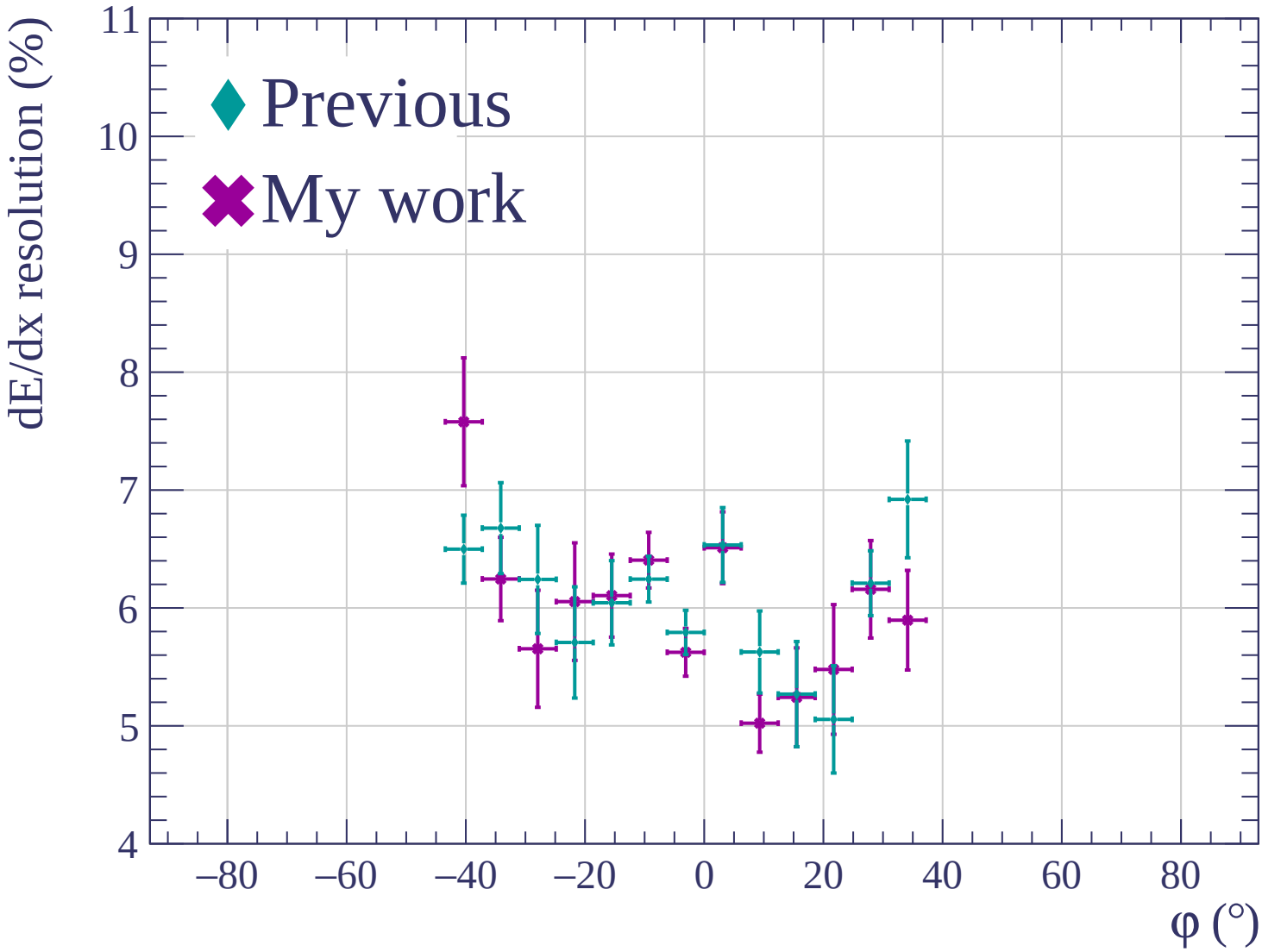


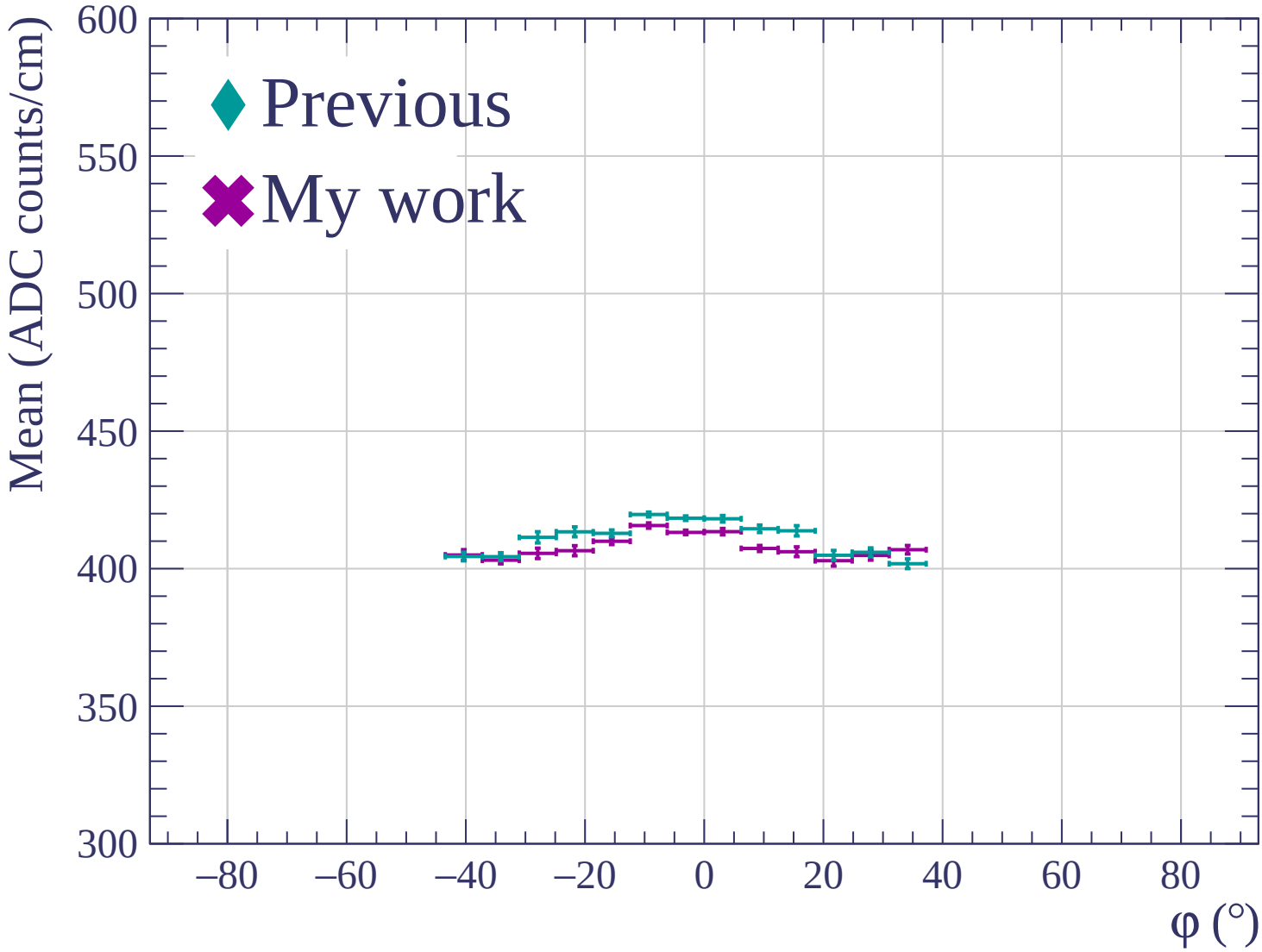


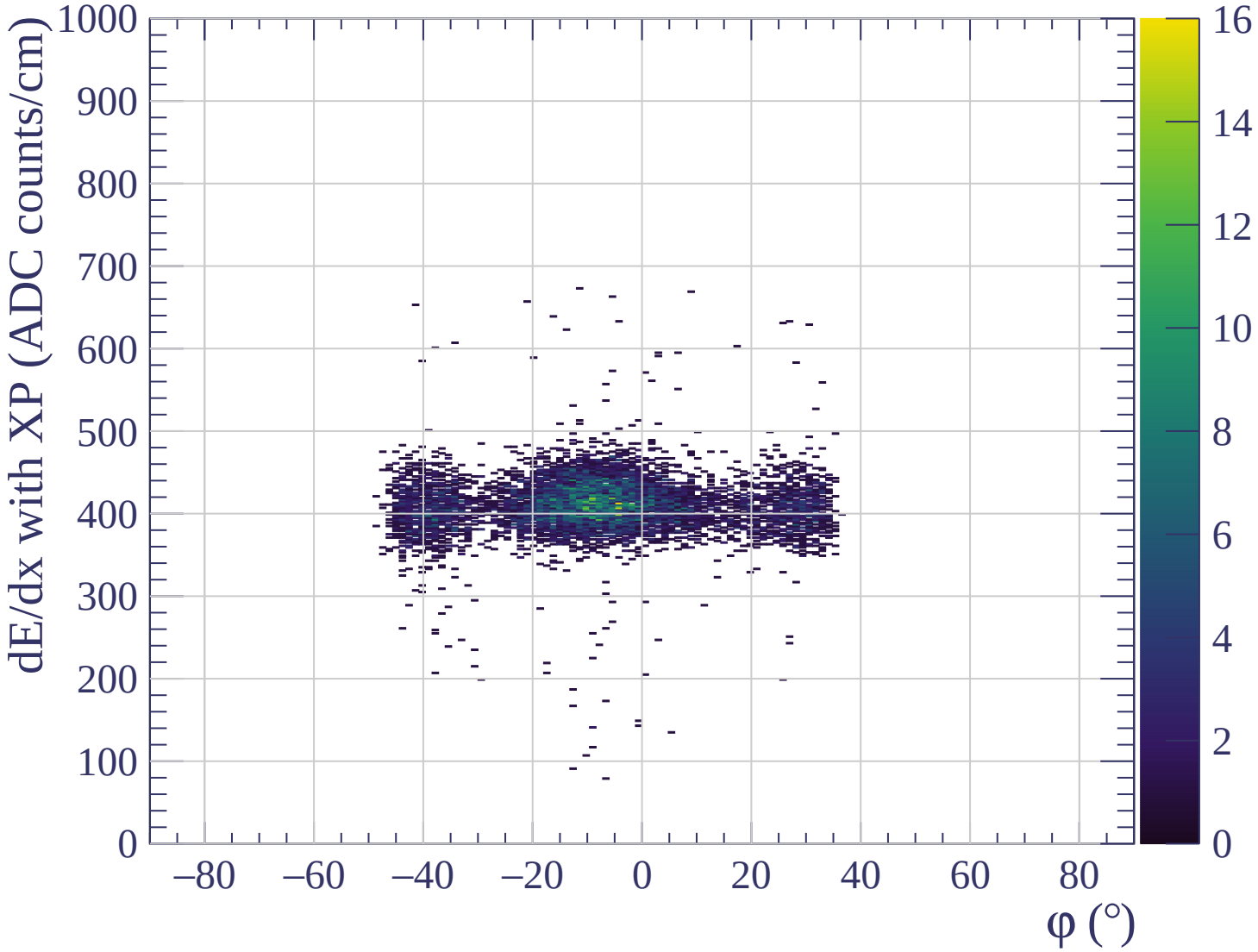


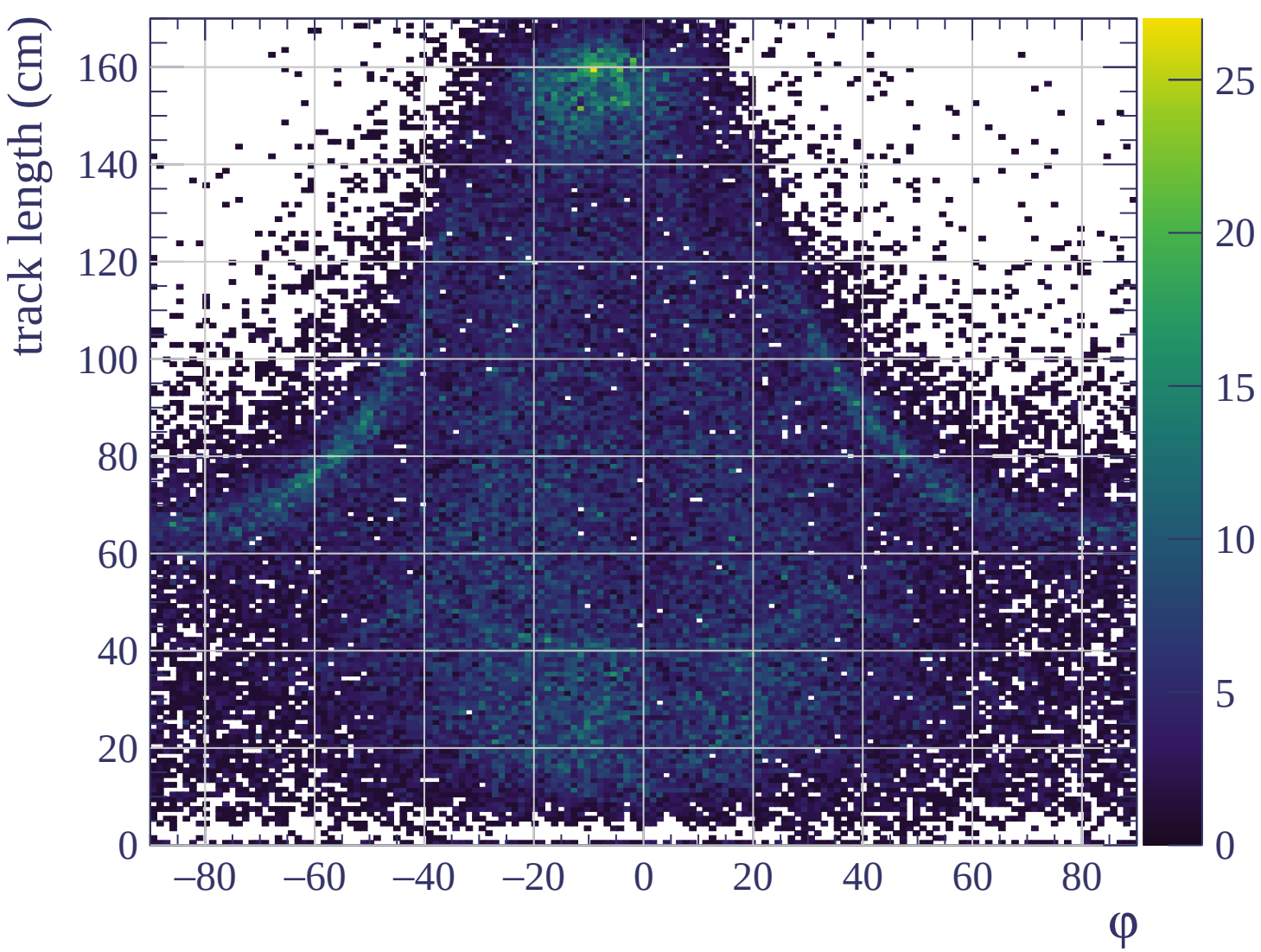


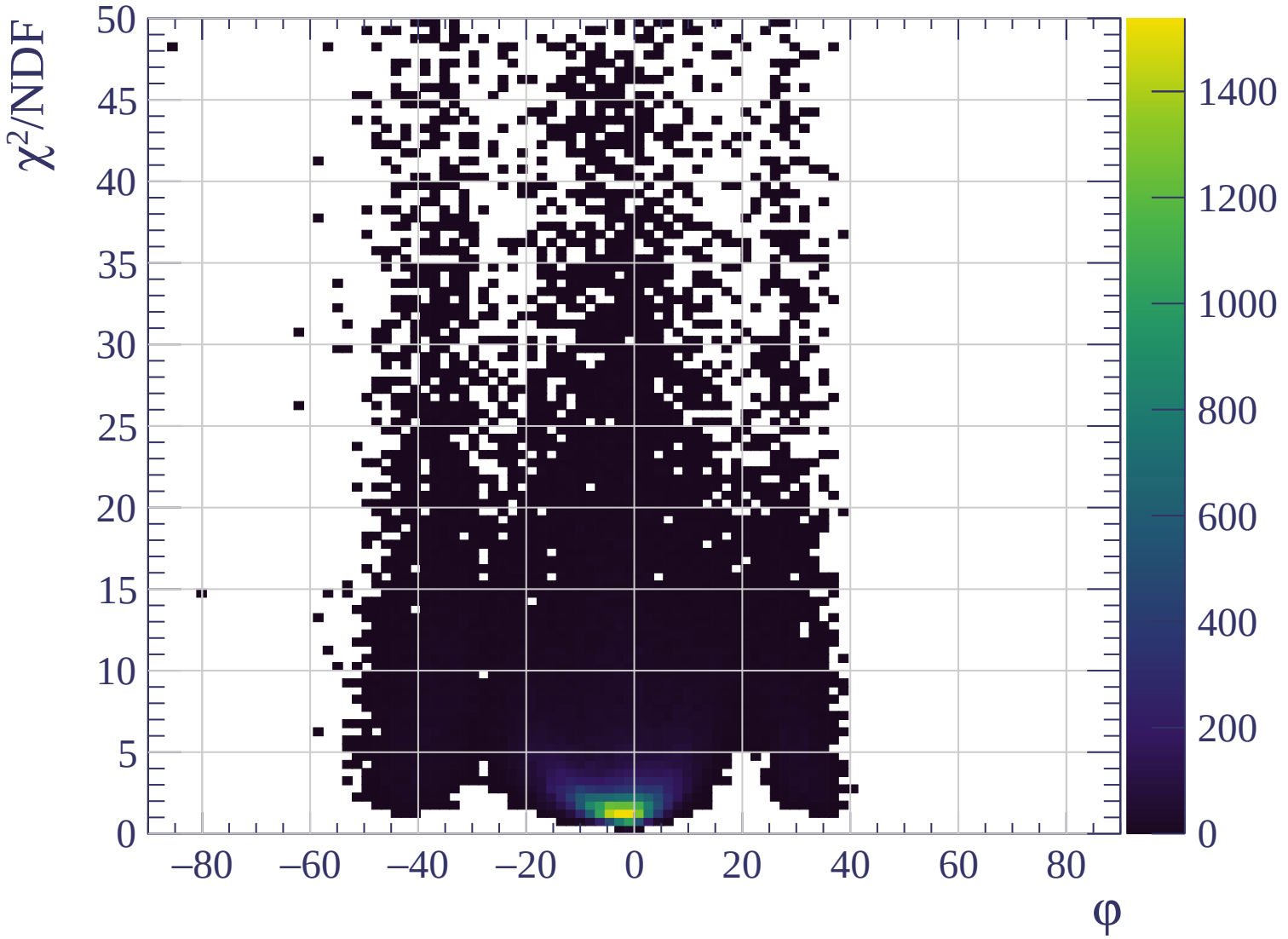


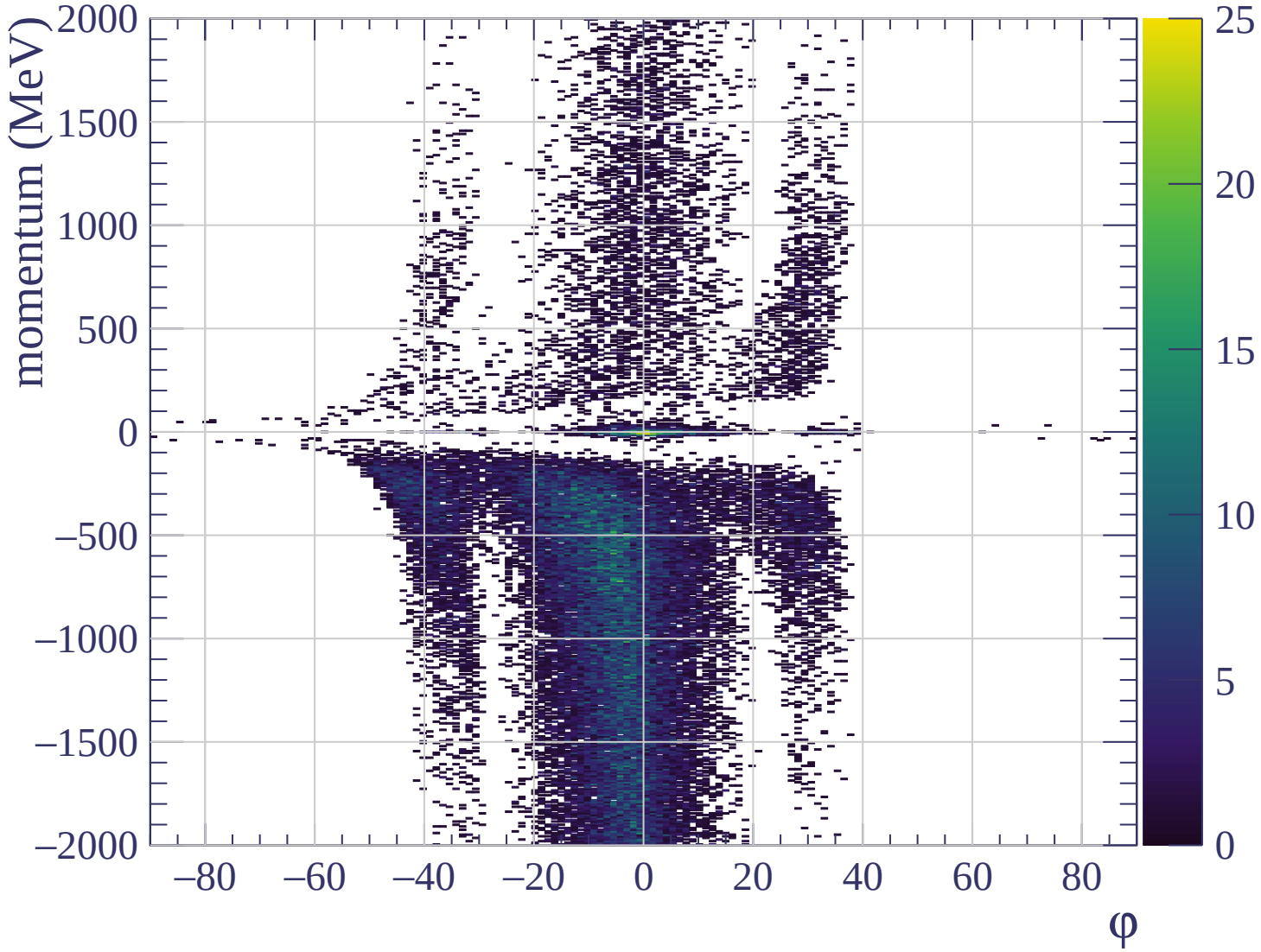


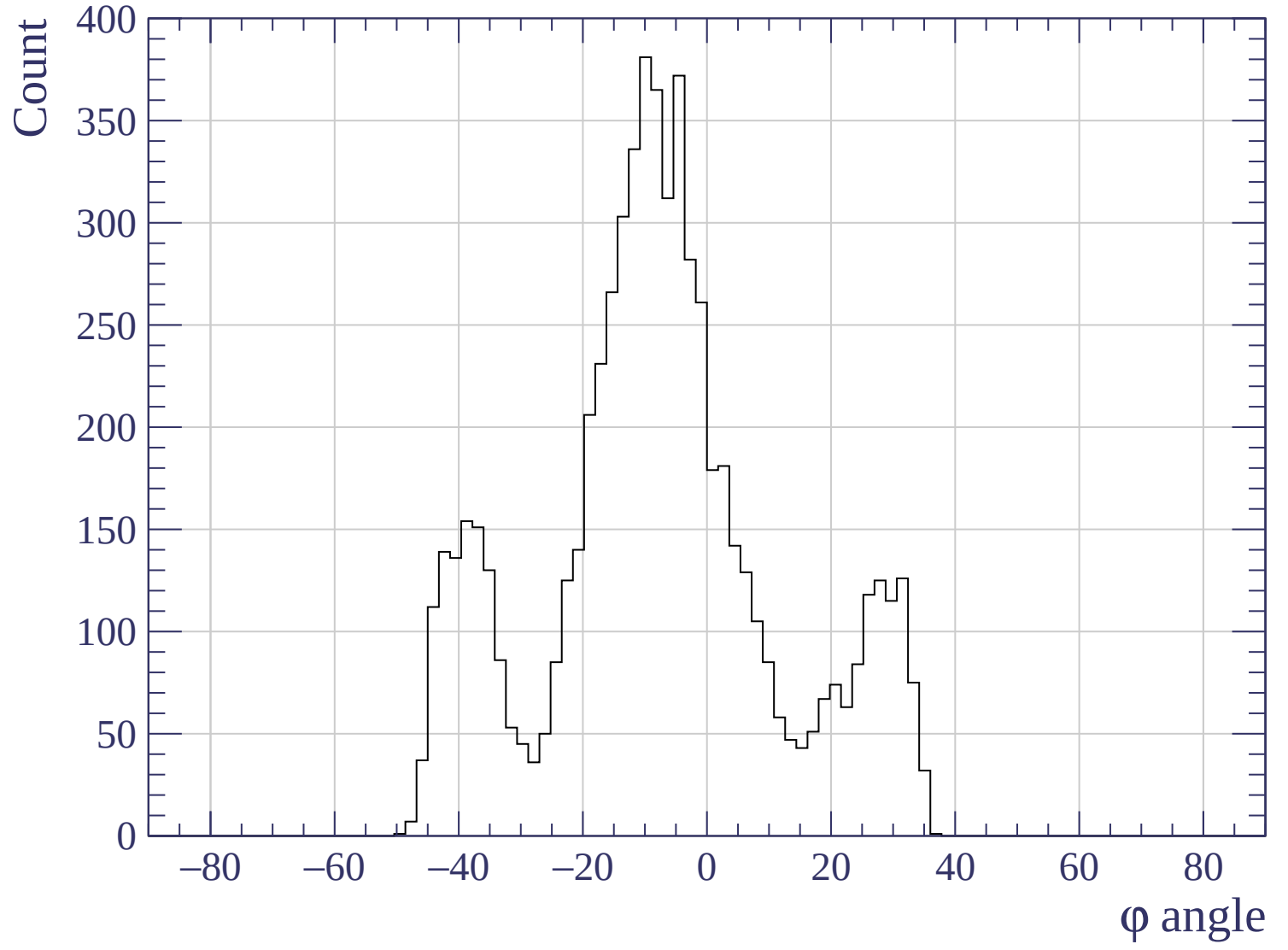


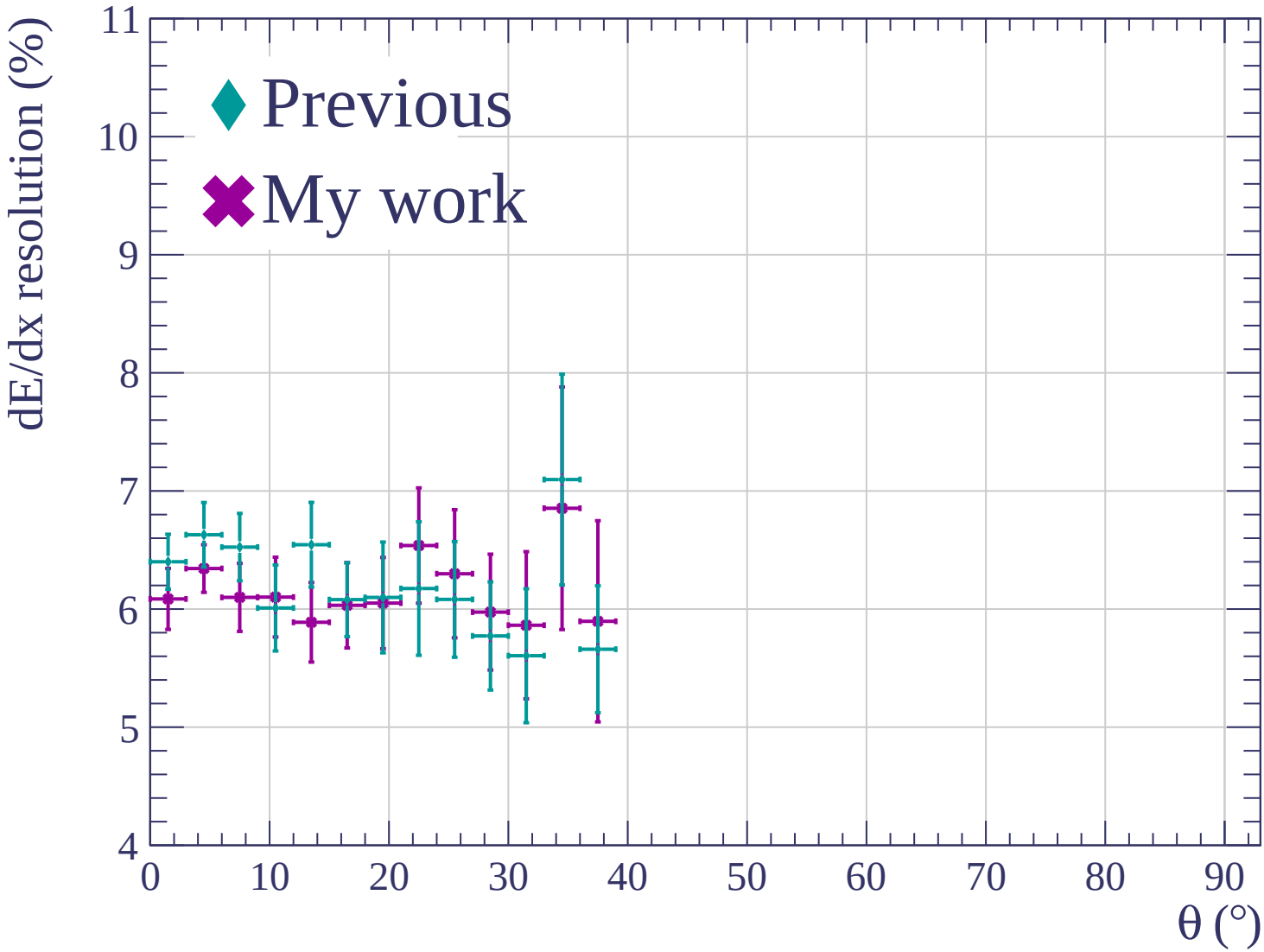


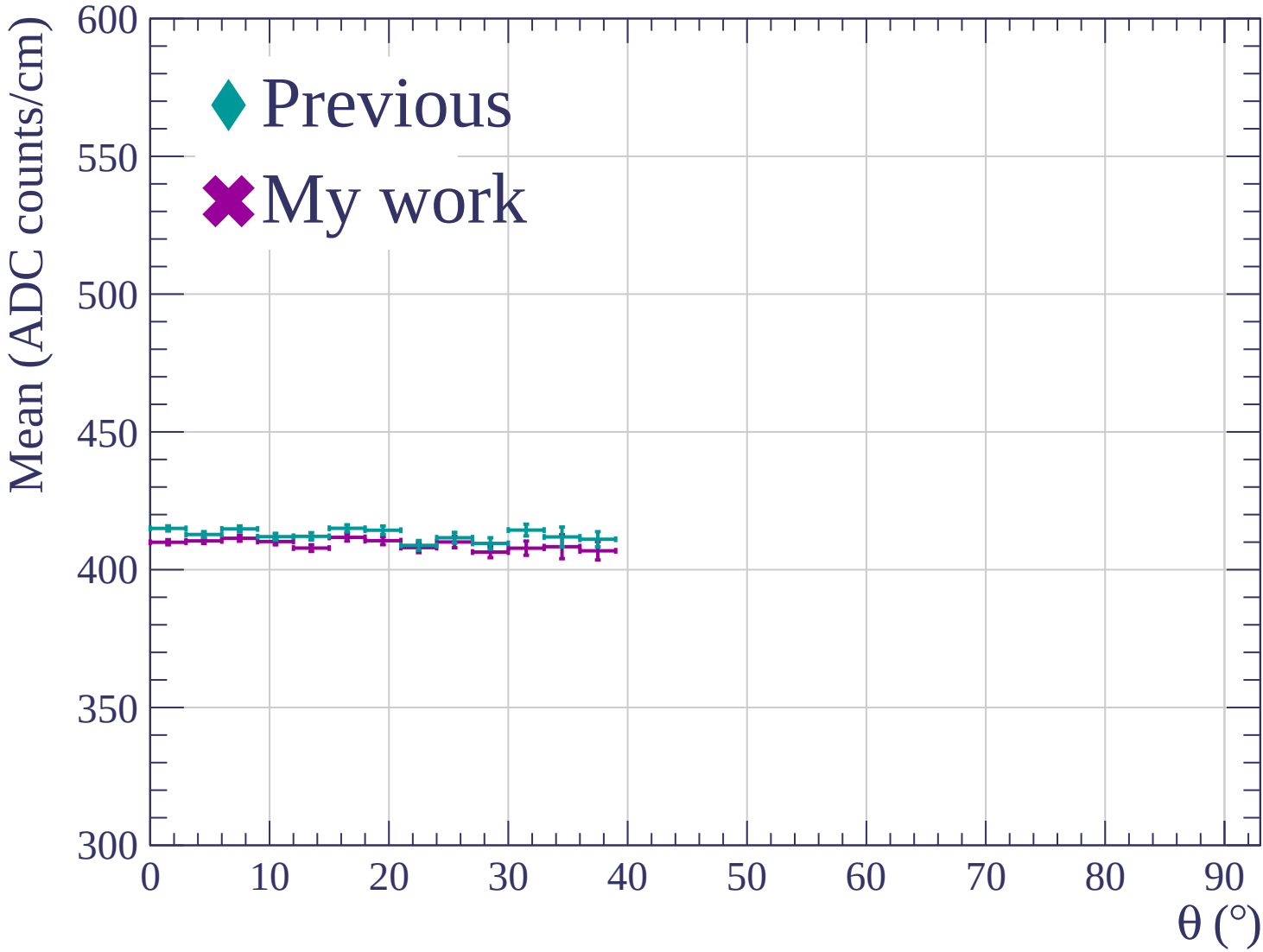


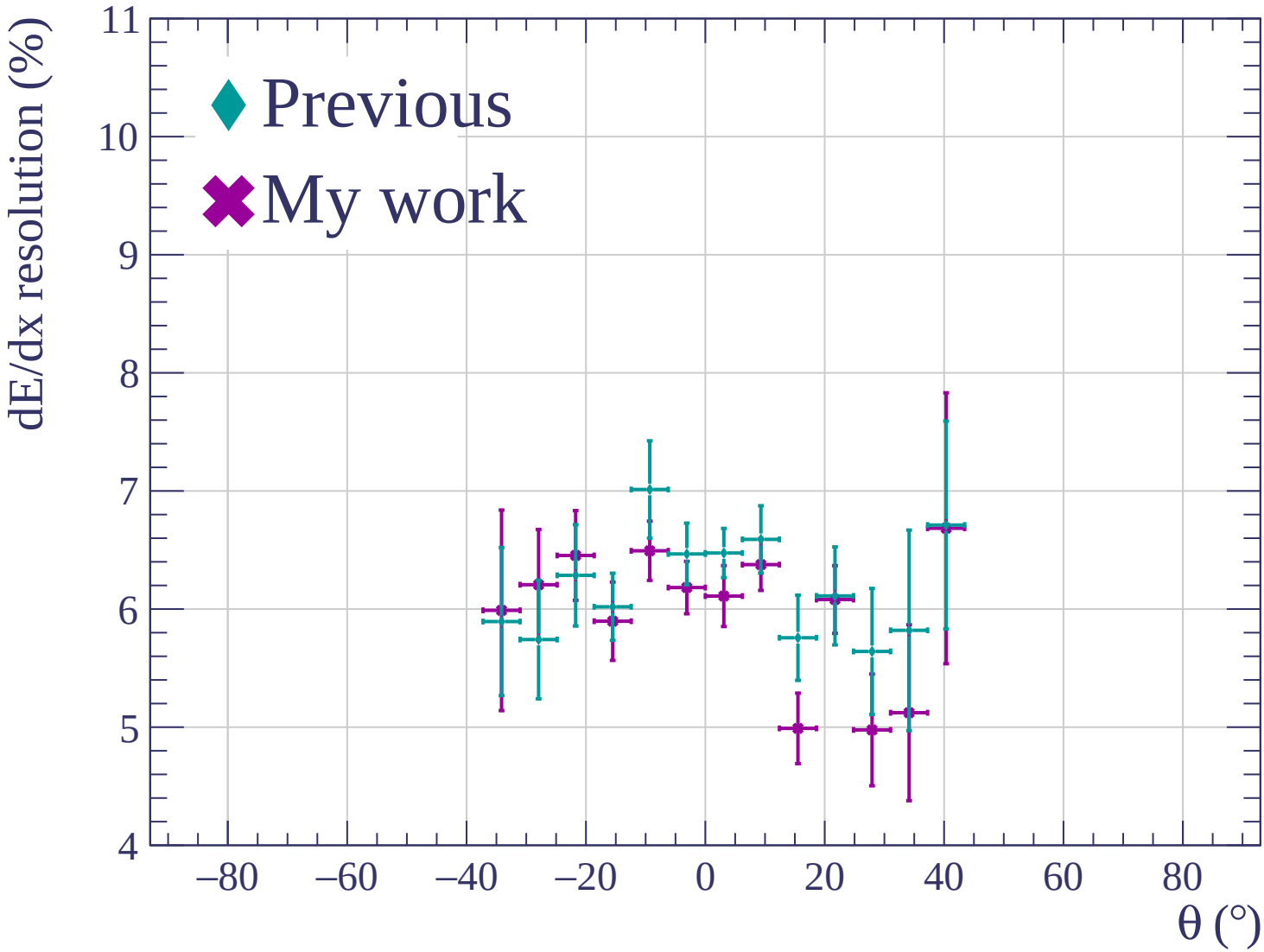


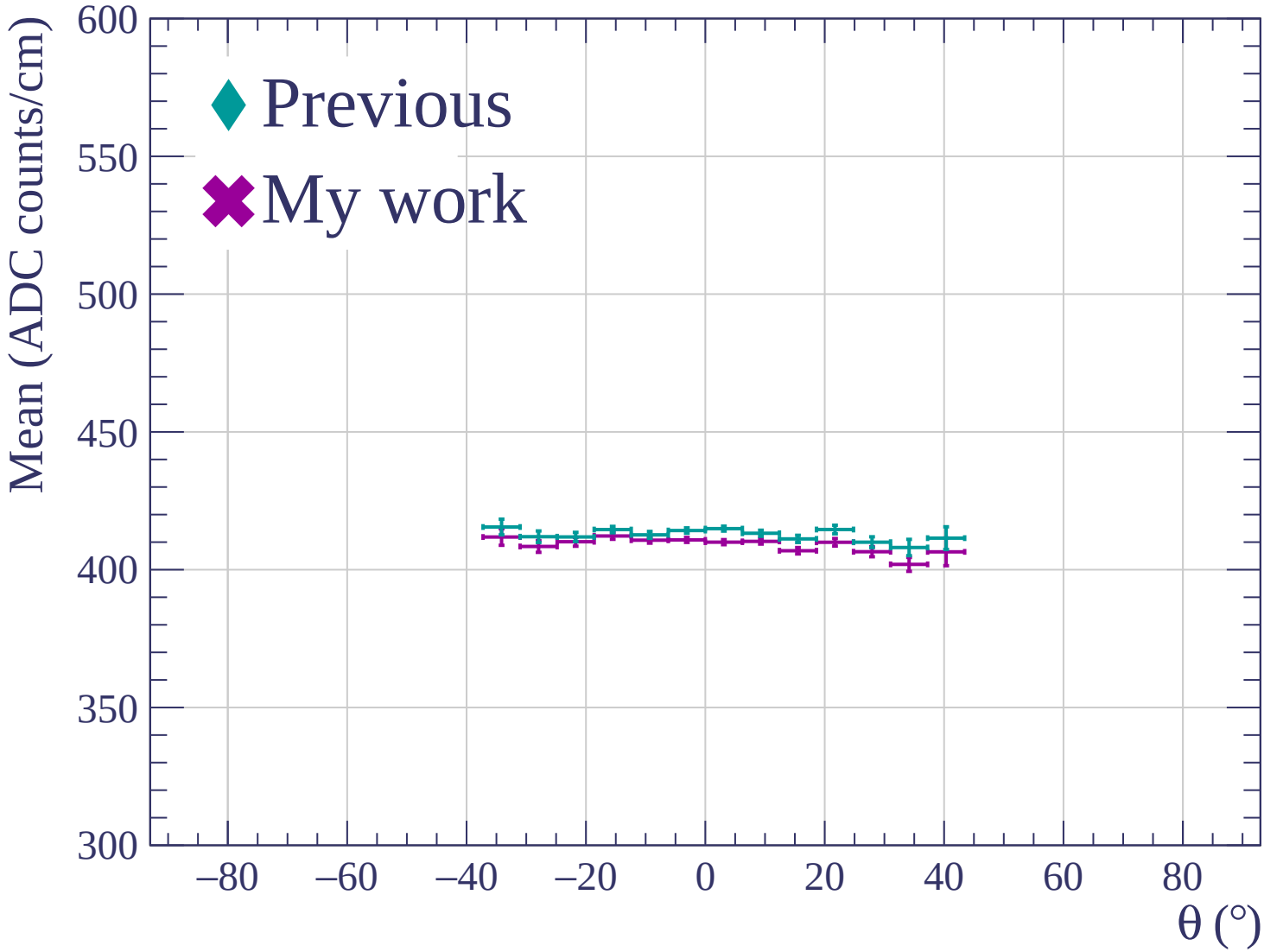


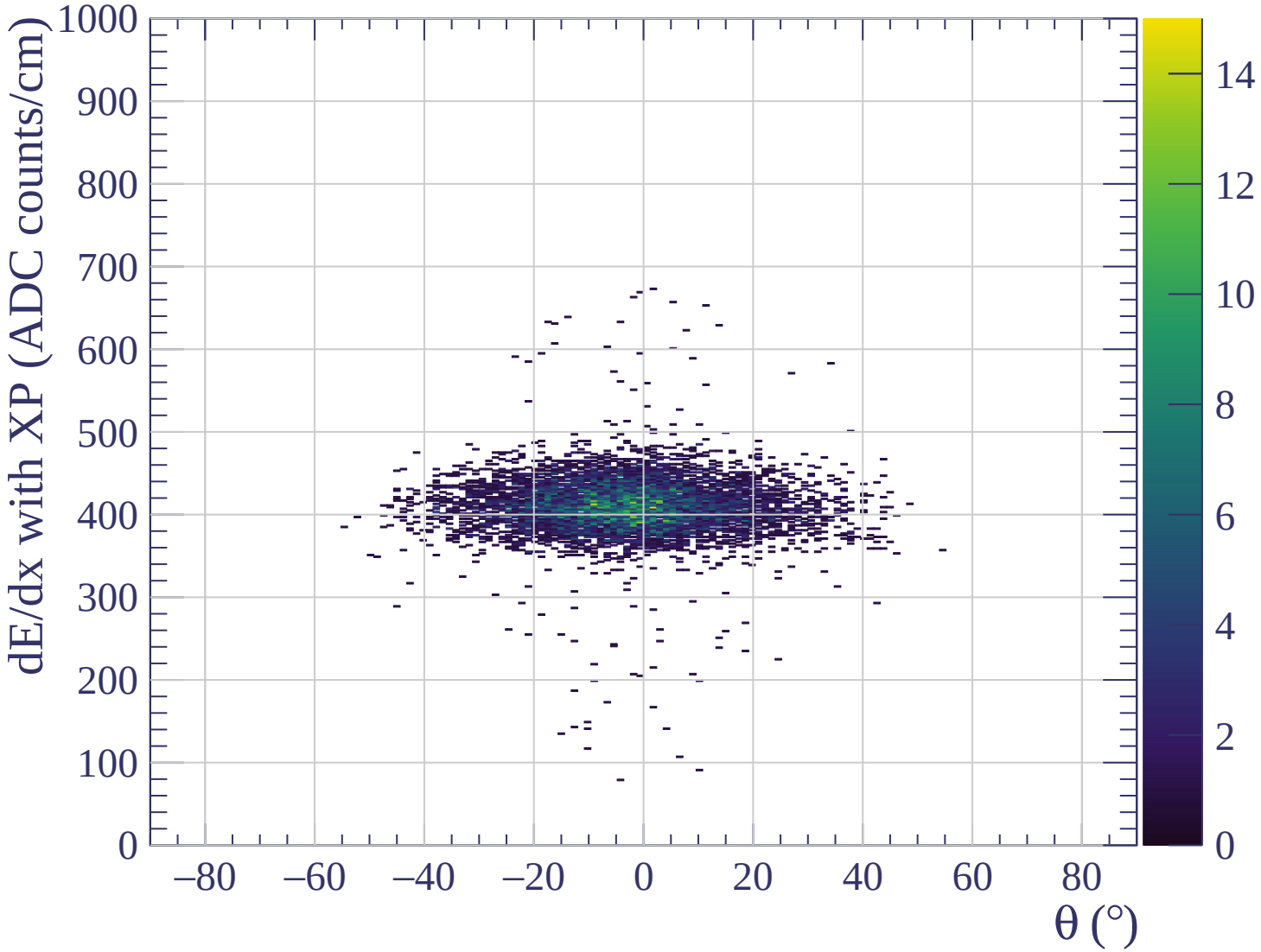


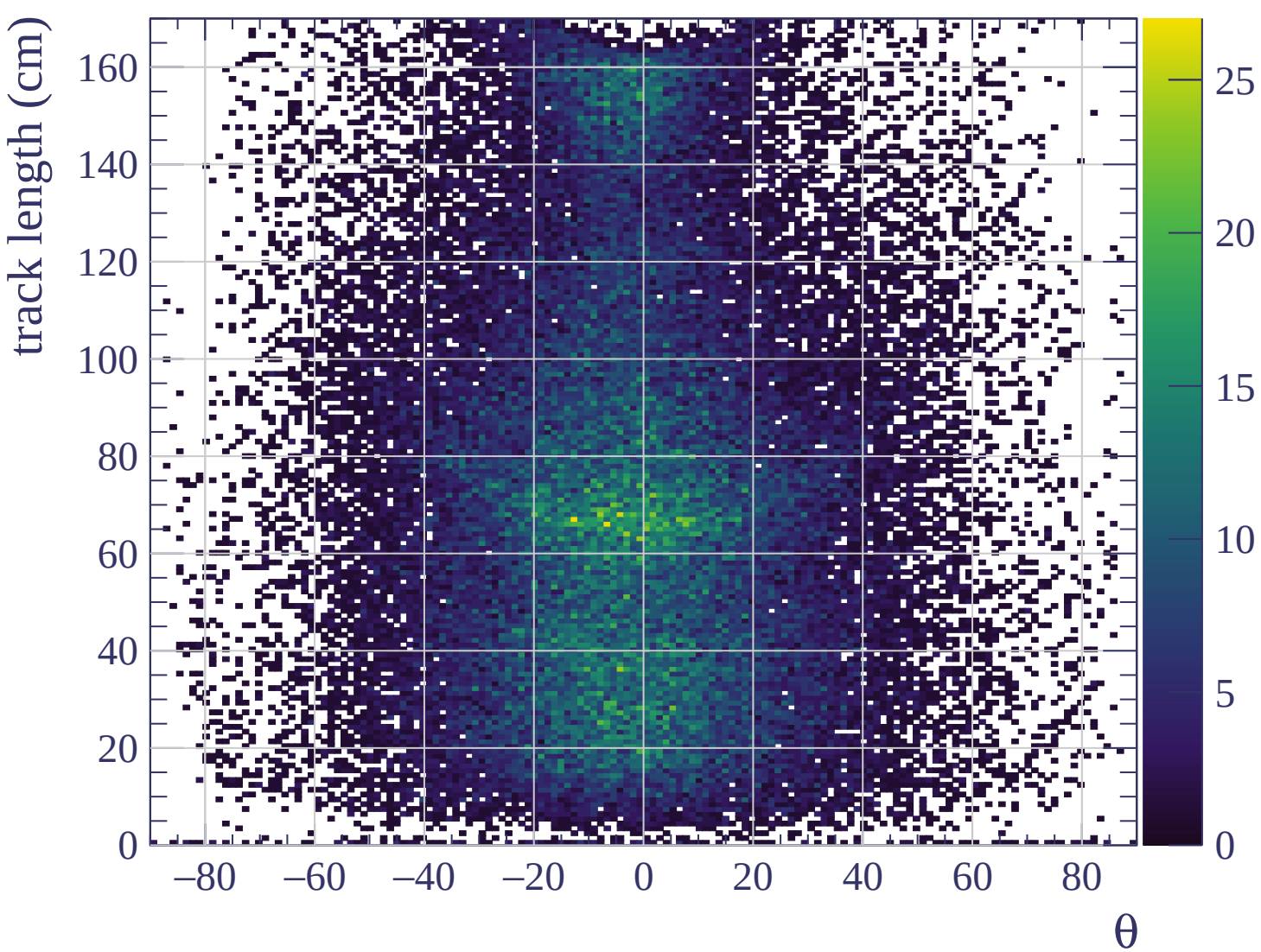


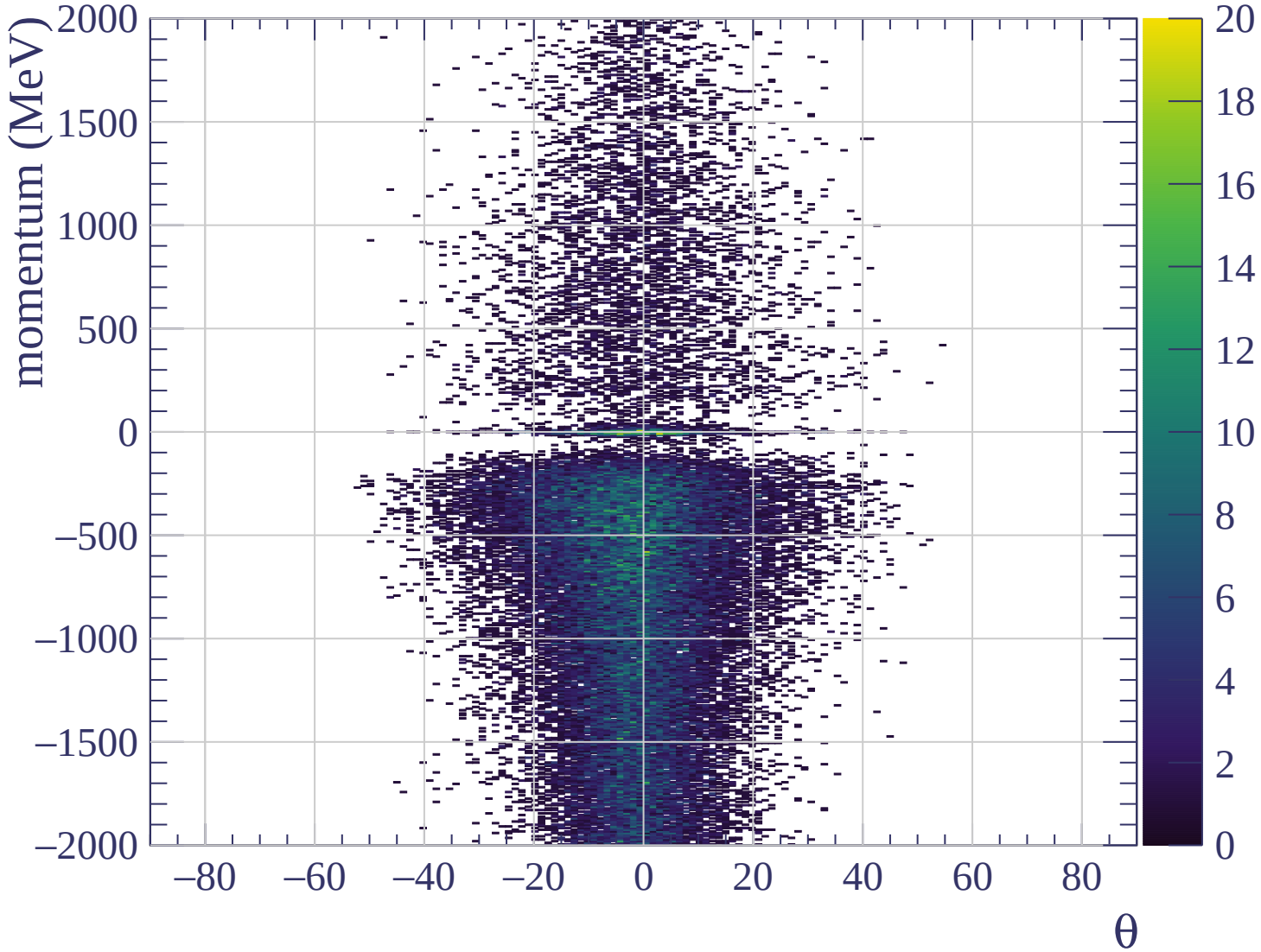


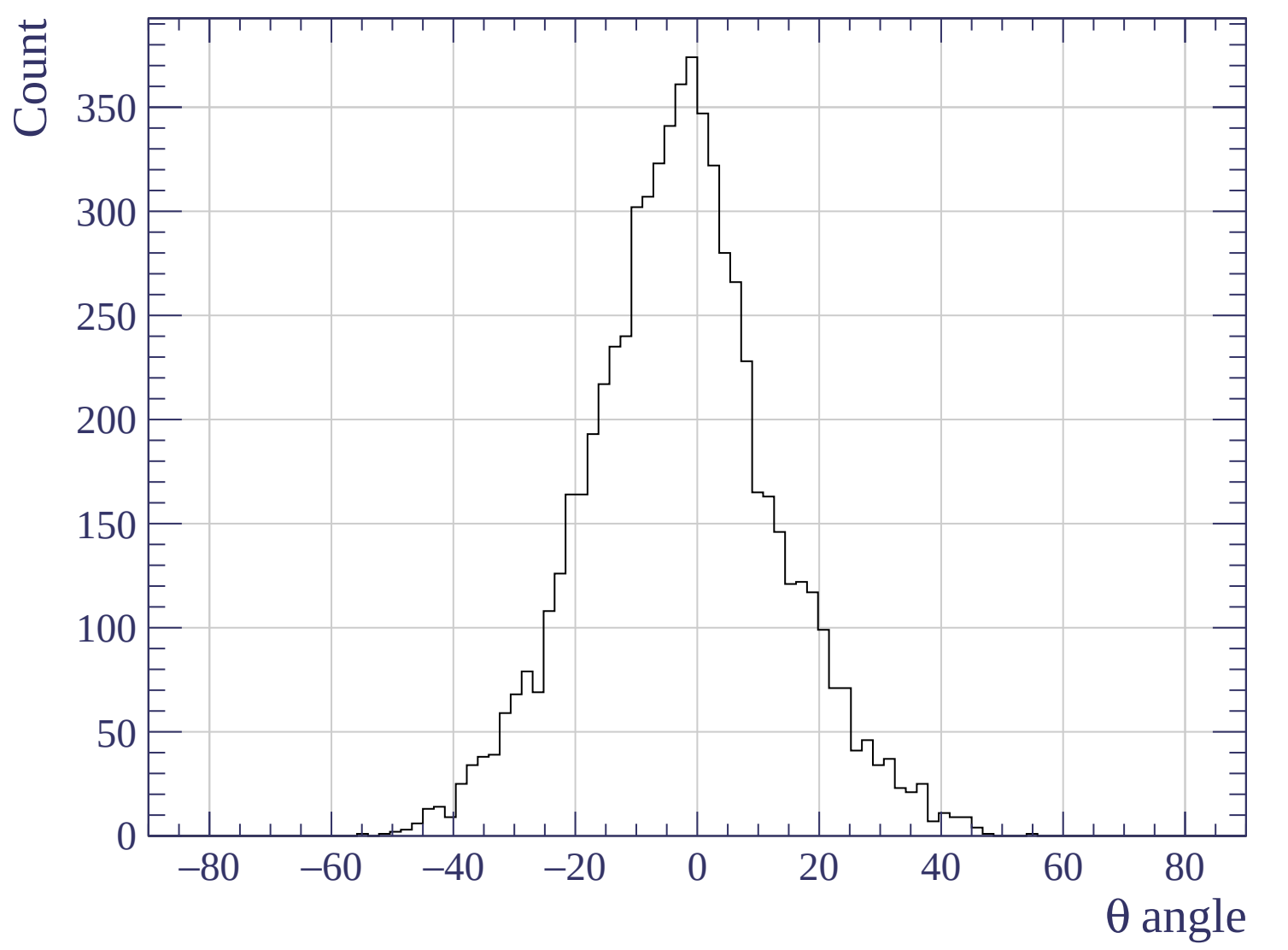


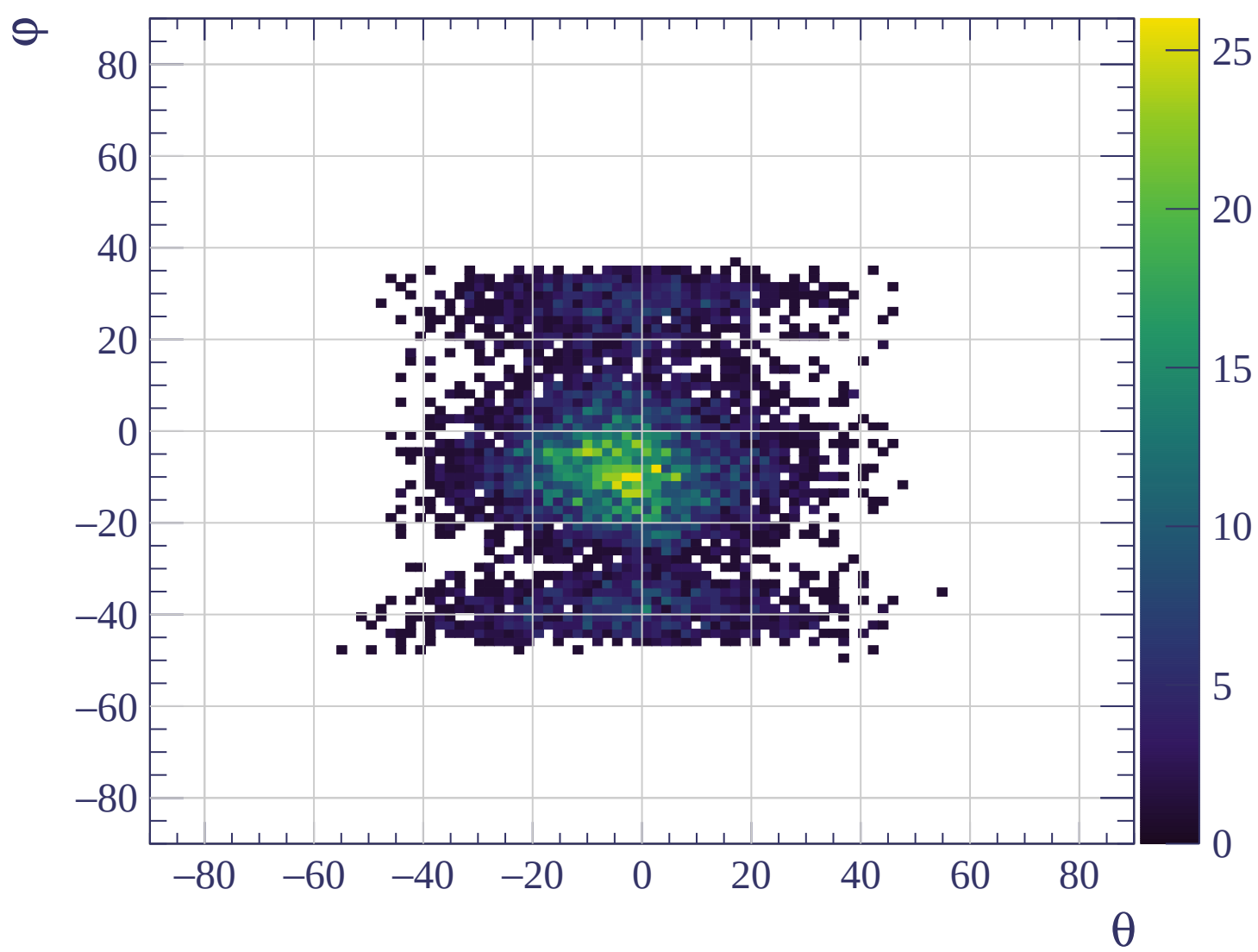


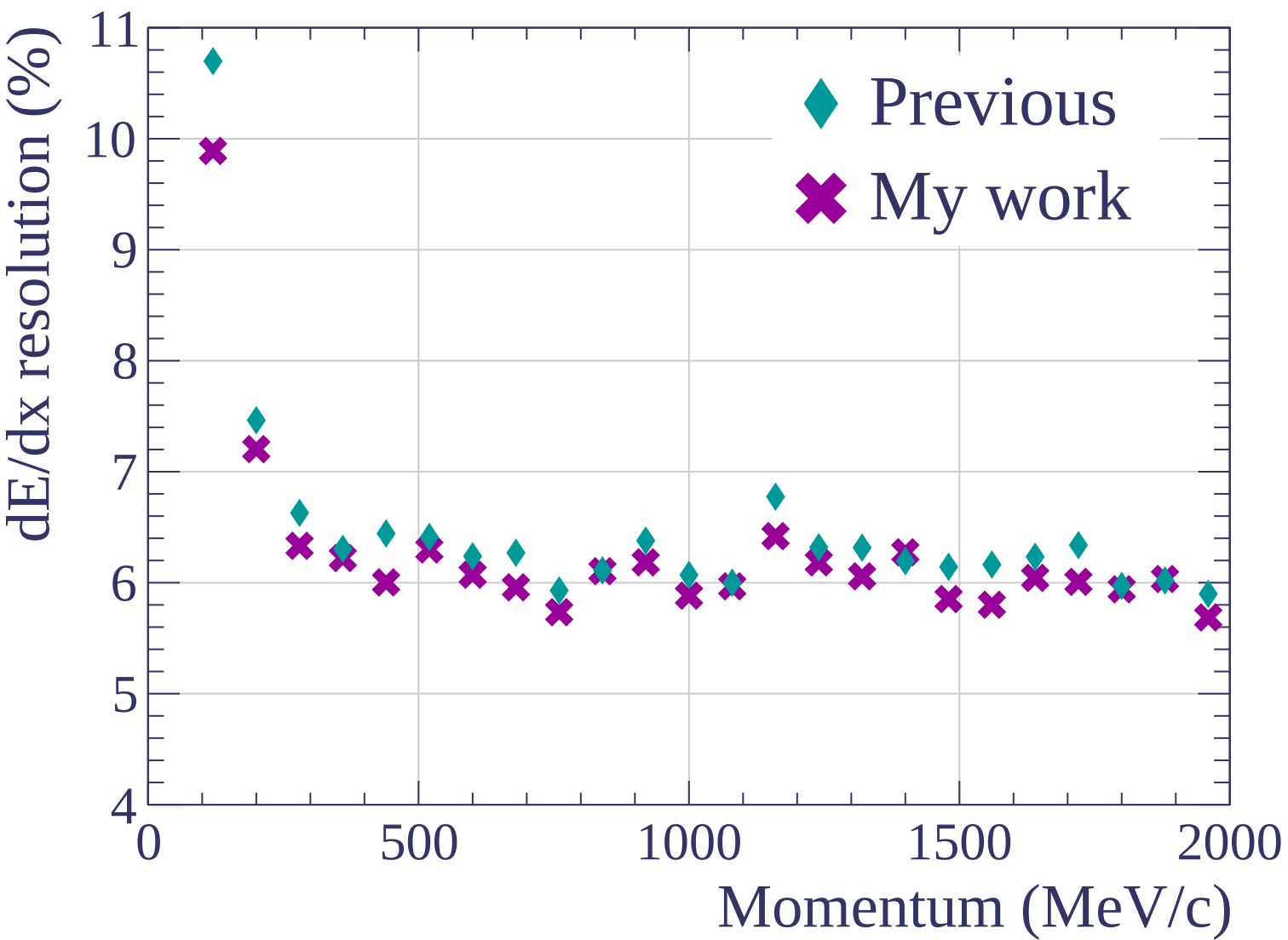


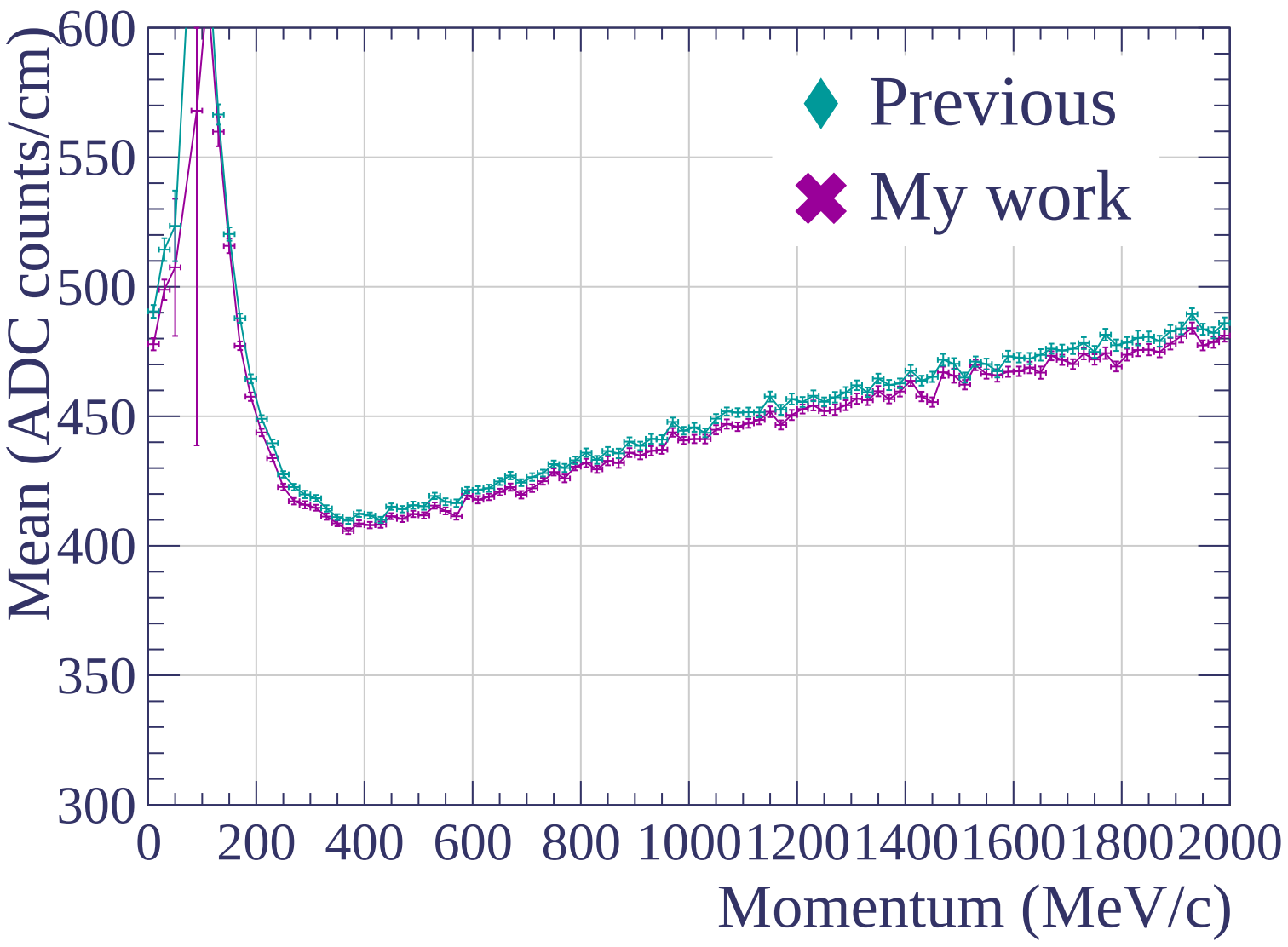


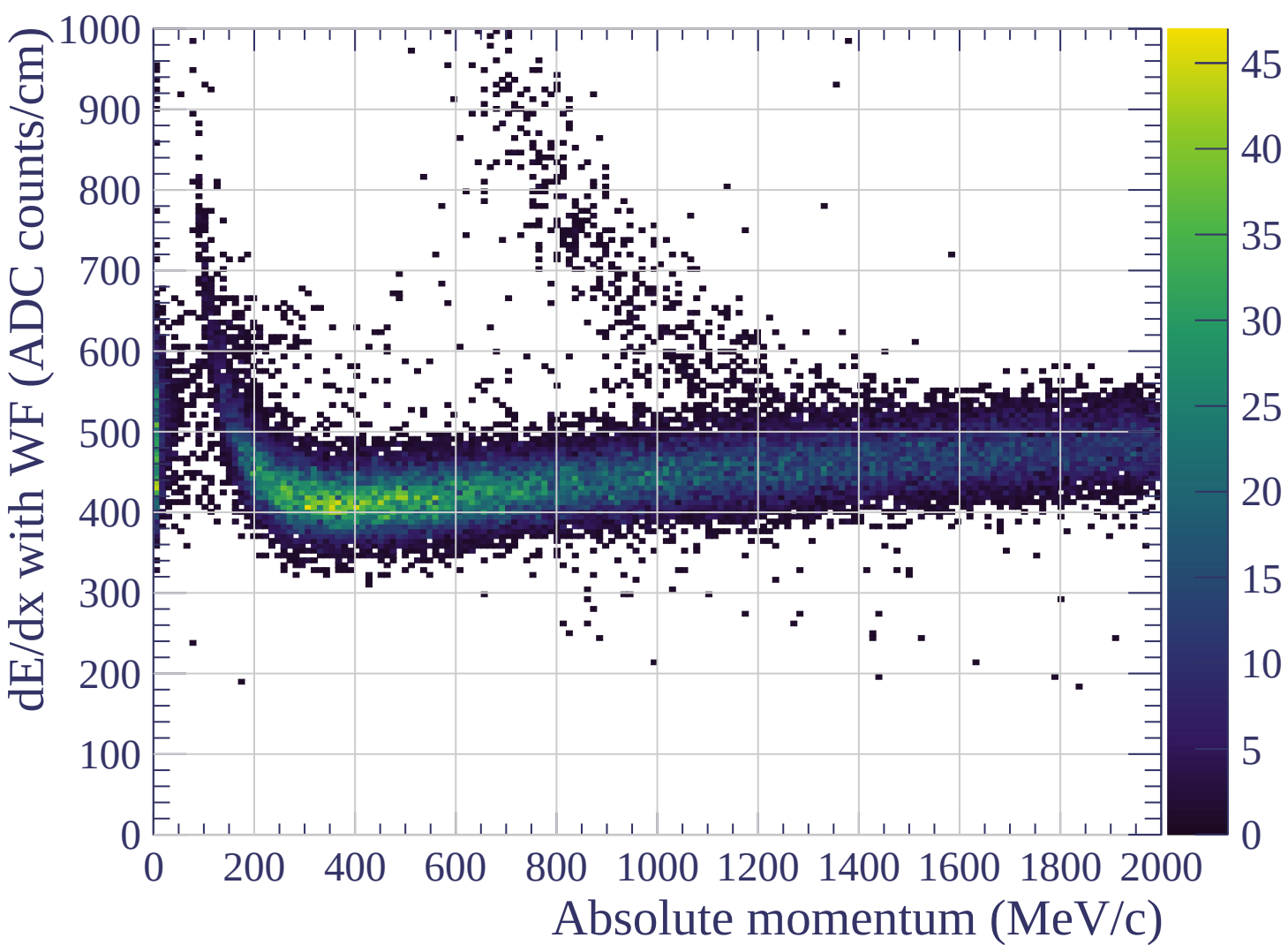


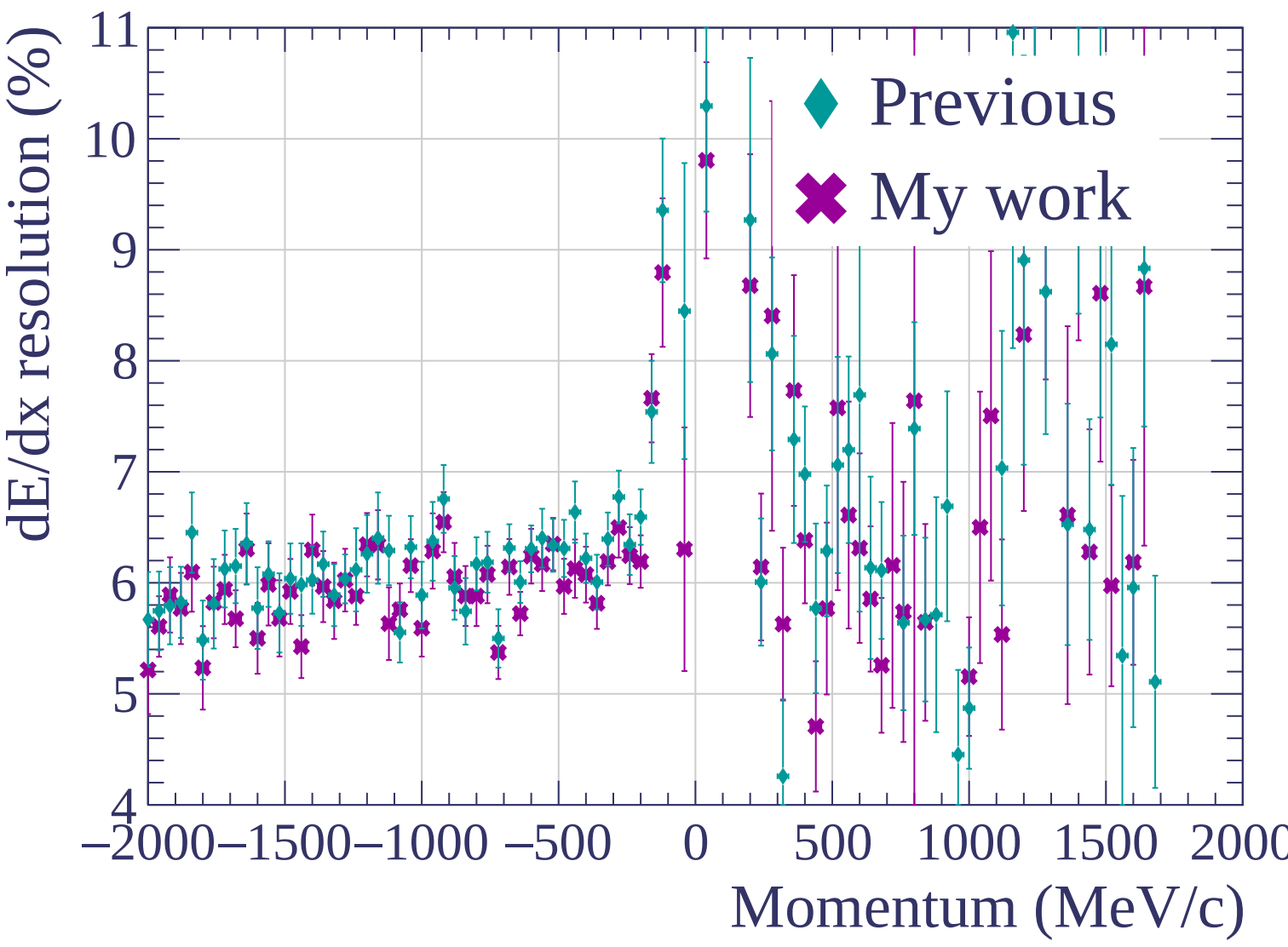


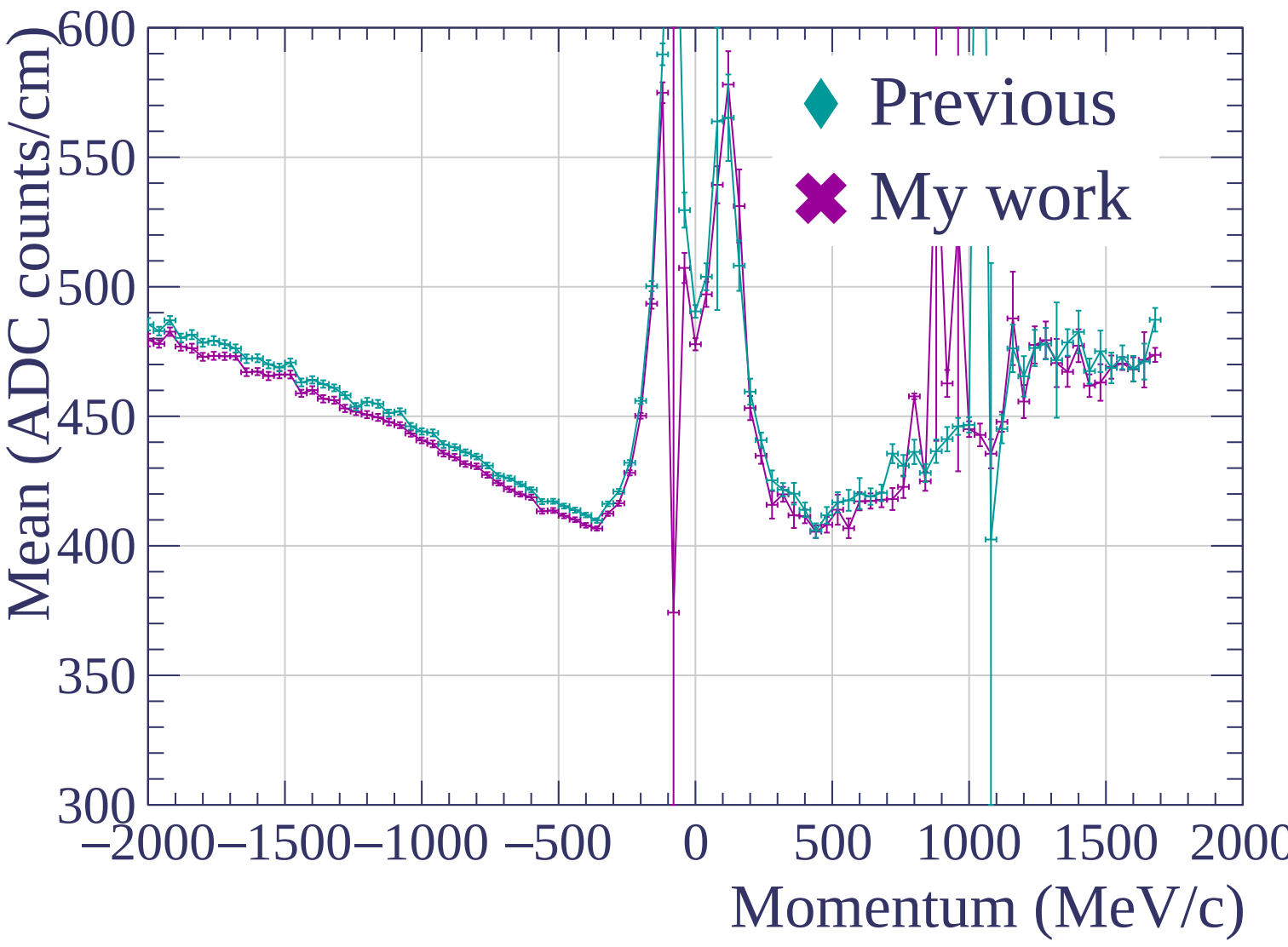


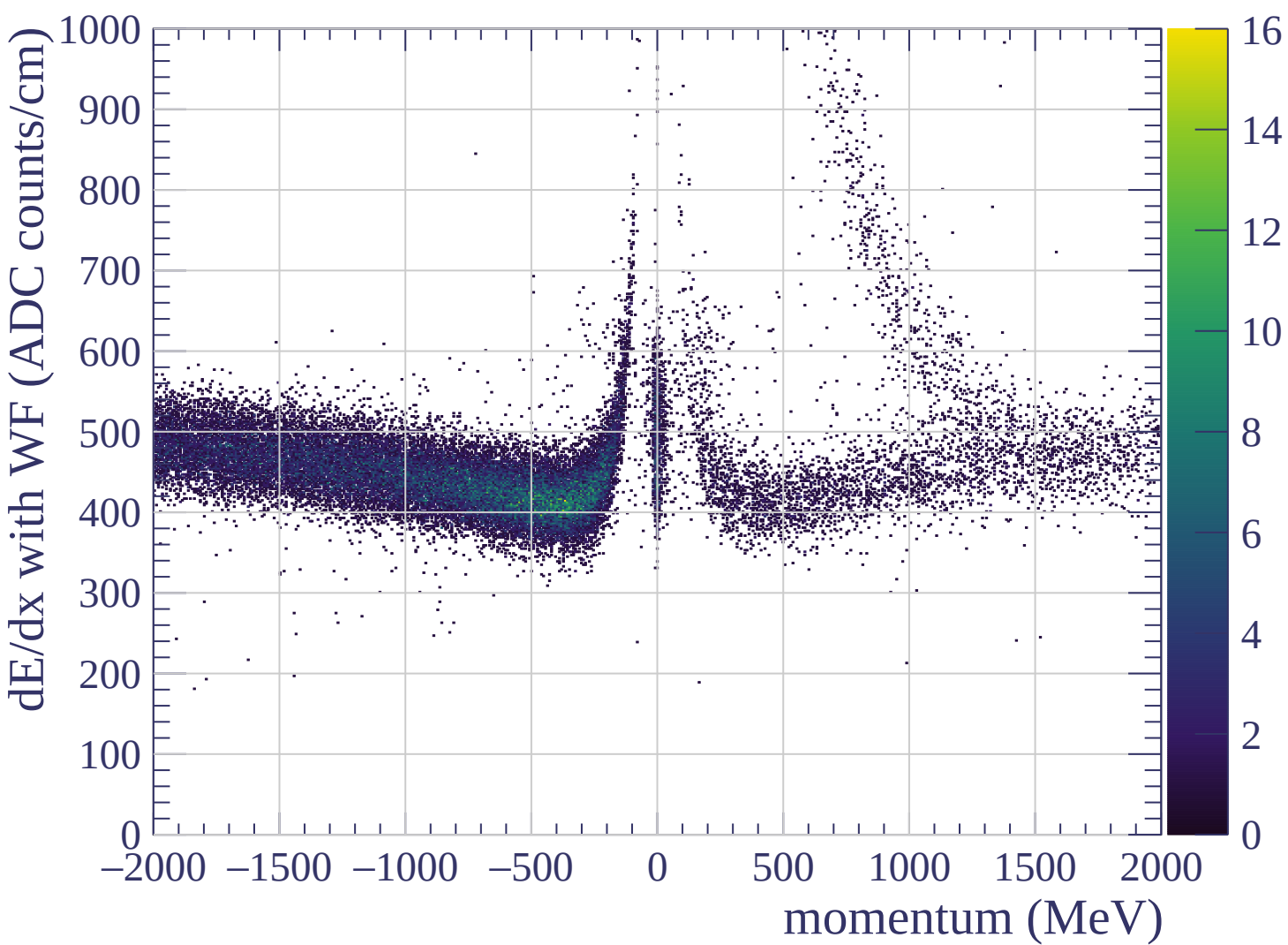


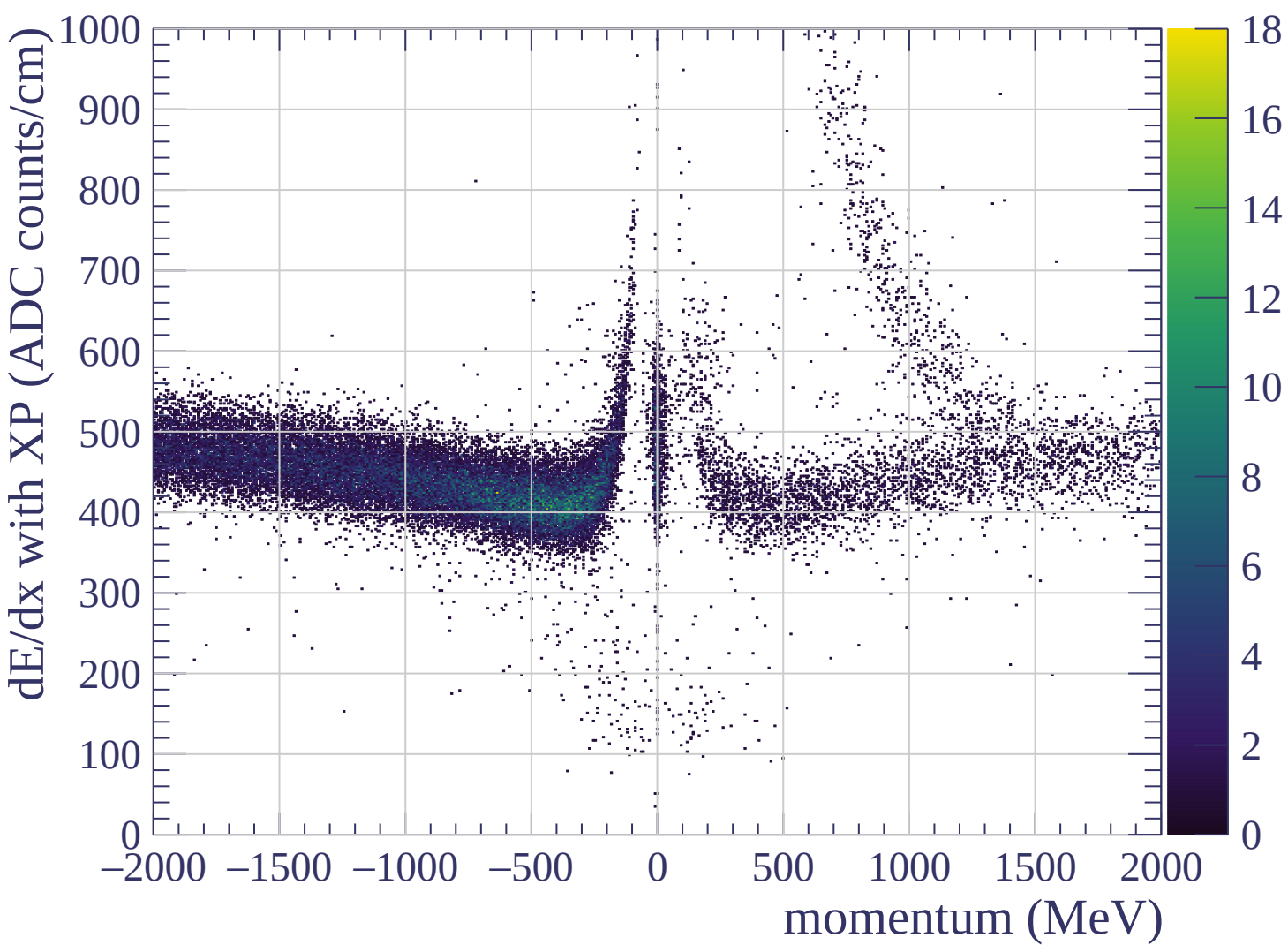


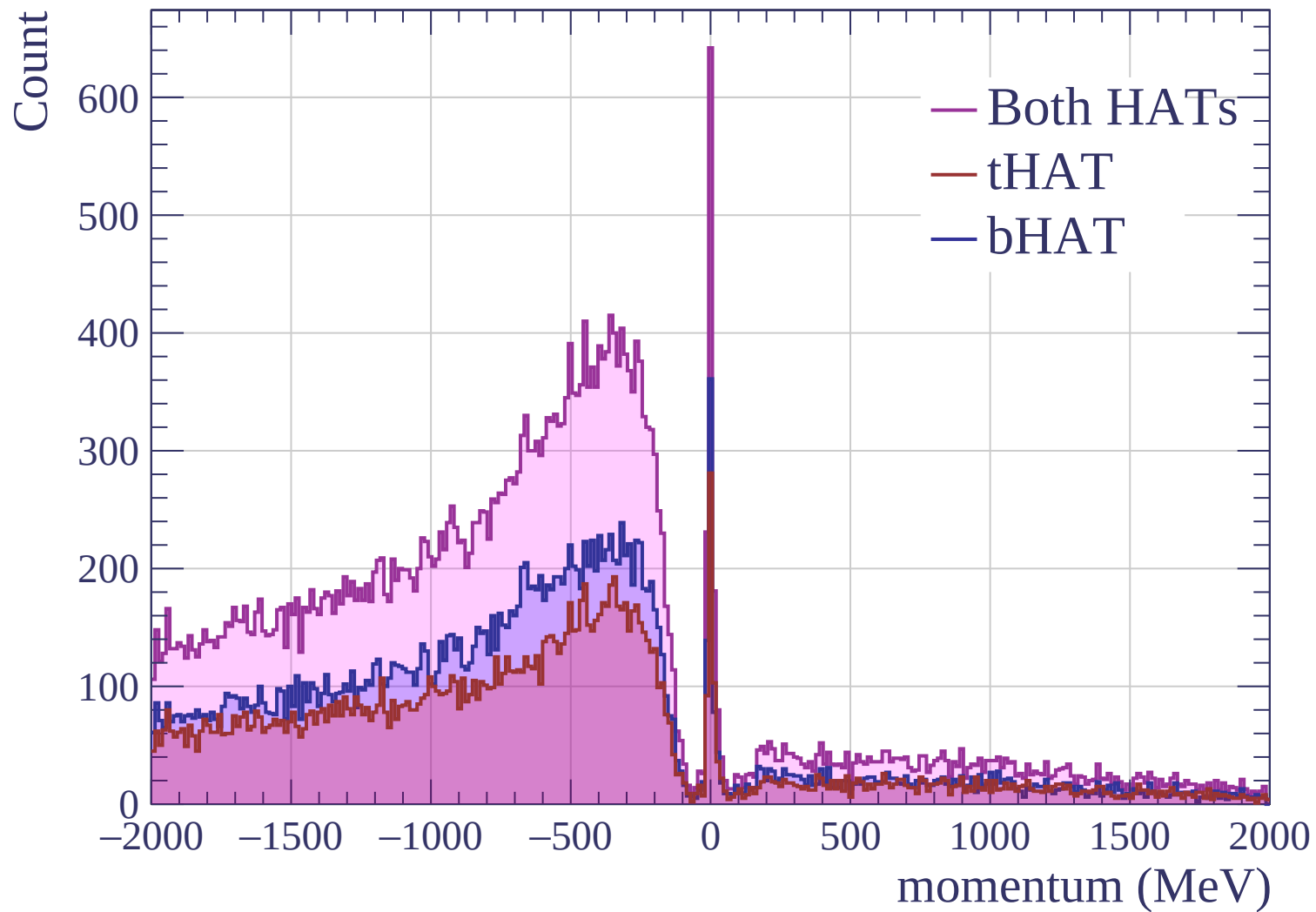


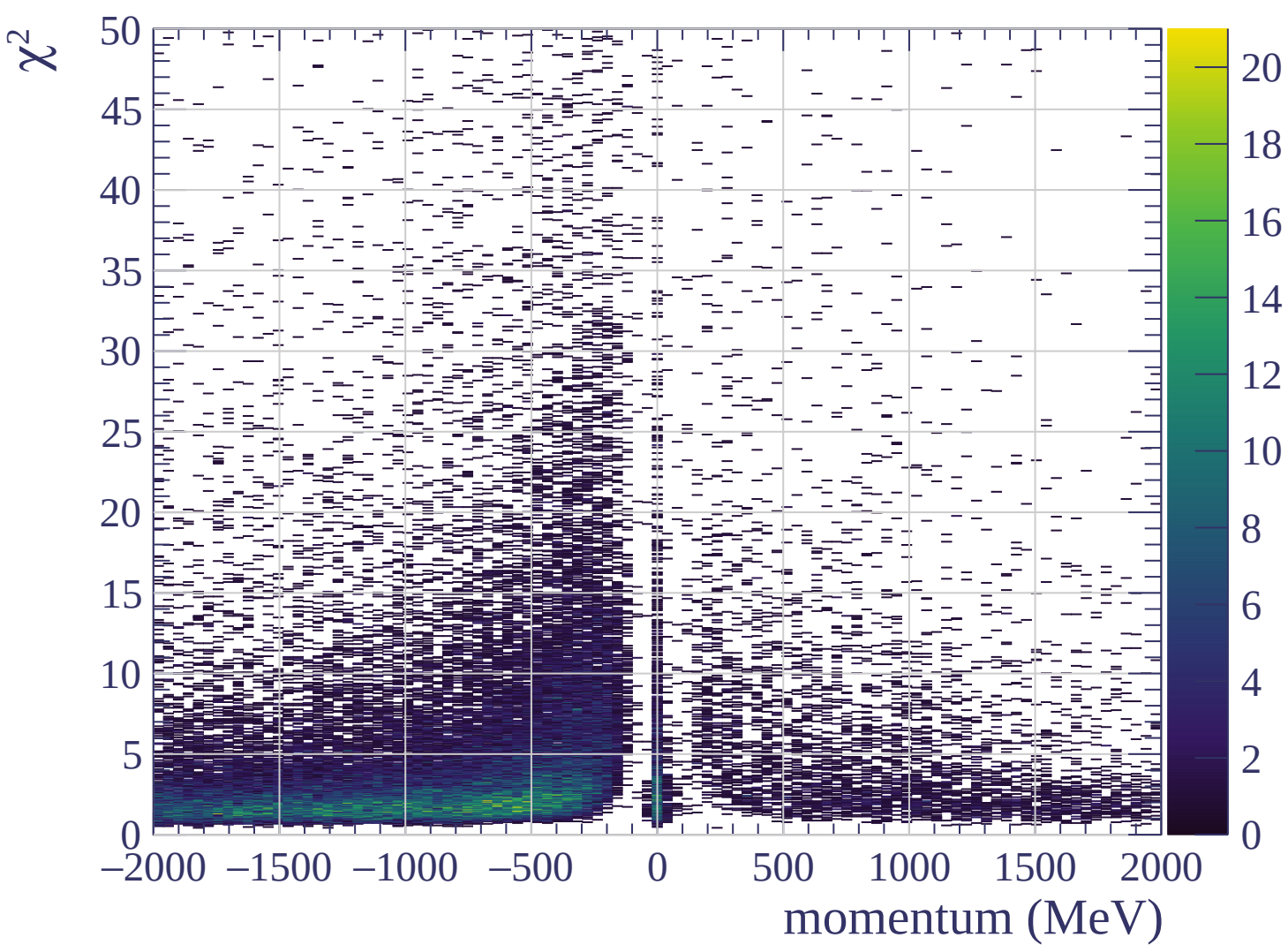


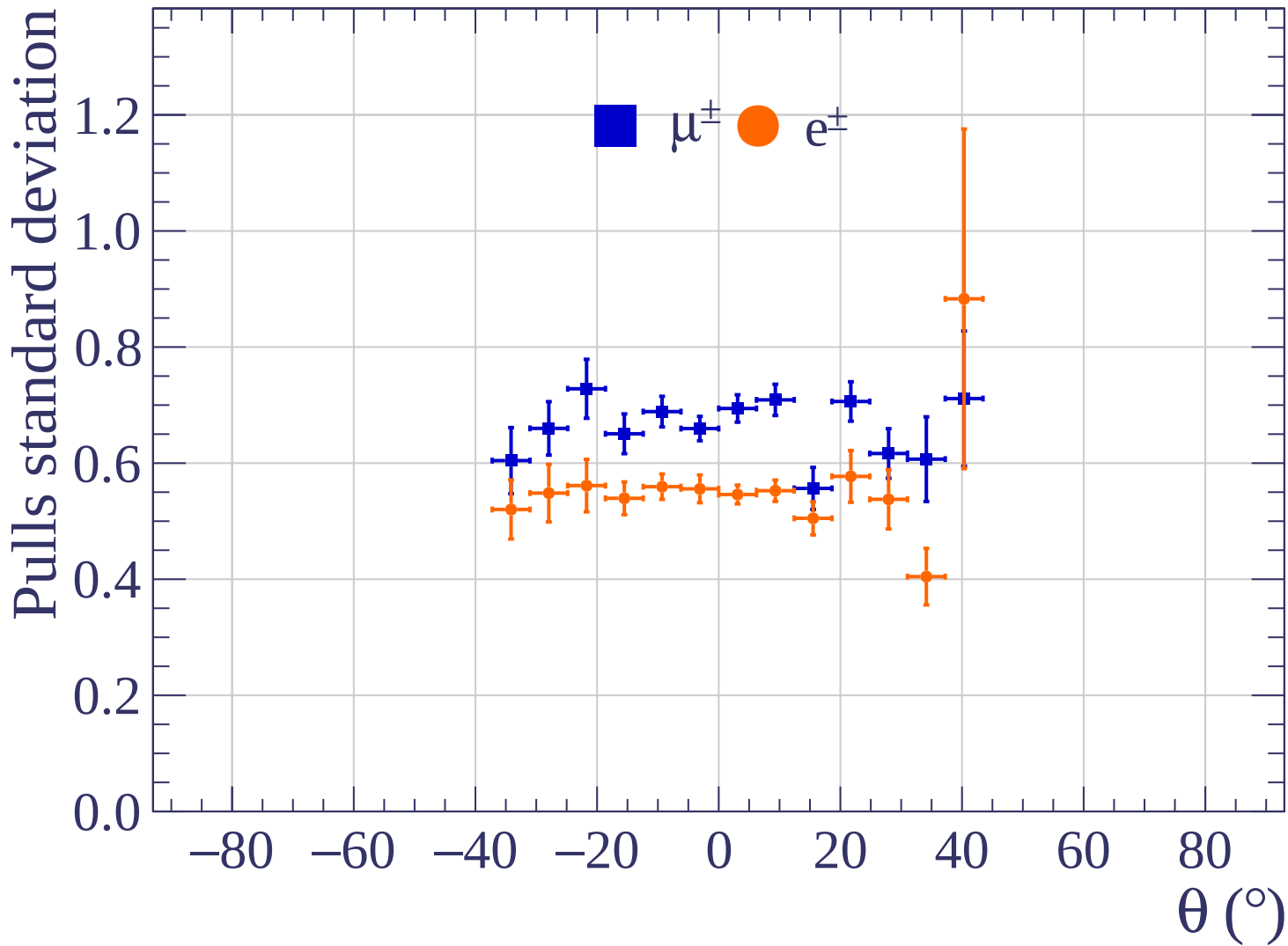


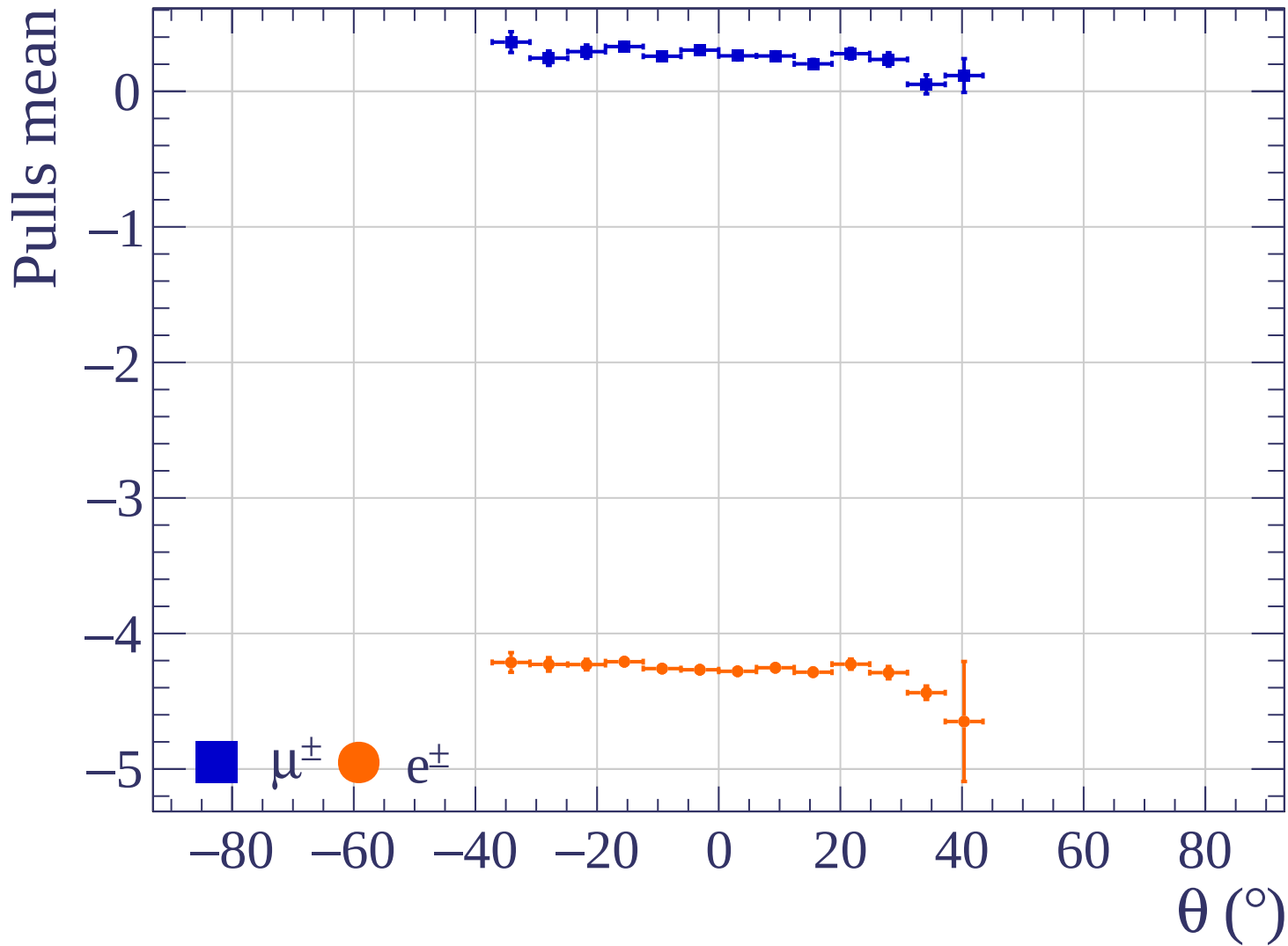


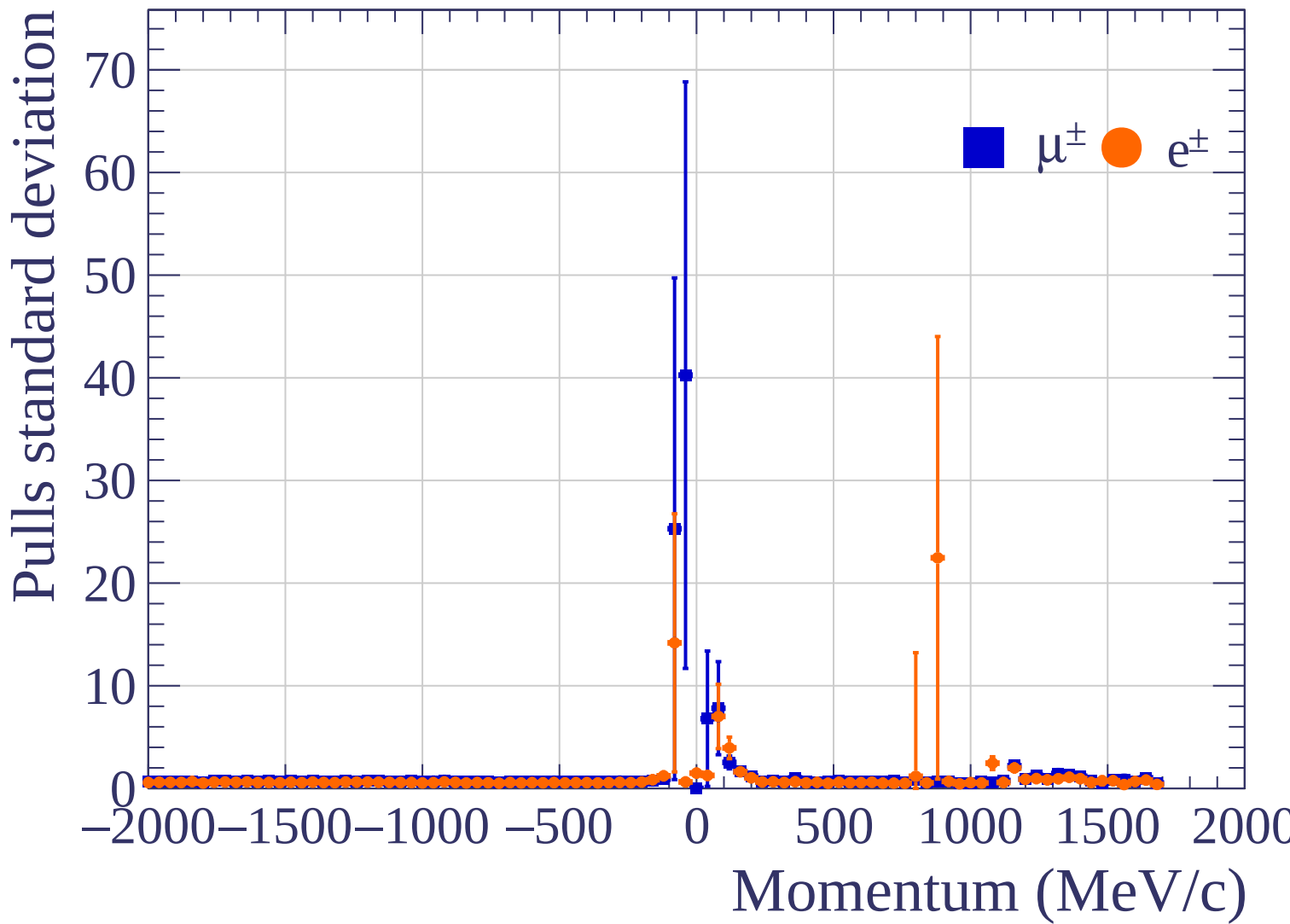


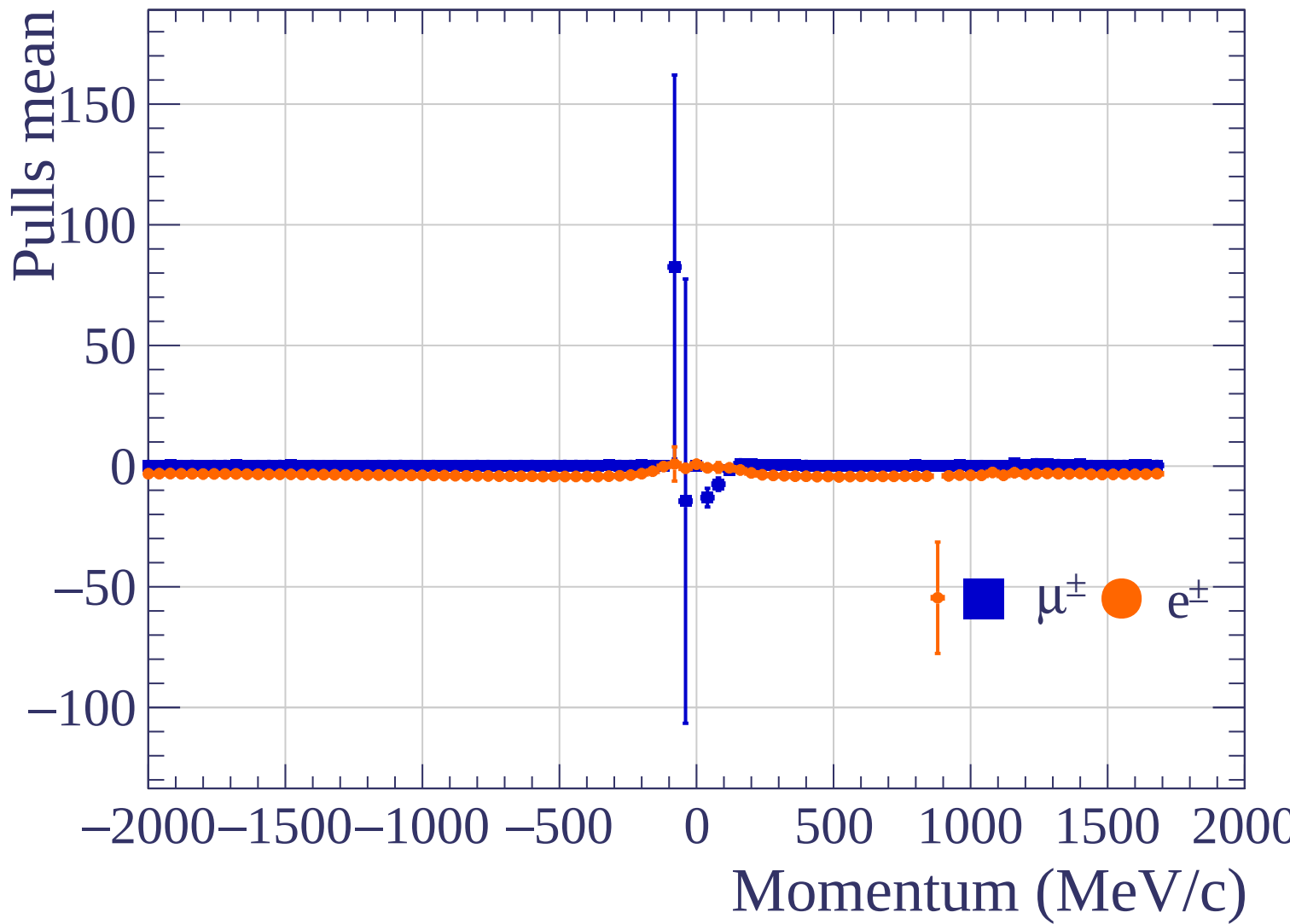


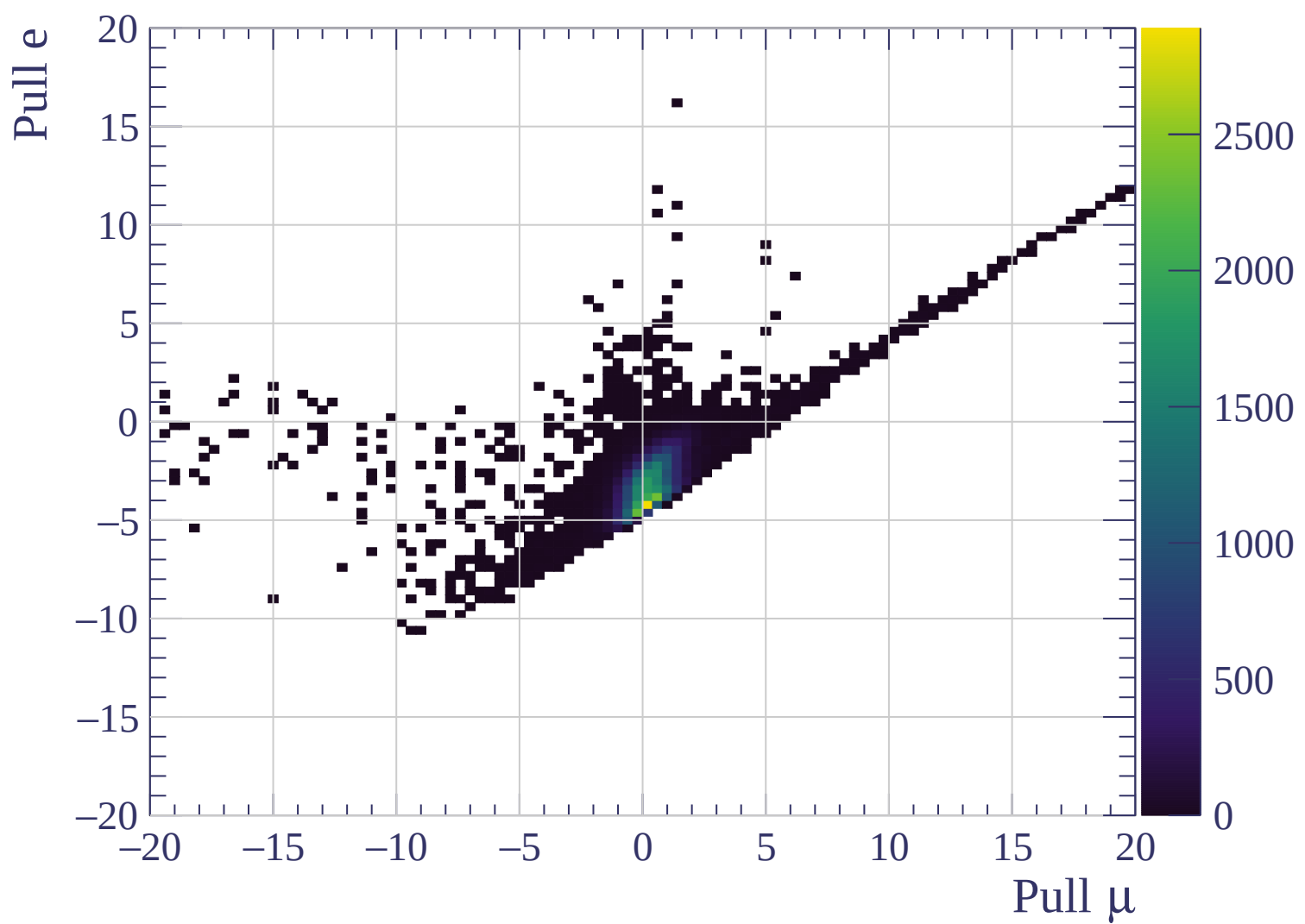


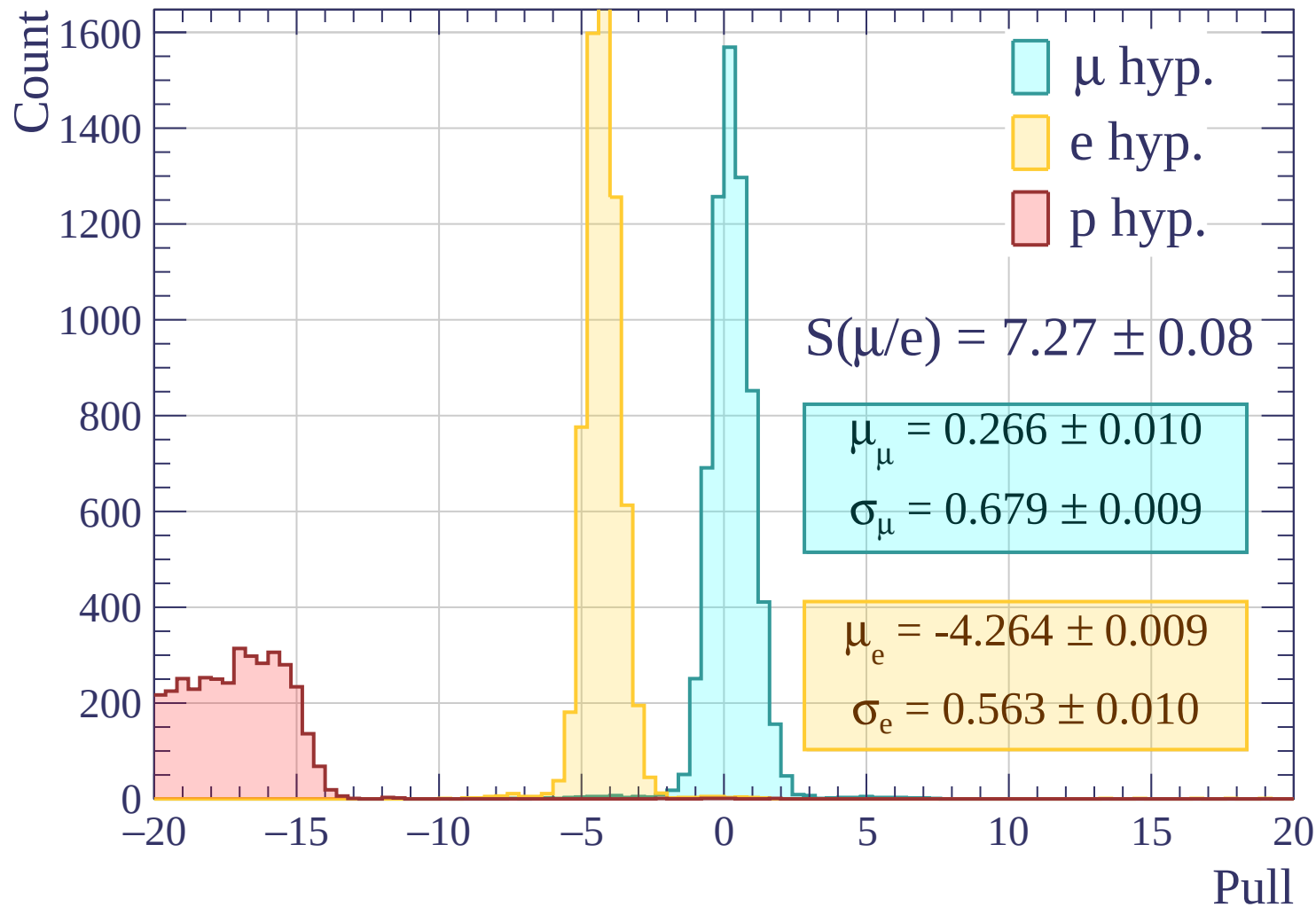


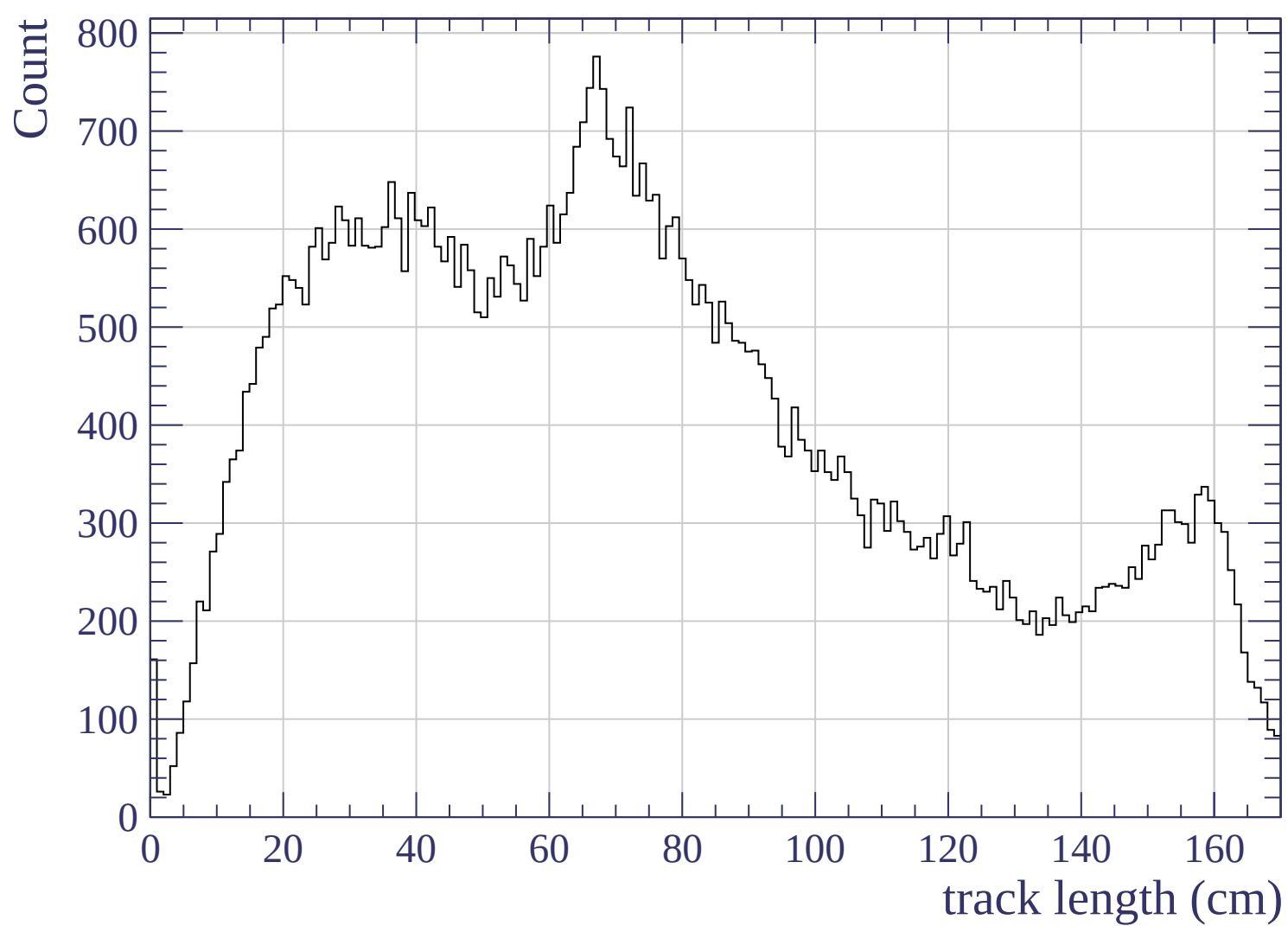


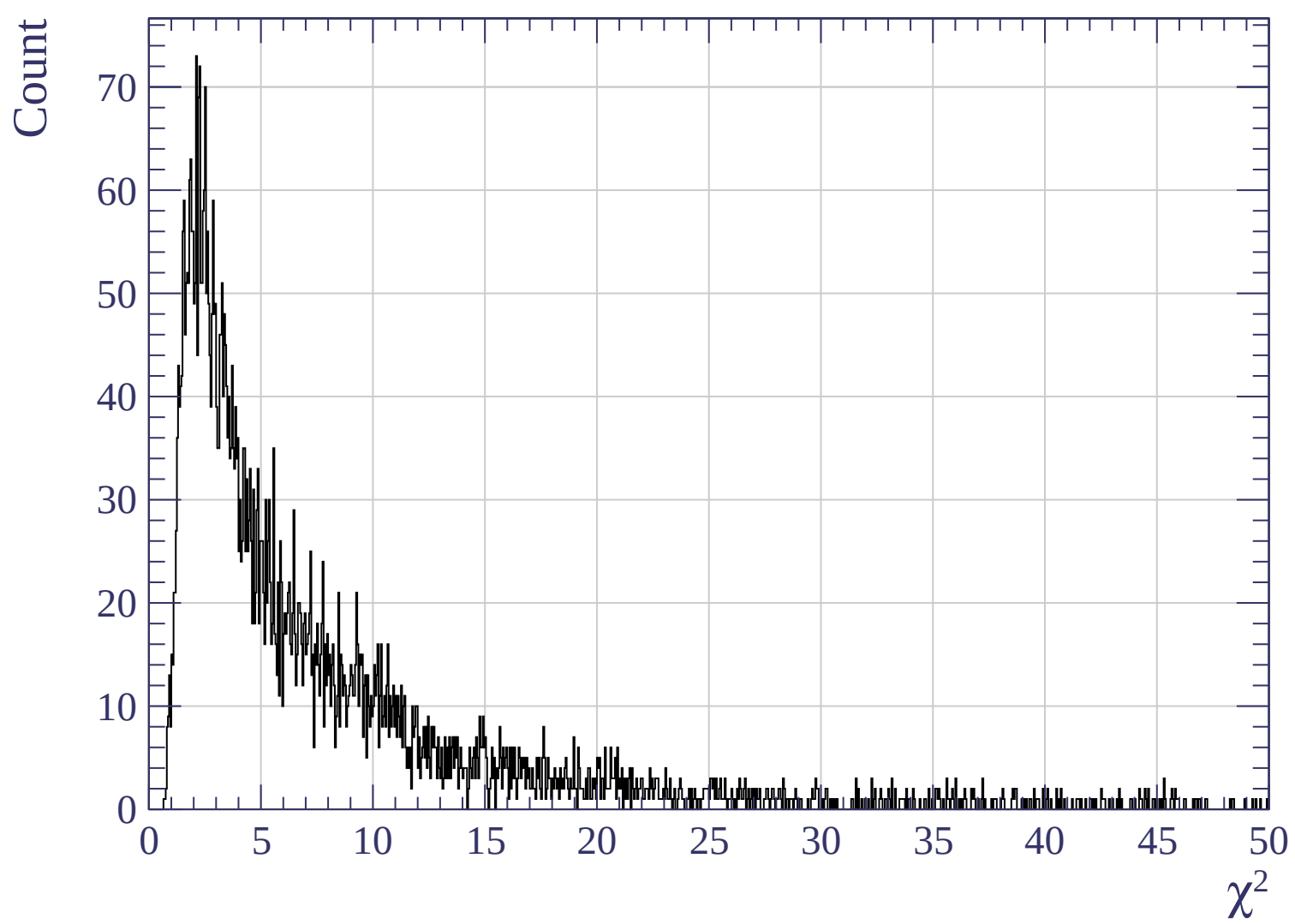


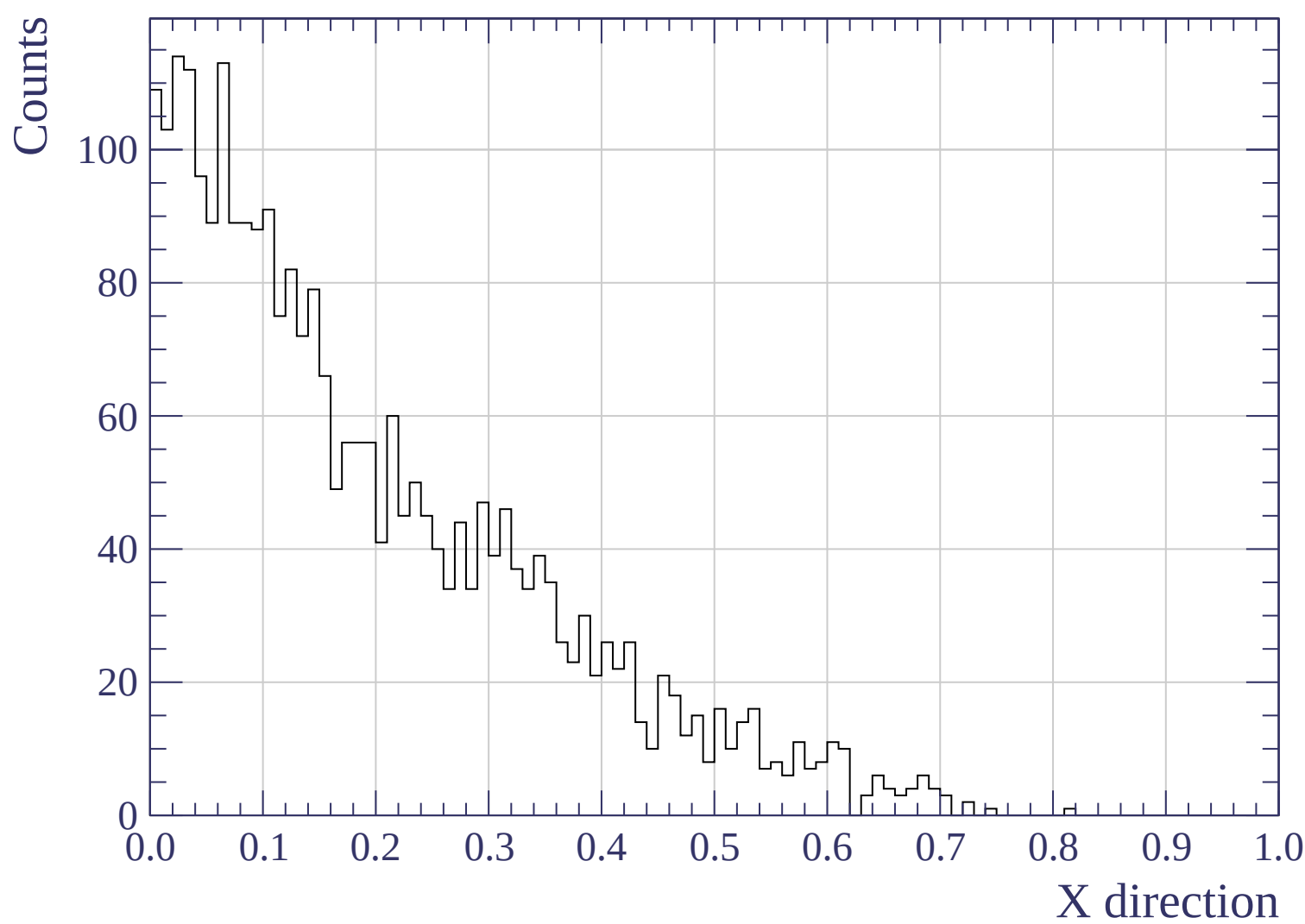


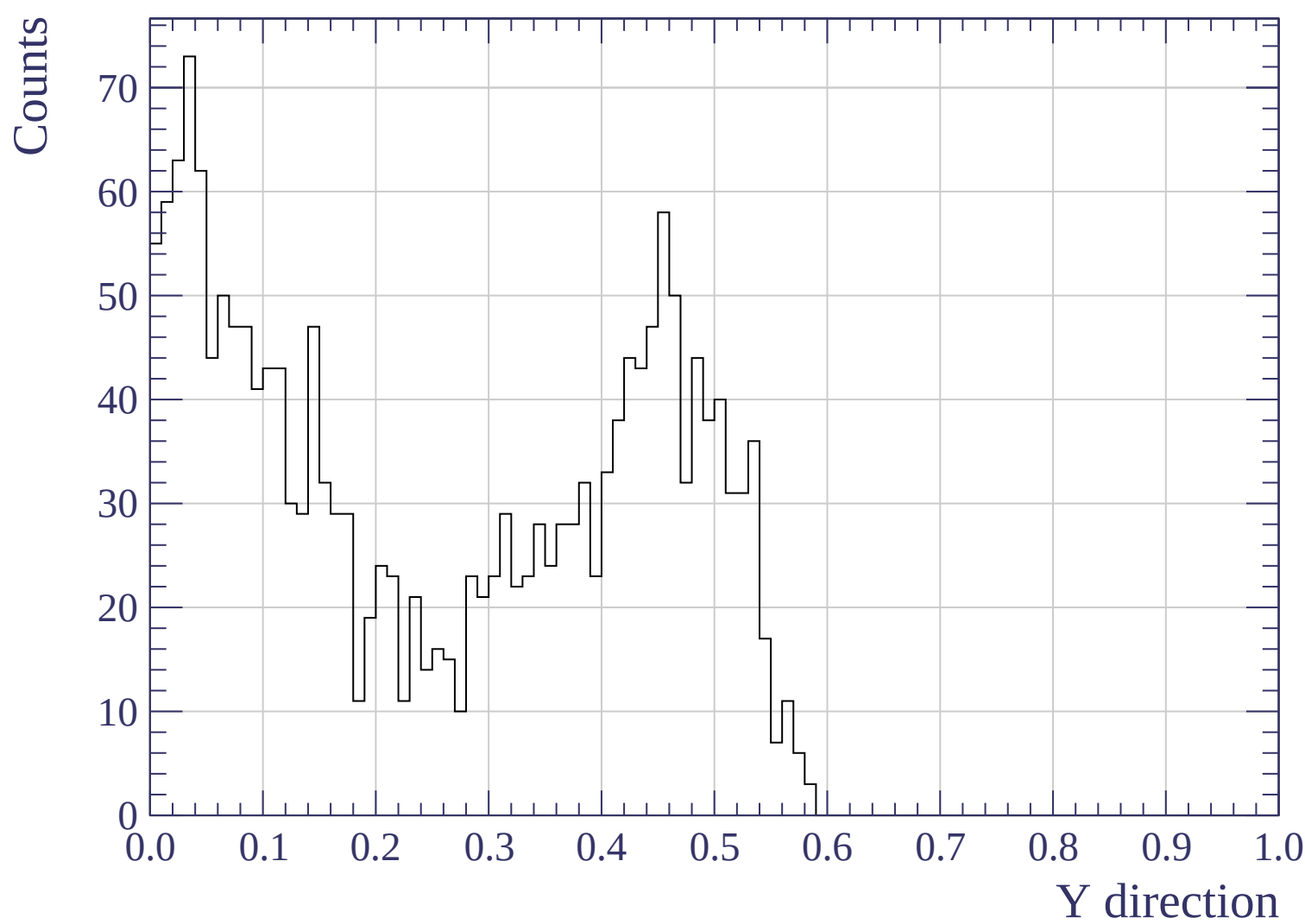


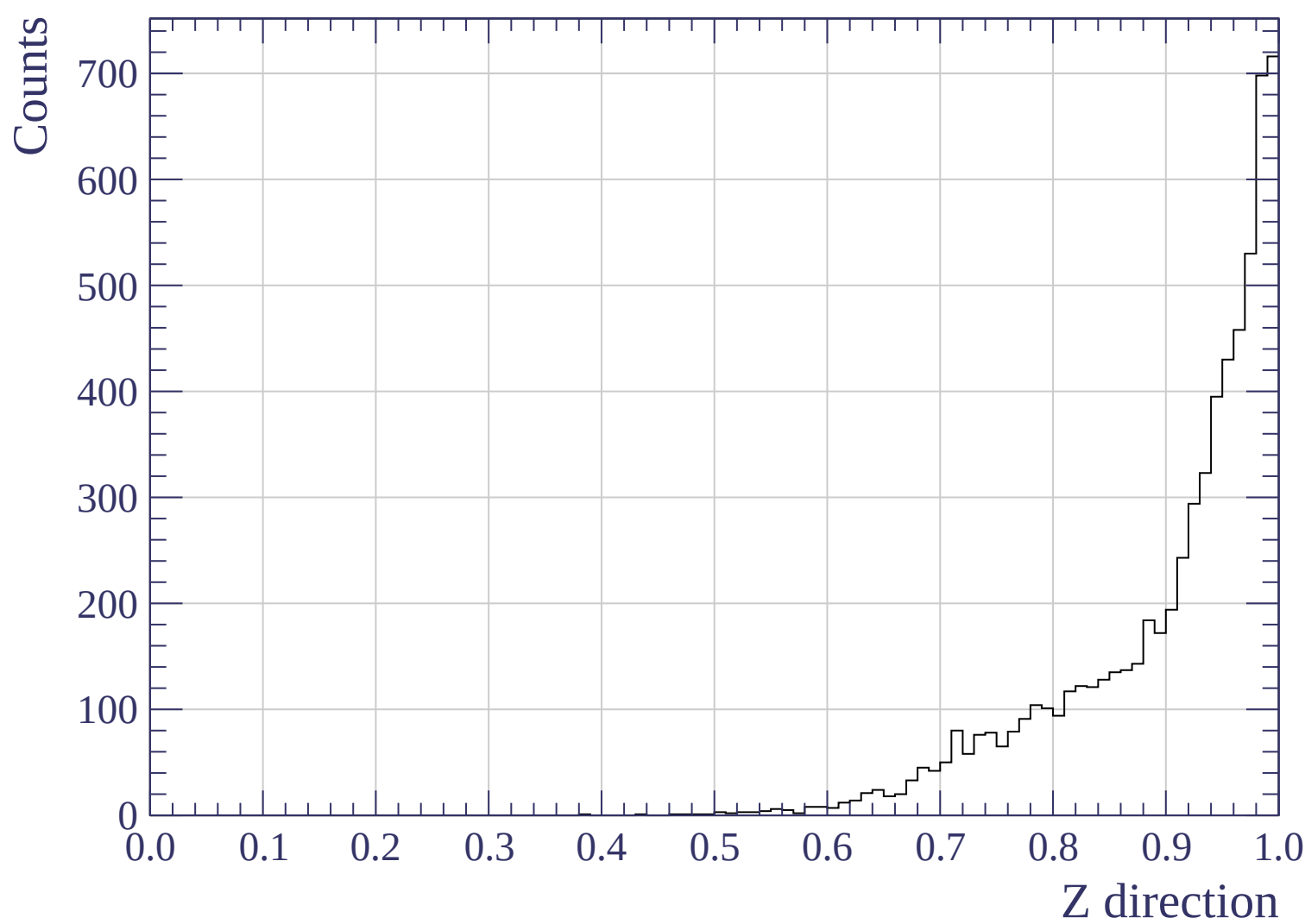


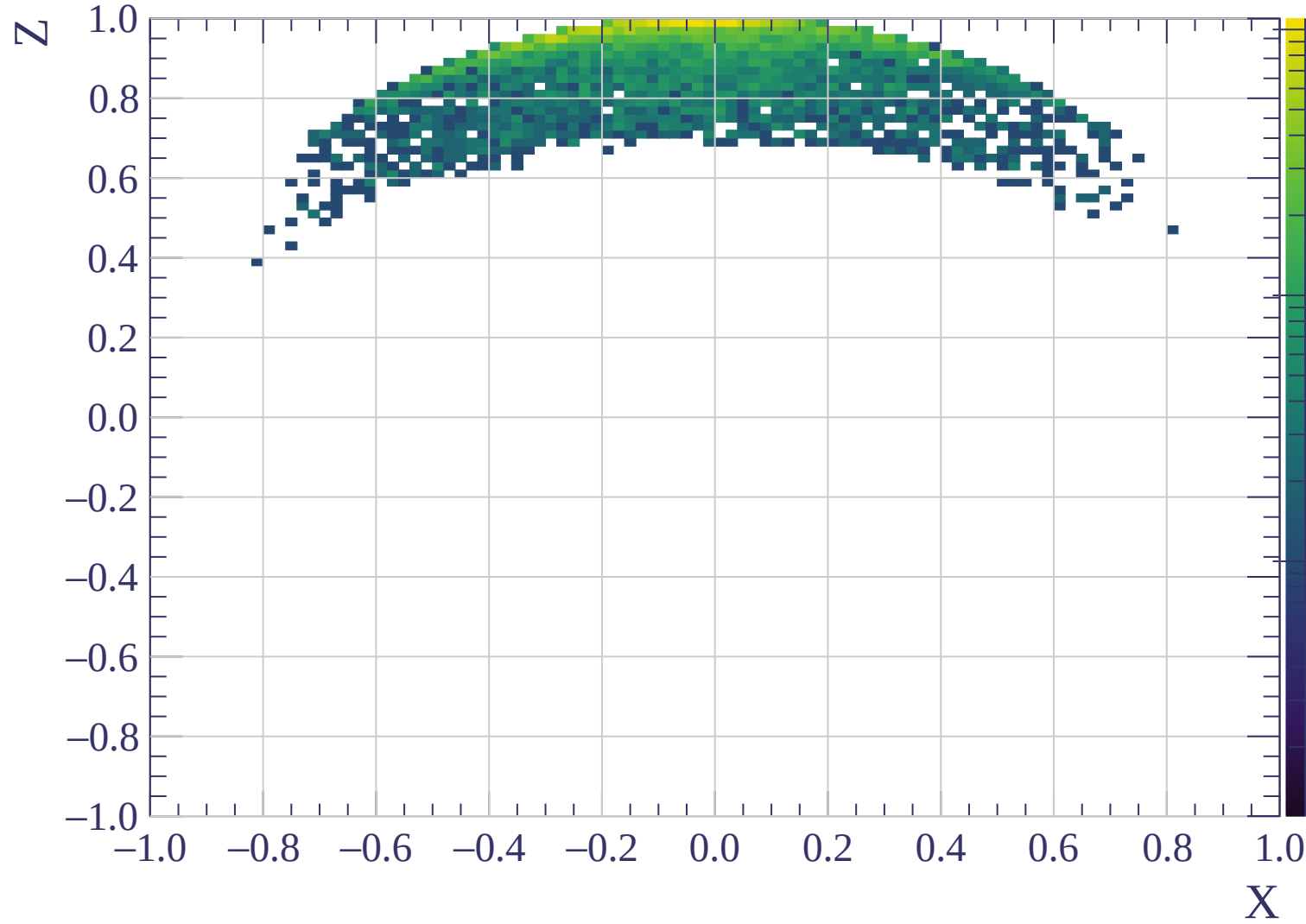


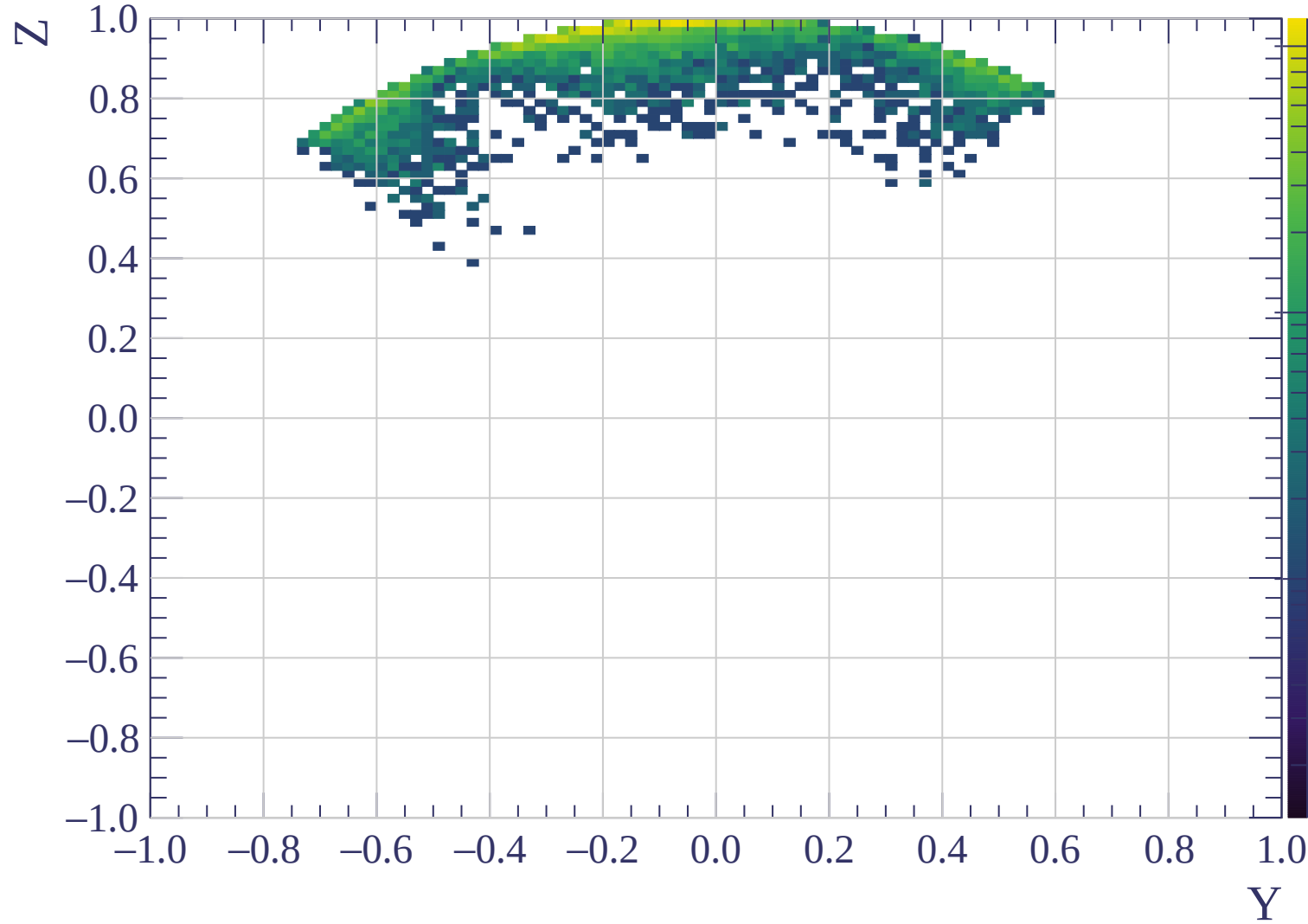


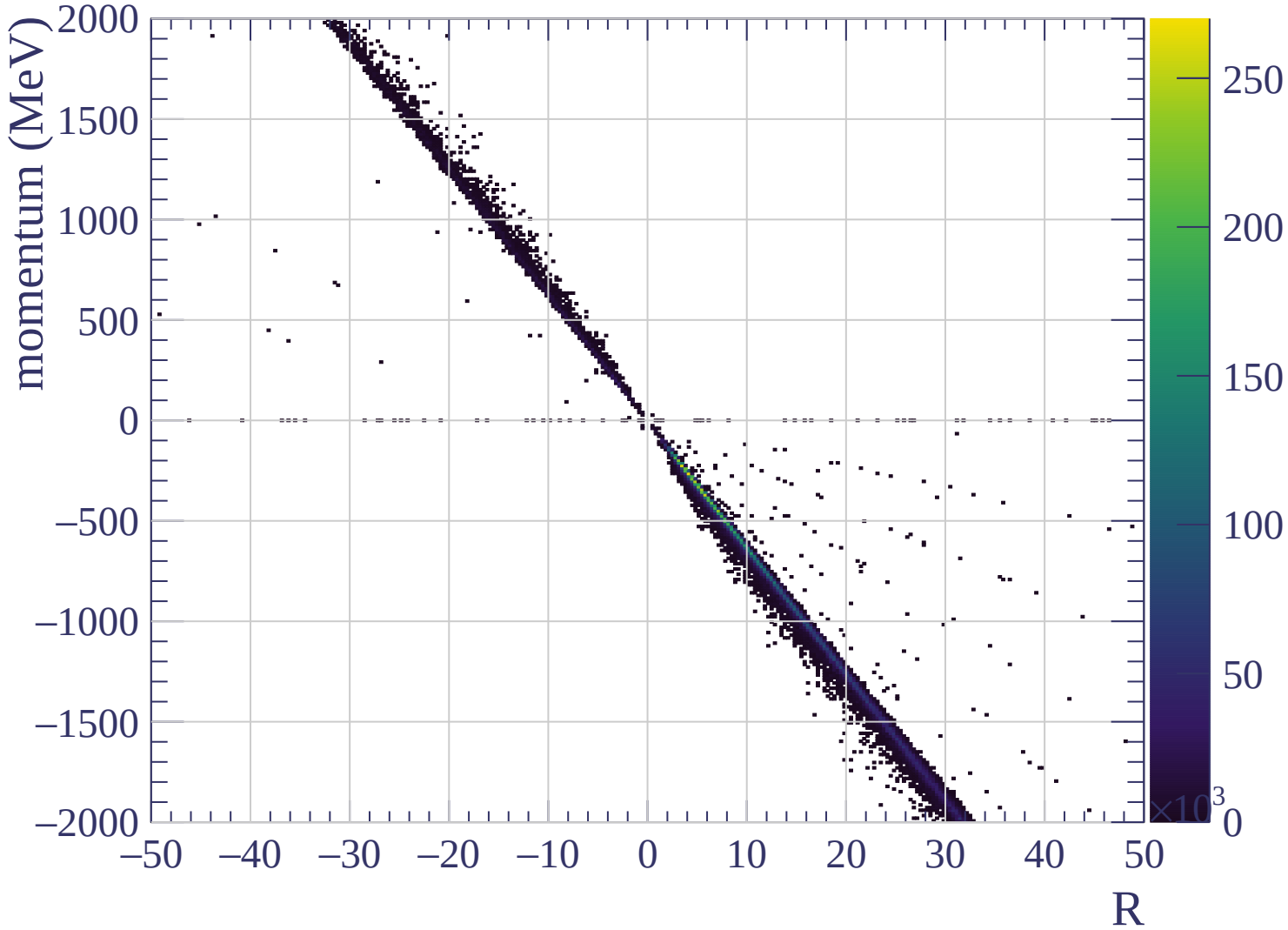


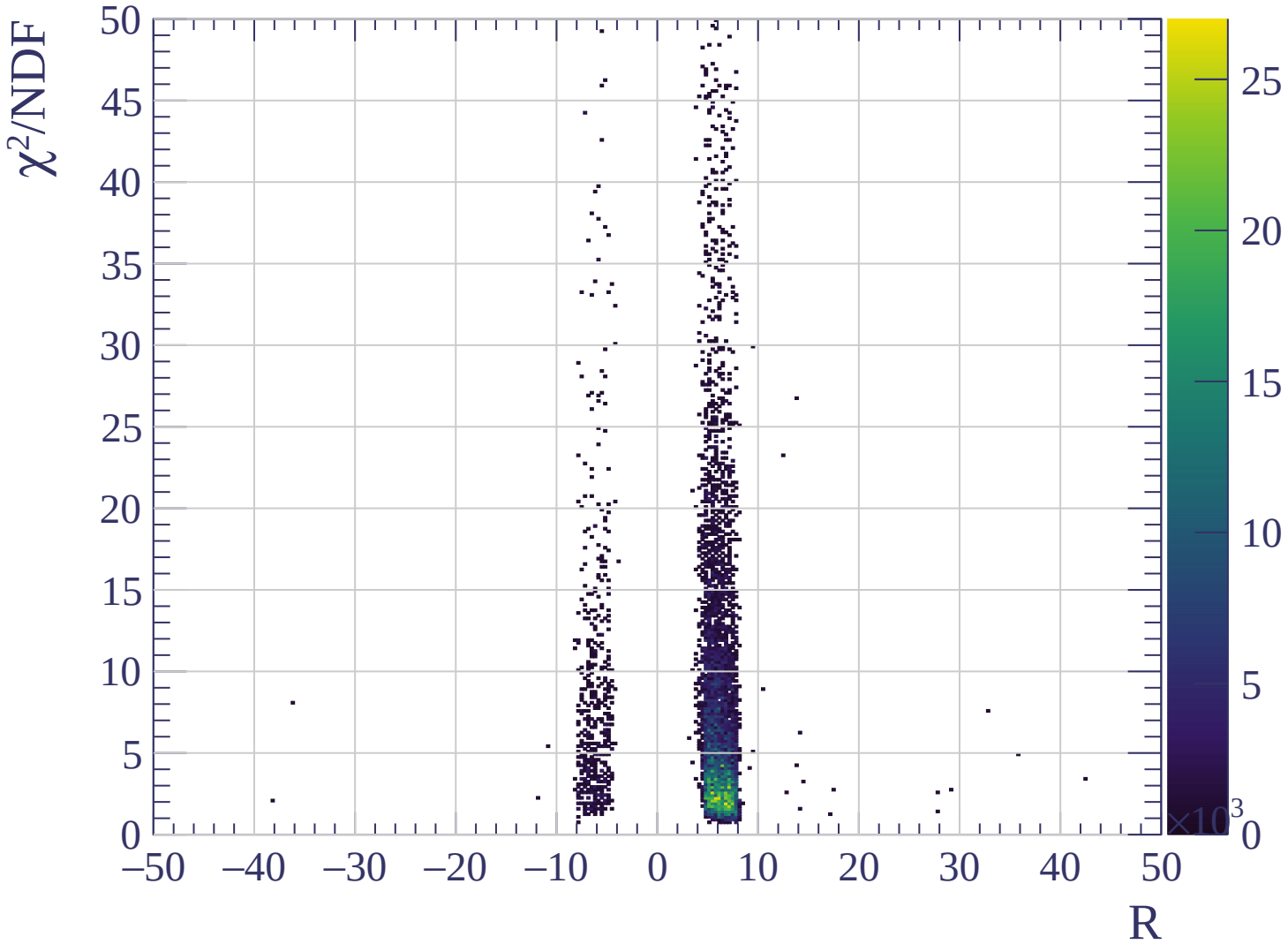






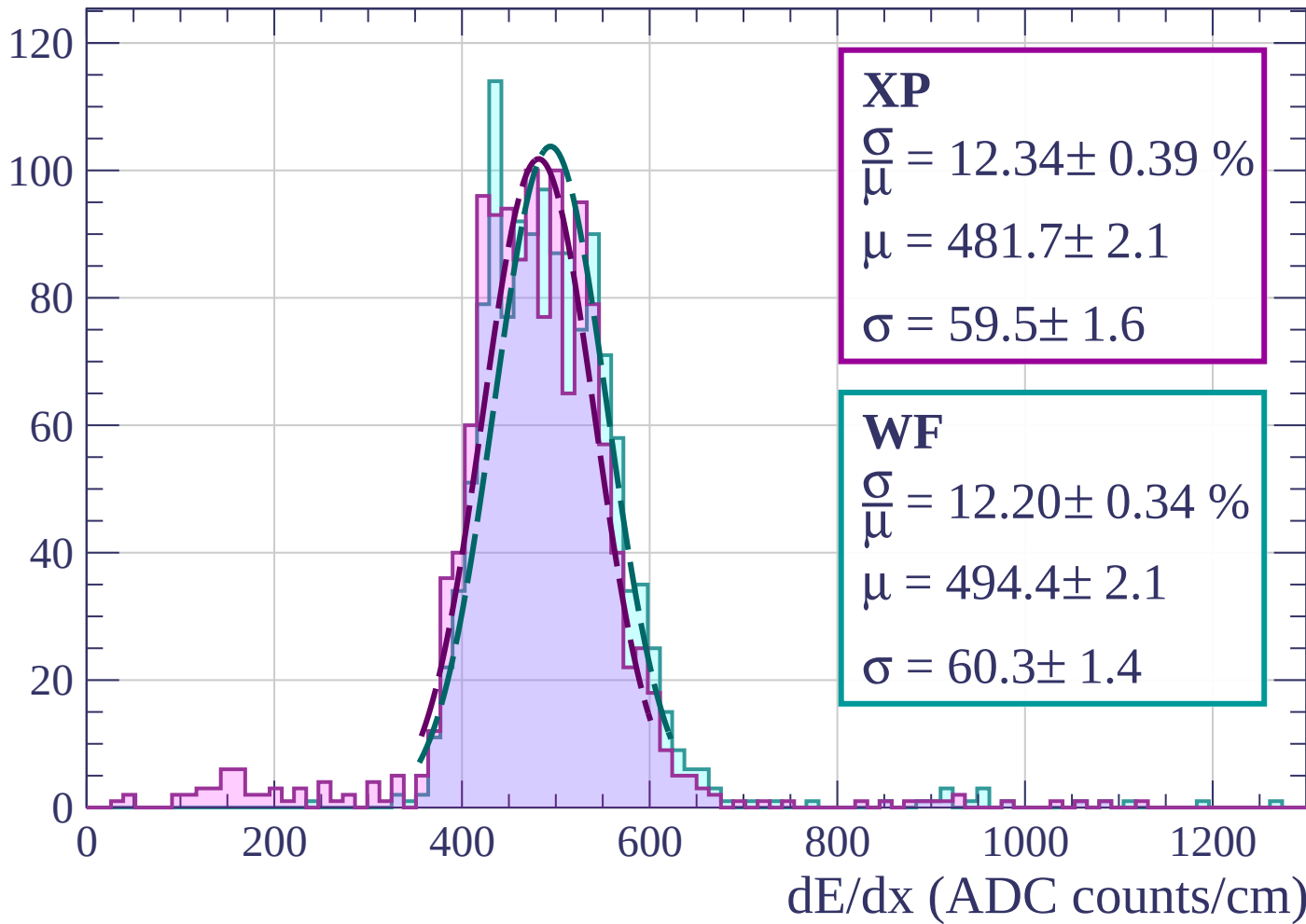






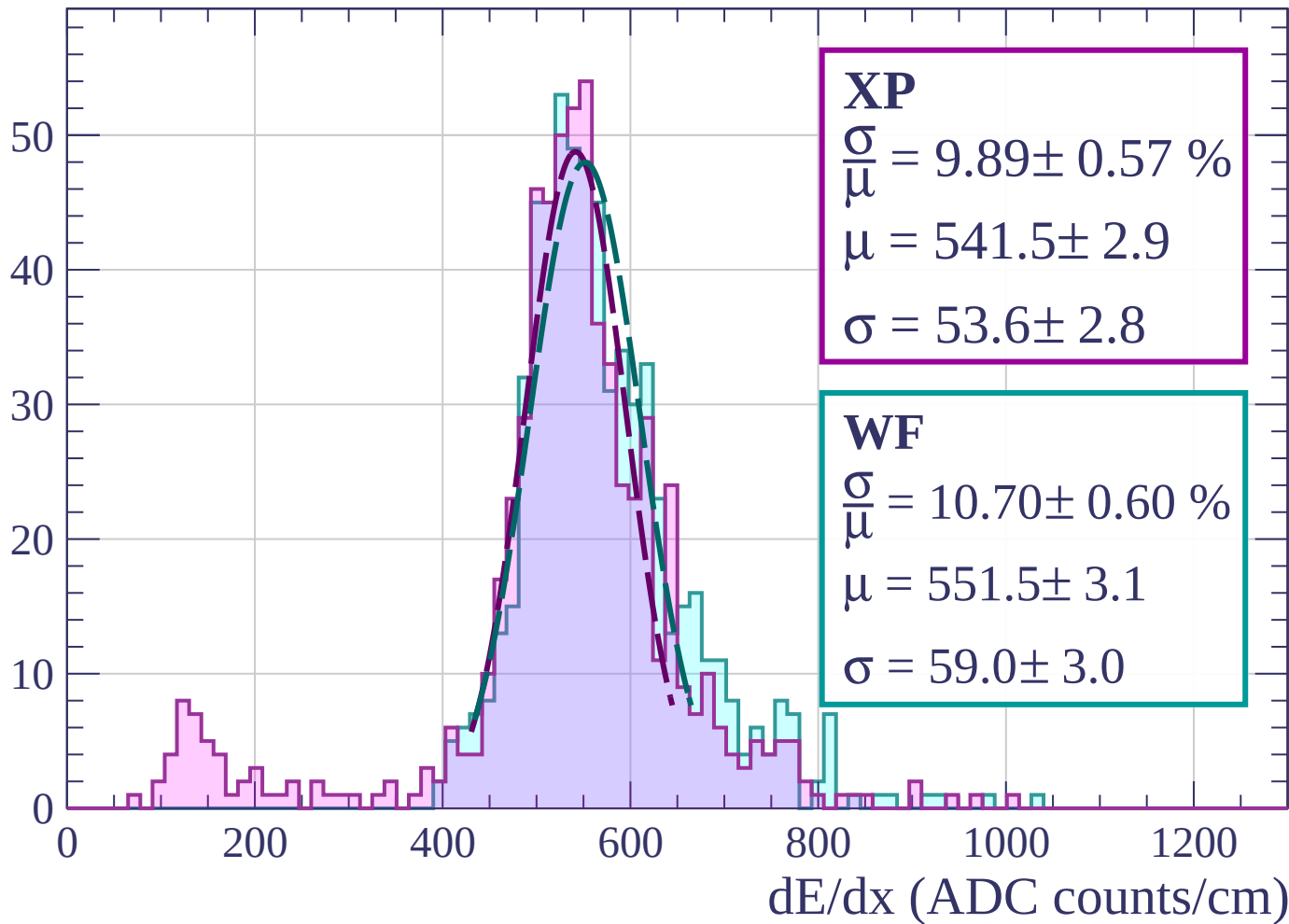
Energy loss | $0 < p < 80$

Count



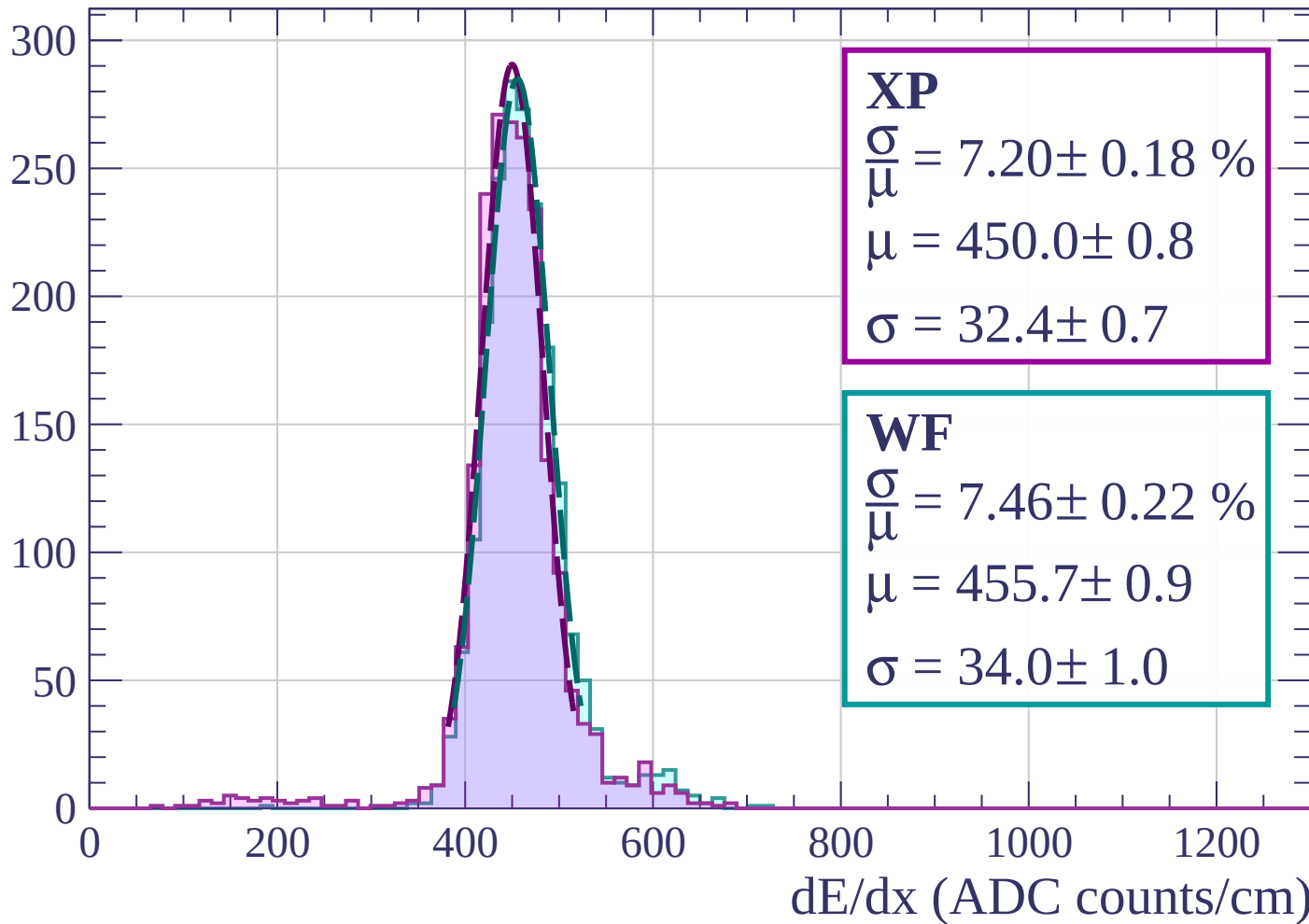
Energy loss | $80 < p < 160$

Count



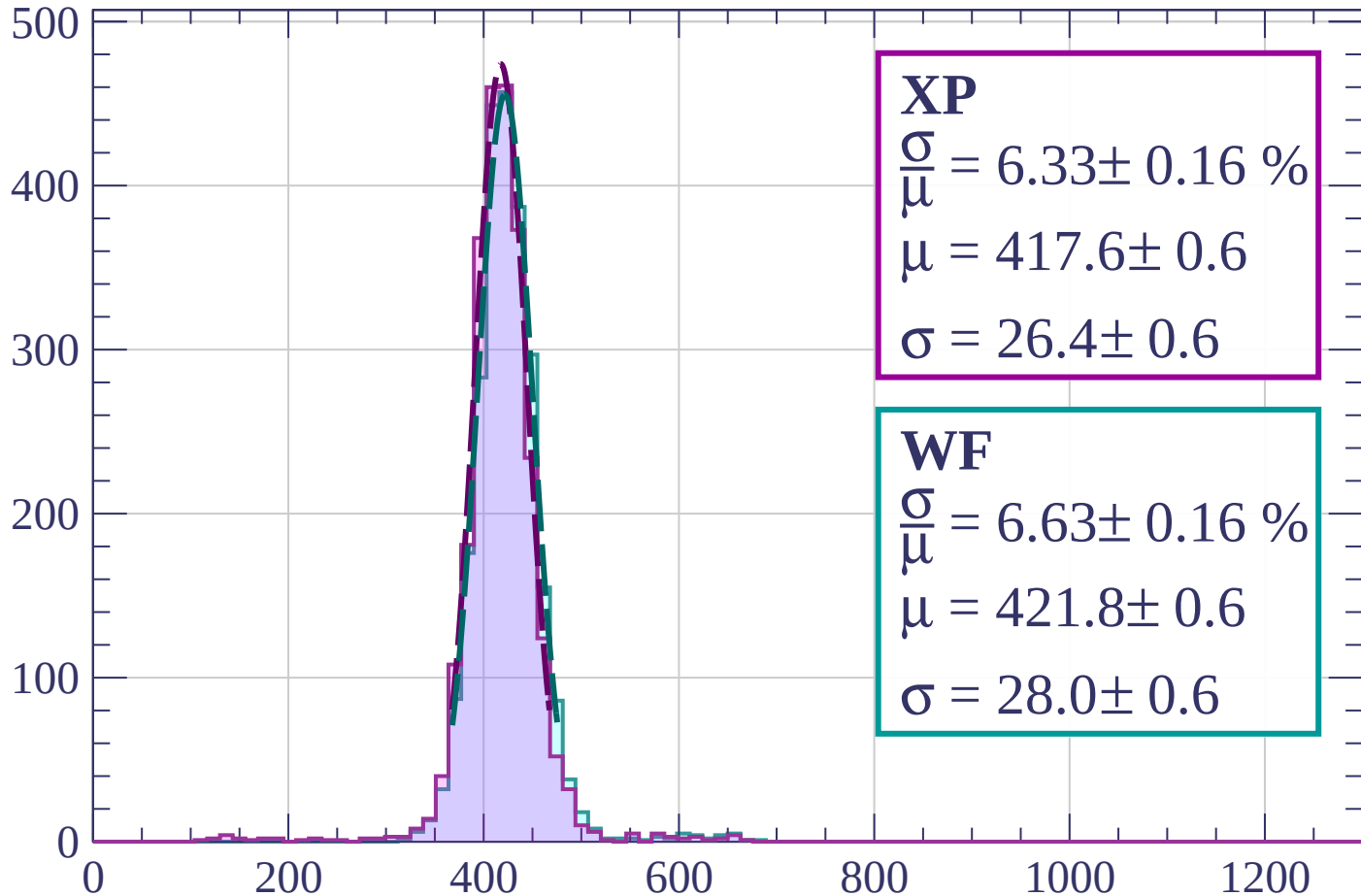
Energy loss | $160 < p < 240$

Count



Energy loss | $240 < p < 320$

Count



XP

$$\frac{\sigma}{\mu} = 6.33 \pm 0.16 \%$$

$$\mu = 417.6 \pm 0.6$$

$$\sigma = 26.4 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.63 \pm 0.16 \%$$

$$\mu = 421.8 \pm 0.6$$

$$\sigma = 28.0 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $320 < p < 400$

Count

500

400

300

200

100

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

XP

$$\frac{\sigma}{\mu} = 6.22 \pm 0.19 \%$$

$$\mu = 408.6 \pm 0.6$$

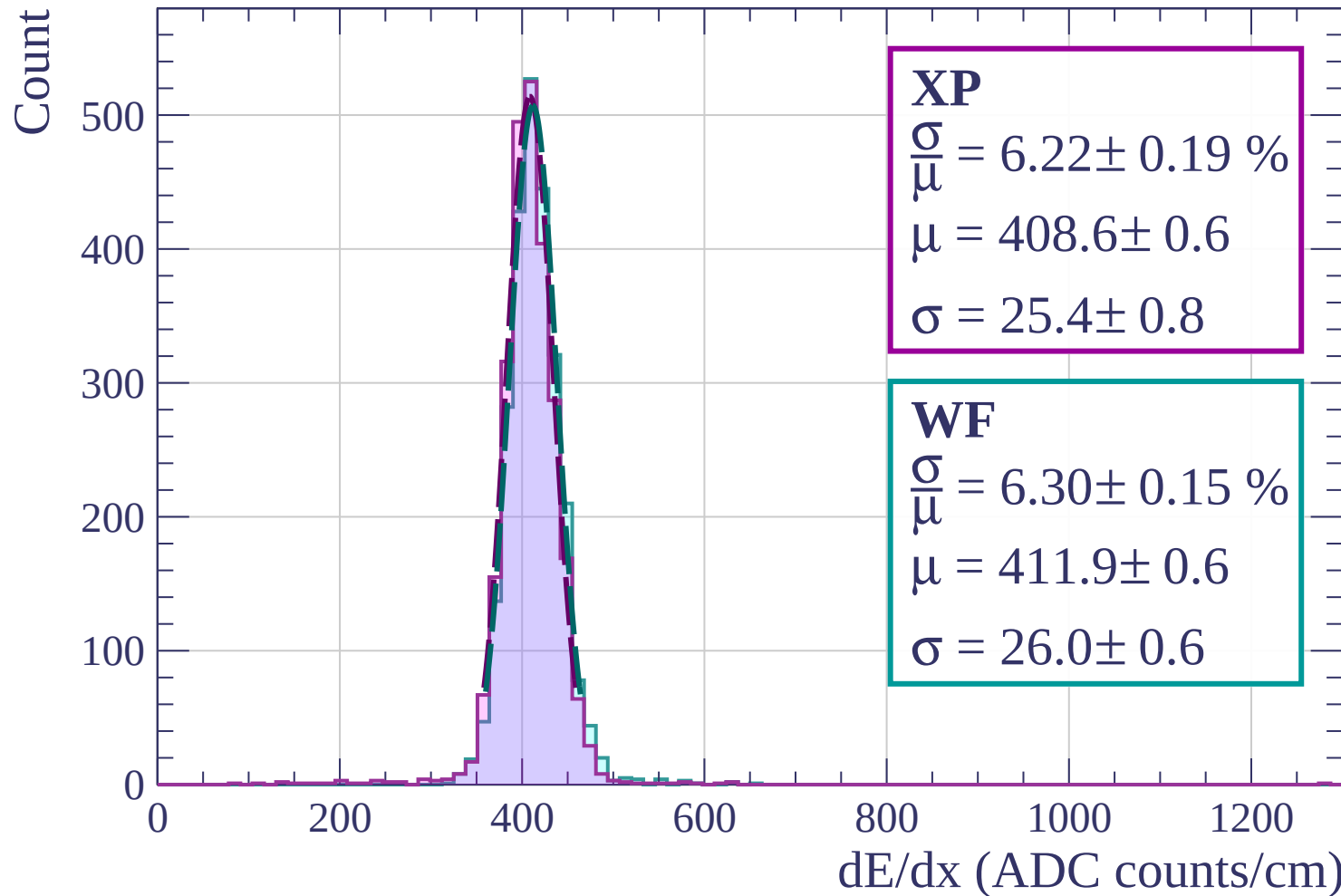
$$\sigma = 25.4 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.30 \pm 0.15 \%$$

$$\mu = 411.9 \pm 0.6$$

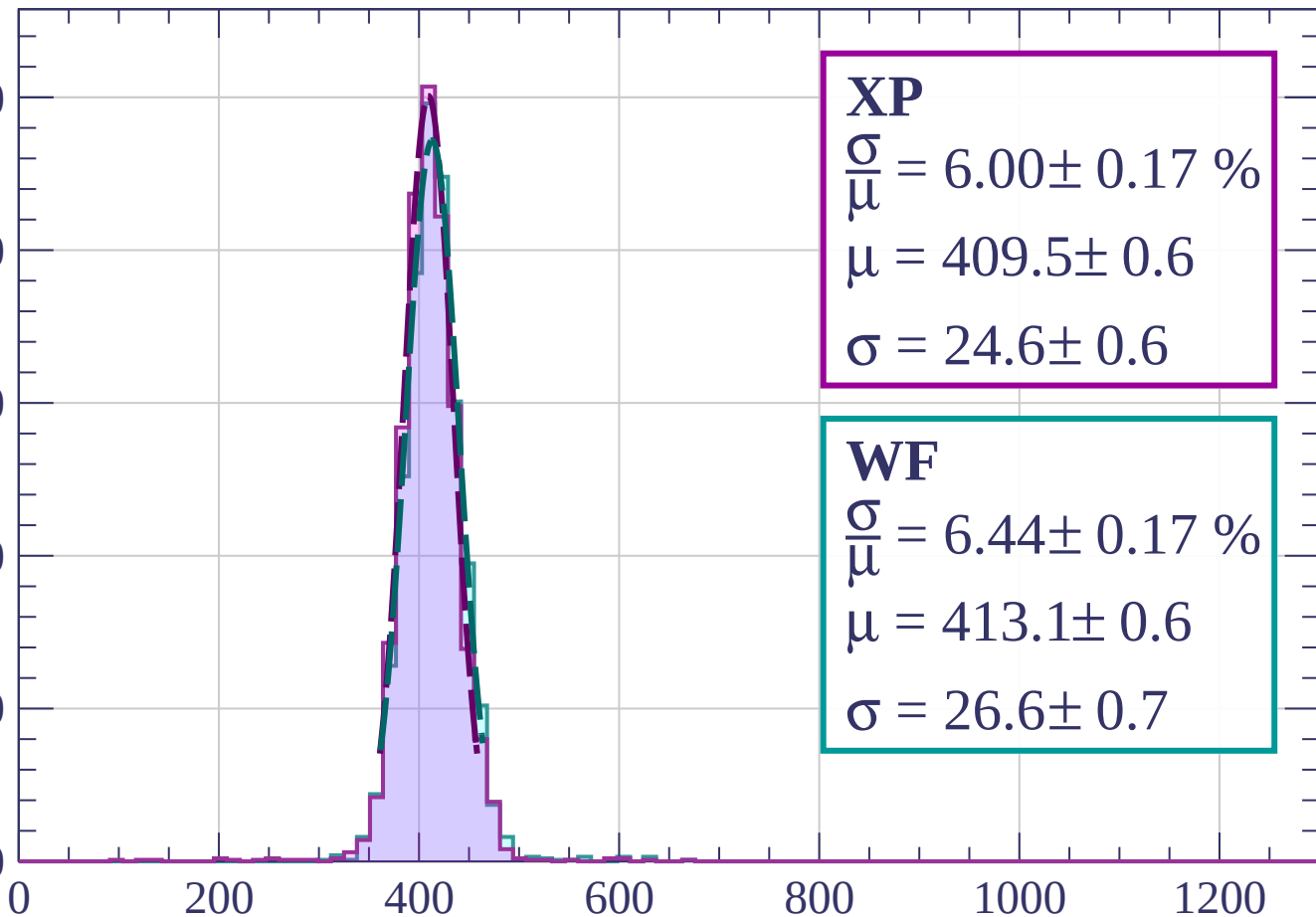
$$\sigma = 26.0 \pm 0.6$$



Energy loss | $400 < p < 480$

Count

500
400
300
200
100
0



XP

$$\frac{\sigma}{\mu} = 6.00 \pm 0.17 \%$$

$$\mu = 409.5 \pm 0.6$$

$$\sigma = 24.6 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.44 \pm 0.17 \%$$

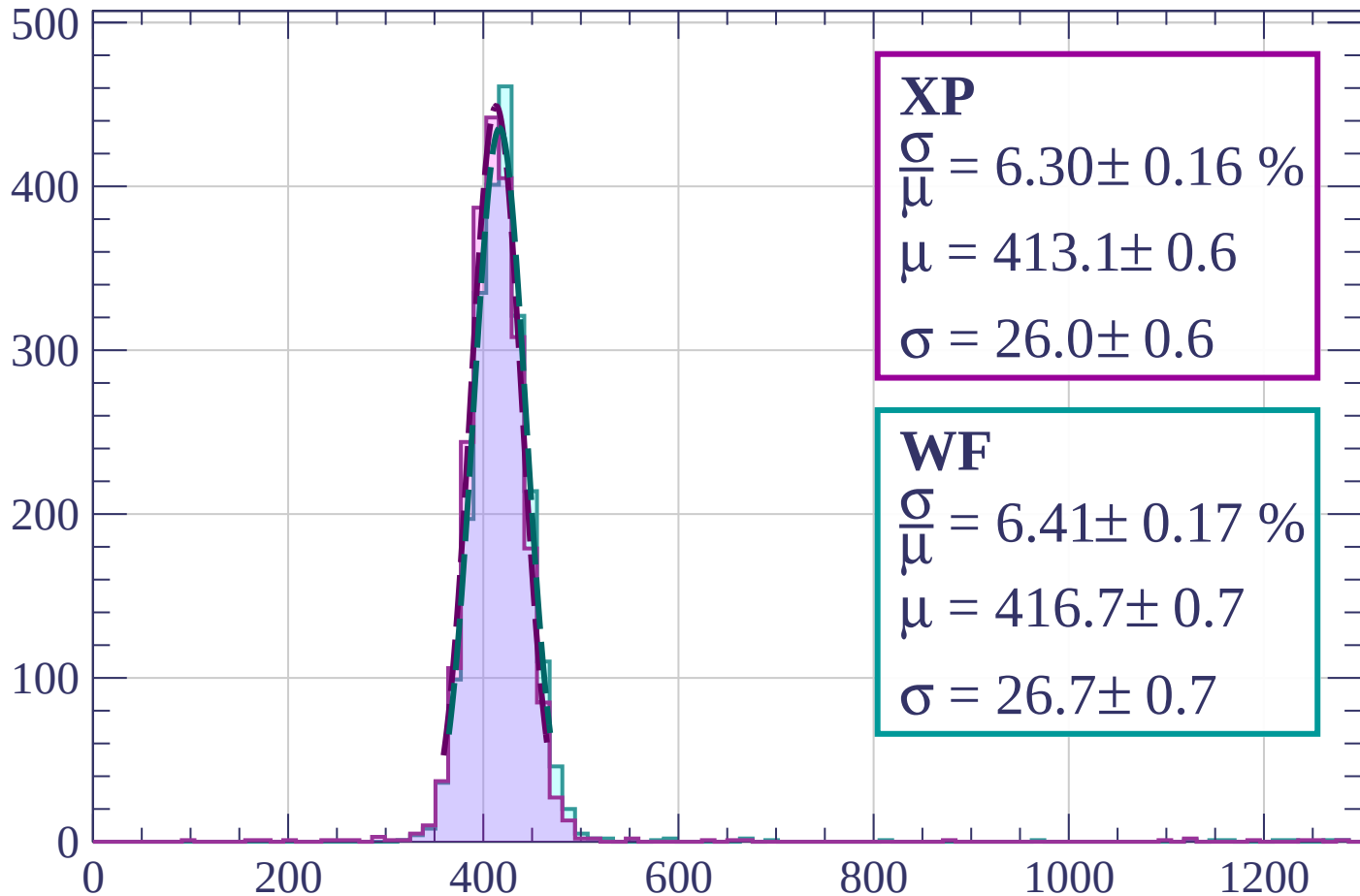
$$\mu = 413.1 \pm 0.6$$

$$\sigma = 26.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\mu} = 6.30 \pm 0.16 \%$$

$$\mu = 413.1 \pm 0.6$$

$$\sigma = 26.0 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.41 \pm 0.17 \%$$

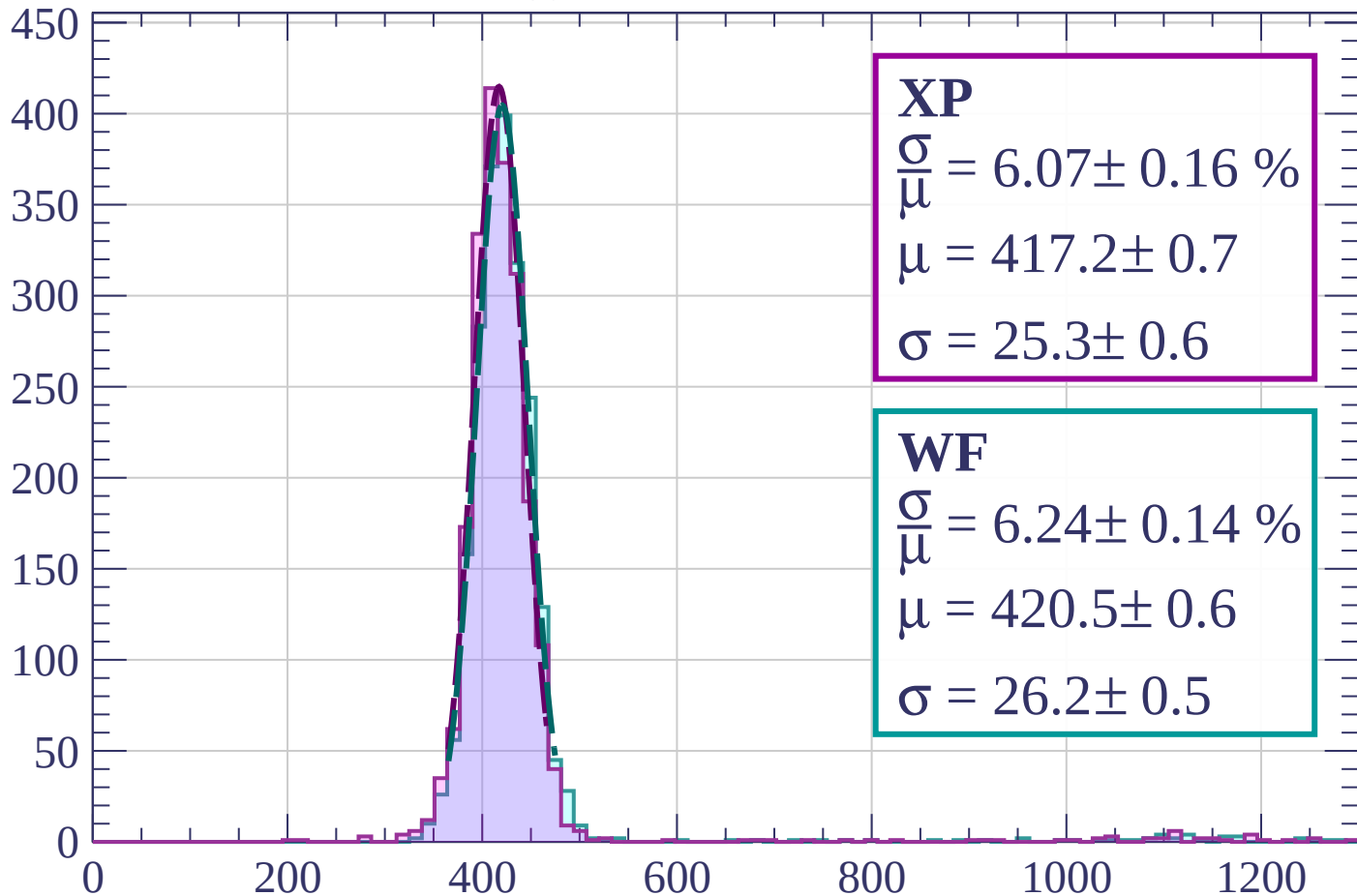
$$\mu = 416.7 \pm 0.7$$

$$\sigma = 26.7 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $560 < p < 640$

Count



XP

$$\frac{\sigma}{\mu} = 6.07 \pm 0.16 \%$$

$$\mu = 417.2 \pm 0.7$$

$$\sigma = 25.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.24 \pm 0.14 \%$$

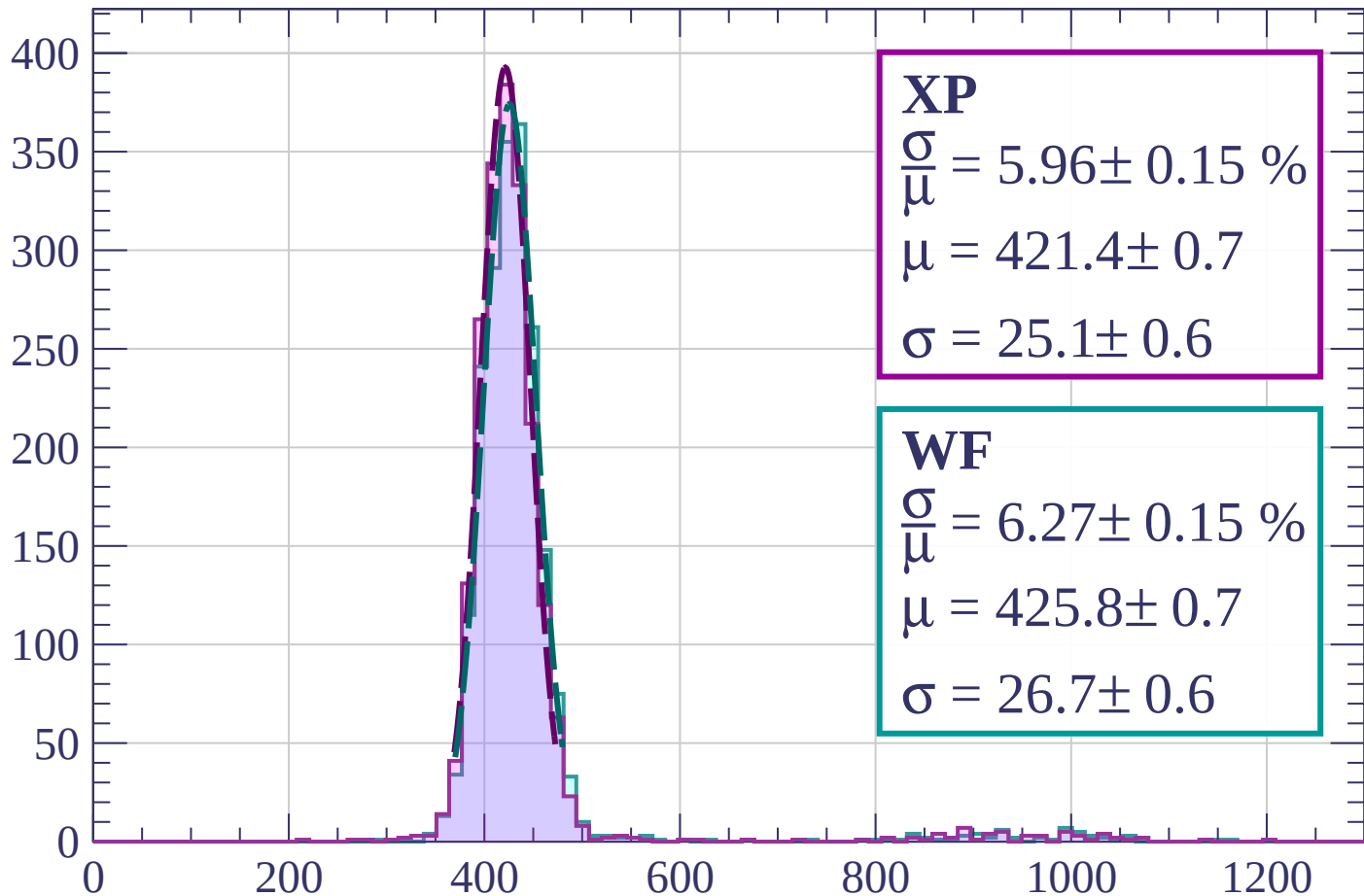
$$\mu = 420.5 \pm 0.6$$

$$\sigma = 26.2 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 5.96 \pm 0.15 \%$$

$$\mu = 421.4 \pm 0.7$$

$$\sigma = 25.1 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 6.27 \pm 0.15 \%$$

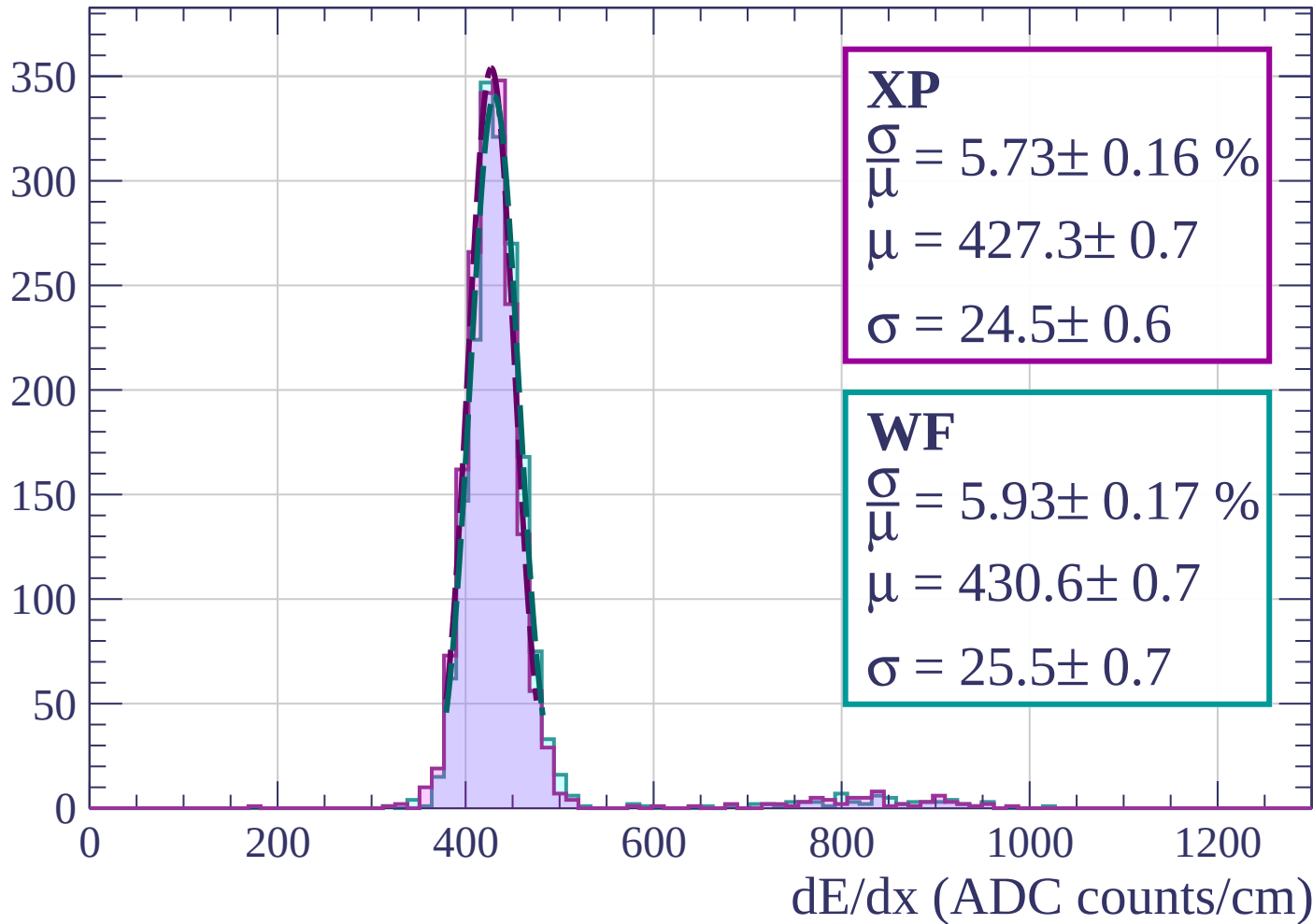
$$\mu = 425.8 \pm 0.7$$

$$\sigma = 26.7 \pm 0.6$$

dE/dx (ADC counts/cm)

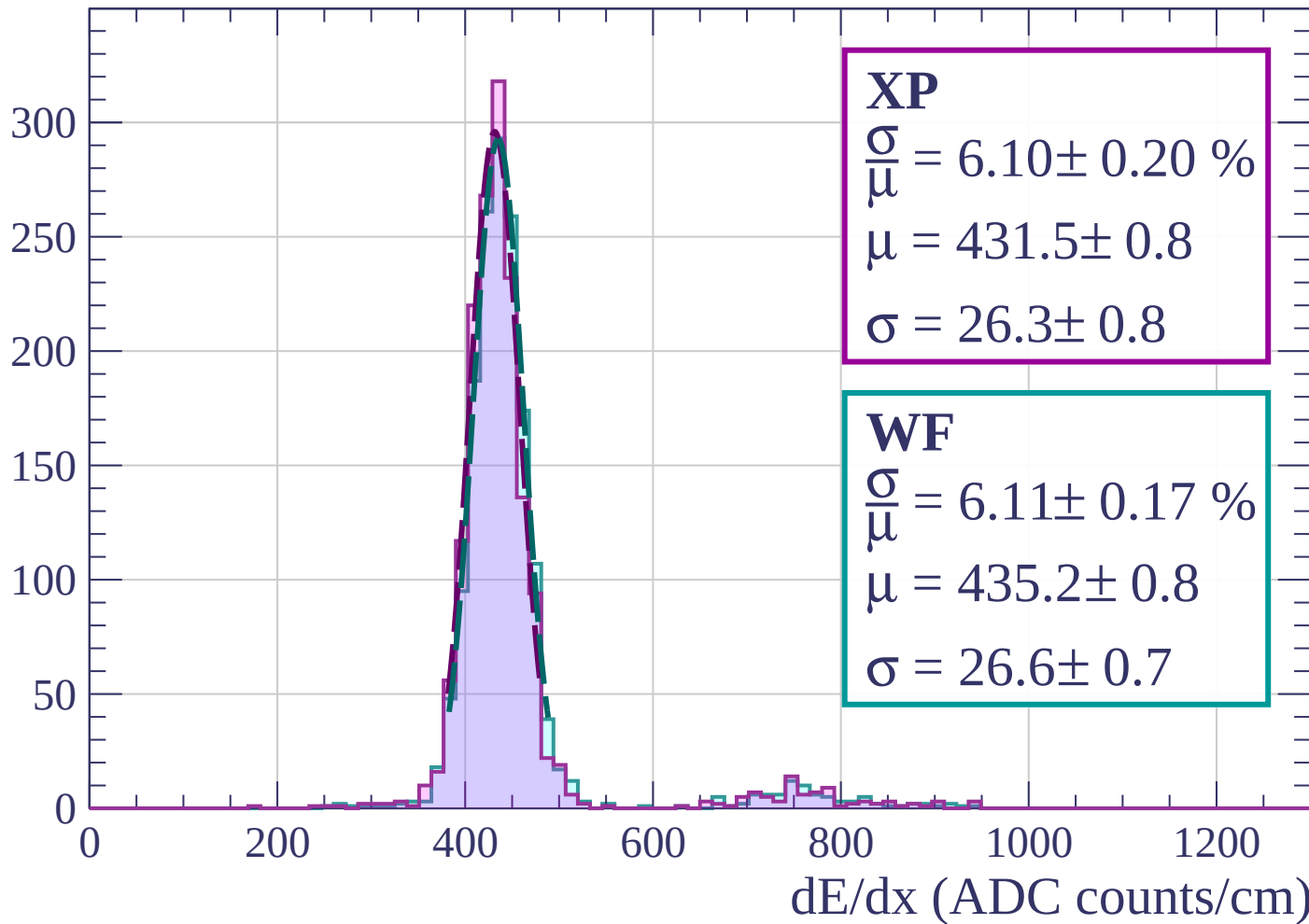
Energy loss | $720 < p < 800$

Count

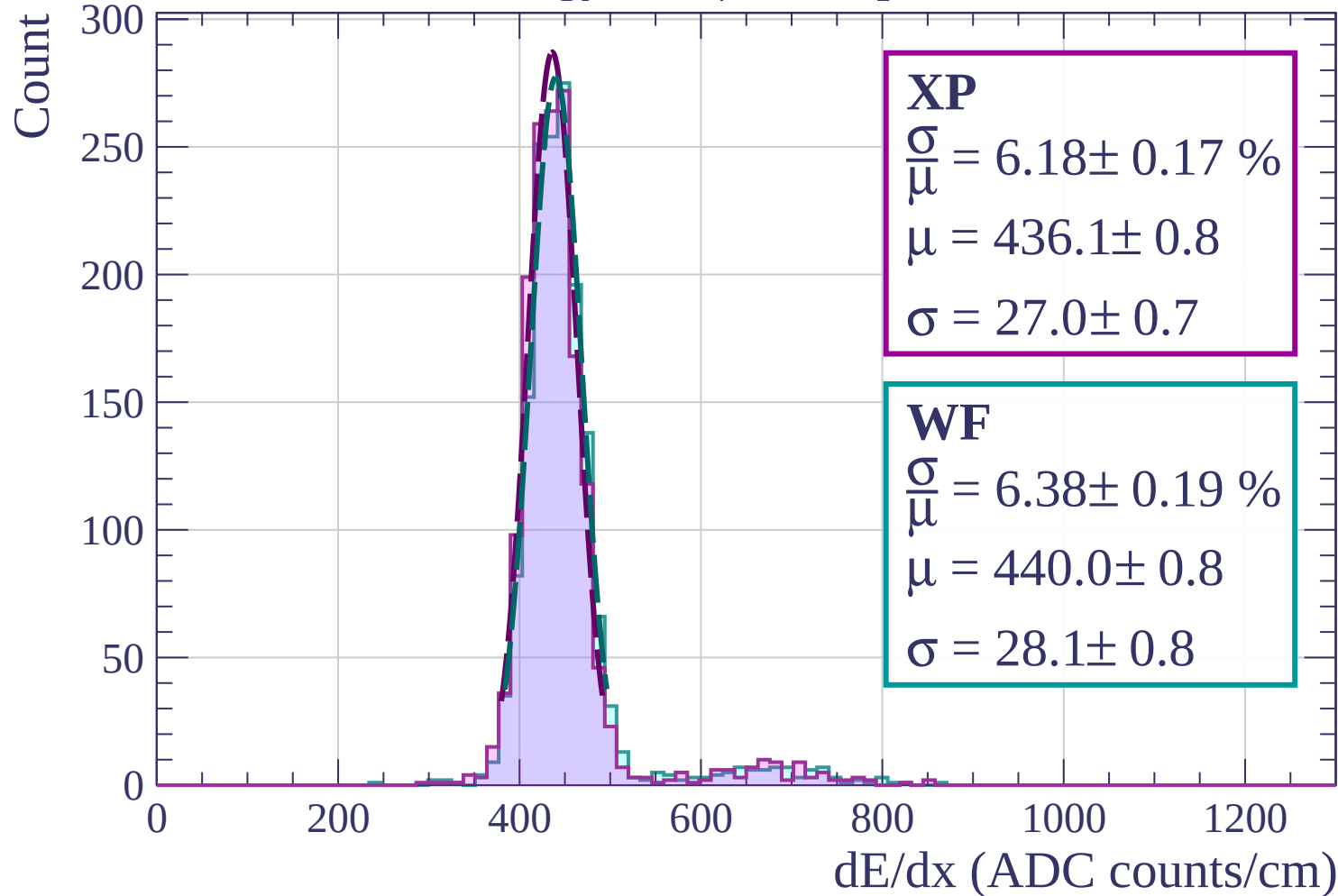


Energy loss | $800 < p < 880$

Count

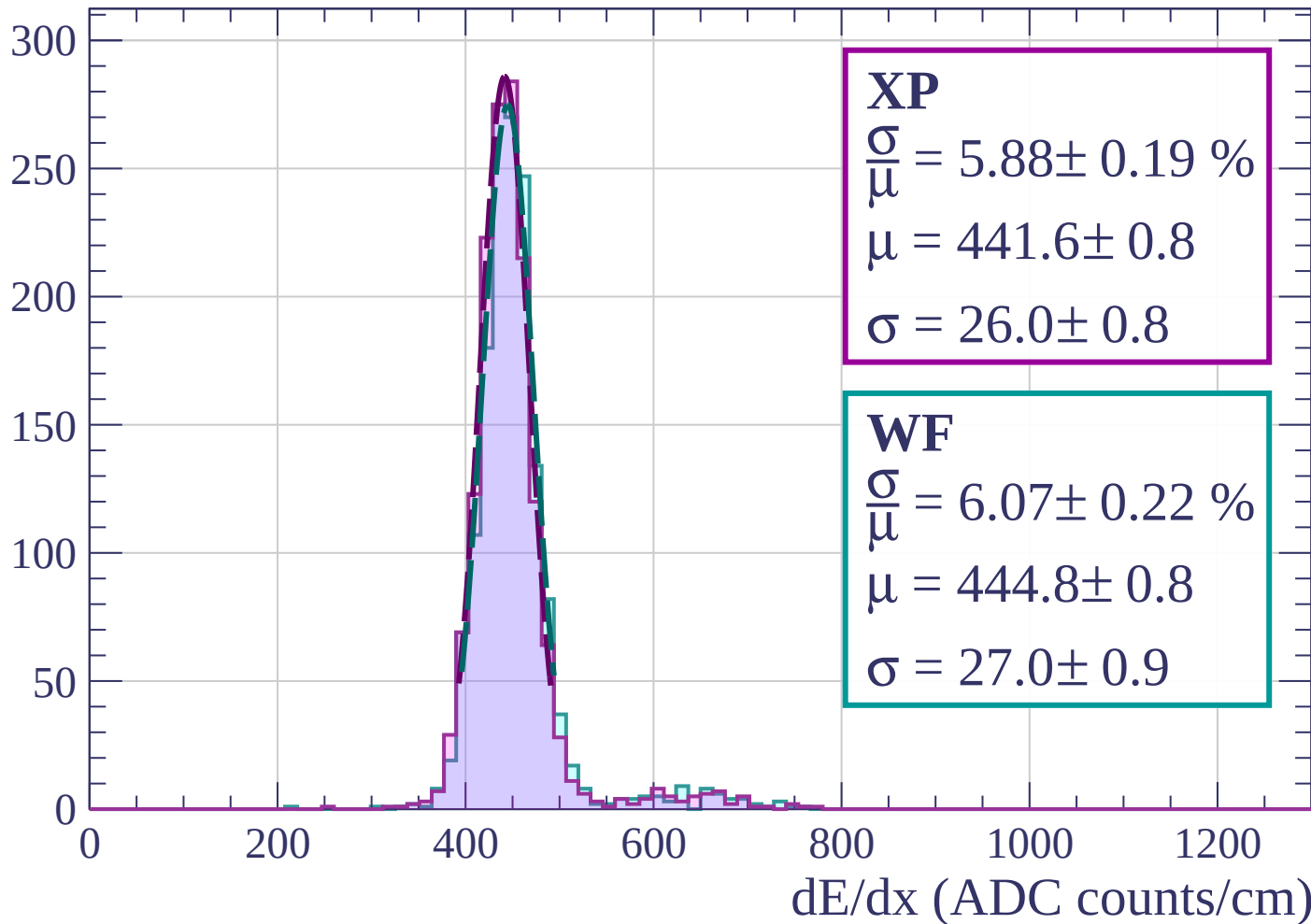


Energy loss | $880 < p < 960$



Energy loss | $960 < p < 1040$

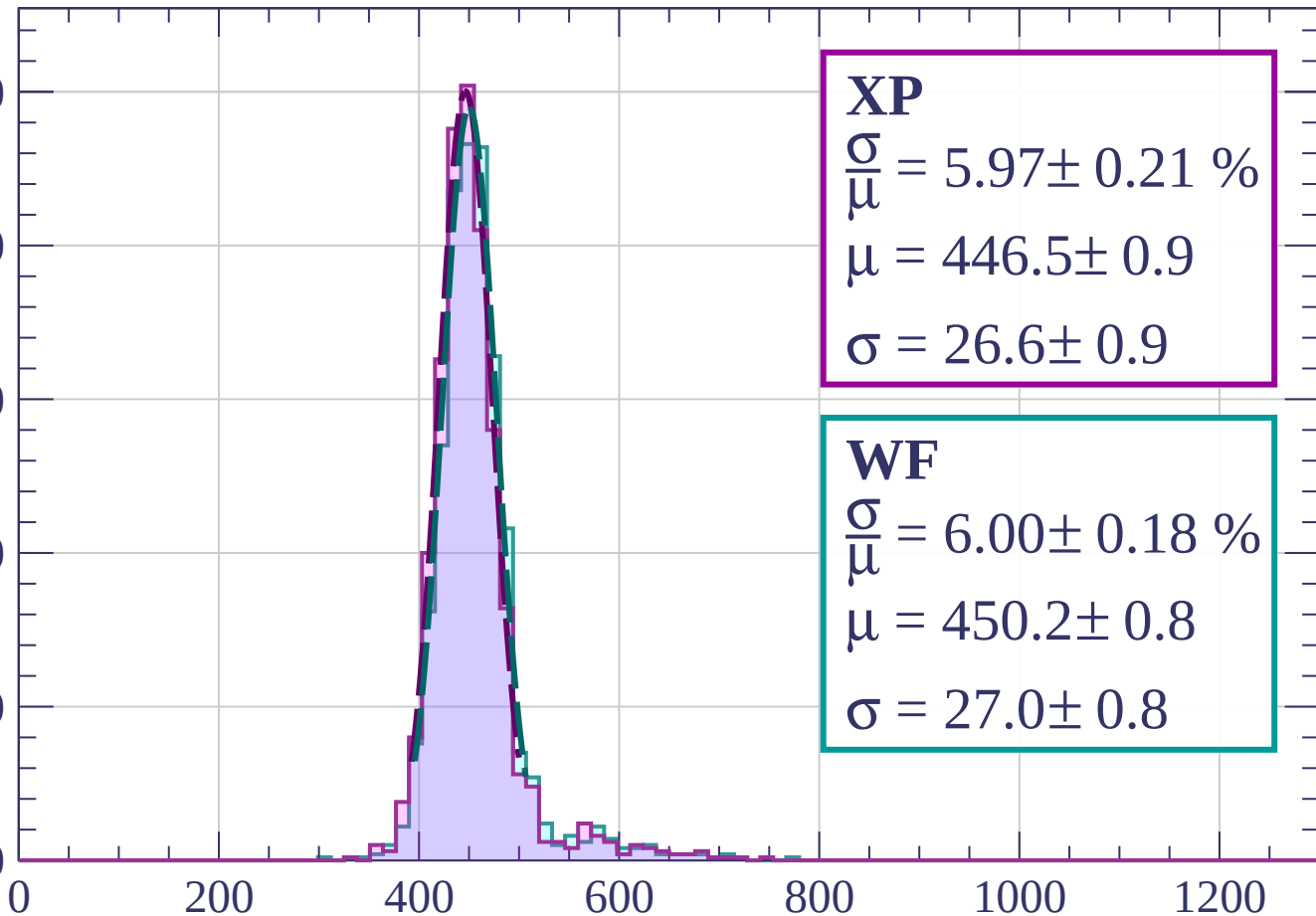
Count



Energy loss | $1040 < p < 1120$

Count

250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 5.97 \pm 0.21 \%$$

$$\mu = 446.5 \pm 0.9$$

$$\sigma = 26.6 \pm 0.9$$

WF

$$\frac{\sigma}{\mu} = 6.00 \pm 0.18 \%$$

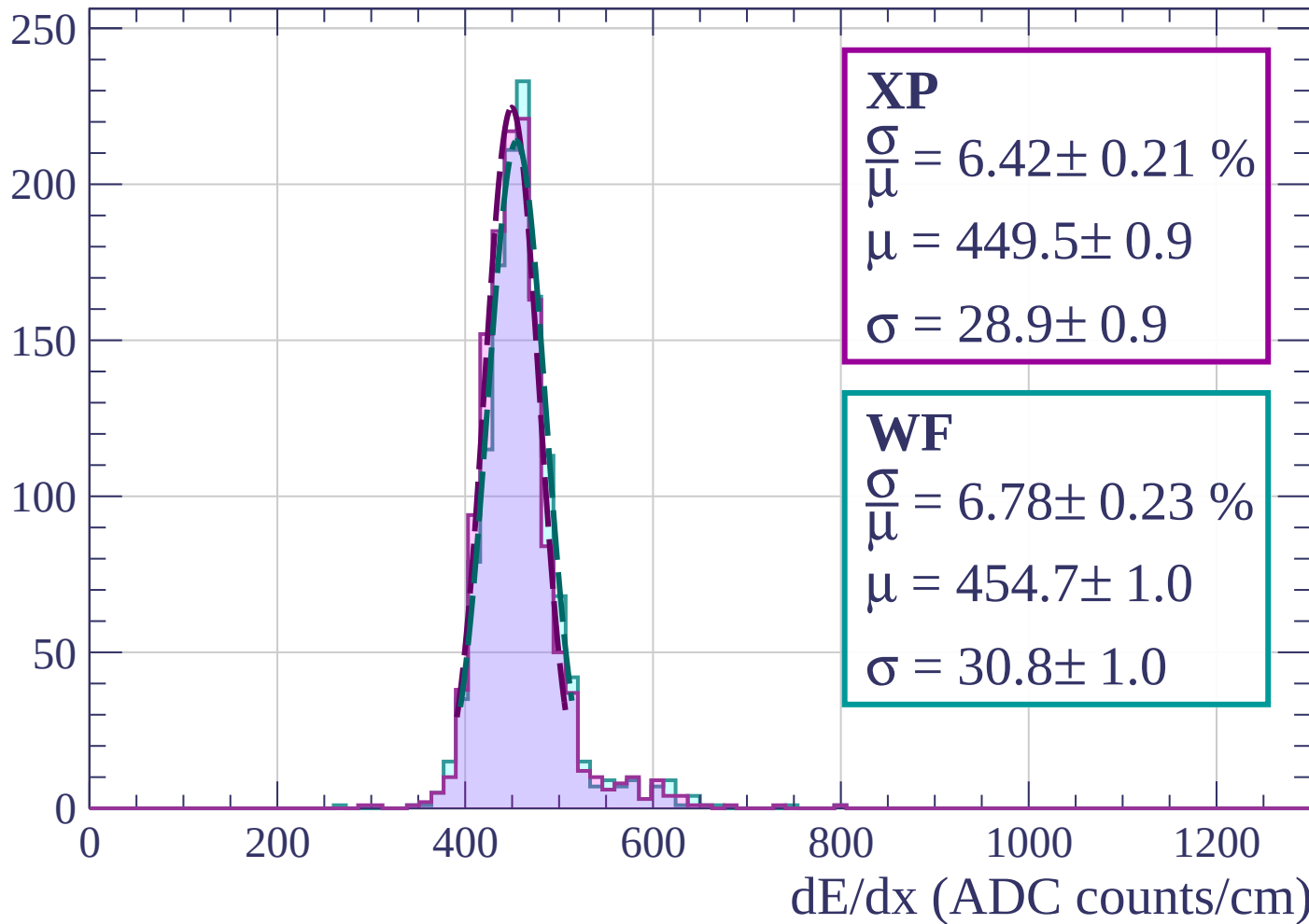
$$\mu = 450.2 \pm 0.8$$

$$\sigma = 27.0 \pm 0.8$$

dE/dx (ADC counts/cm)

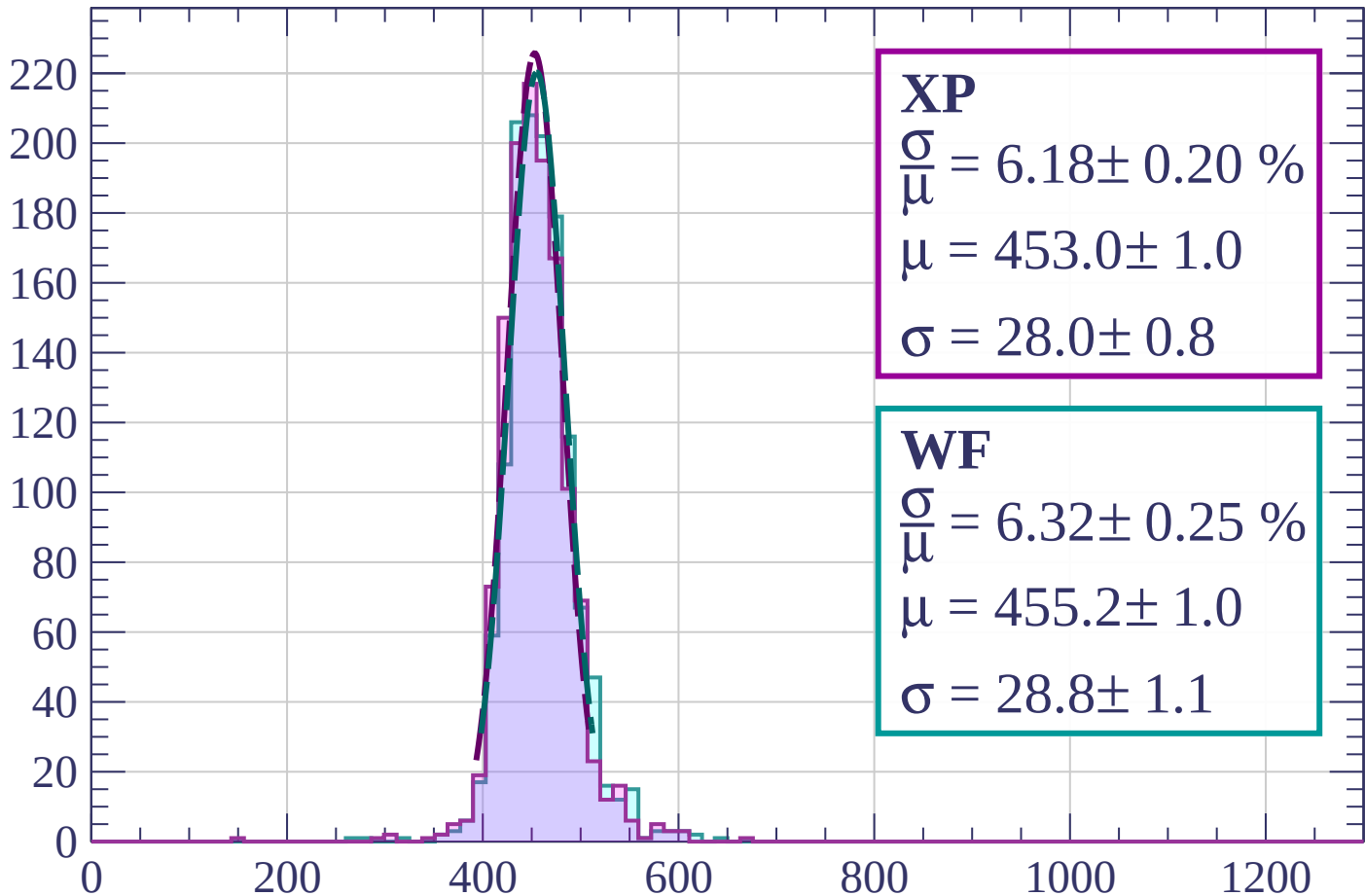
Energy loss | $1120 < p < 1200$

Count



Energy loss | $1200 < p < 1280$

Count



XP

$$\frac{\sigma}{\mu} = 6.18 \pm 0.20 \%$$

$$\mu = 453.0 \pm 1.0$$

$$\sigma = 28.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 6.32 \pm 0.25 \%$$

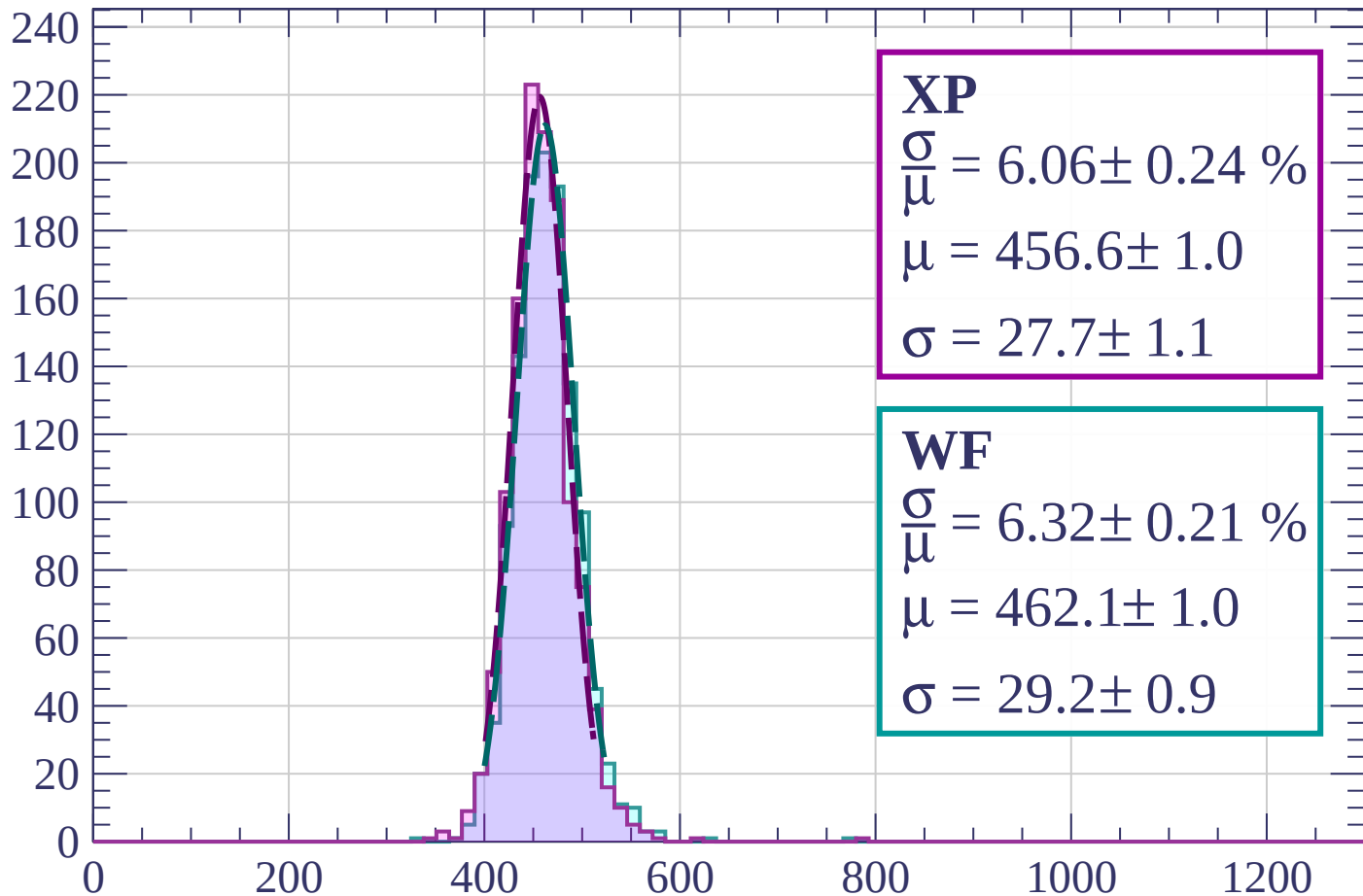
$$\mu = 455.2 \pm 1.0$$

$$\sigma = 28.8 \pm 1.1$$

dE/dx (ADC counts/cm)

Energy loss | $1280 < p < 1360$

Count



XP

$$\frac{\sigma}{\mu} = 6.06 \pm 0.24 \%$$

$$\mu = 456.6 \pm 1.0$$

$$\sigma = 27.7 \pm 1.1$$

WF

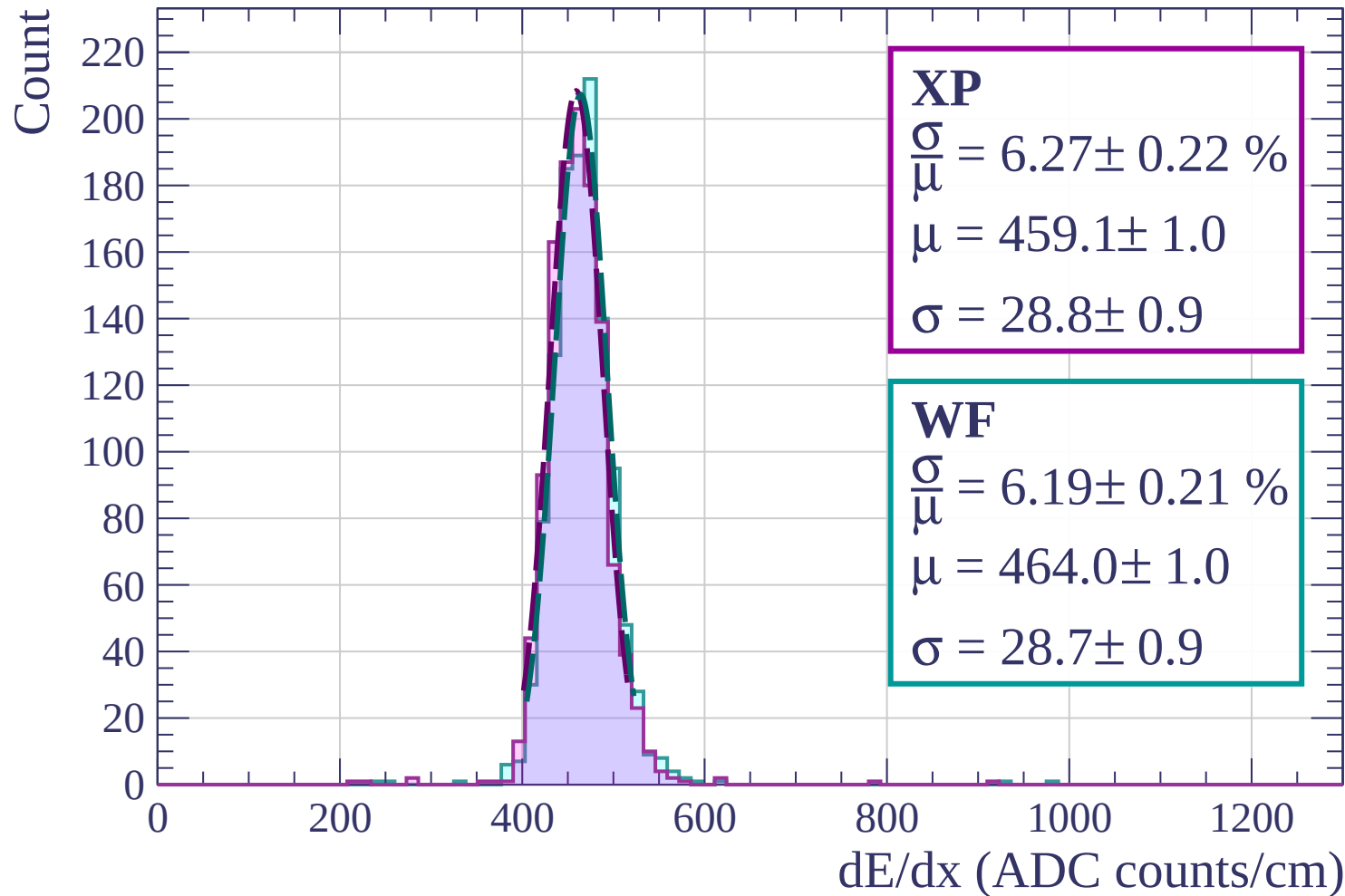
$$\frac{\sigma}{\mu} = 6.32 \pm 0.21 \%$$

$$\mu = 462.1 \pm 1.0$$

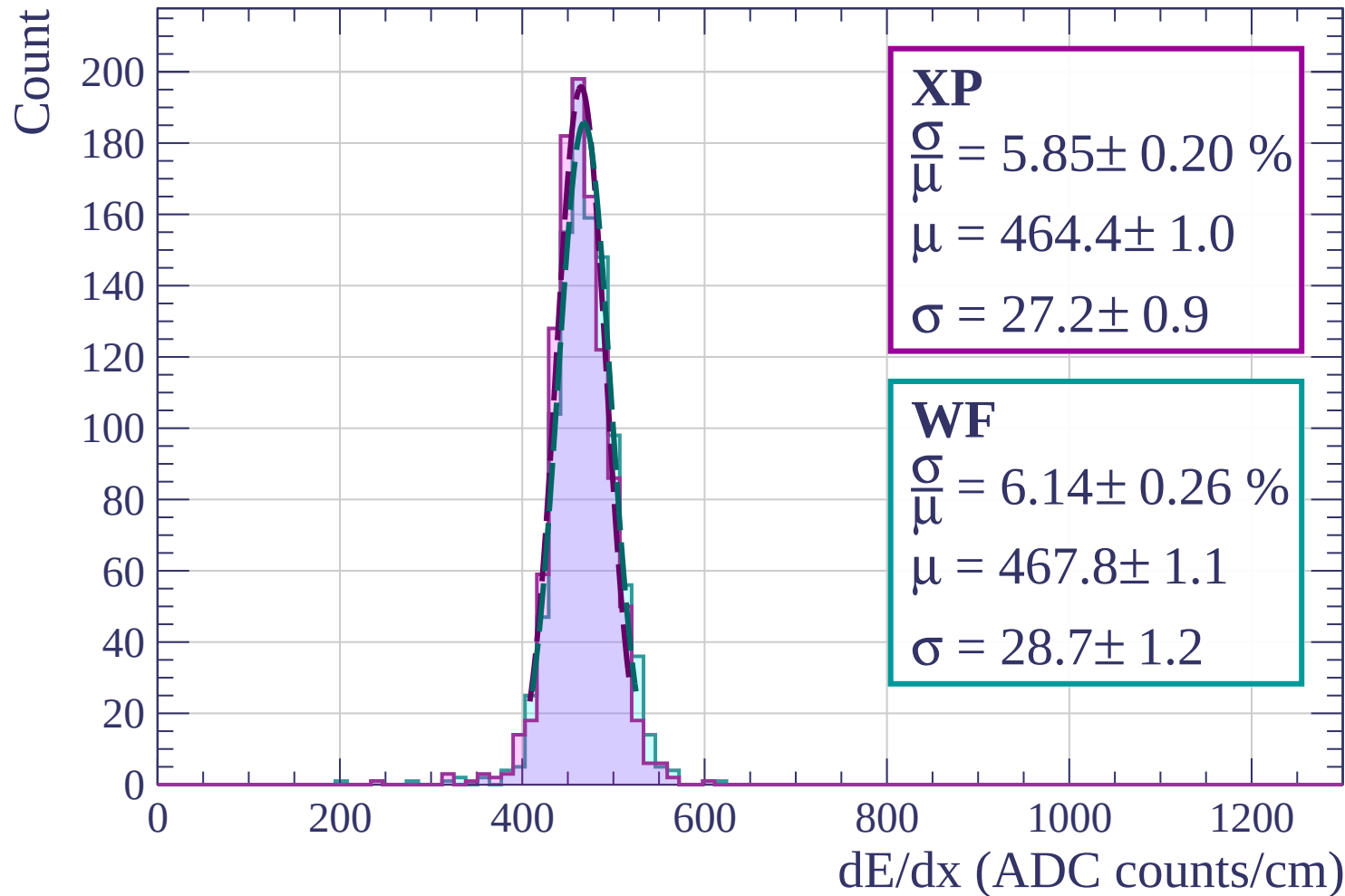
$$\sigma = 29.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1360 < p < 1440$

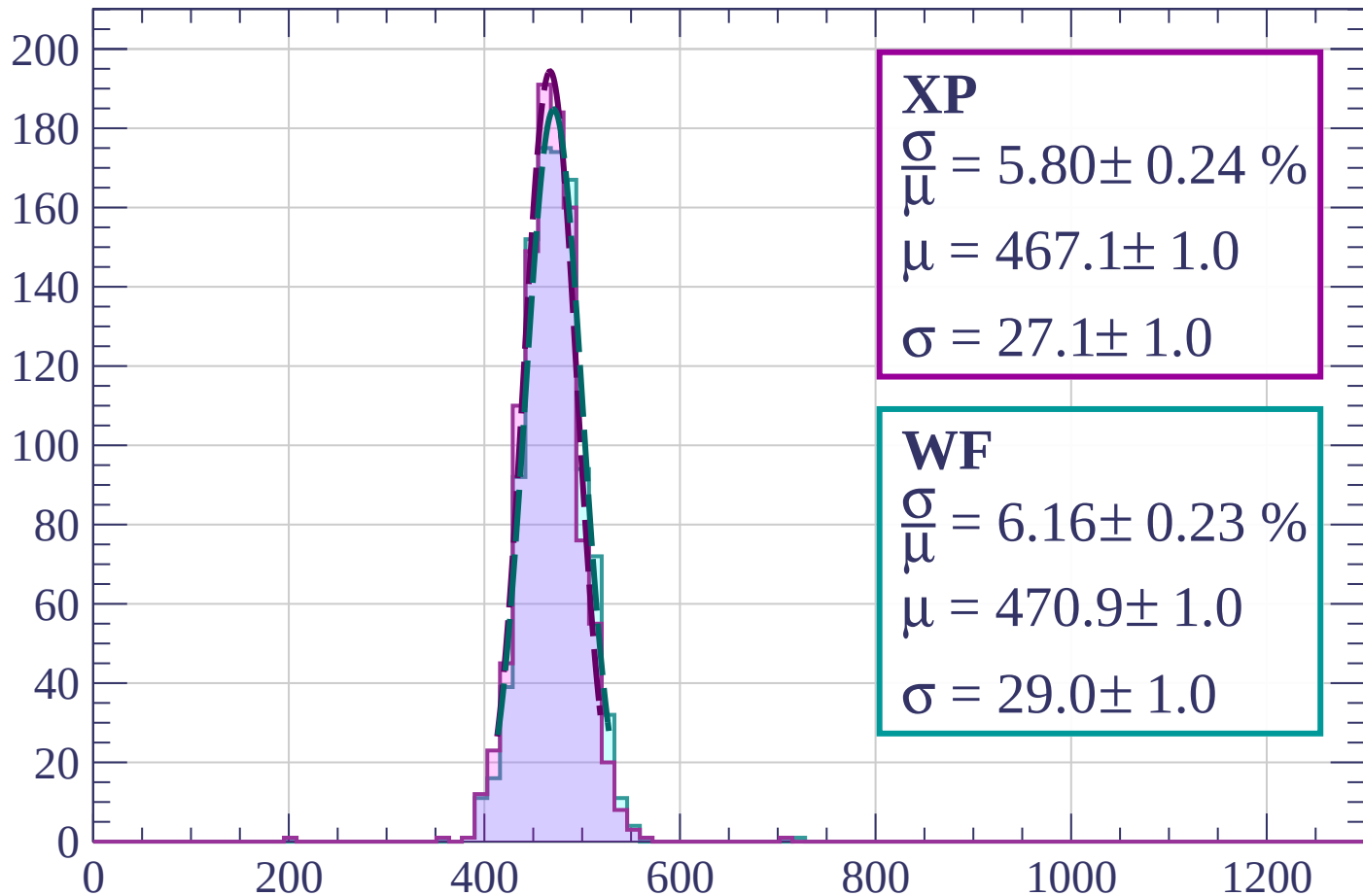


Energy loss | $1440 < p < 1520$



Energy loss | $1520 < p < 1600$

Count



XP

$$\frac{\sigma}{\mu} = 5.80 \pm 0.24 \%$$

$$\mu = 467.1 \pm 1.0$$

$$\sigma = 27.1 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.16 \pm 0.23 \%$$

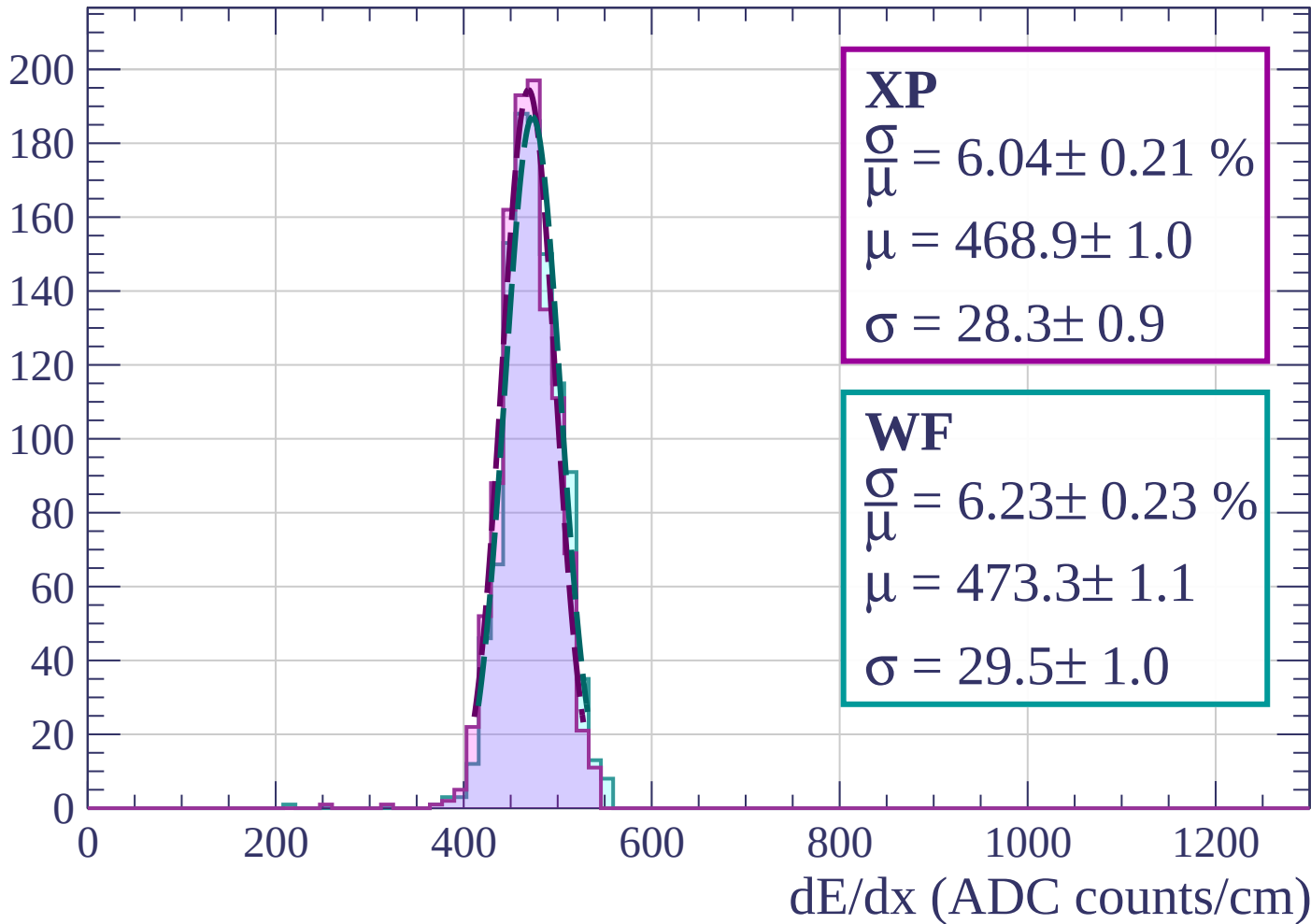
$$\mu = 470.9 \pm 1.0$$

$$\sigma = 29.0 \pm 1.0$$

dE/dx (ADC counts/cm)

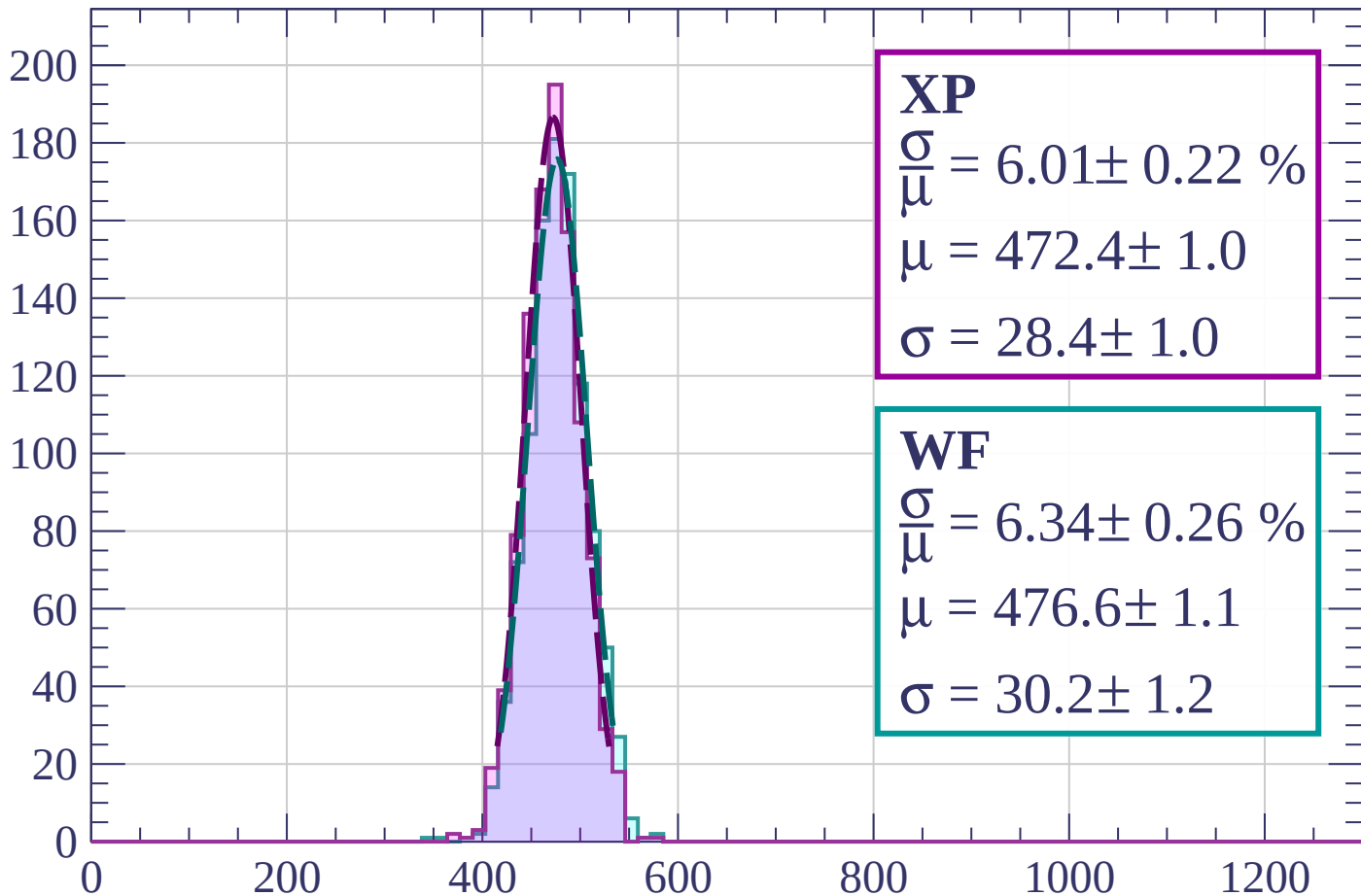
Energy loss | $1600 < p < 1680$

Count



Energy loss | $1680 < p < 1760$

Count



XP

$$\frac{\sigma}{\mu} = 6.01 \pm 0.22 \%$$

$$\mu = 472.4 \pm 1.0$$

$$\sigma = 28.4 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 6.34 \pm 0.26 \%$$

$$\mu = 476.6 \pm 1.1$$

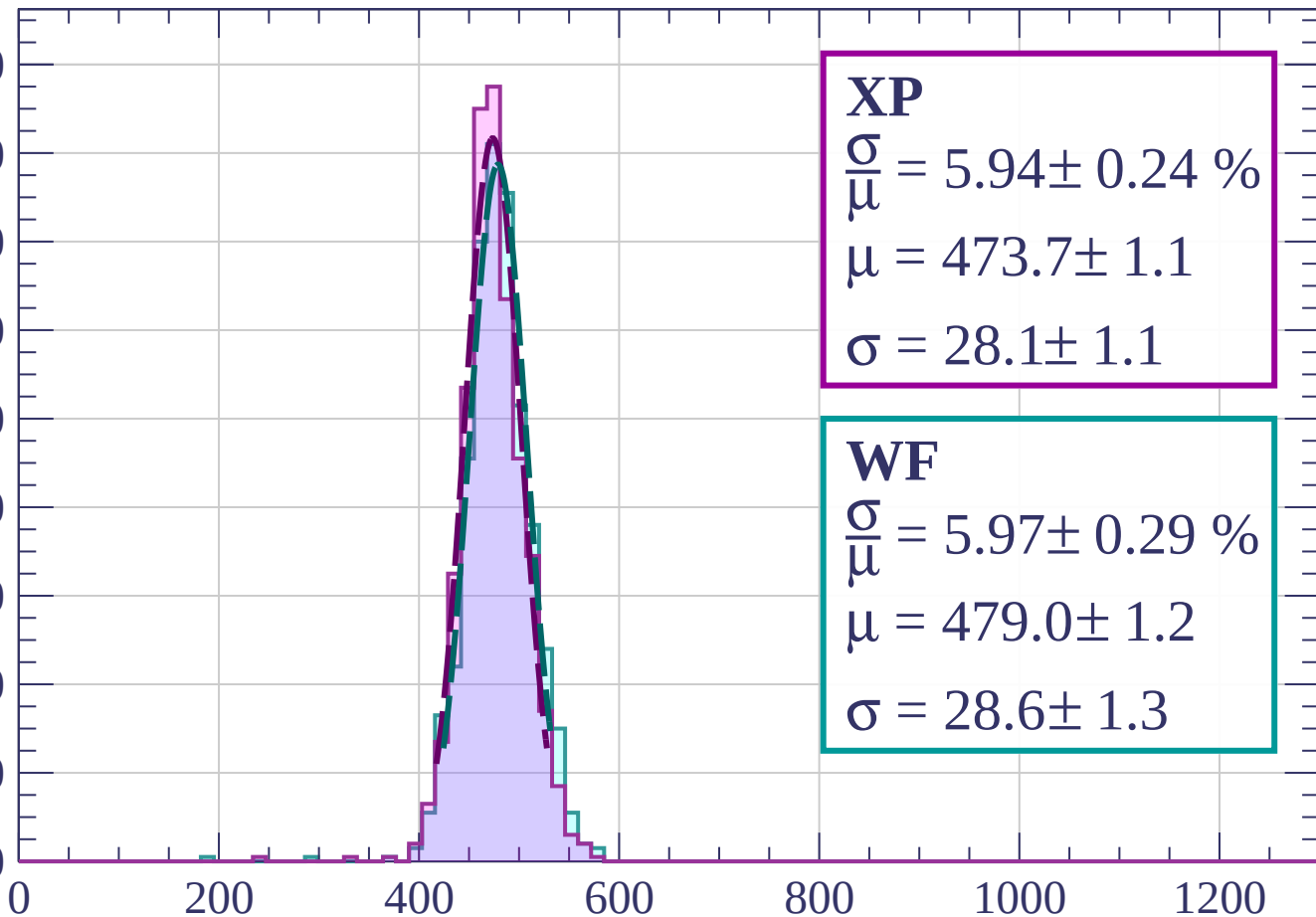
$$\sigma = 30.2 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $1760 < p < 1840$

Count

180
160
140
120
100
80
60
40
20
0



XP

$$\frac{\sigma}{\mu} = 5.94 \pm 0.24 \%$$

$$\mu = 473.7 \pm 1.1$$

$$\sigma = 28.1 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 5.97 \pm 0.29 \%$$

$$\mu = 479.0 \pm 1.2$$

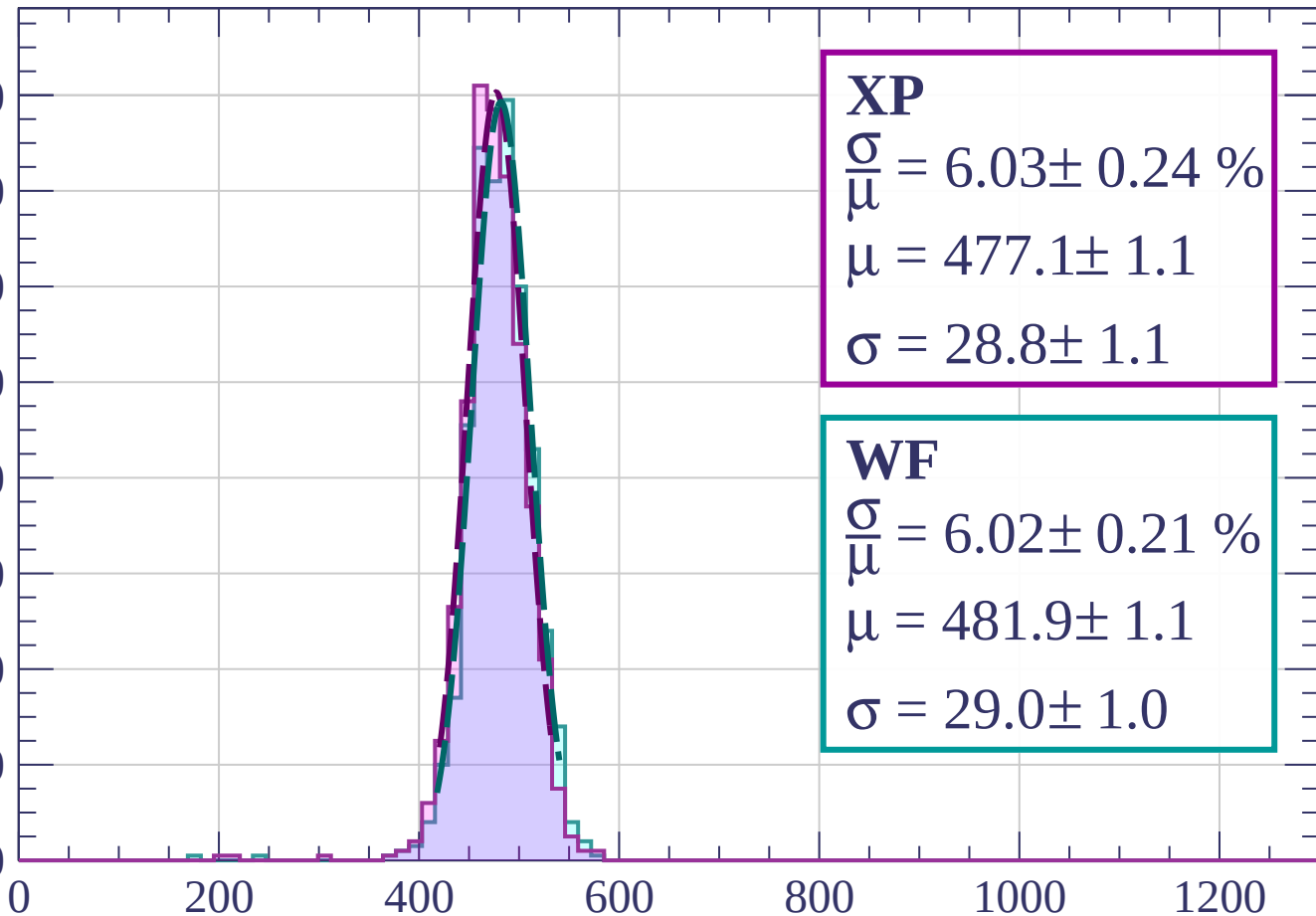
$$\sigma = 28.6 \pm 1.3$$

dE/dx (ADC counts/cm)

Energy loss | $1840 < p < 1920$

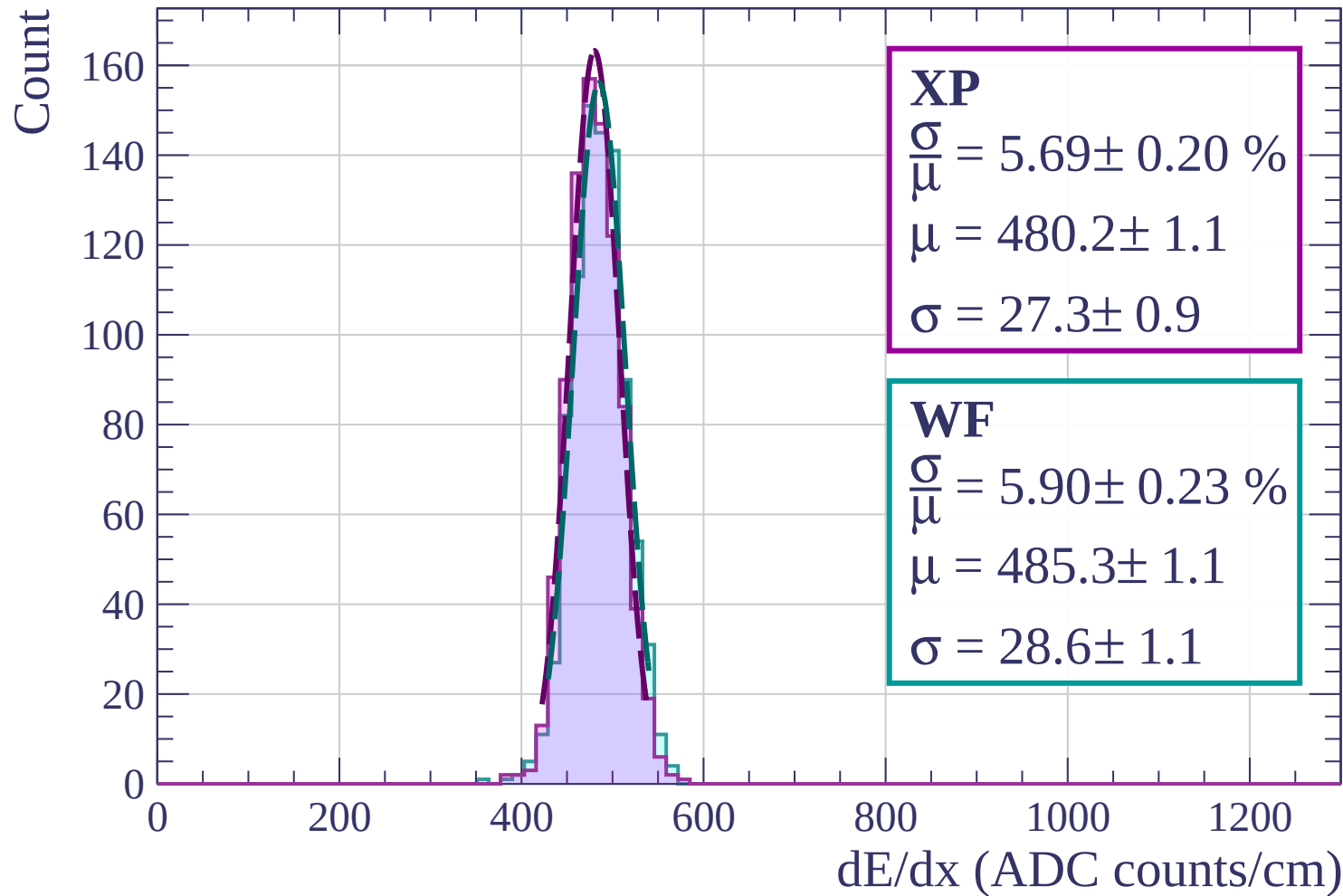
Count

160
140
120
100
80
60
40
20
0



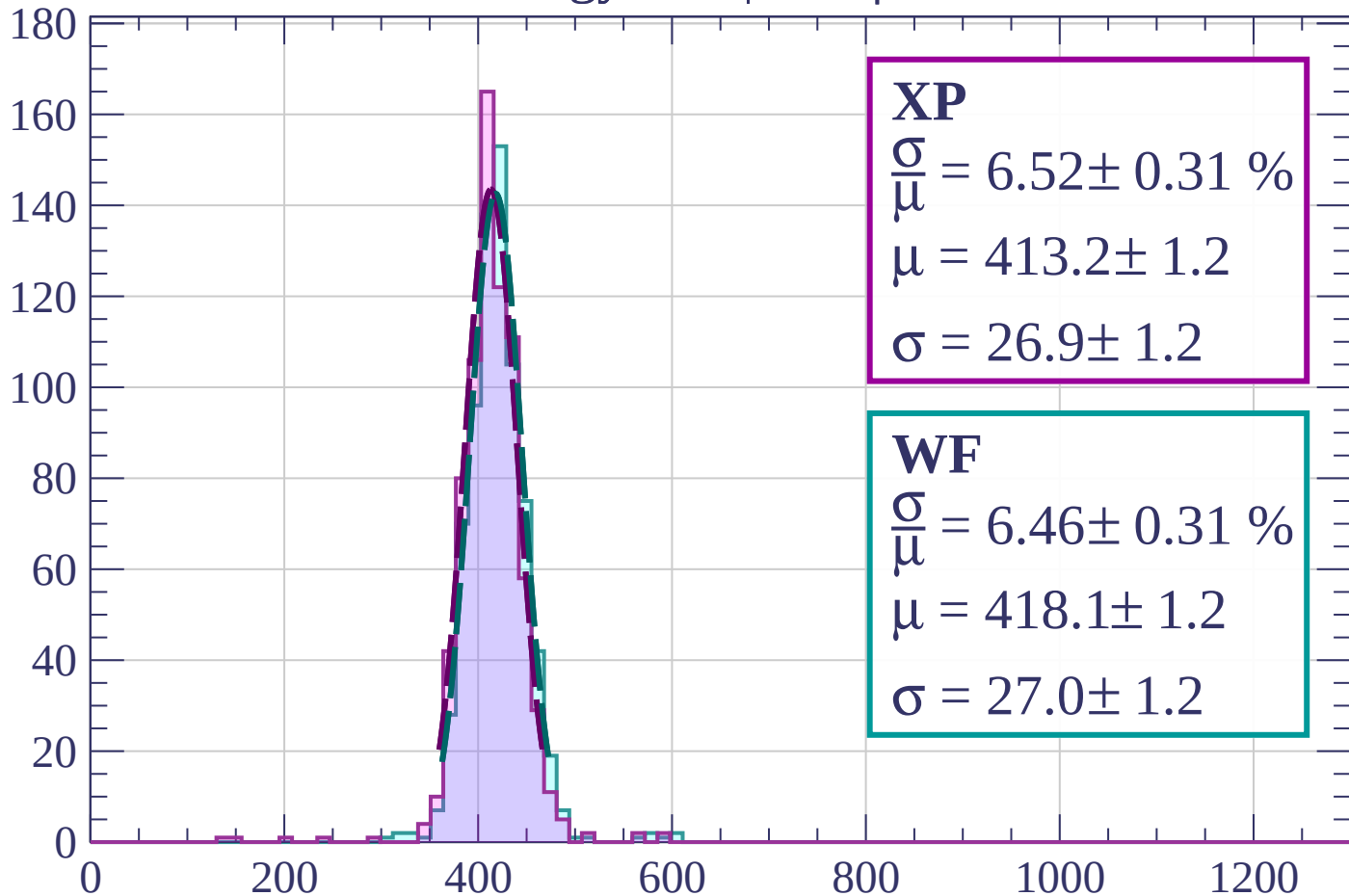
dE/dx (ADC counts/cm)

Energy loss | $1920 < p < 2000$



Energy loss | $0 < \phi < 3$

Count



XP

$$\frac{\sigma}{\mu} = 6.52 \pm 0.31 \%$$

$$\mu = 413.2 \pm 1.2$$

$$\sigma = 26.9 \pm 1.2$$

WF

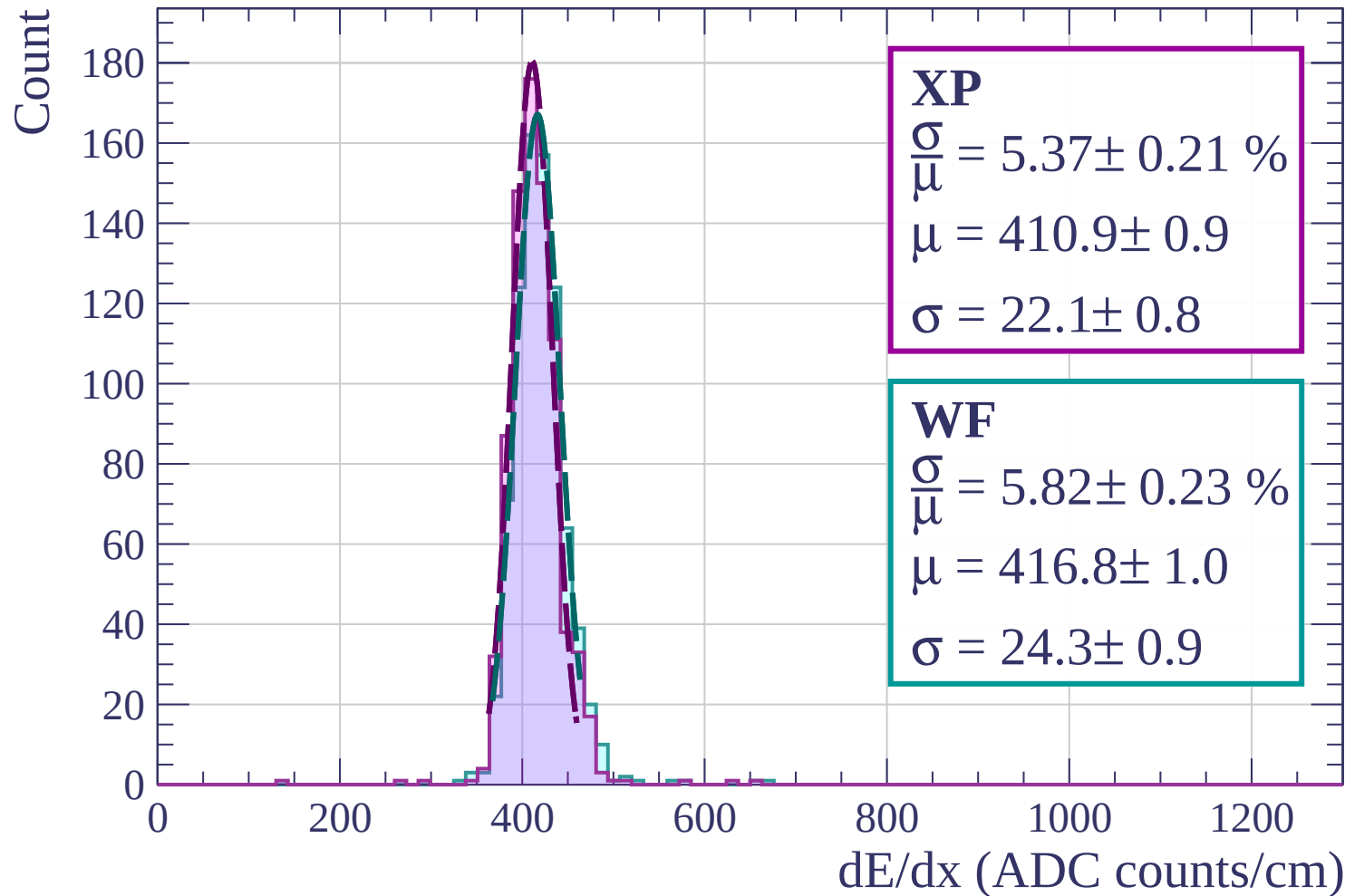
$$\frac{\sigma}{\mu} = 6.46 \pm 0.31 \%$$

$$\mu = 418.1 \pm 1.2$$

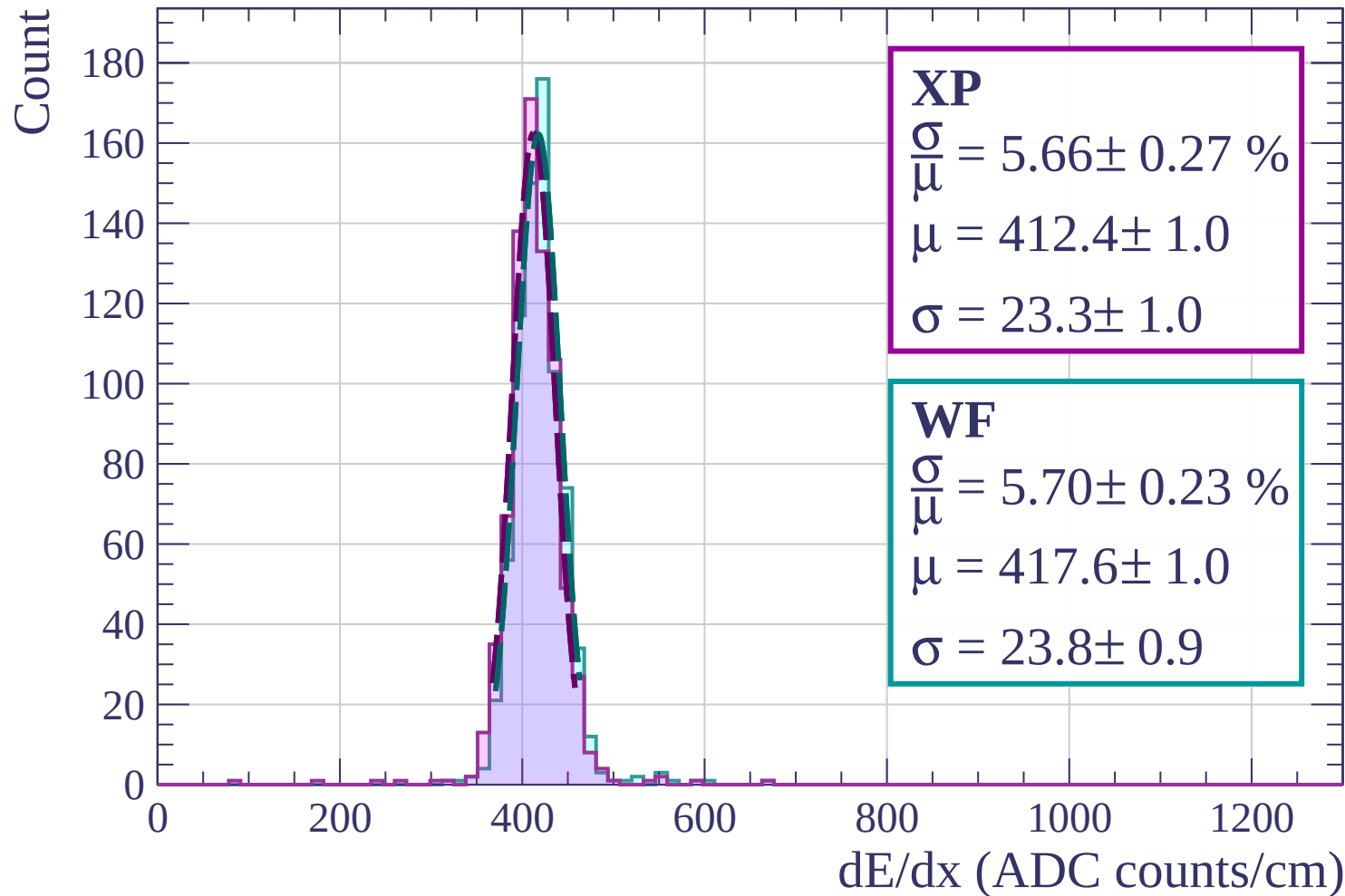
$$\sigma = 27.0 \pm 1.2$$

dE/dx (ADC counts/cm)

Energy loss | $3 < \phi < 6$

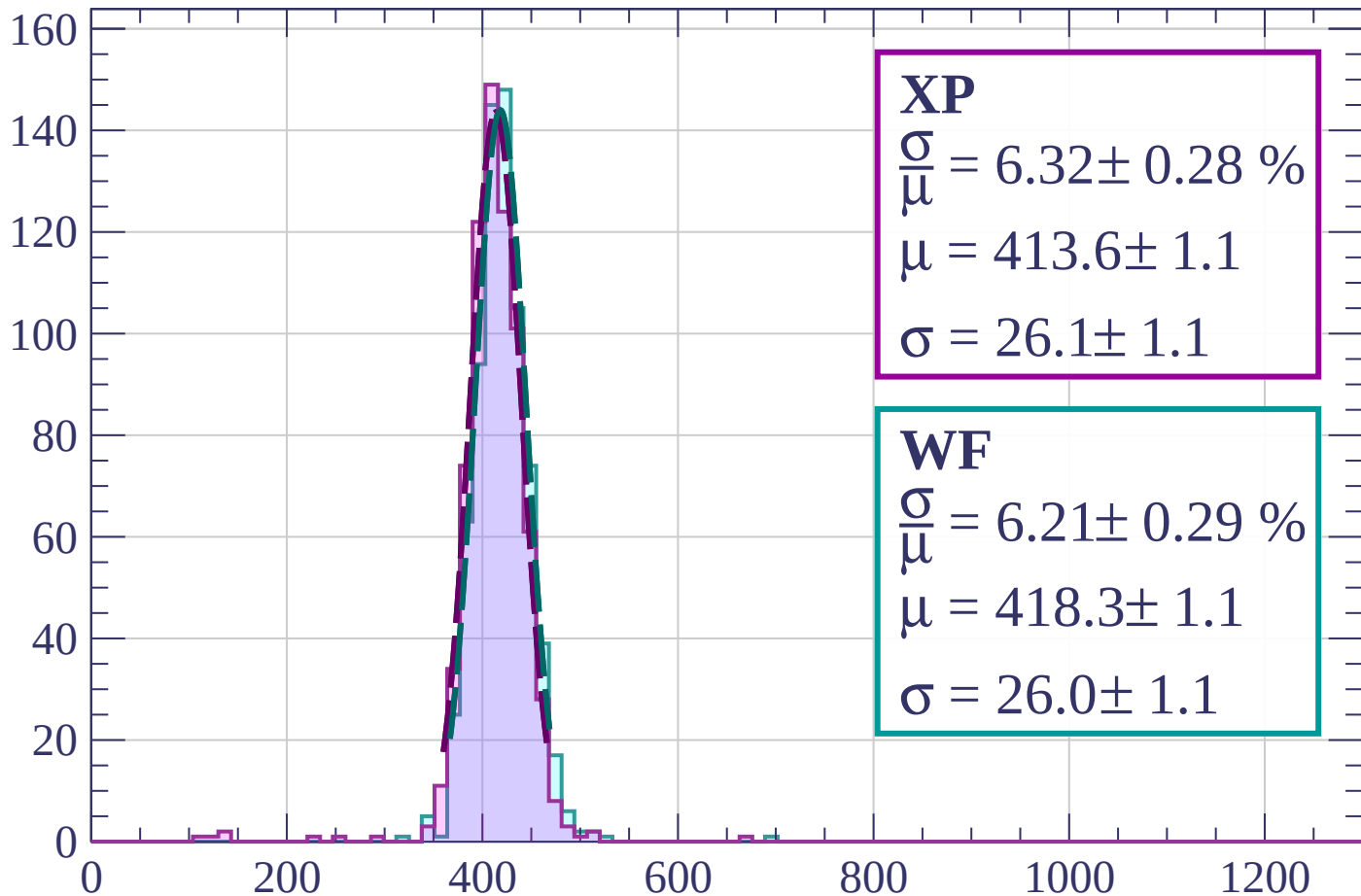


Energy loss | $6 < \phi < 9$



Energy loss | $9 < \phi < 12$

Count



XP

$$\frac{\sigma}{\mu} = 6.32 \pm 0.28 \%$$

$$\mu = 413.6 \pm 1.1$$

$$\sigma = 26.1 \pm 1.1$$

WF

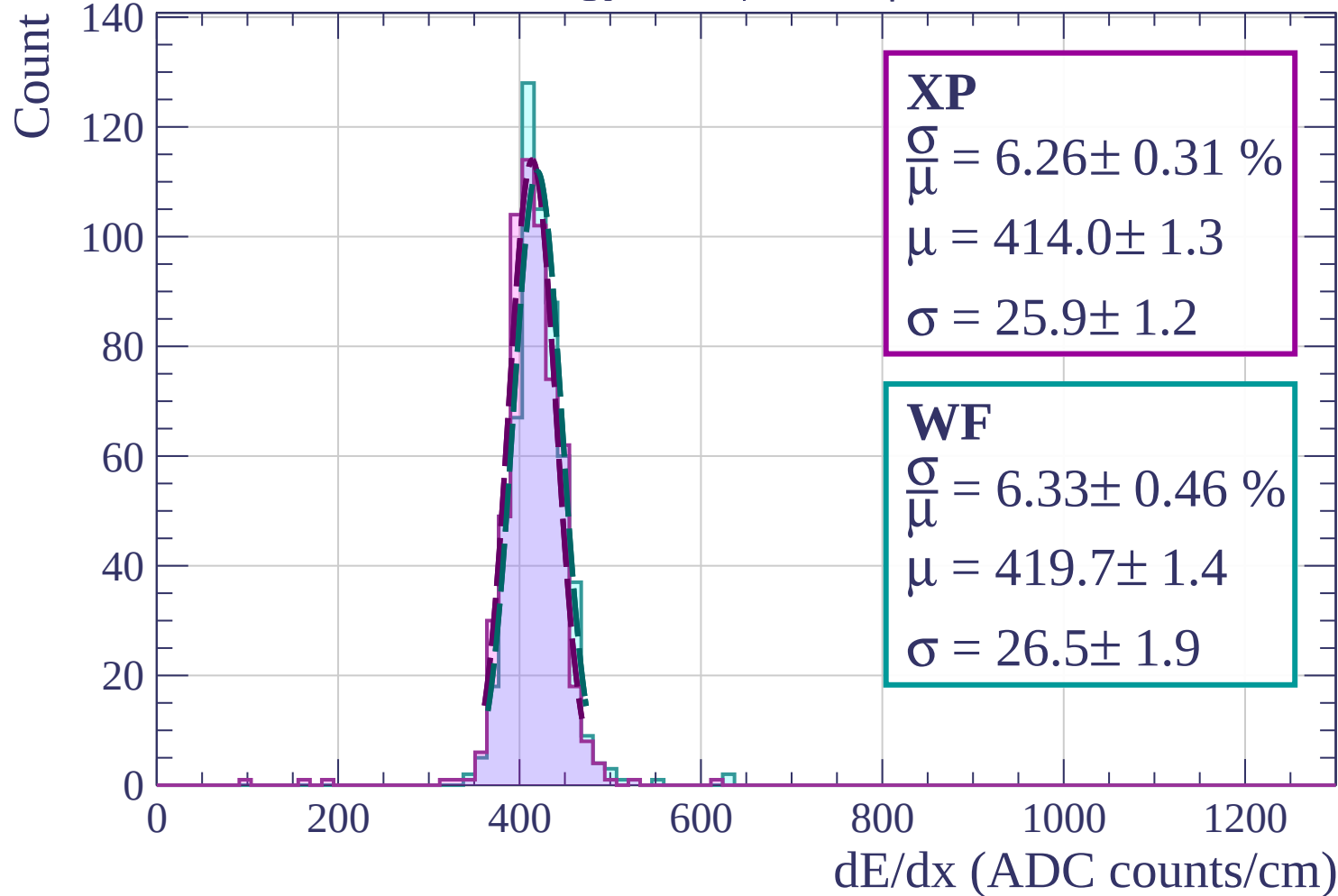
$$\frac{\sigma}{\mu} = 6.21 \pm 0.29 \%$$

$$\mu = 418.3 \pm 1.1$$

$$\sigma = 26.0 \pm 1.1$$

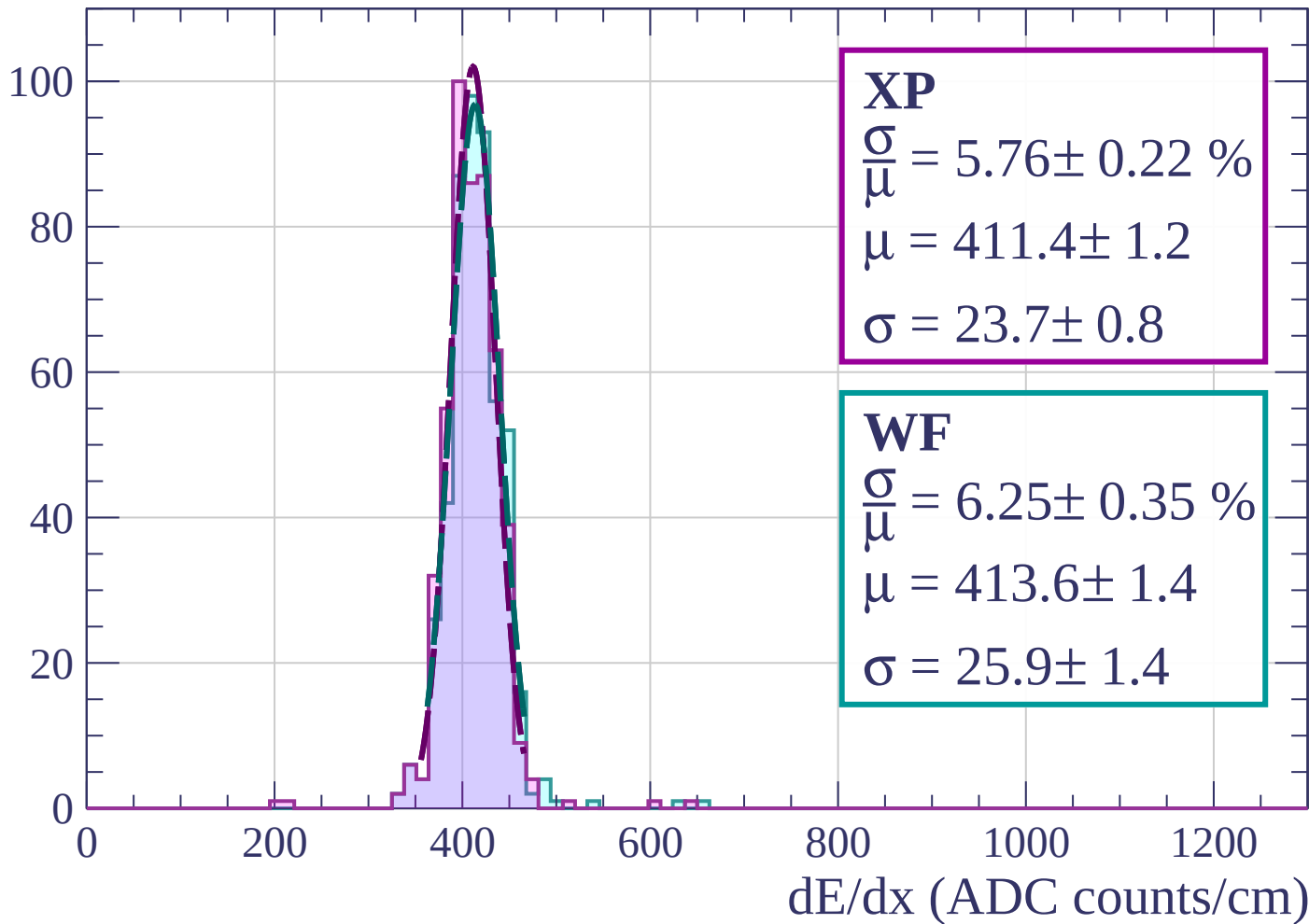
dE/dx (ADC counts/cm)

Energy loss | $12 < \phi < 15$



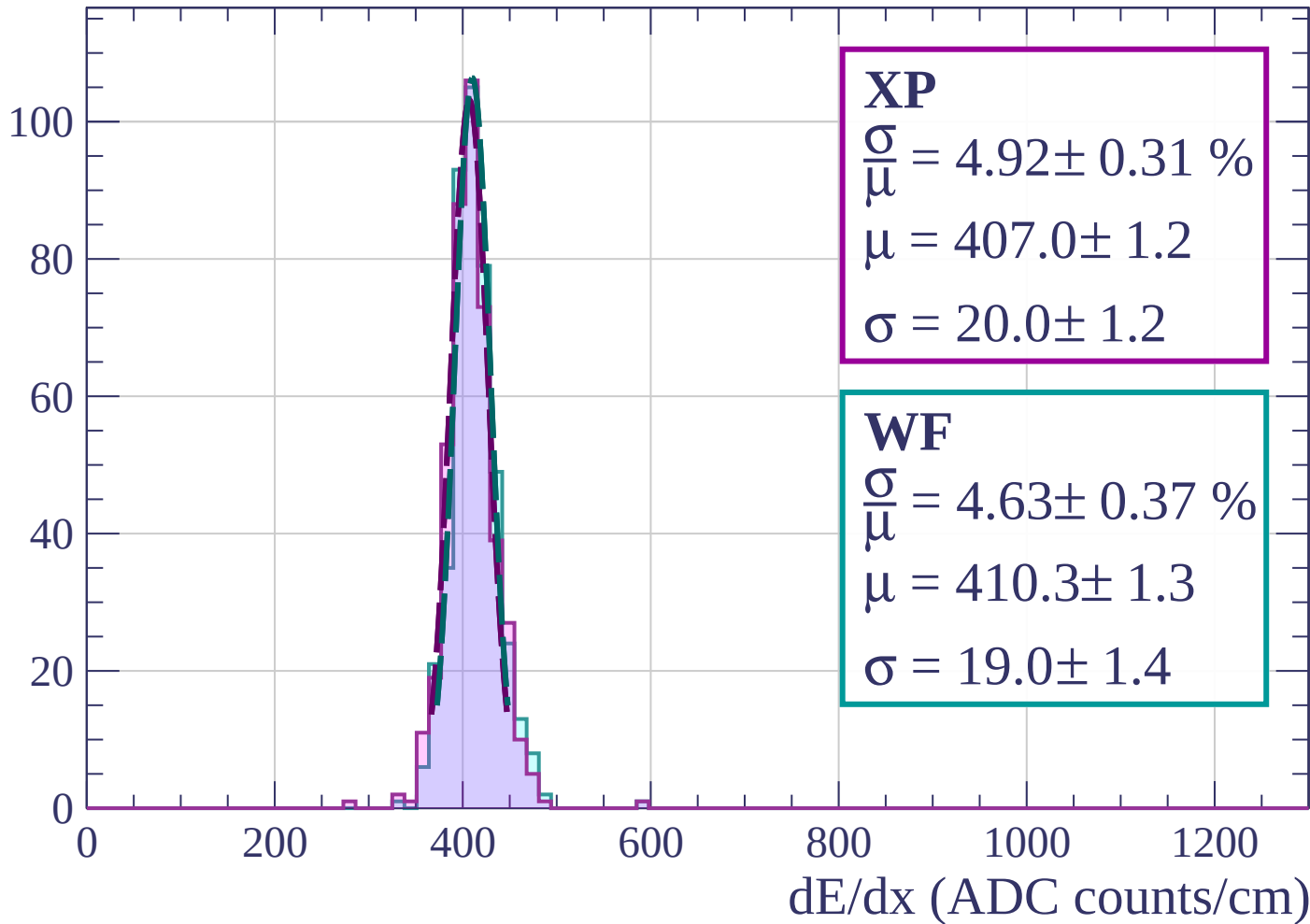
Energy loss | $15 < \phi < 18$

Count



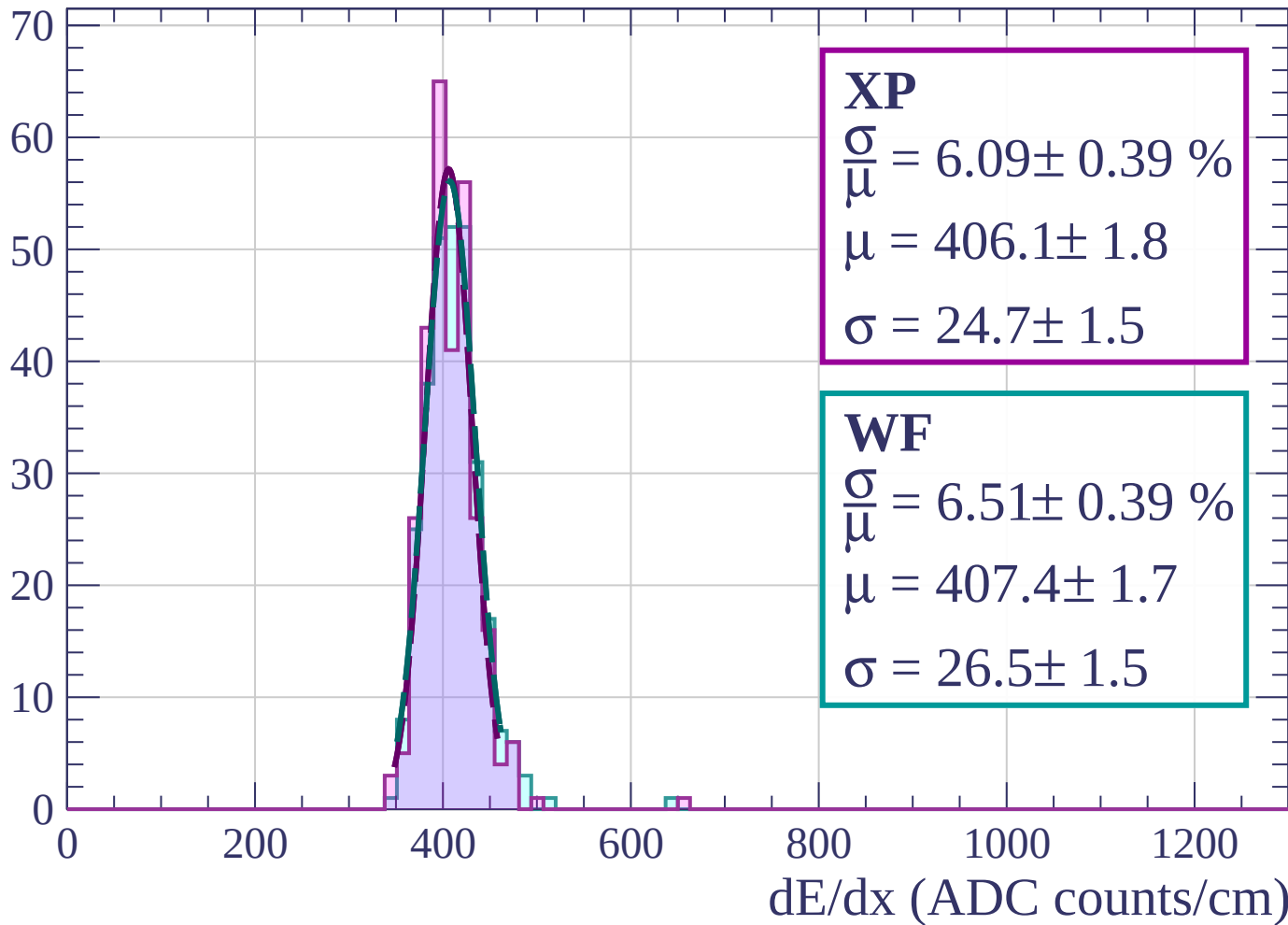
Energy loss | $18 < \phi < 21$

Count



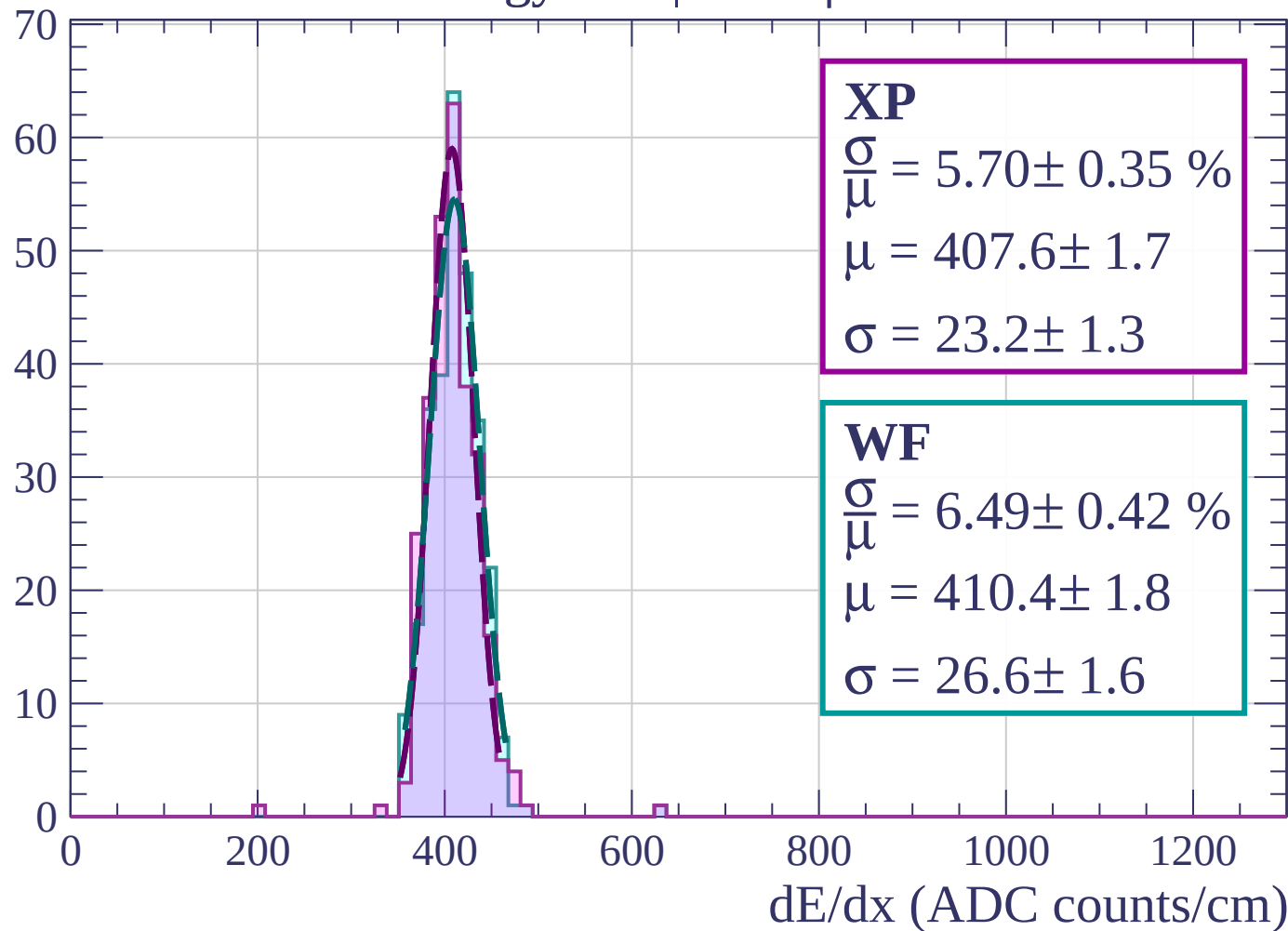
Energy loss | $21 < \phi < 24$

Count



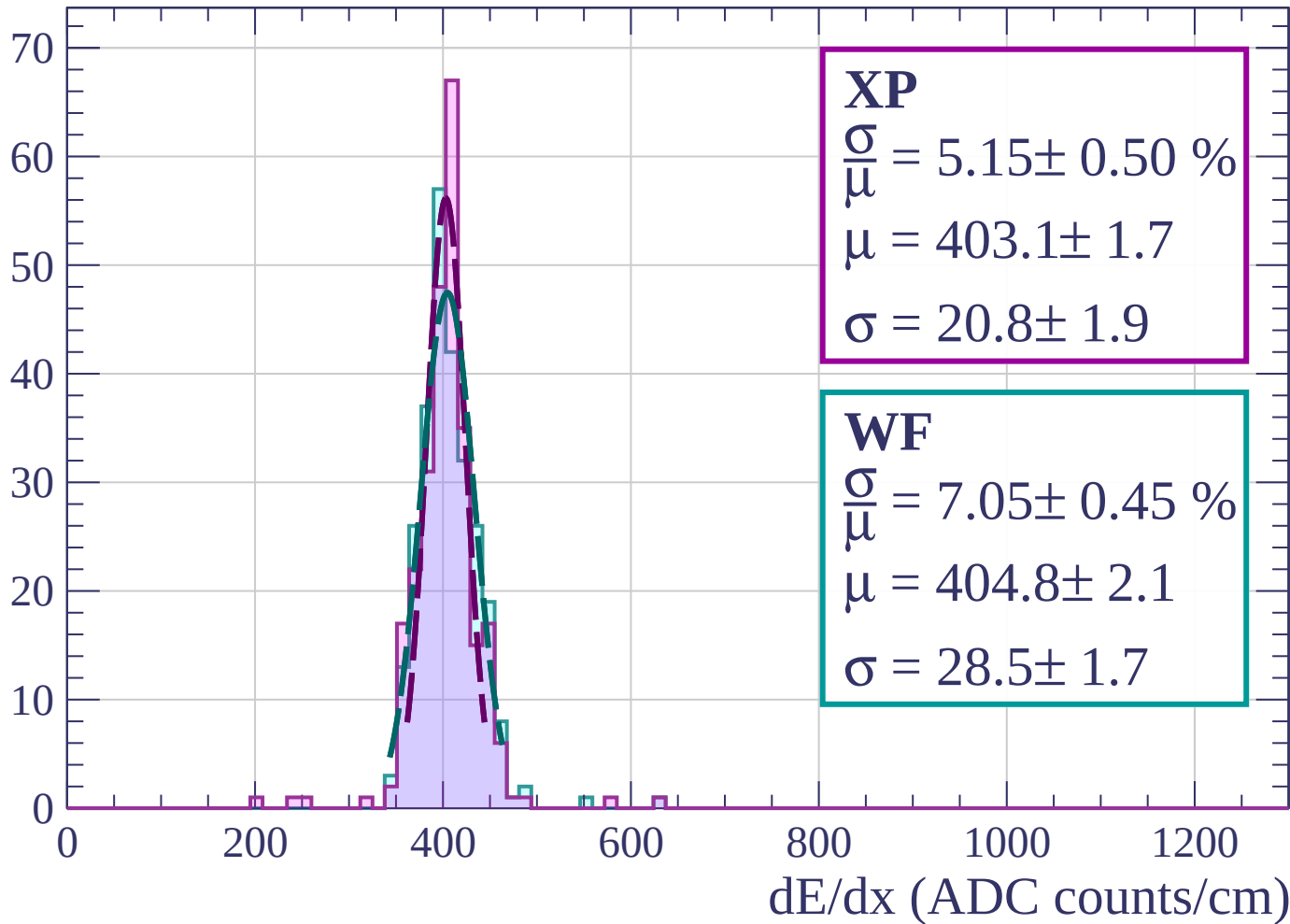
Energy loss | $24 < \phi < 27$

Count



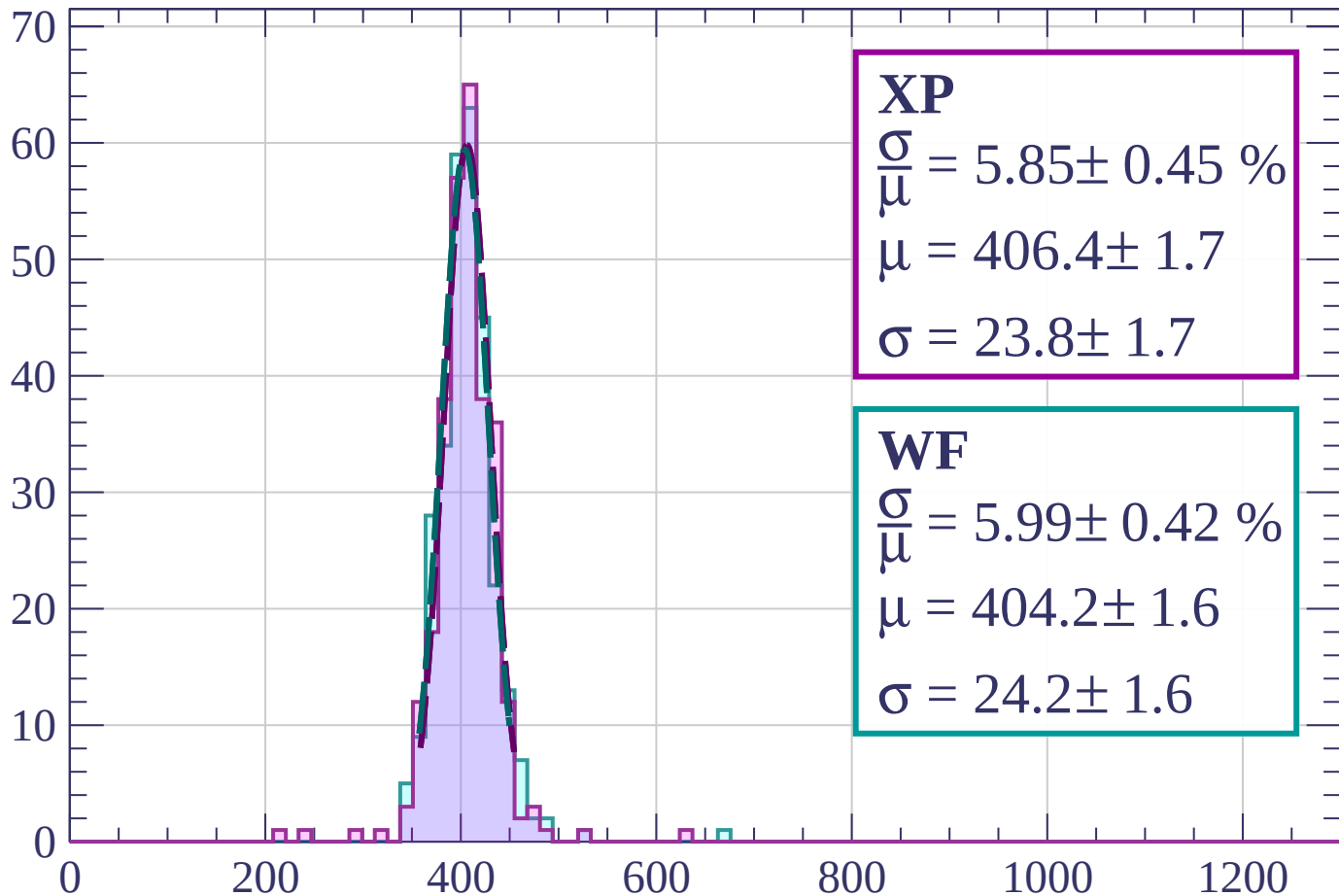
Energy loss | $27 < \phi < 30$

Count



Energy loss | $30 < \phi < 33$

Count



XP

$$\frac{\sigma}{\mu} = 5.85 \pm 0.45 \%$$

$$\mu = 406.4 \pm 1.7$$

$$\sigma = 23.8 \pm 1.7$$

WF

$$\frac{\sigma}{\mu} = 5.99 \pm 0.42 \%$$

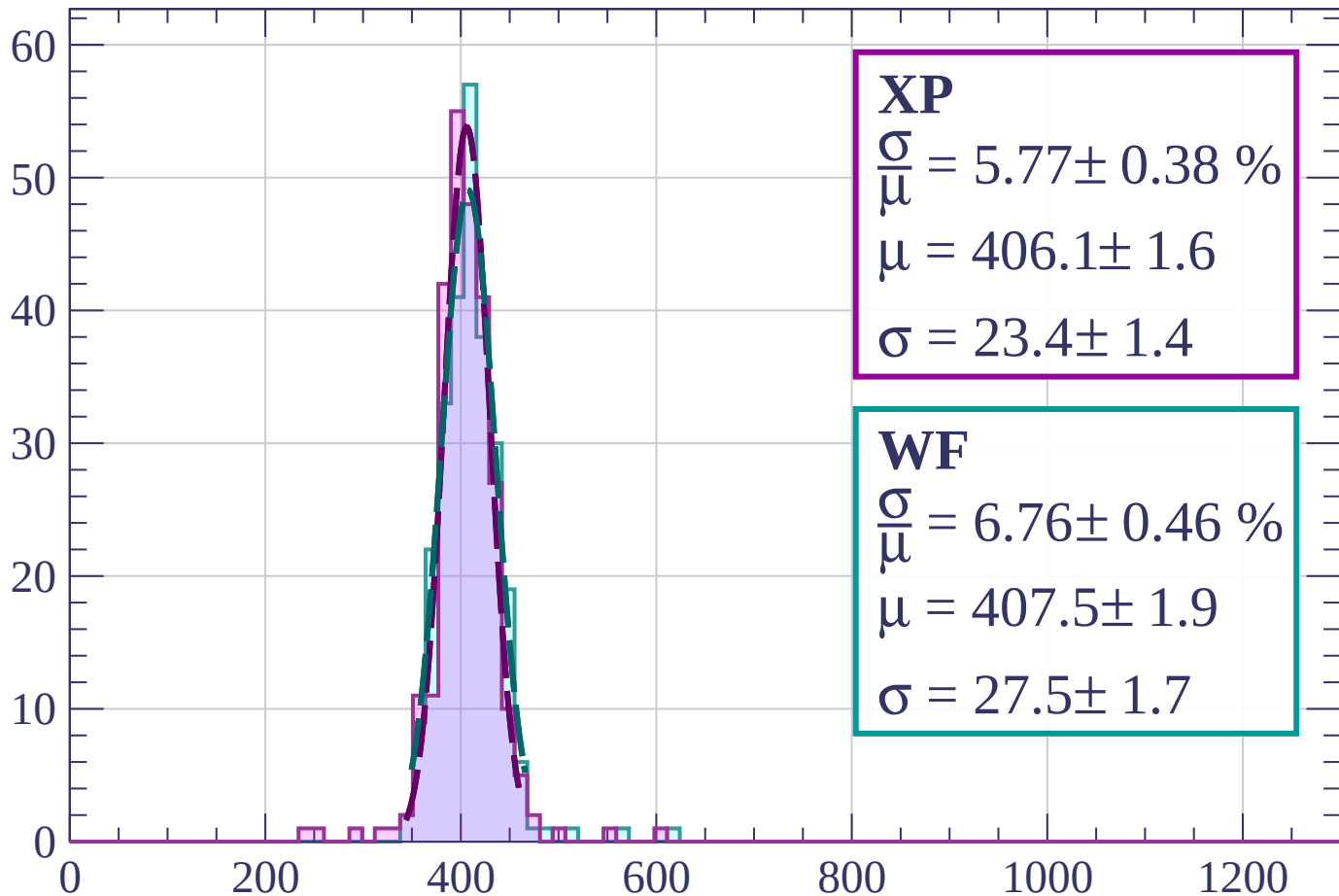
$$\mu = 404.2 \pm 1.6$$

$$\sigma = 24.2 \pm 1.6$$

dE/dx (ADC counts/cm)

Energy loss | $33 < \phi < 36$

Count



XP

$$\frac{\sigma}{\mu} = 5.77 \pm 0.38 \%$$

$$\mu = 406.1 \pm 1.6$$

$$\sigma = 23.4 \pm 1.4$$

WF

$$\frac{\sigma}{\mu} = 6.76 \pm 0.46 \%$$

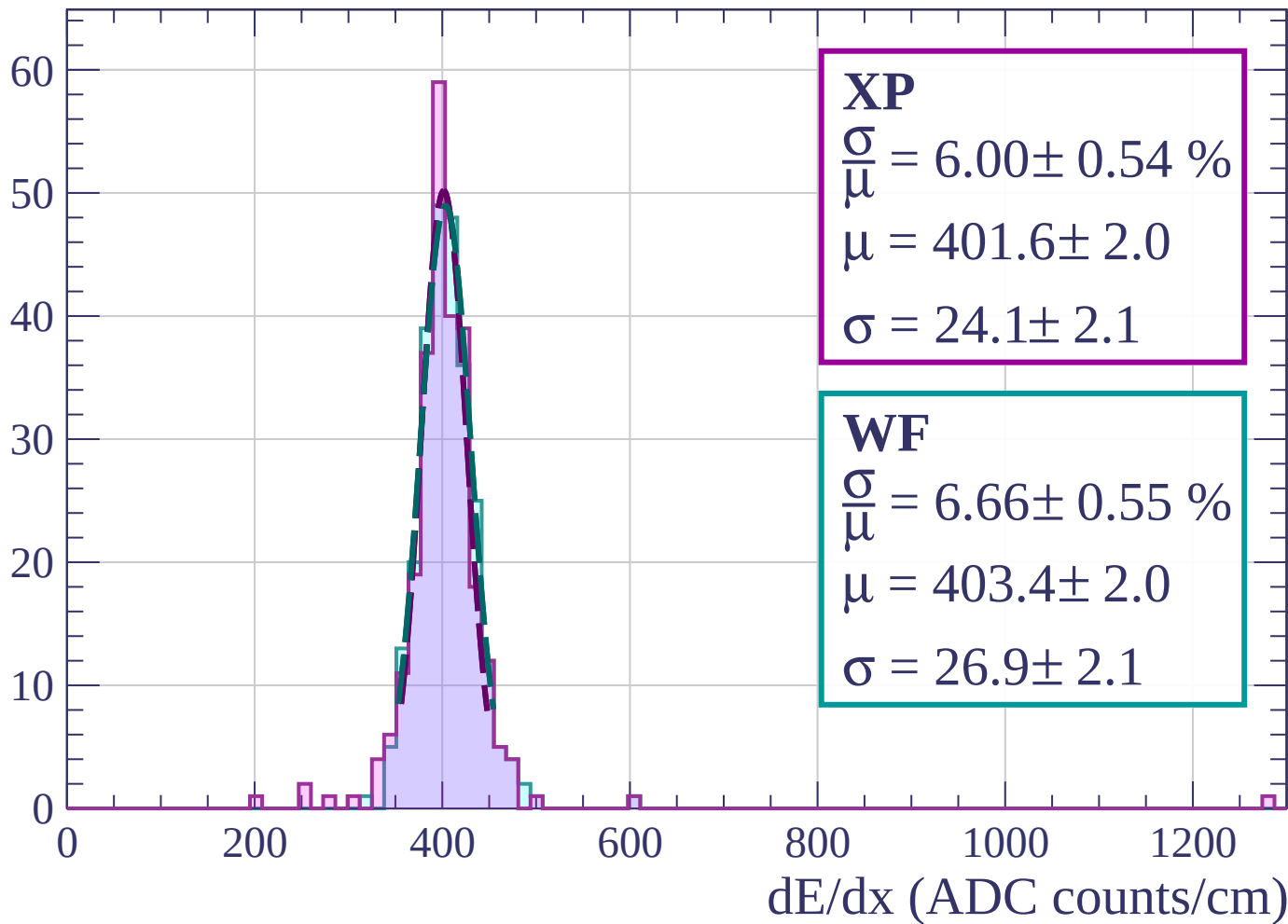
$$\mu = 407.5 \pm 1.9$$

$$\sigma = 27.5 \pm 1.7$$

dE/dx (ADC counts/cm)

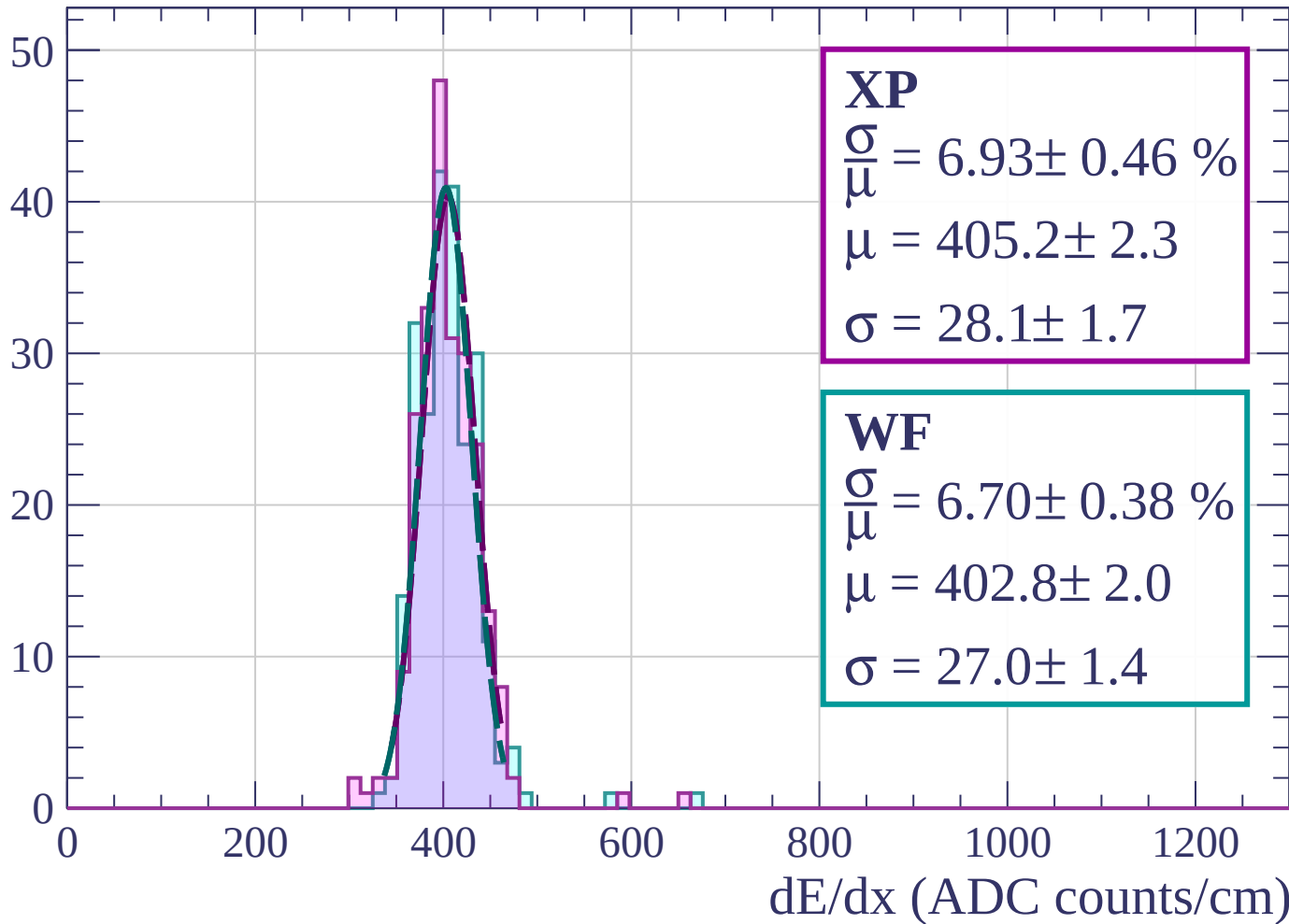
Energy loss | $36 < \phi < 39$

Count



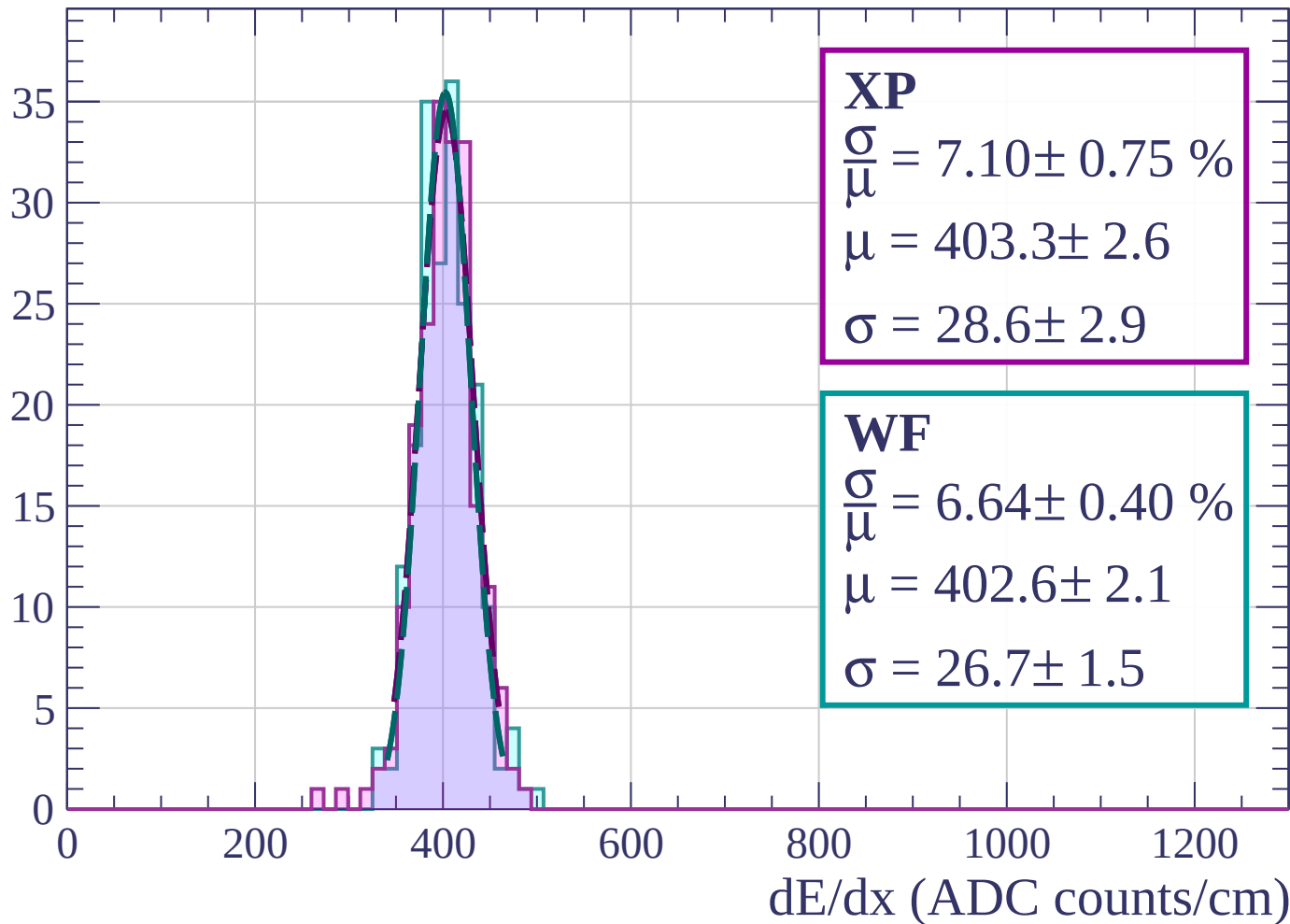
Energy loss | $39 < \phi < 42$

Count



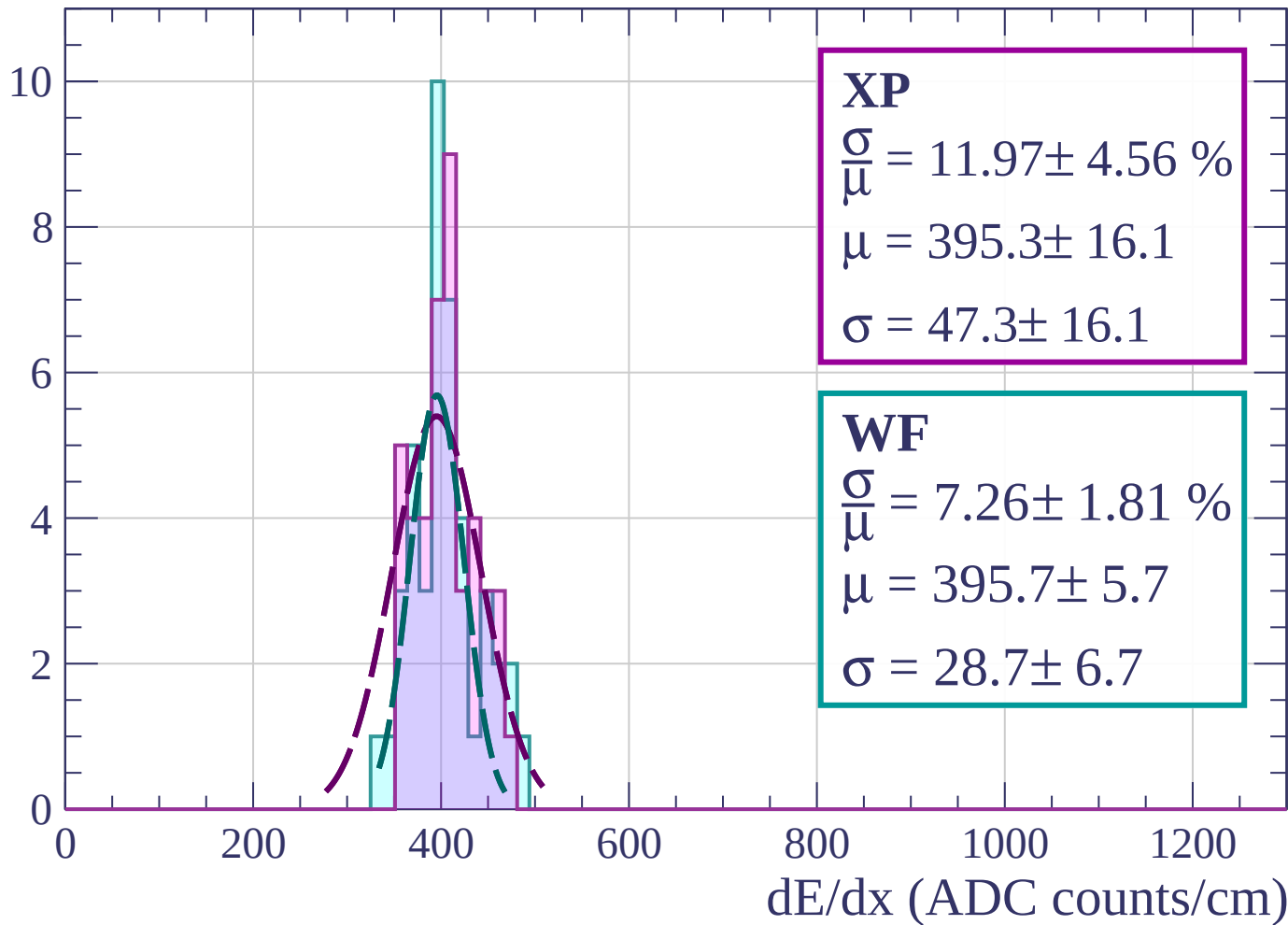
Energy loss | $42 < \phi < 45$

Count



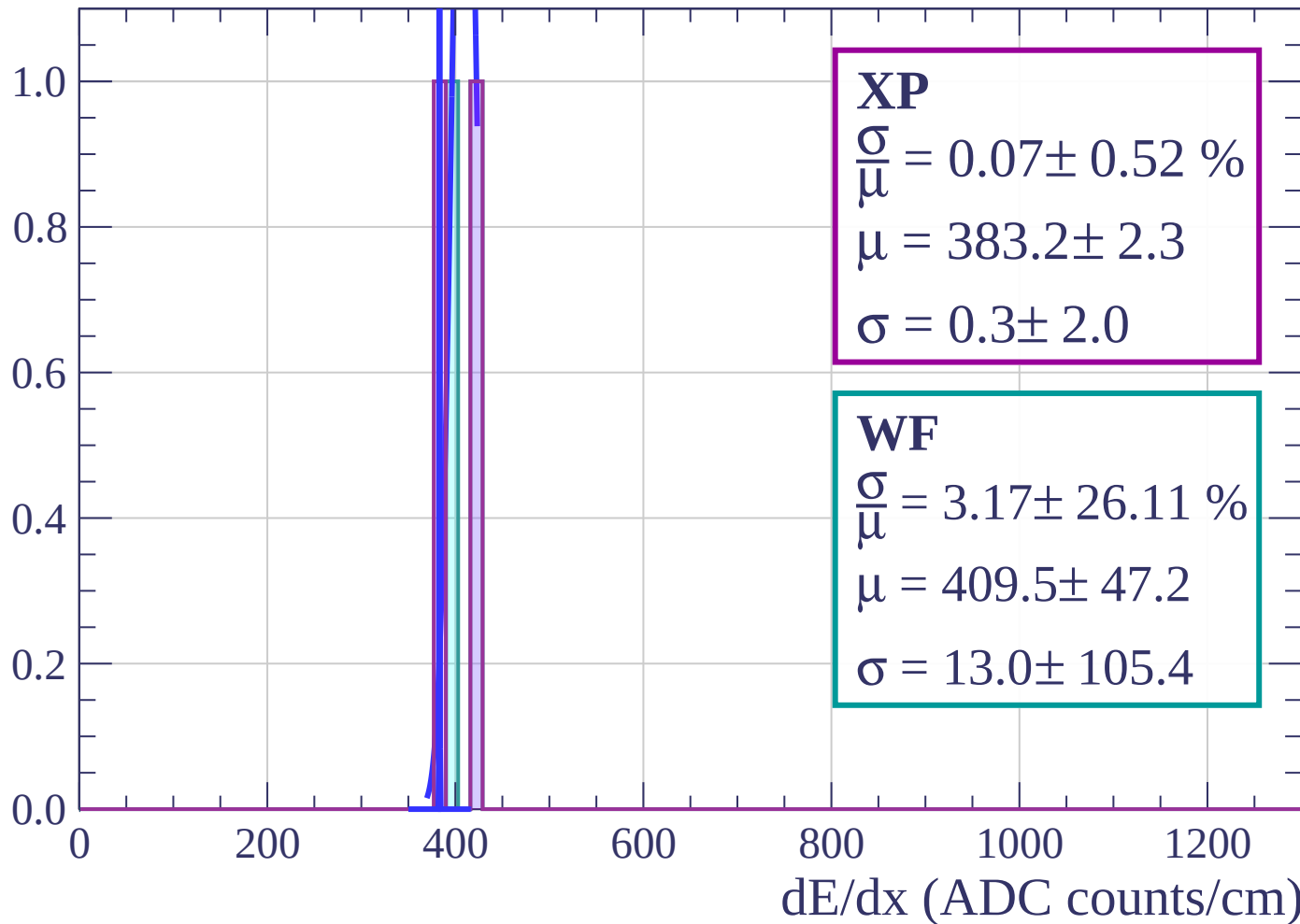
Energy loss | $45 < \phi < 48$

Count



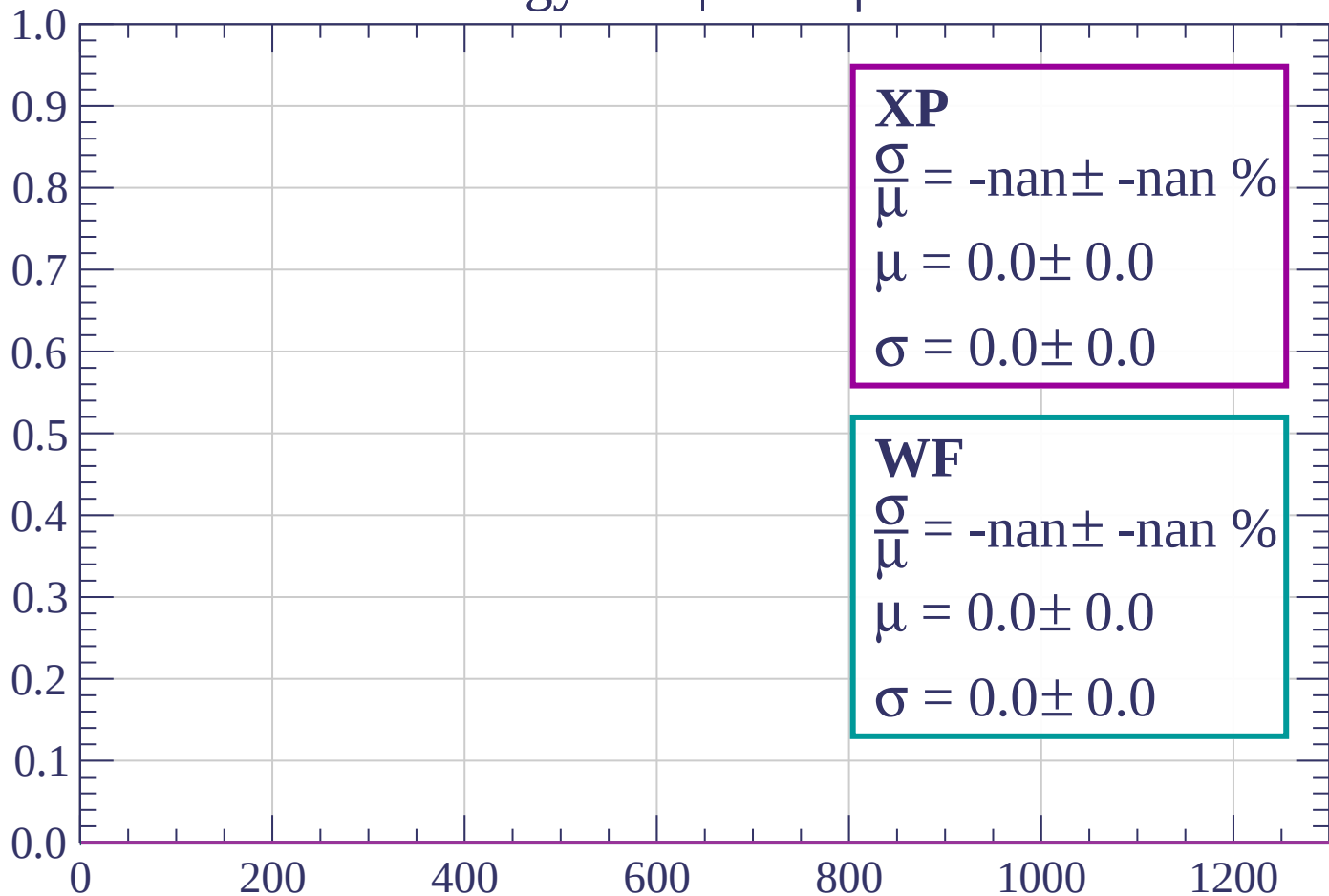
Energy loss | $48 < \phi < 51$

Count



Energy loss | $51 < \phi < 54$

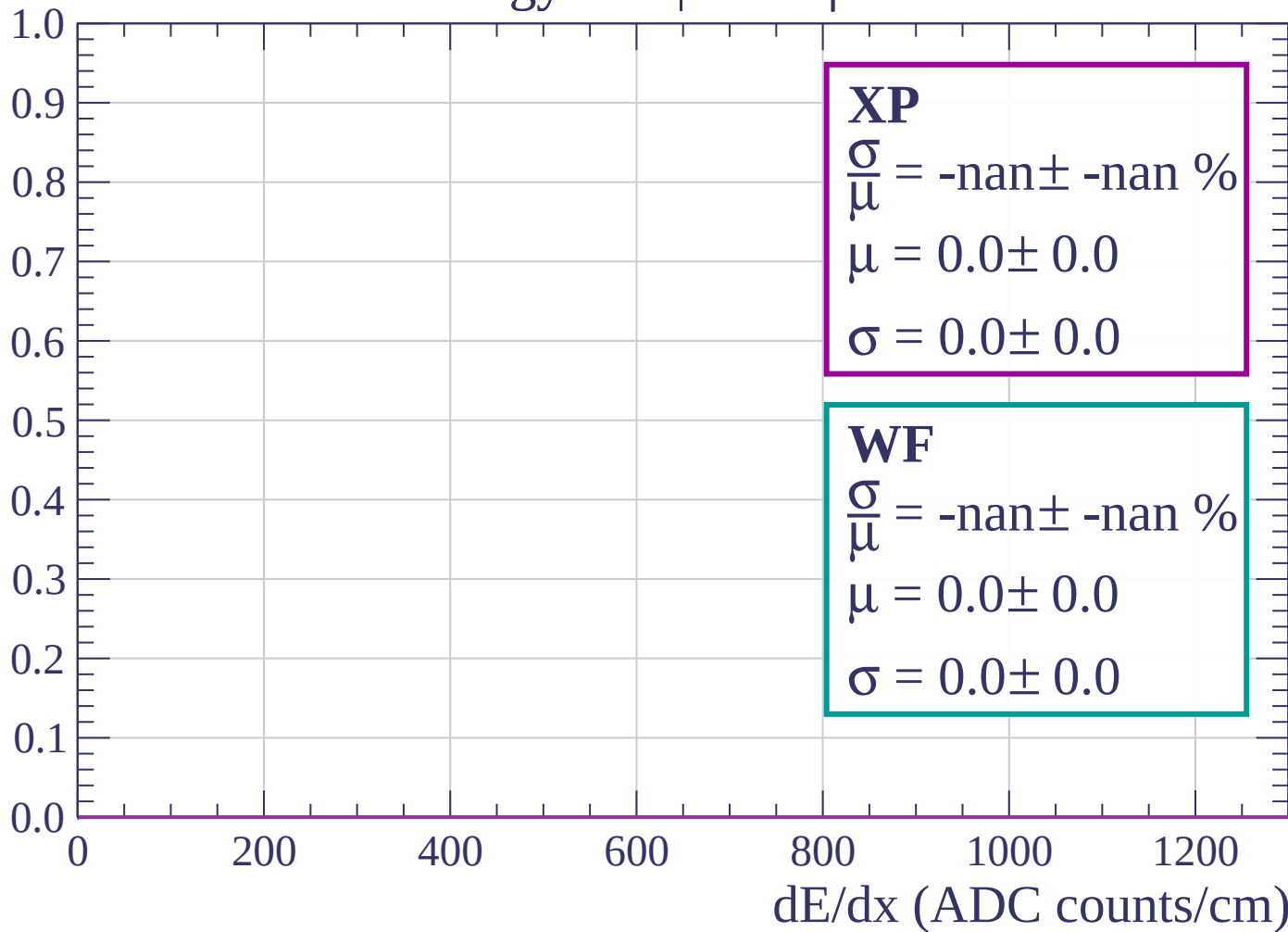
Count



dE/dx (ADC counts/cm)

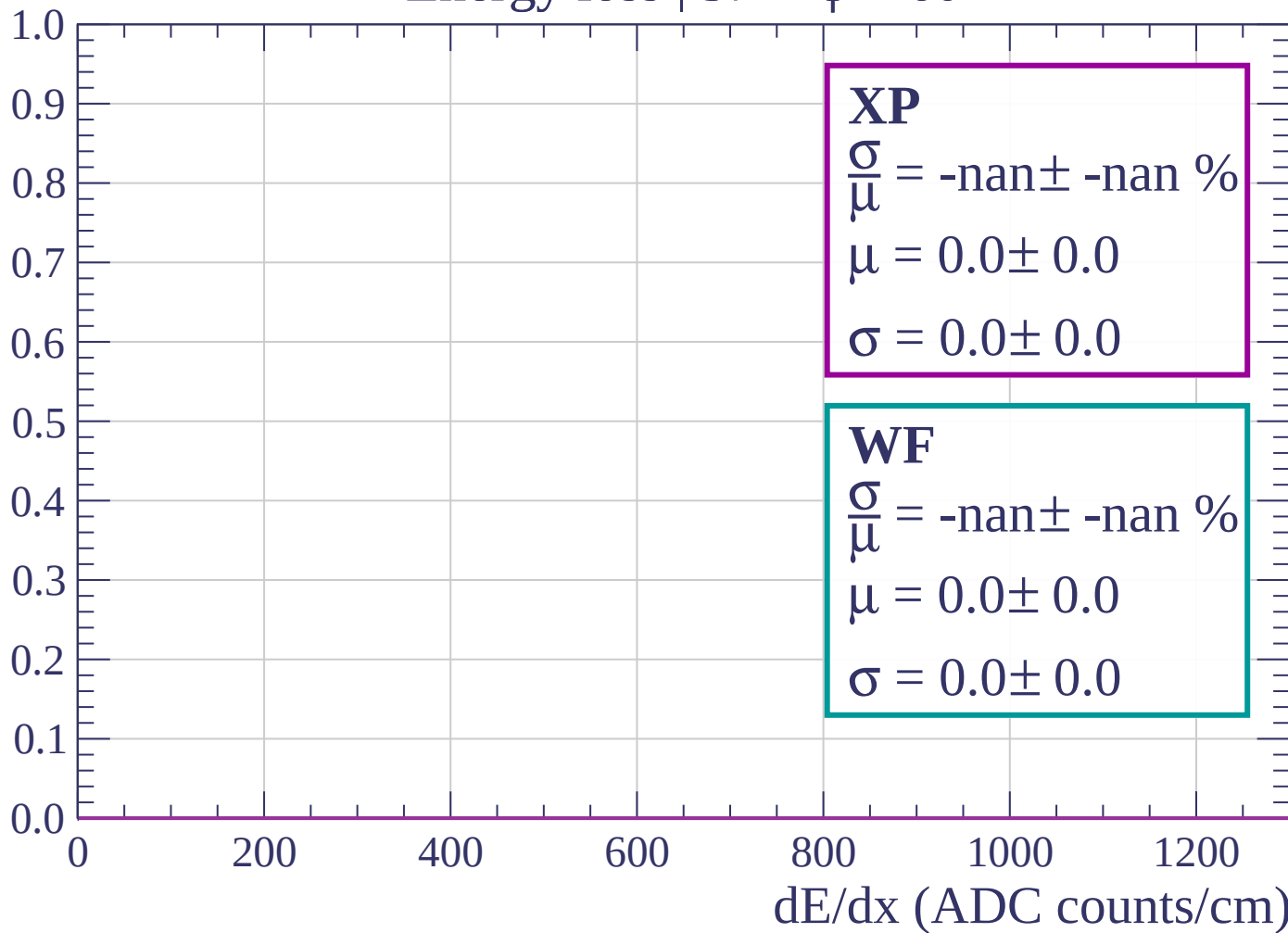
Energy loss | $54 < \phi < 57$

Count



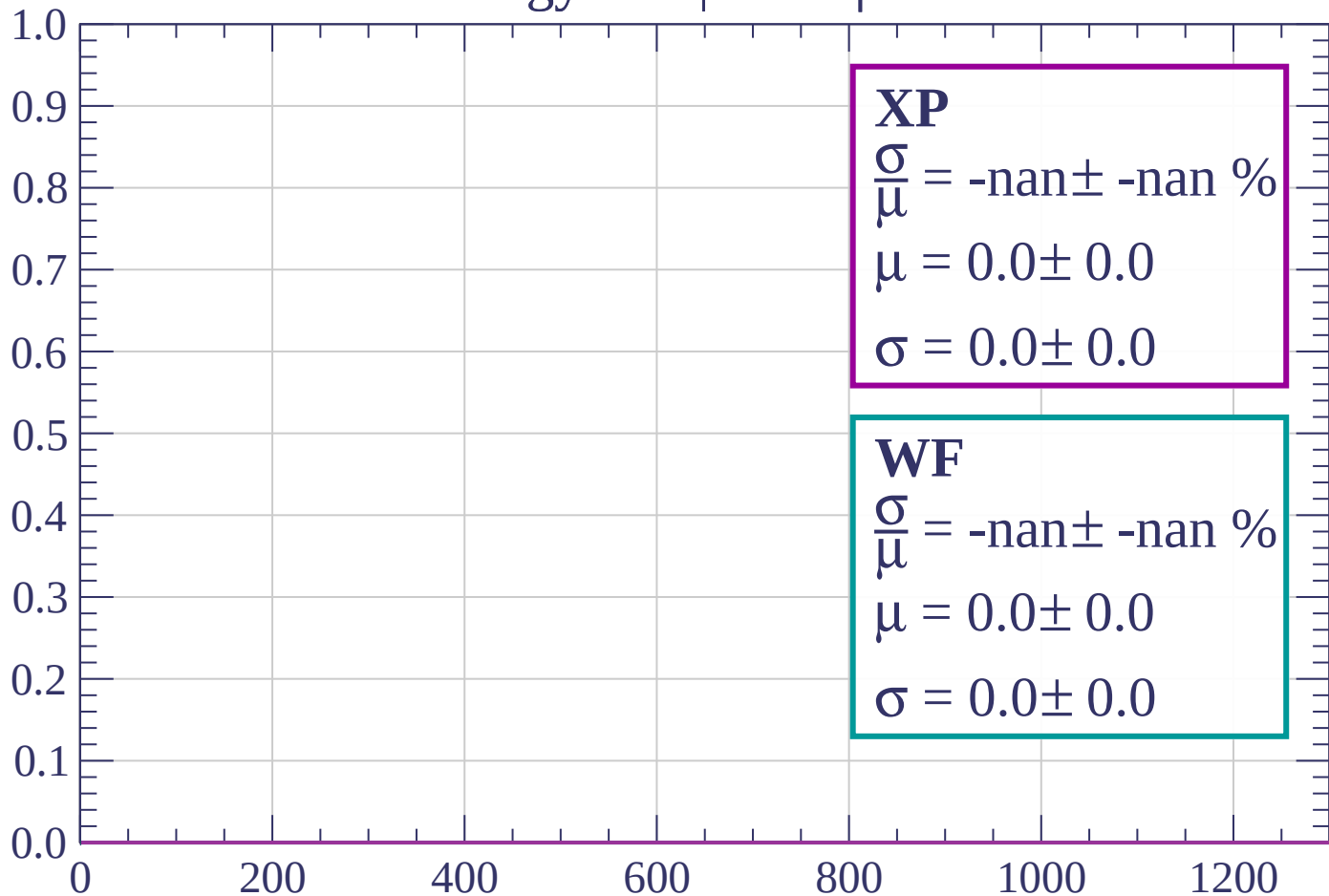
Energy loss | $57 < \phi < 60$

Count



Energy loss | $60 < \phi < 63$

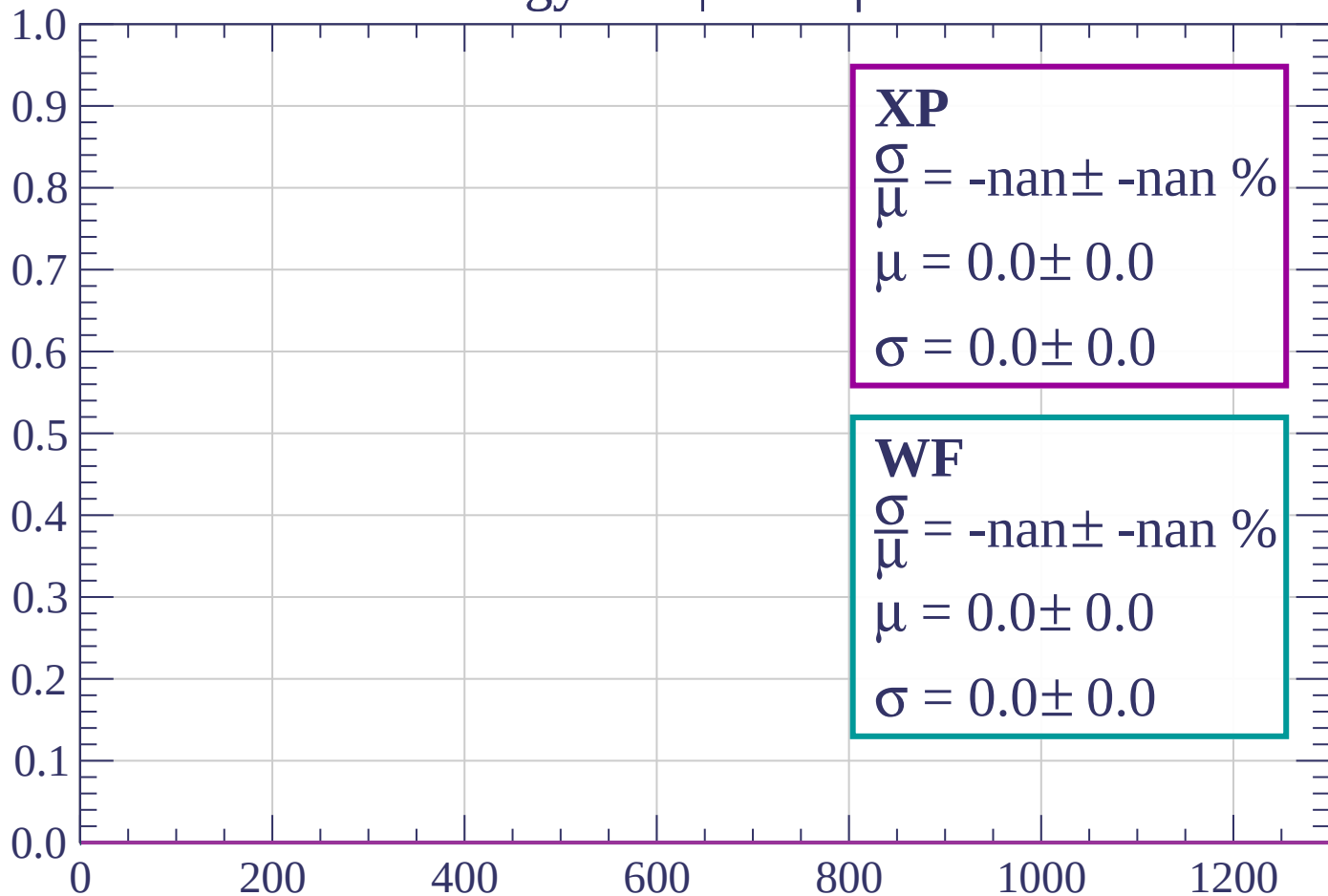
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \phi < 66$

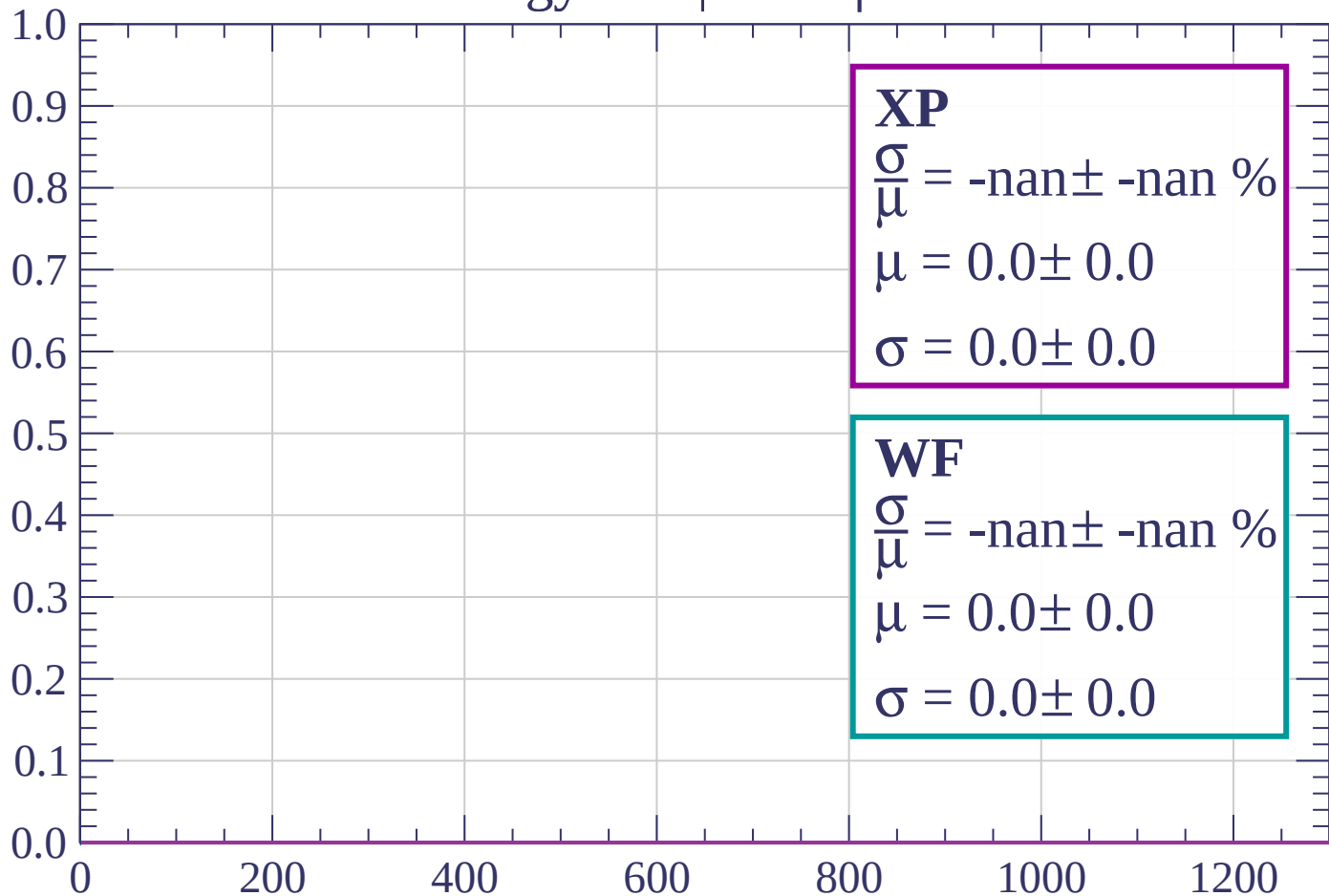
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \phi < 69$

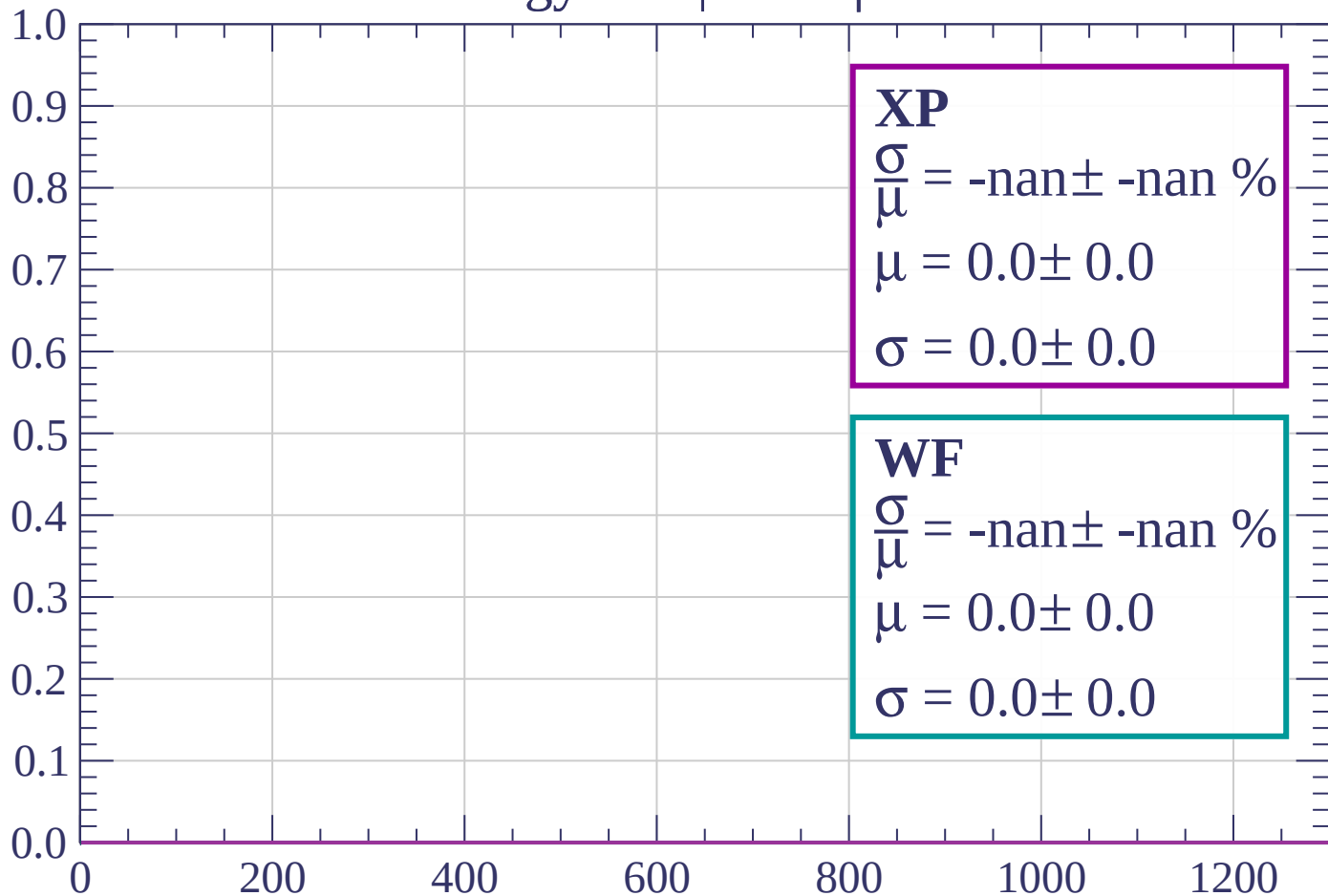
Count



dE/dx (ADC counts/cm)

Energy loss | $69 < \phi < 72$

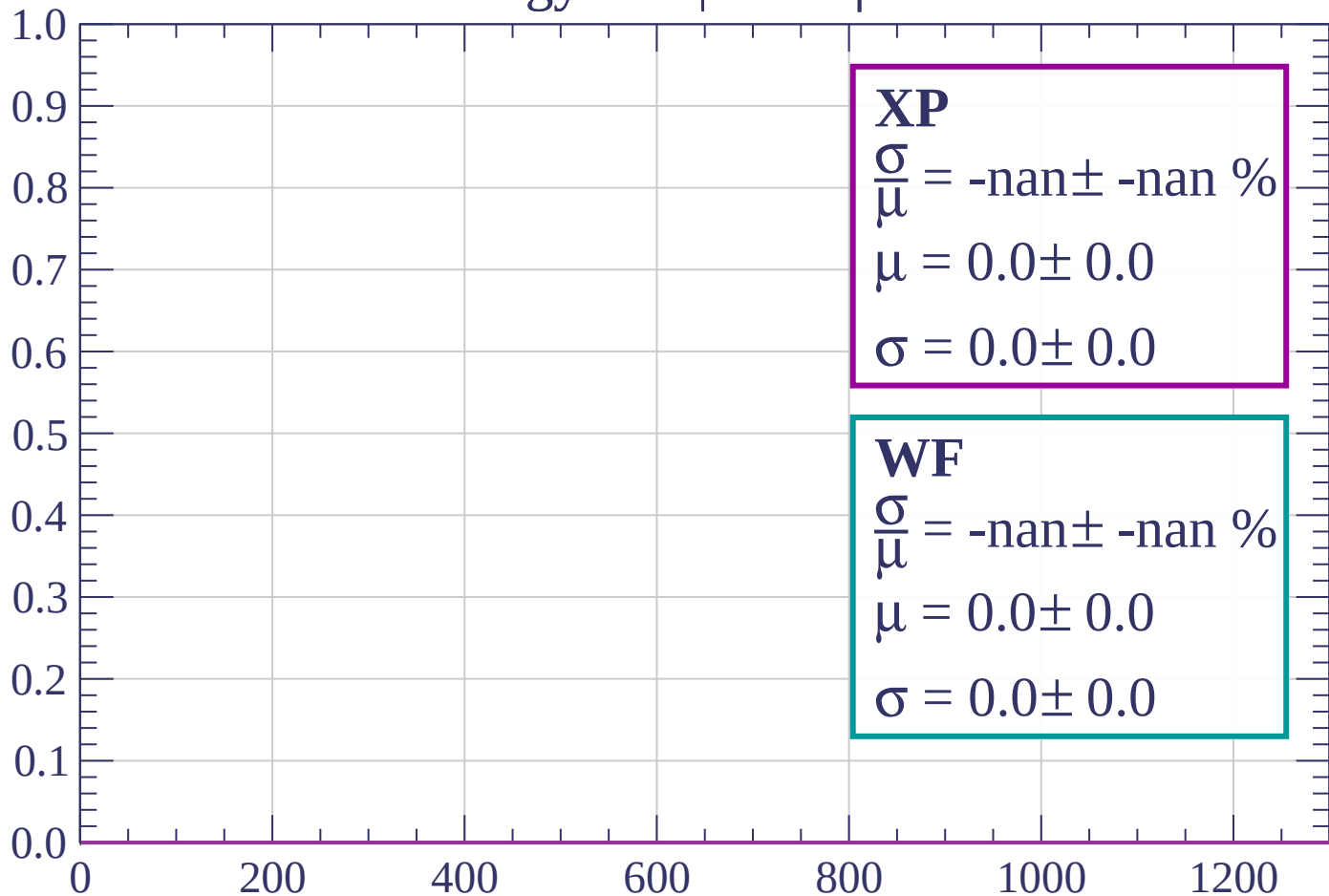
Count



dE/dx (ADC counts/cm)

Energy loss | $72 < \phi < 75$

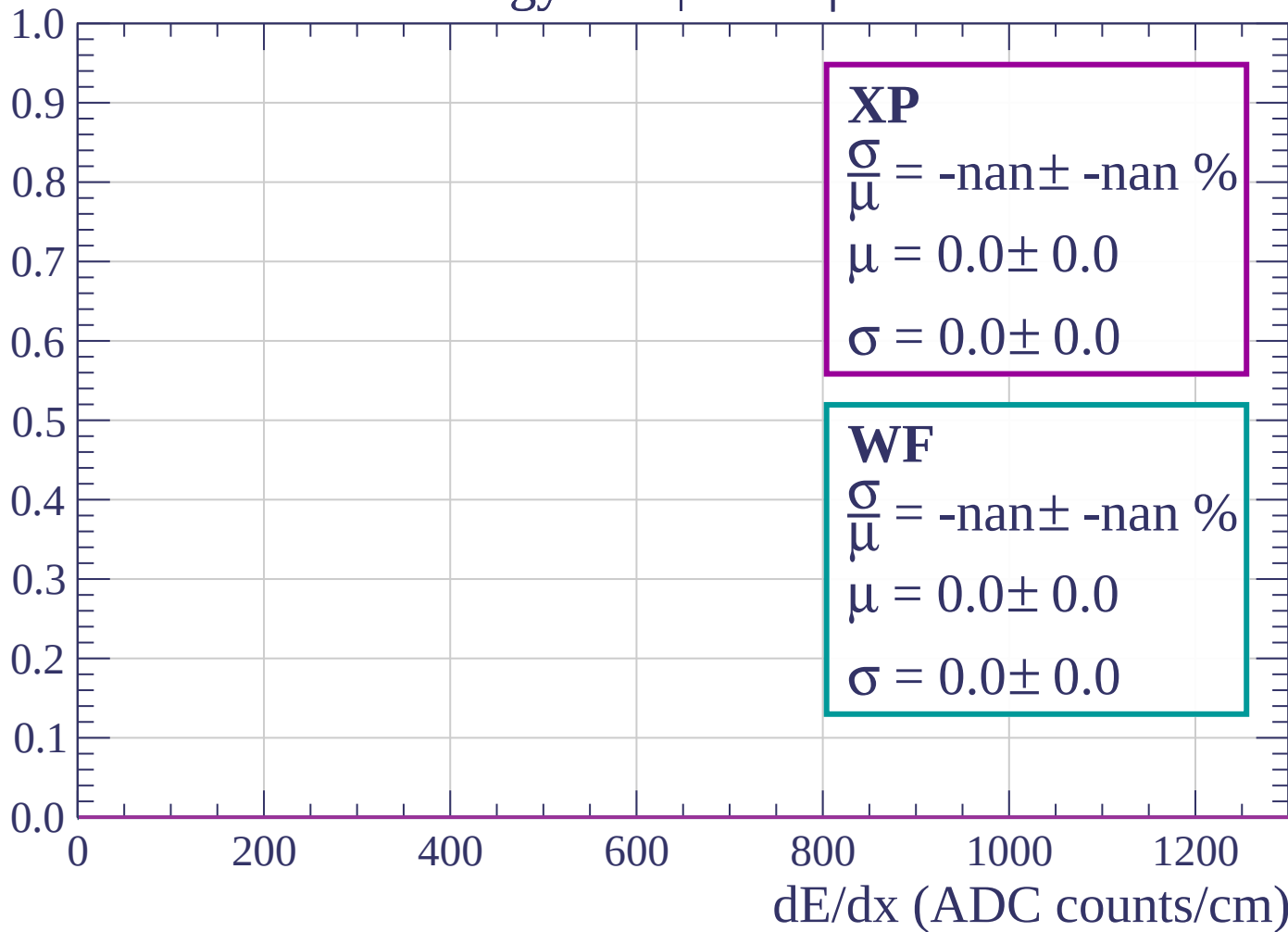
Count



dE/dx (ADC counts/cm)

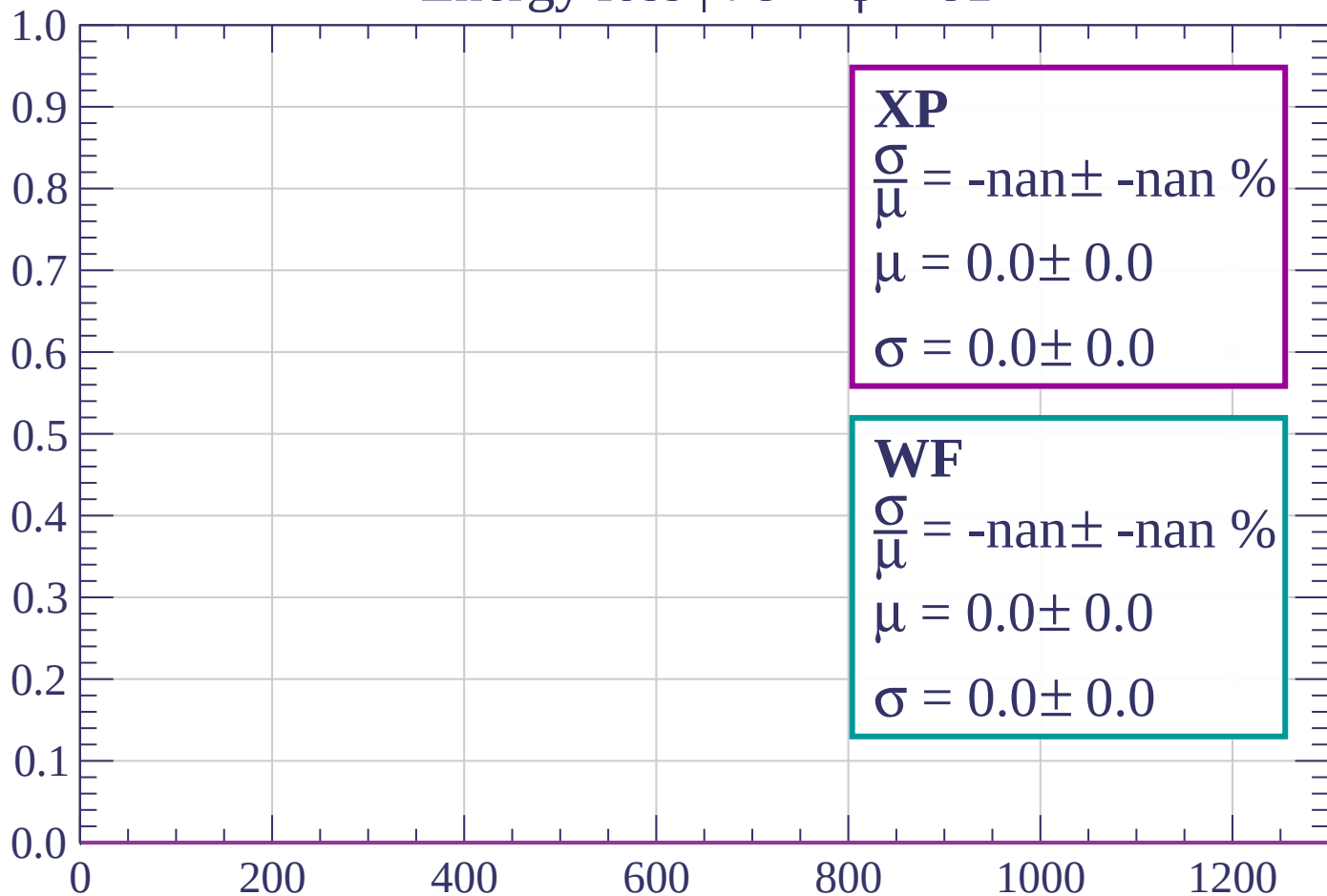
Energy loss | $75 < \phi < 78$

Count



Energy loss | $78 < \phi < 81$

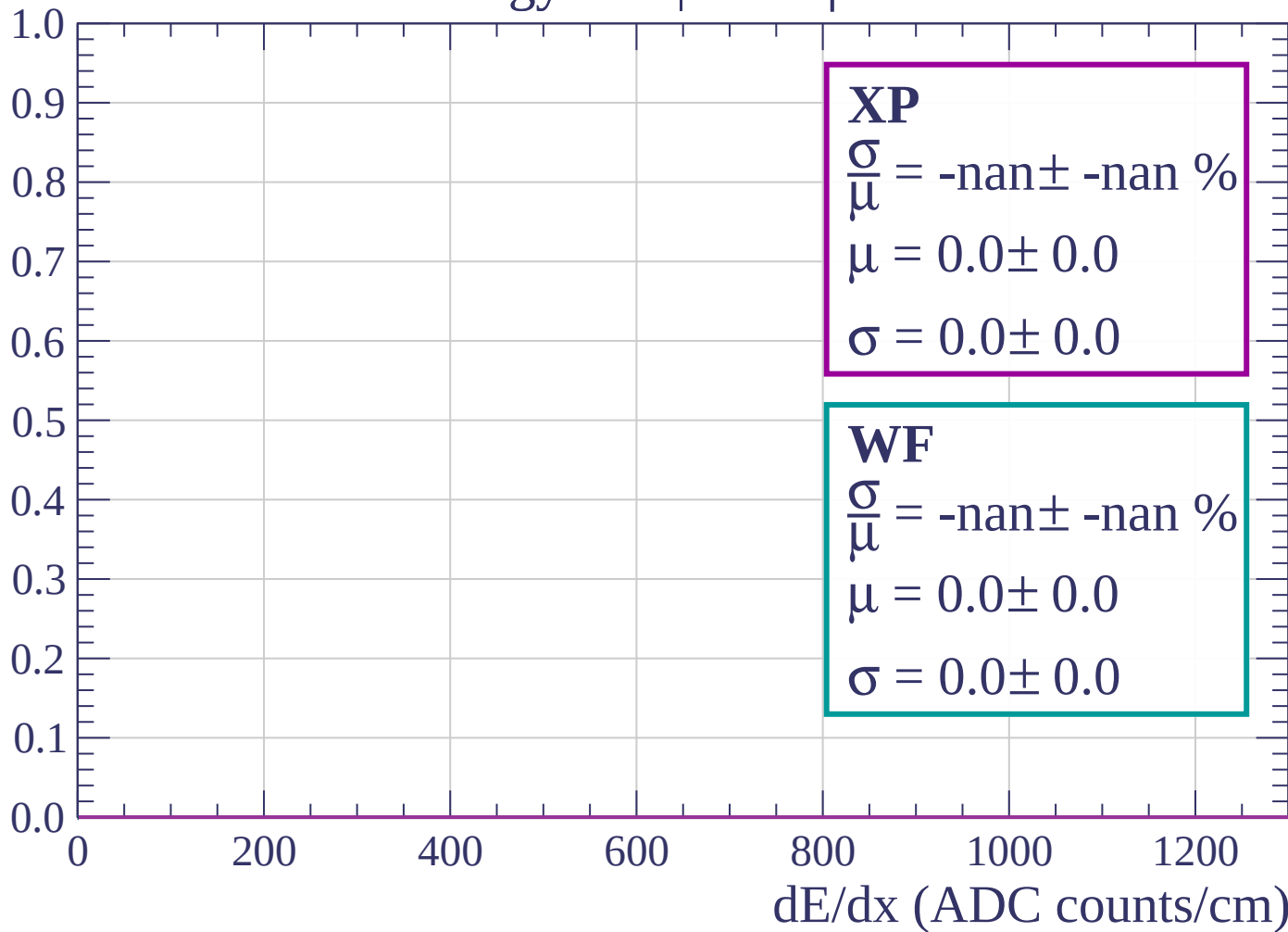
Count



dE/dx (ADC counts/cm)

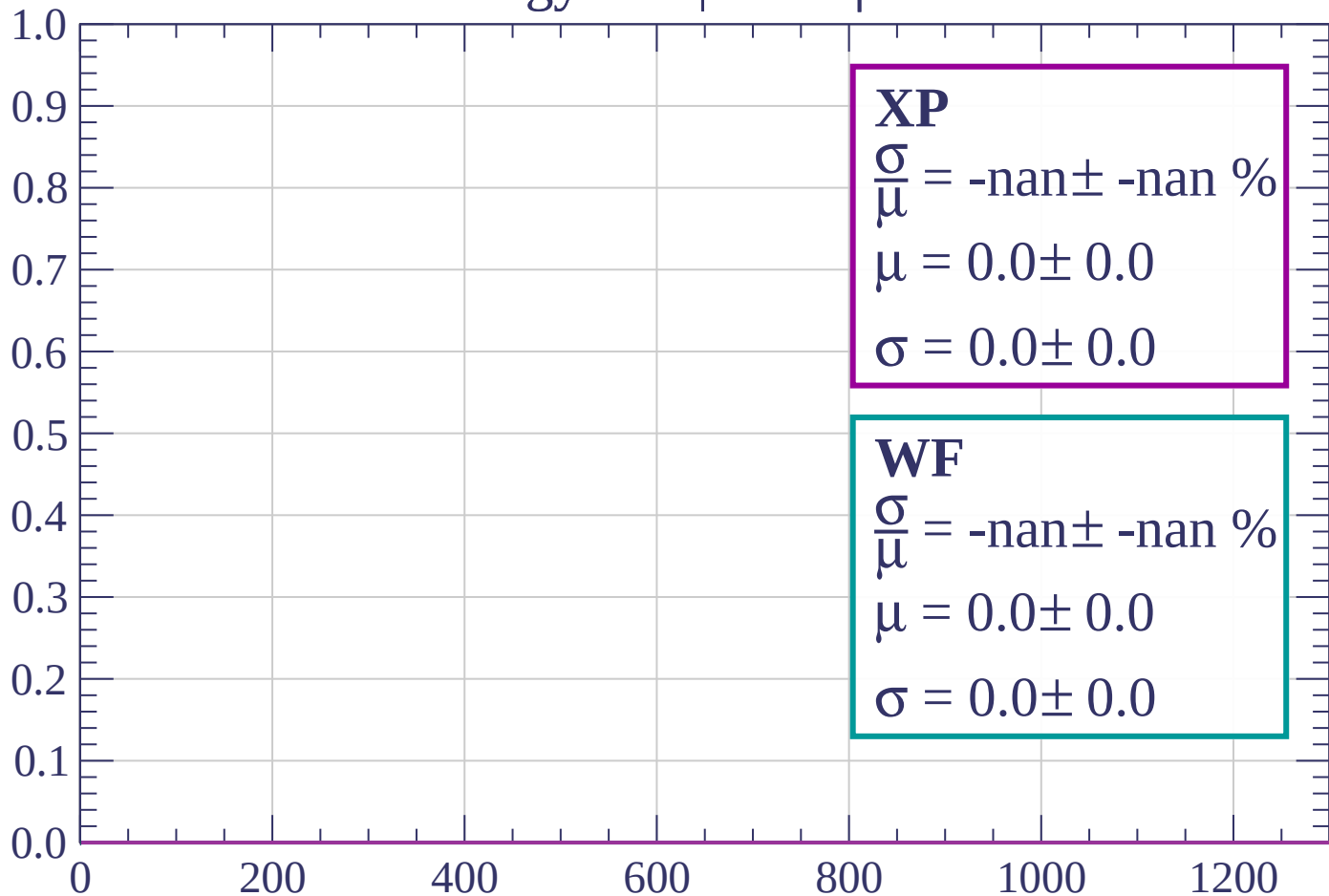
Energy loss | $81 < \phi < 84$

Count



Energy loss | $84 < \phi < 87$

Count



XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

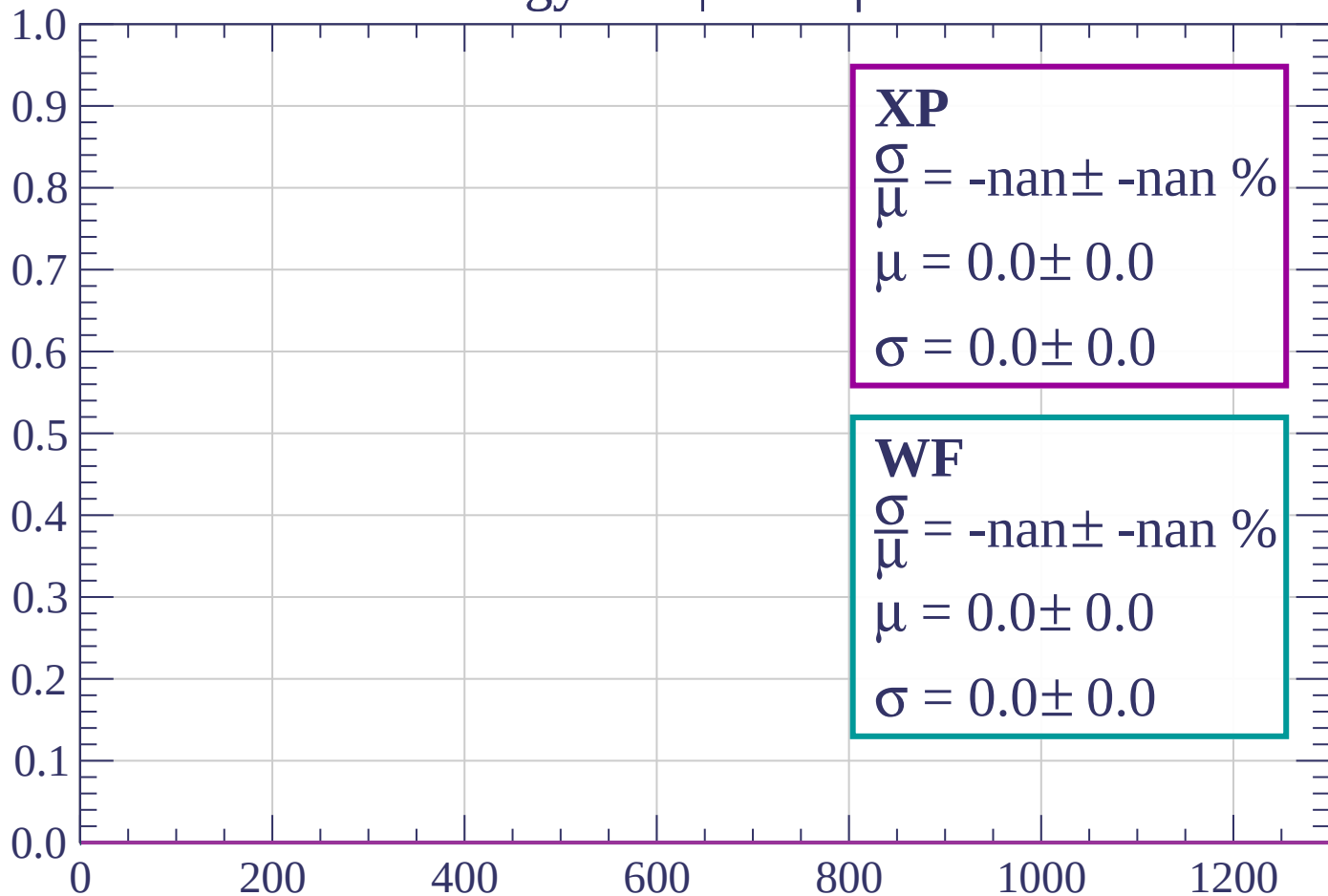
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

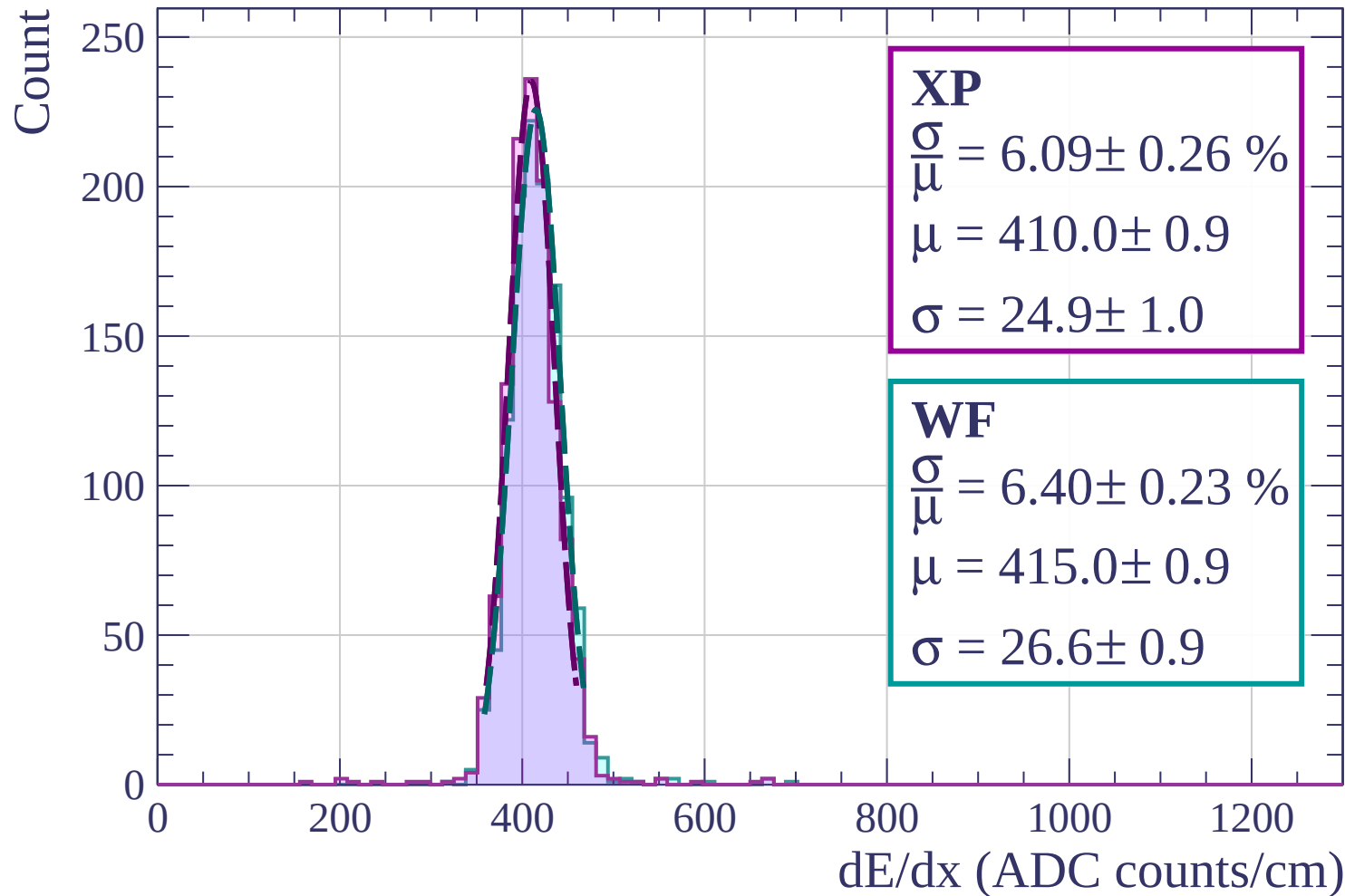
Energy loss | $87 < \phi < 90$

Count

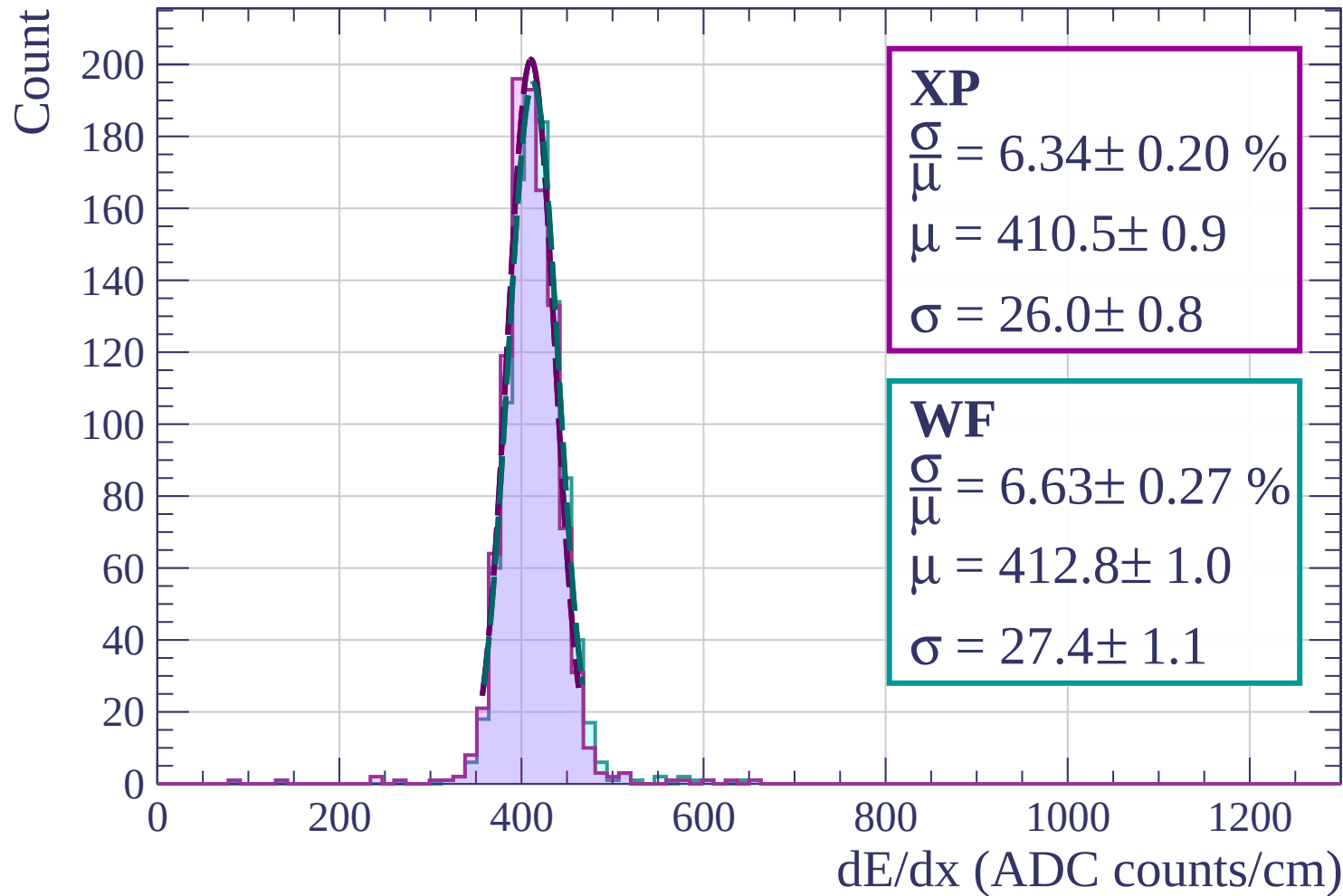


dE/dx (ADC counts/cm)

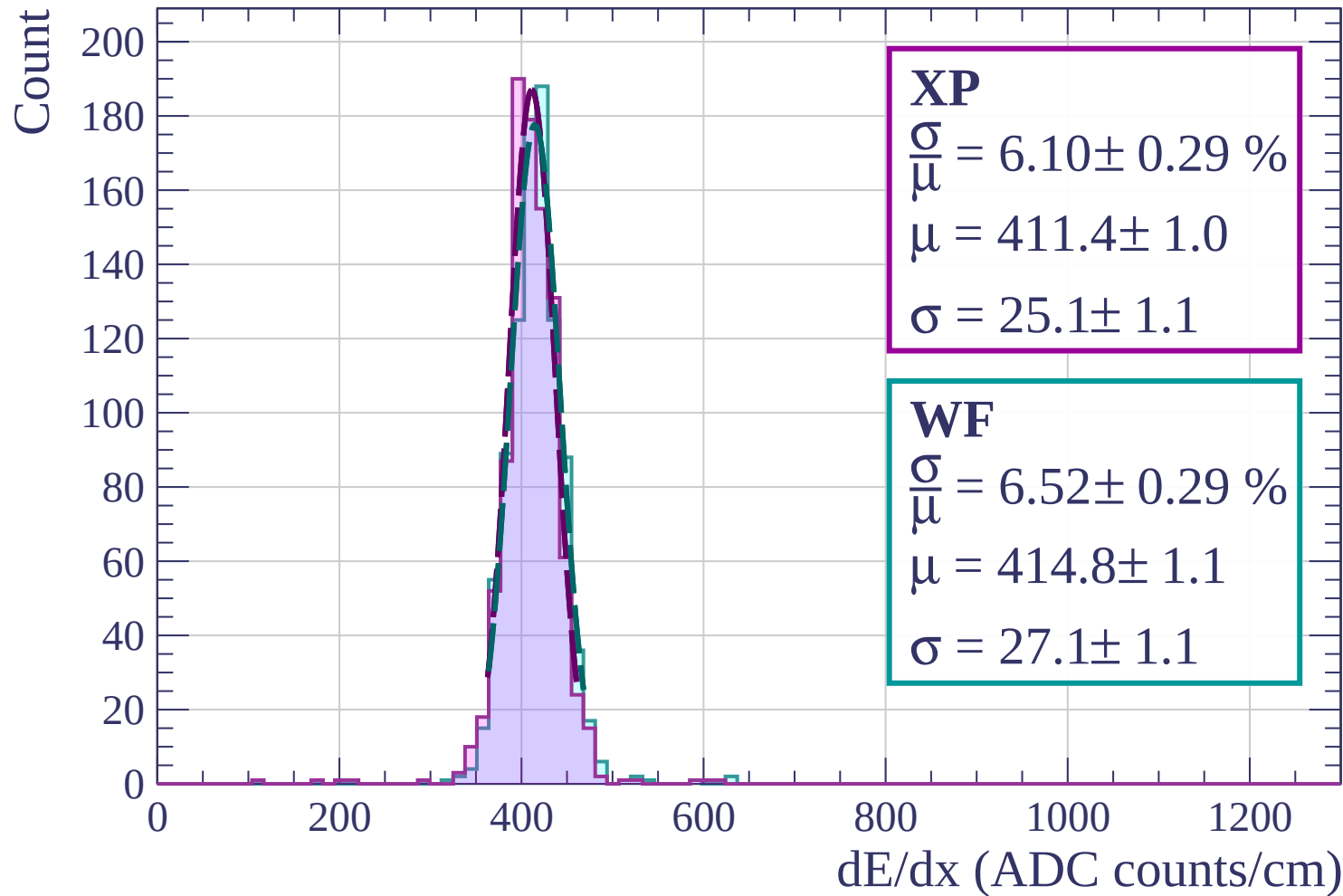
Energy loss | $0 < \theta < 3$



Energy loss | $3 < \theta < 6$

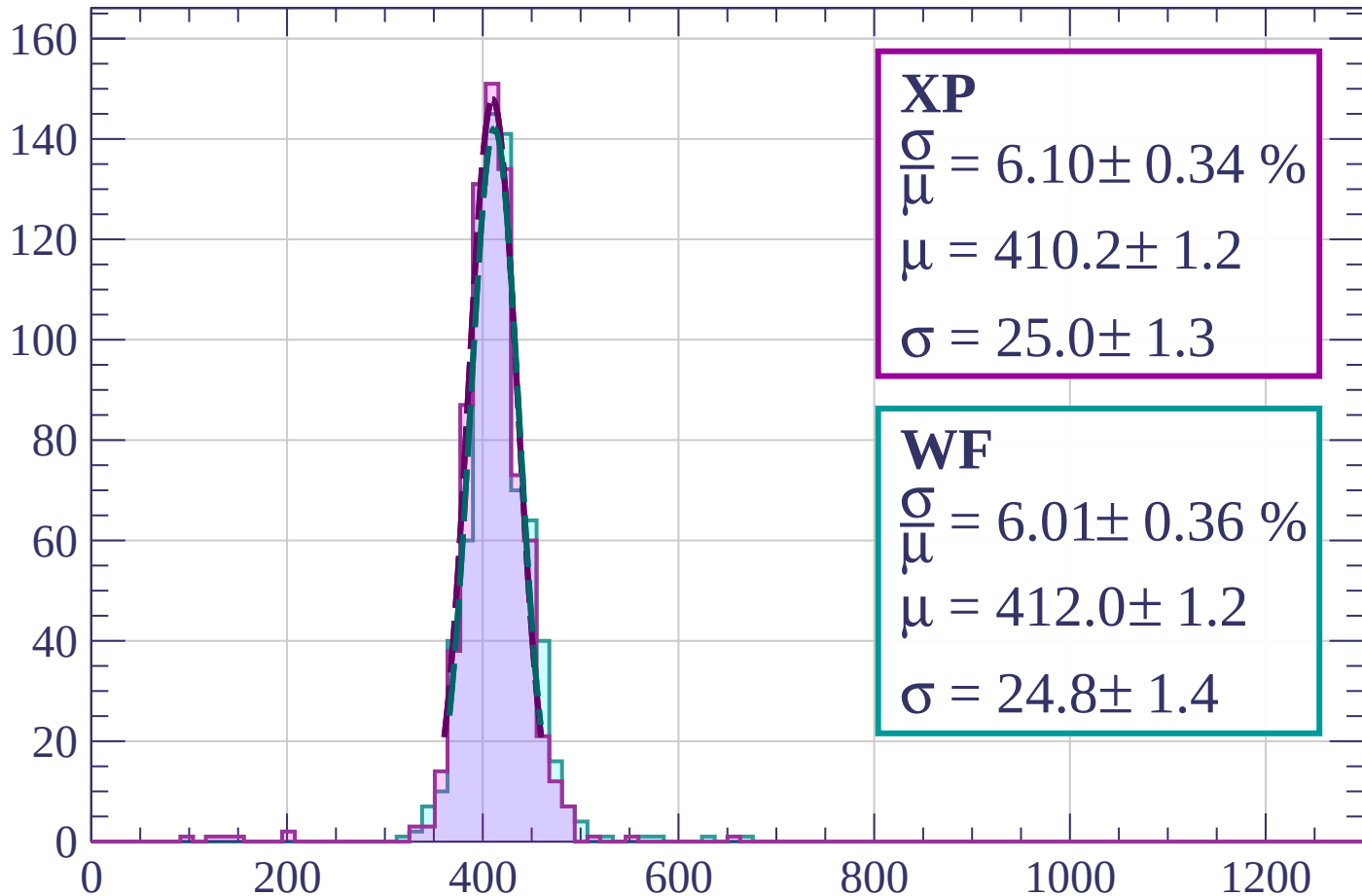


Energy loss | $6 < \theta < 9$



Energy loss | $9 < \theta < 12$

Count



XP

$$\frac{\sigma}{\mu} = 6.10 \pm 0.34 \%$$

$$\mu = 410.2 \pm 1.2$$

$$\sigma = 25.0 \pm 1.3$$

WF

$$\frac{\sigma}{\mu} = 6.01 \pm 0.36 \%$$

$$\mu = 412.0 \pm 1.2$$

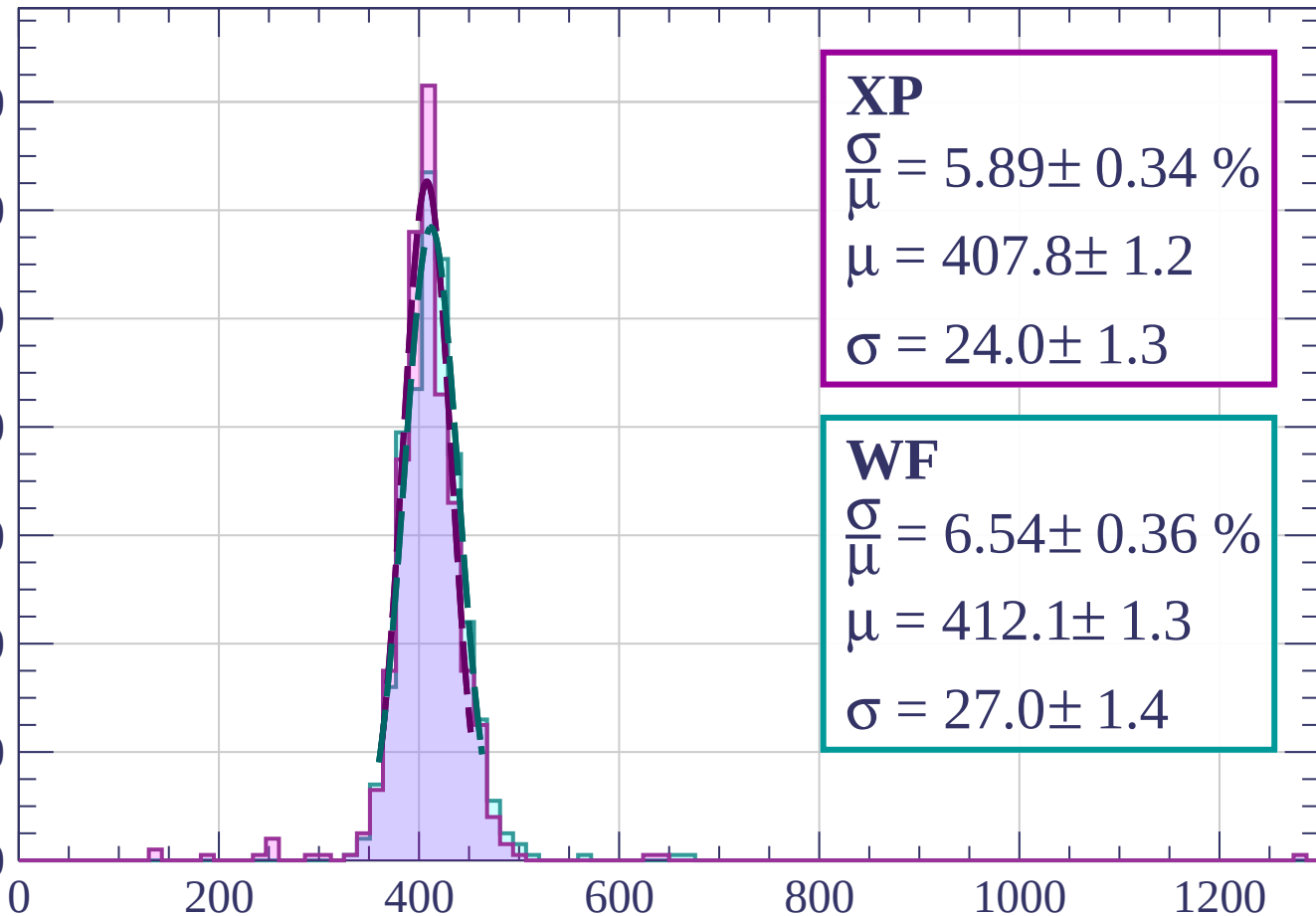
$$\sigma = 24.8 \pm 1.4$$

dE/dx (ADC counts/cm)

Energy loss | $12 < \theta < 15$

Count

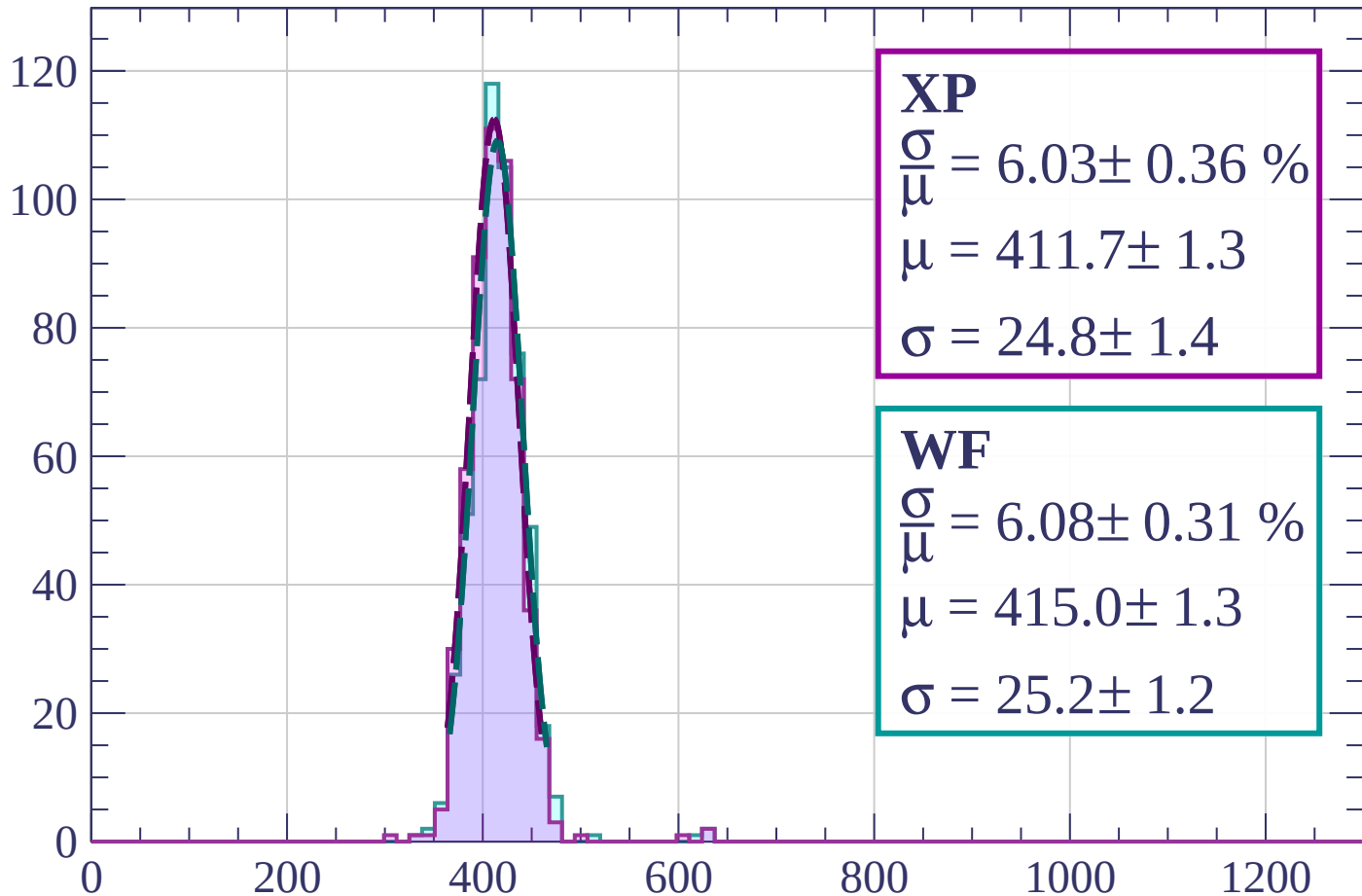
140
120
100
80
60
40
20
0



dE/dx (ADC counts/cm)

Energy loss | $15 < \theta < 18$

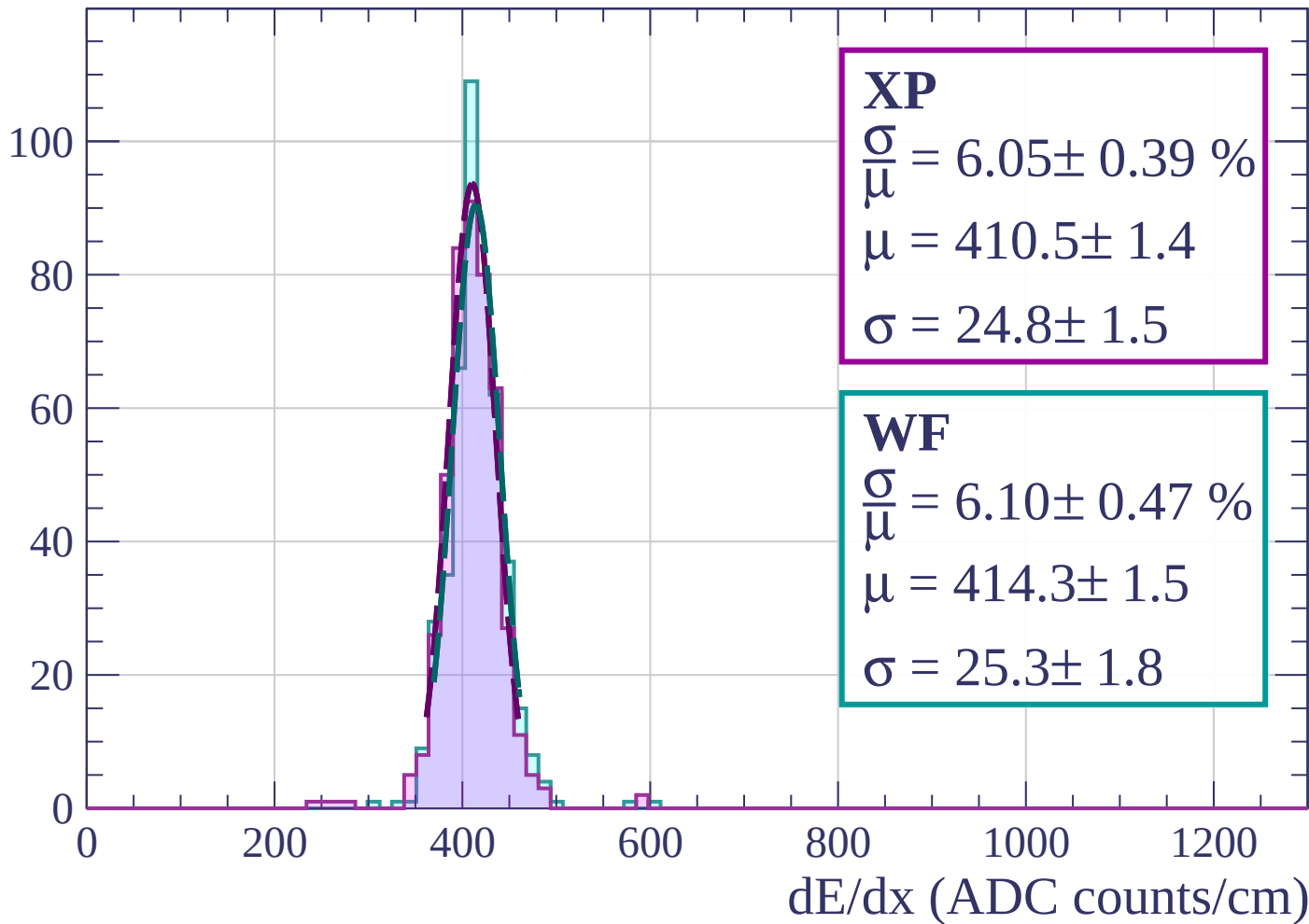
Count



dE/dx (ADC counts/cm)

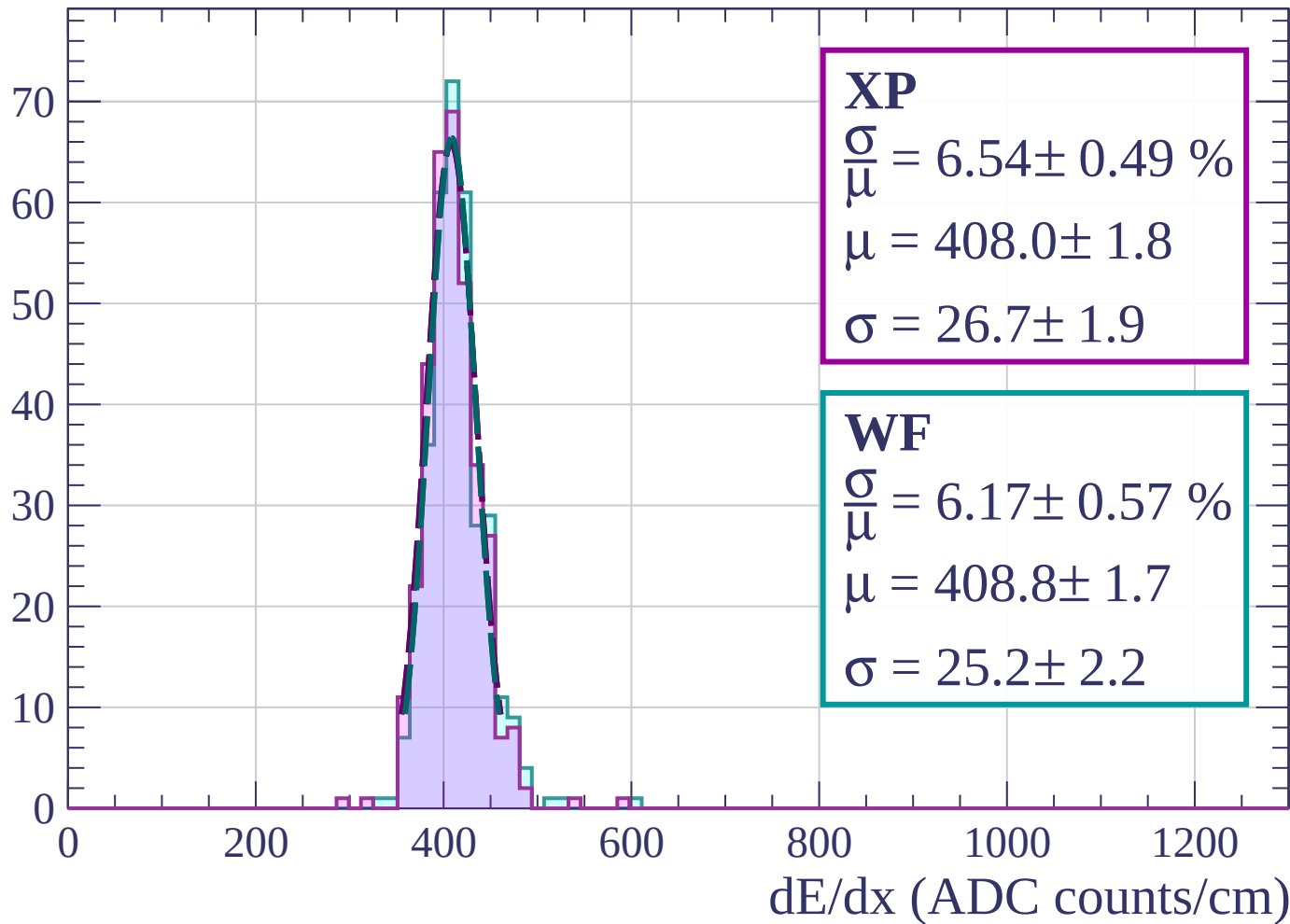
Energy loss | $18 < \theta < 21$

Count



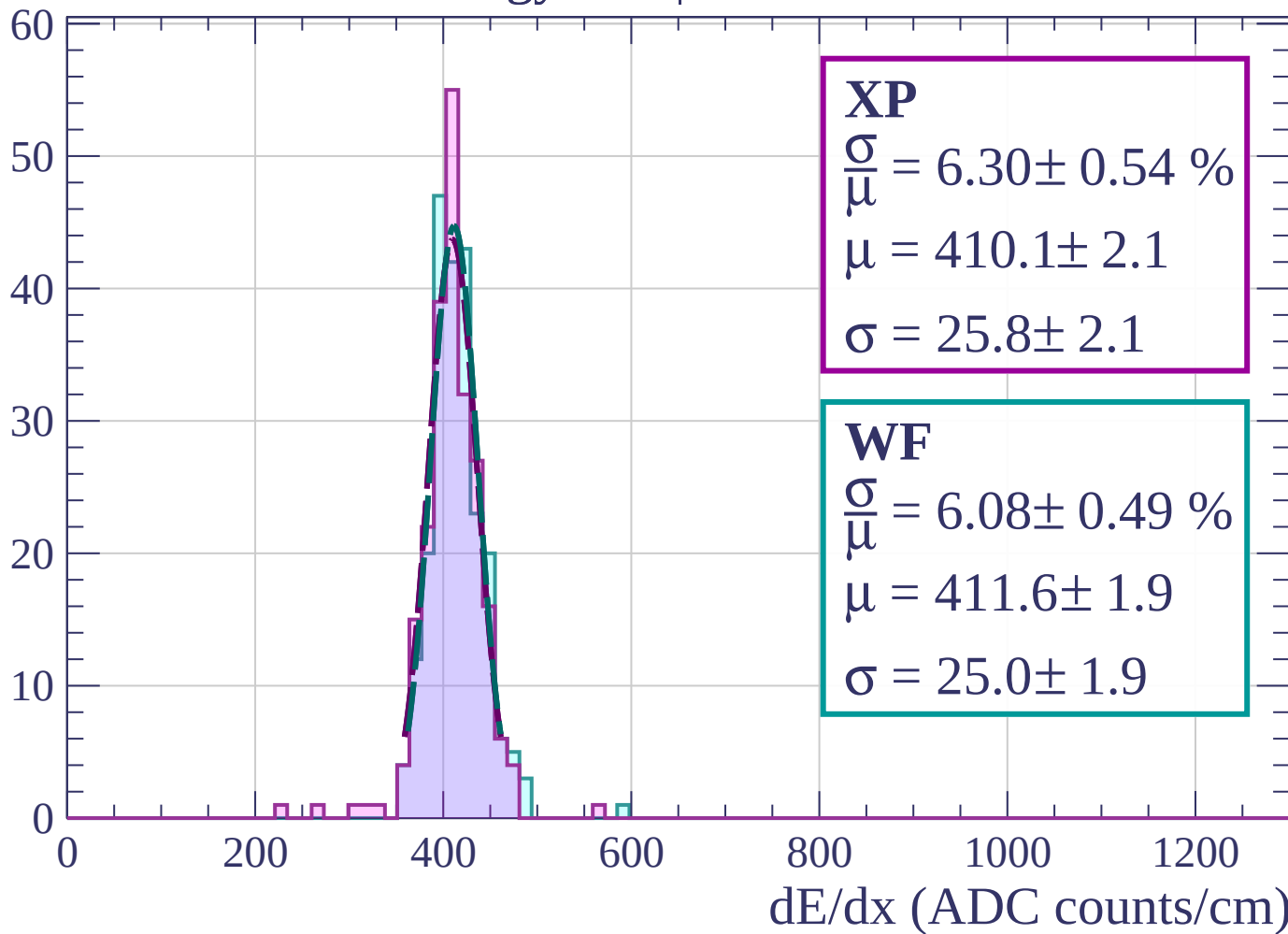
Energy loss | $21 < \theta < 24$

Count



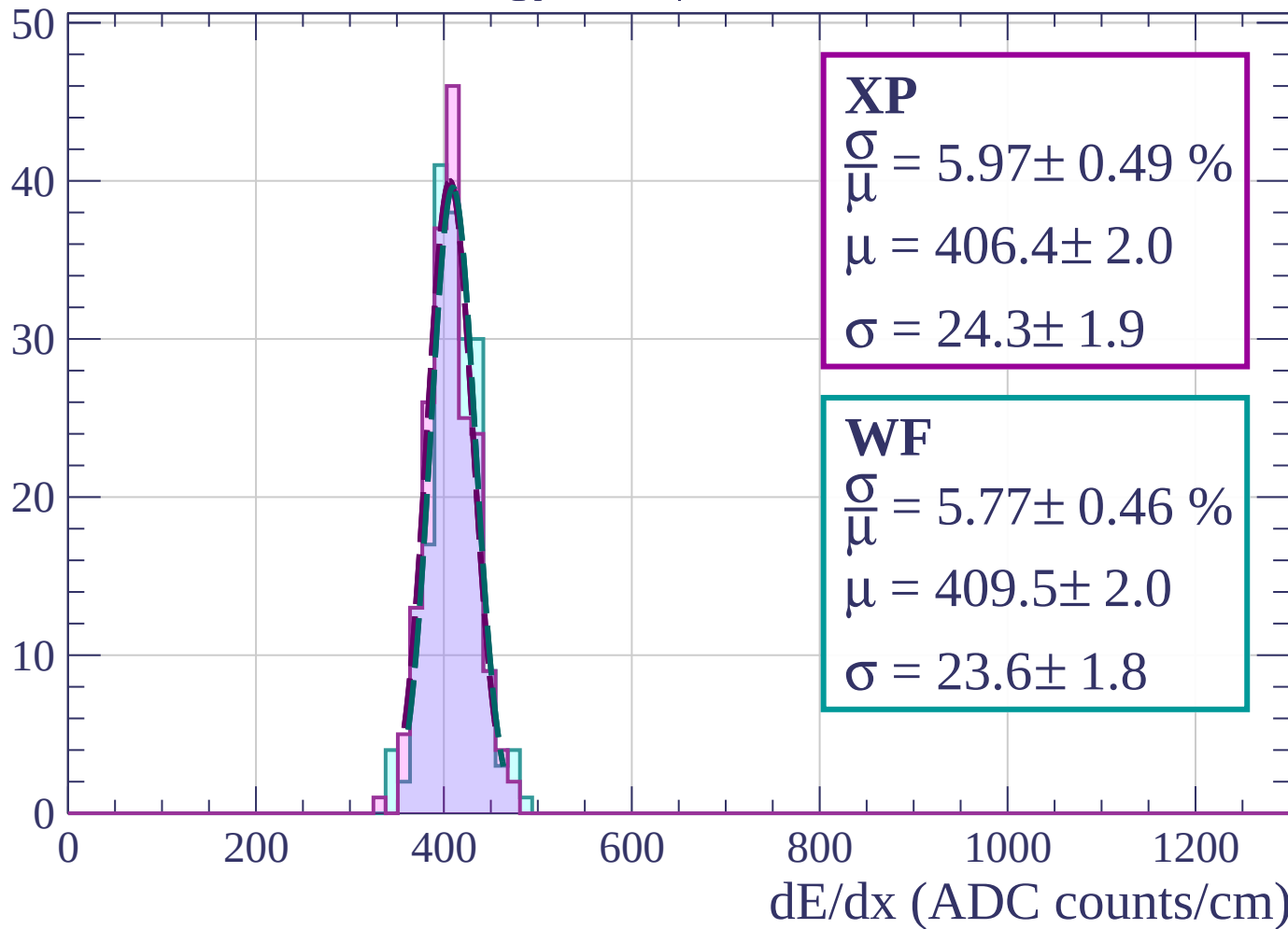
Energy loss | $24 < \theta < 27$

Count



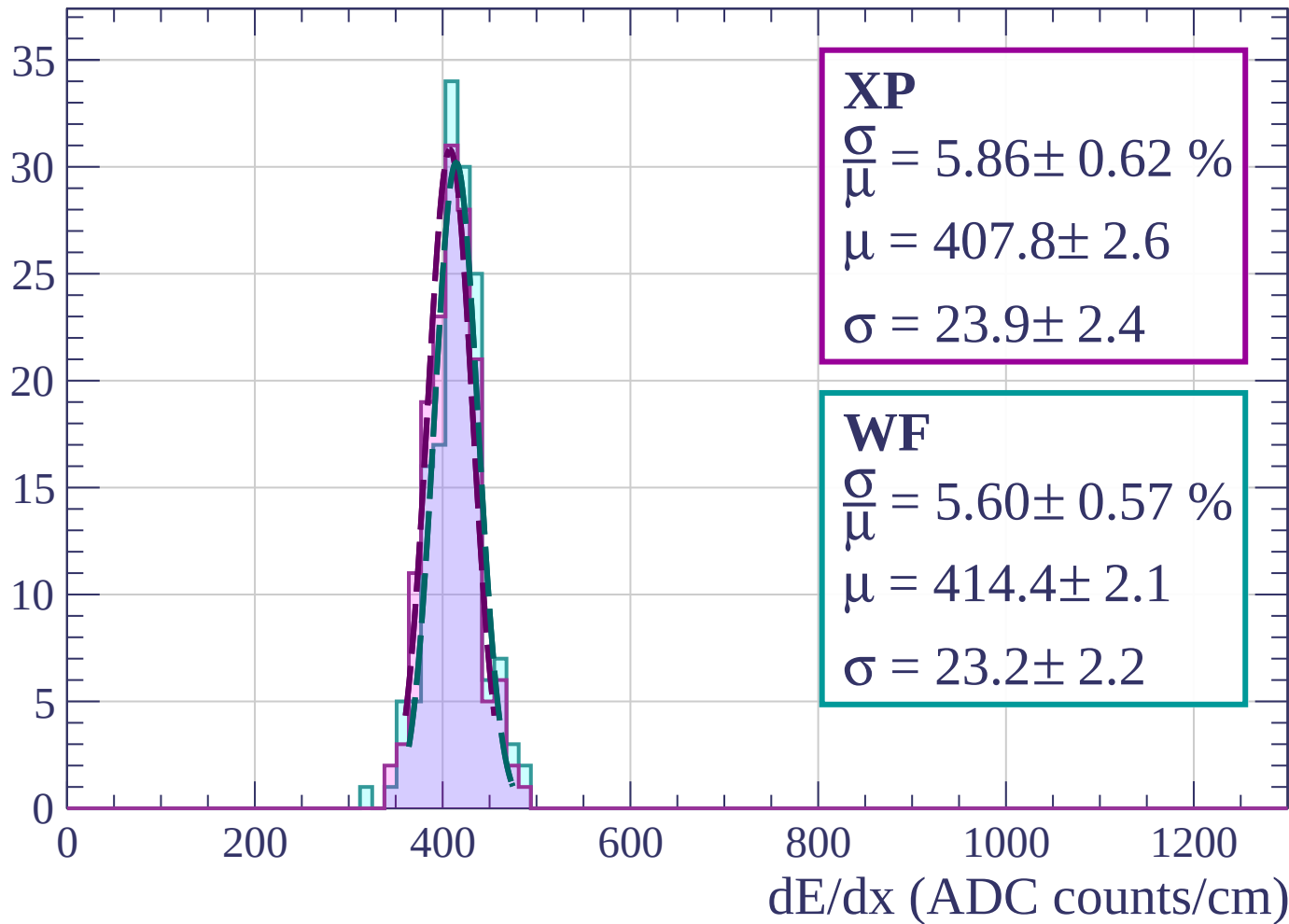
Energy loss | $27 < \theta < 30$

Count



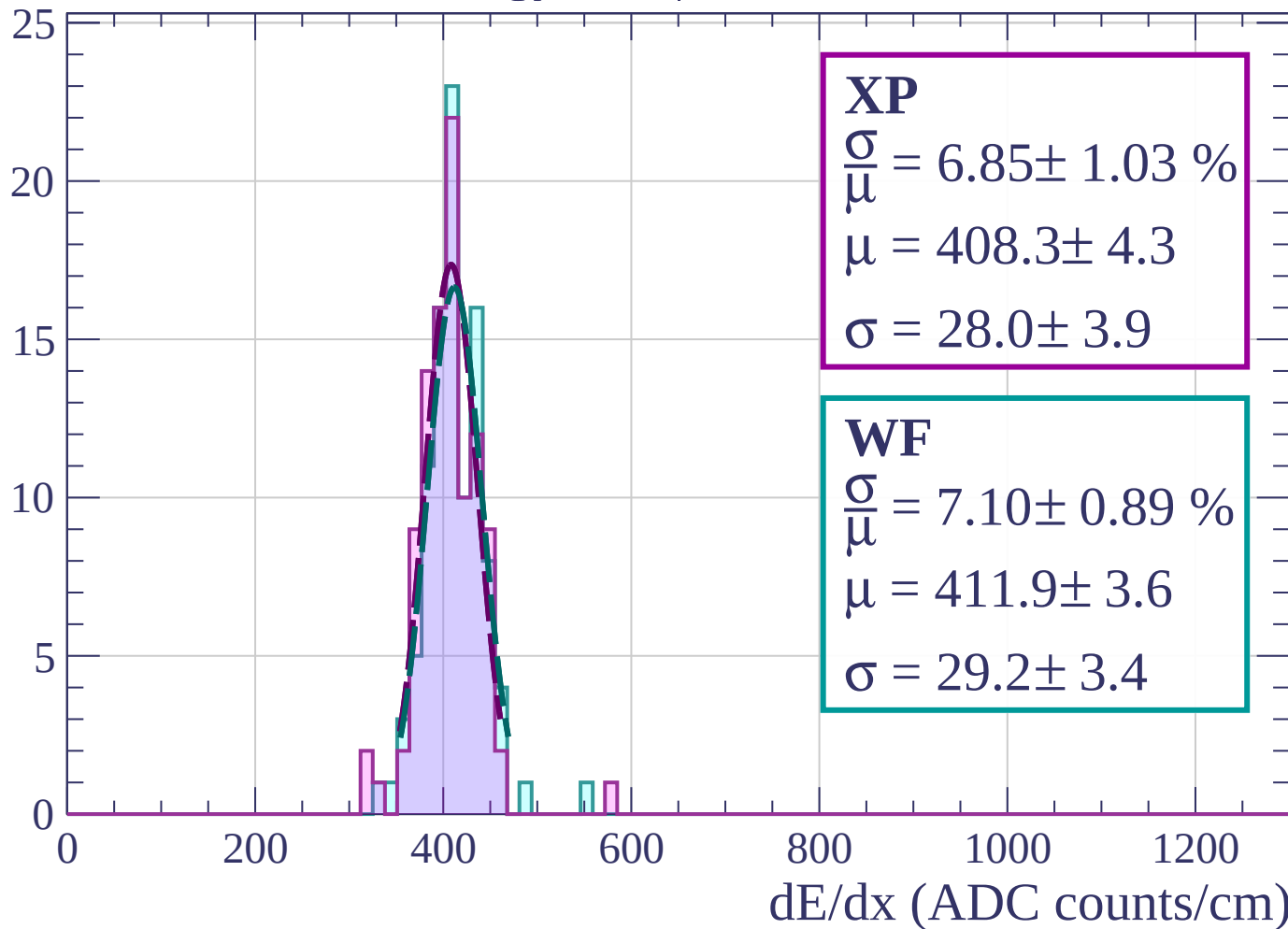
Energy loss | $30 < \theta < 33$

Count



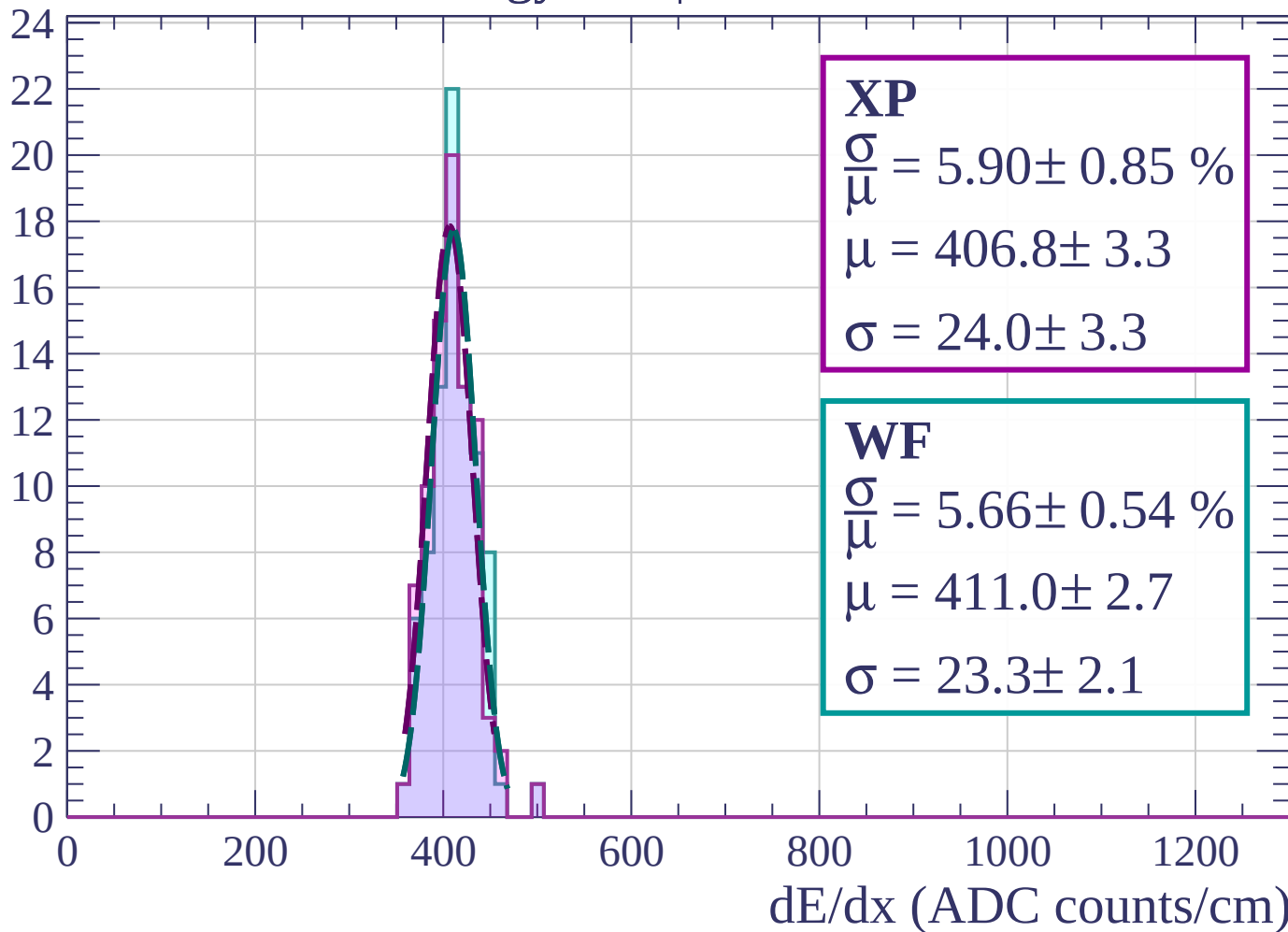
Energy loss | $33 < \theta < 36$

Count



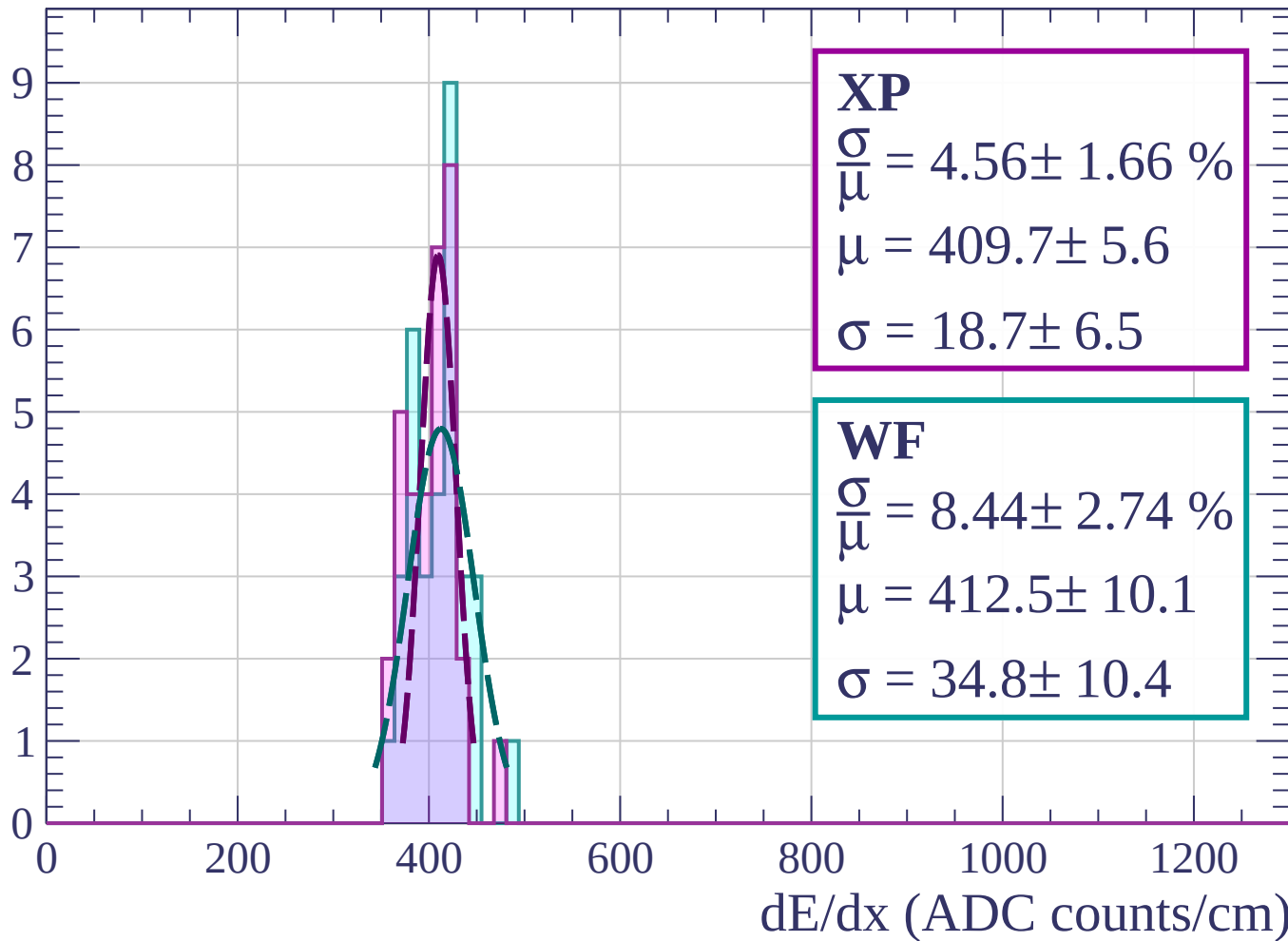
Energy loss | $36 < \theta < 39$

Count



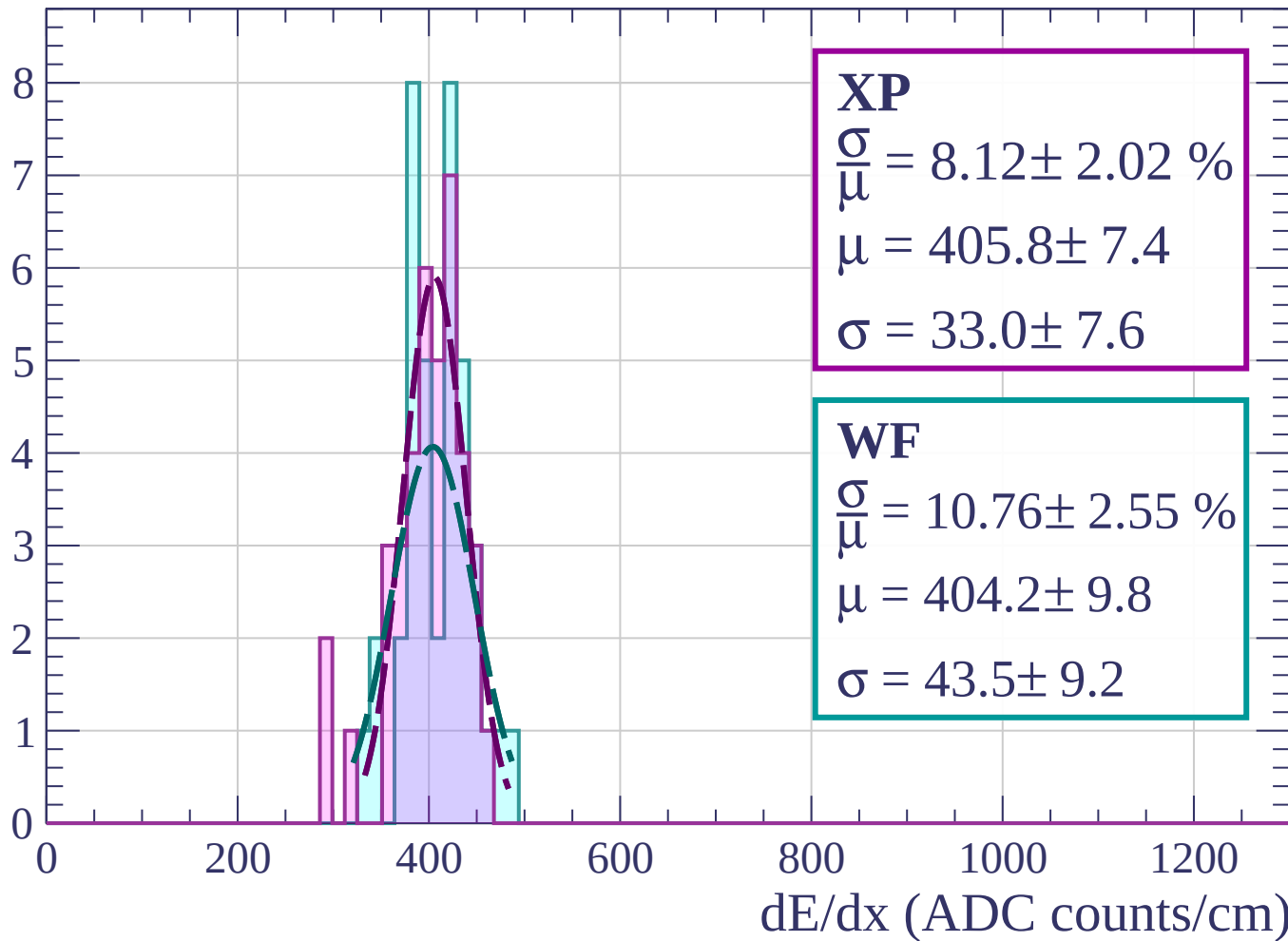
Energy loss | $39 < \theta < 42$

Count



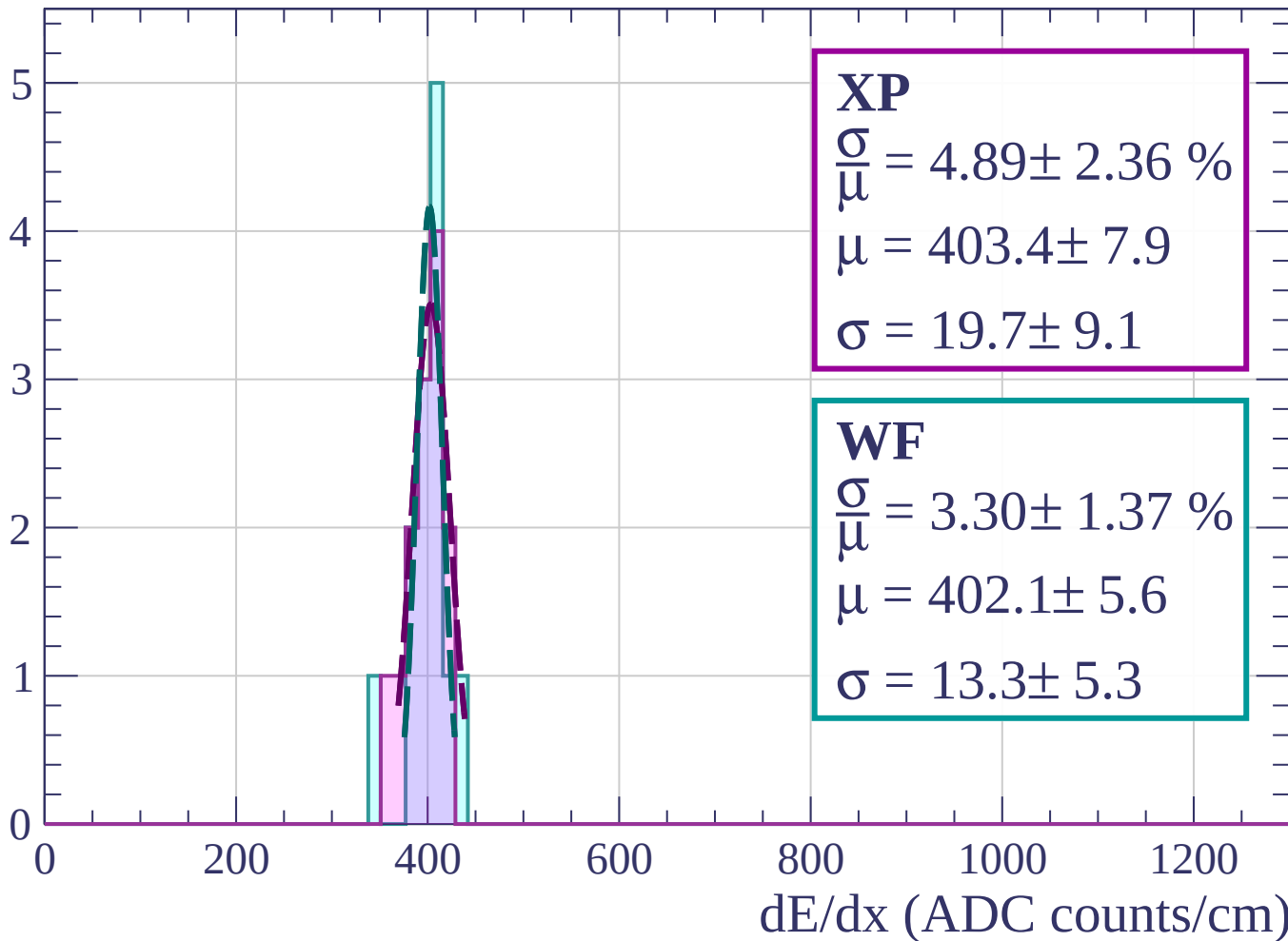
Energy loss | $42 < \theta < 45$

Count



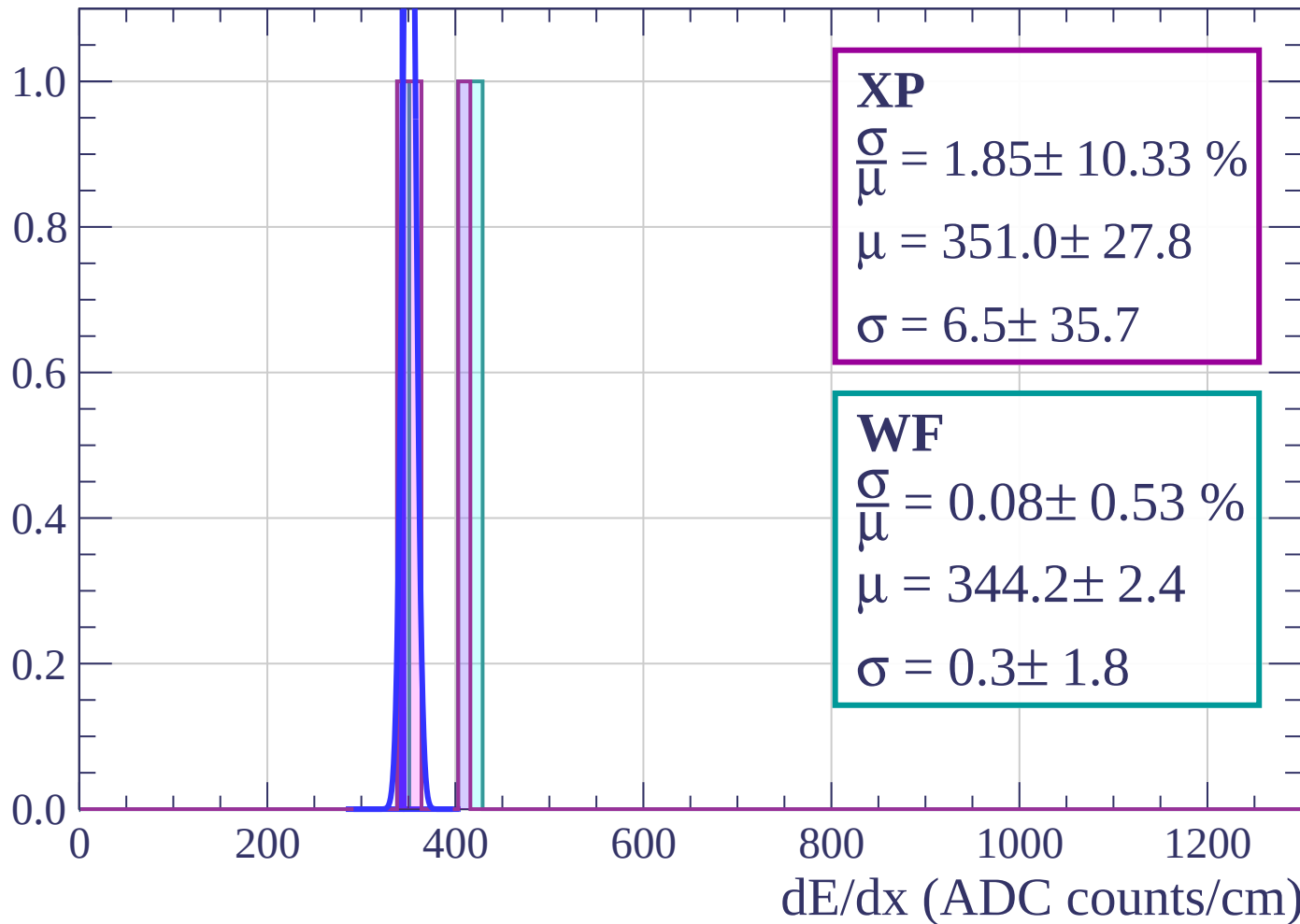
Energy loss | $45 < \theta < 48$

Count



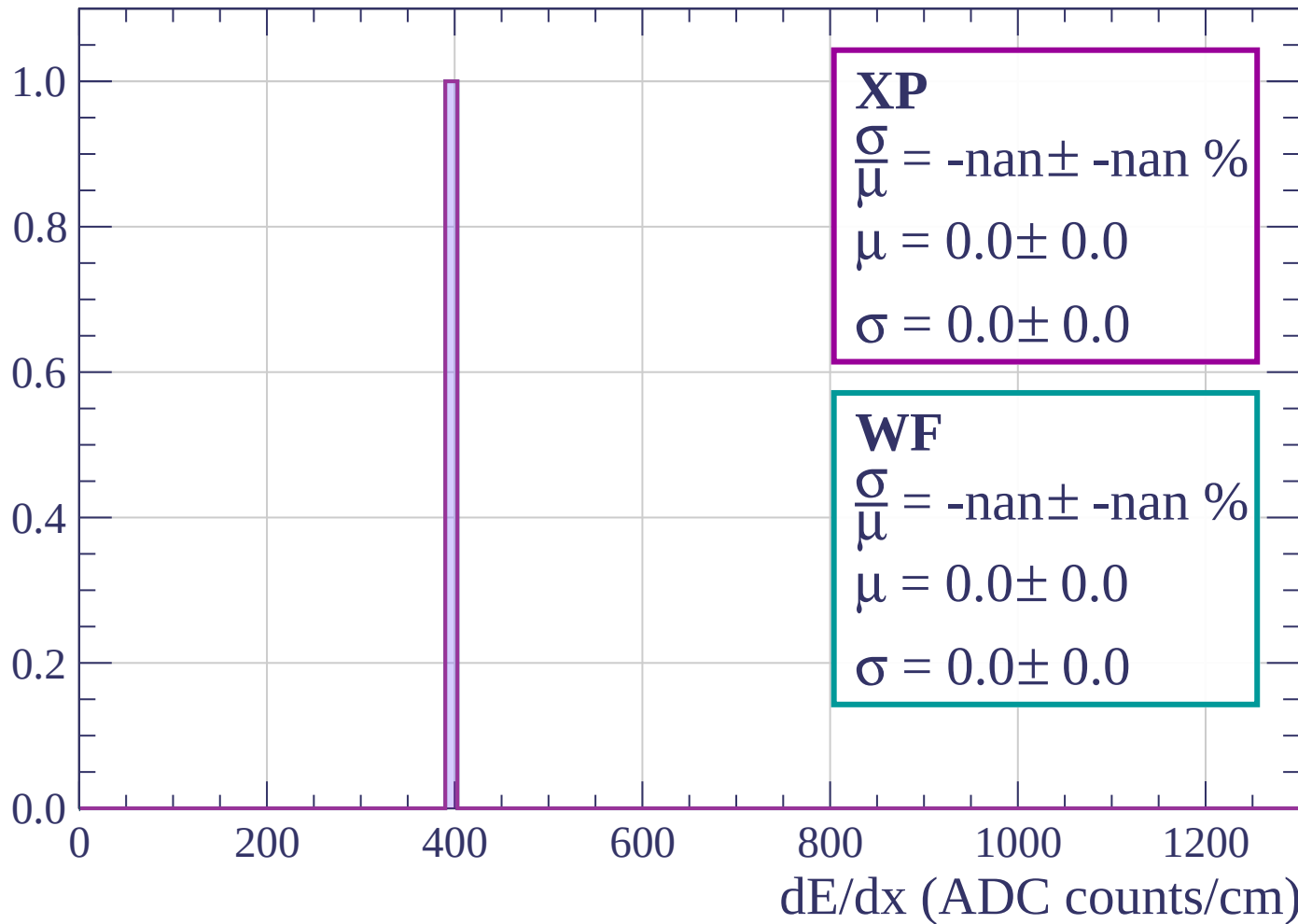
Energy loss | $48 < \theta < 51$

Count



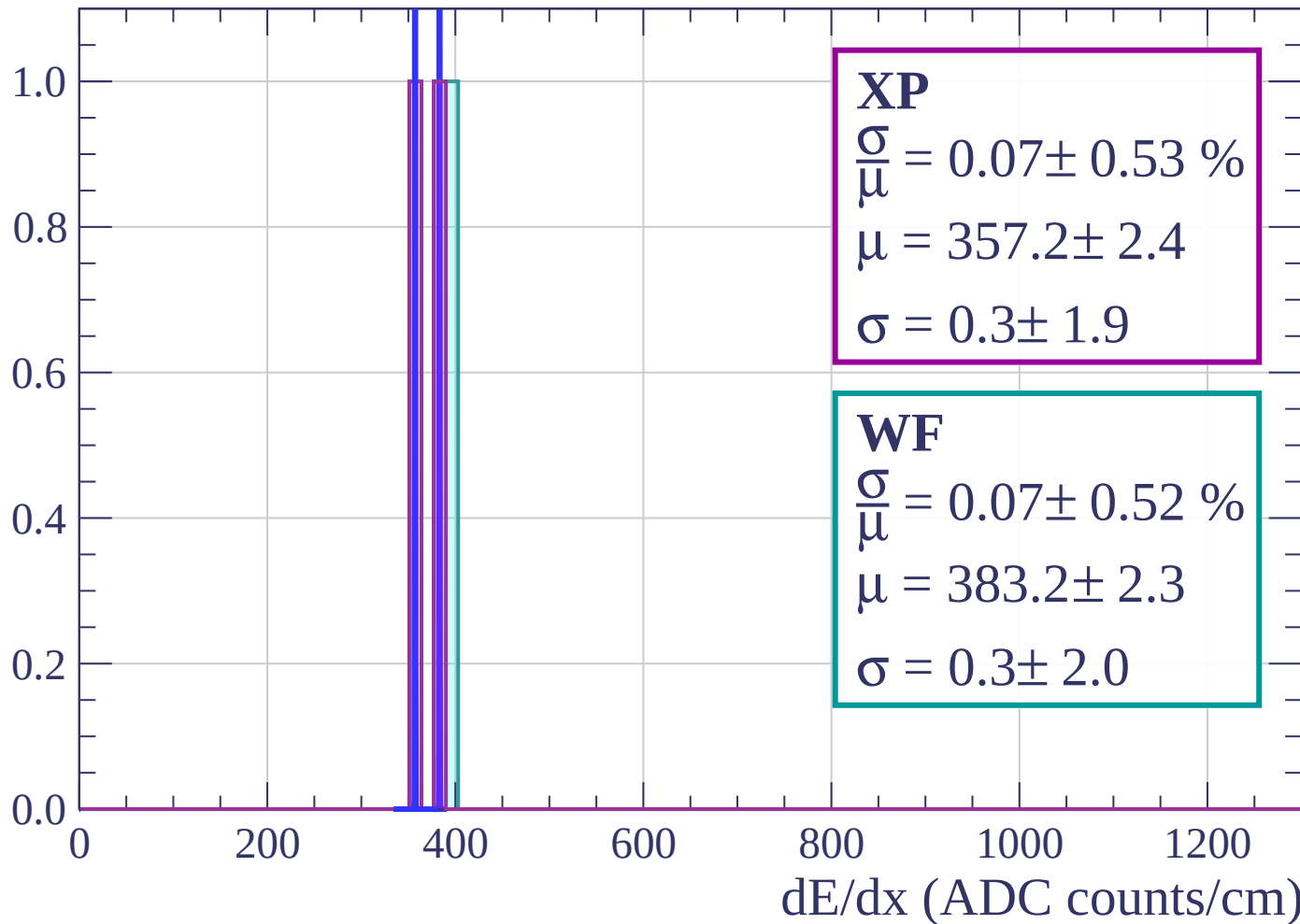
Energy loss | $51 < \theta < 54$

Count



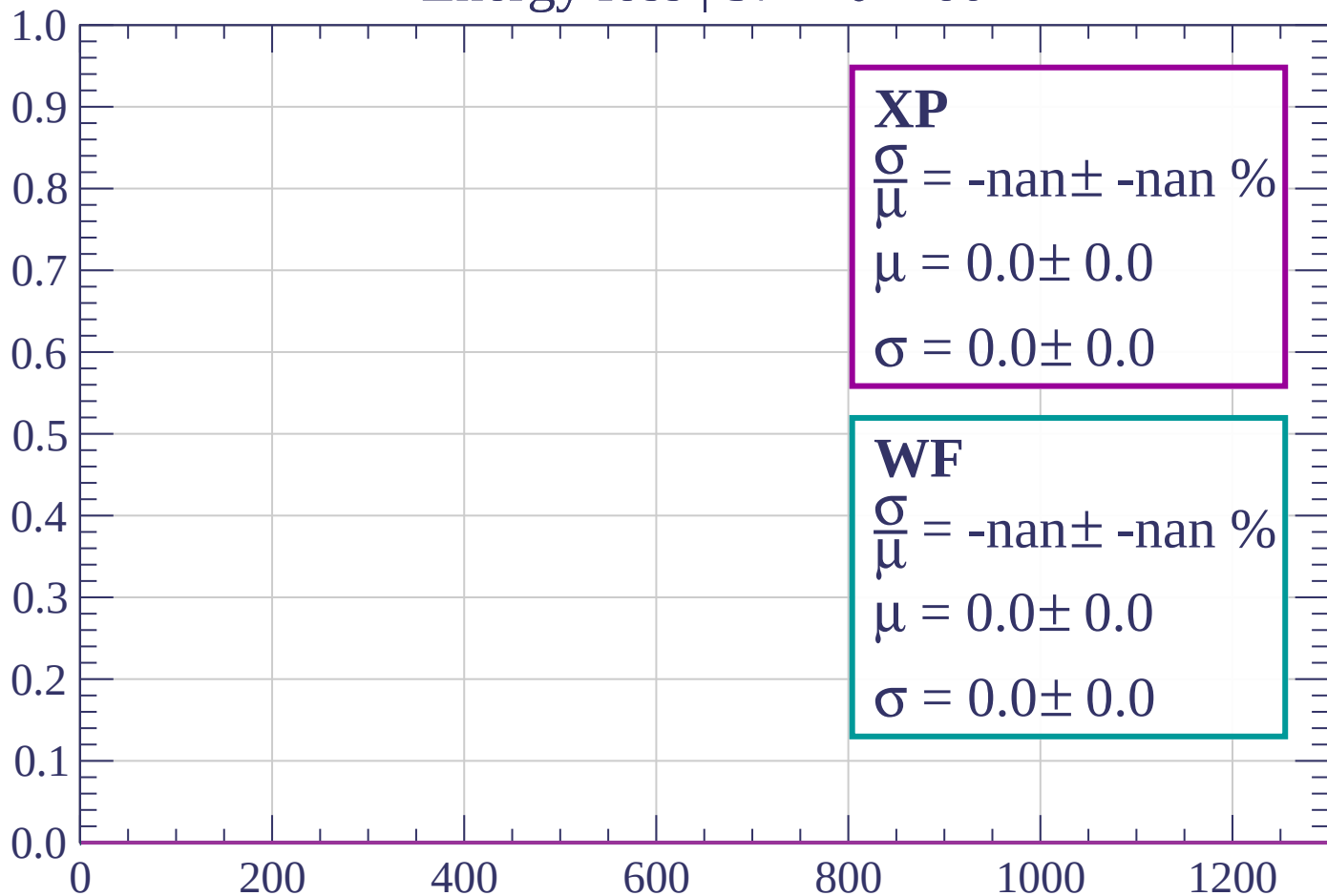
Energy loss | $54 < \theta < 57$

Count



Energy loss | $57 < \theta < 60$

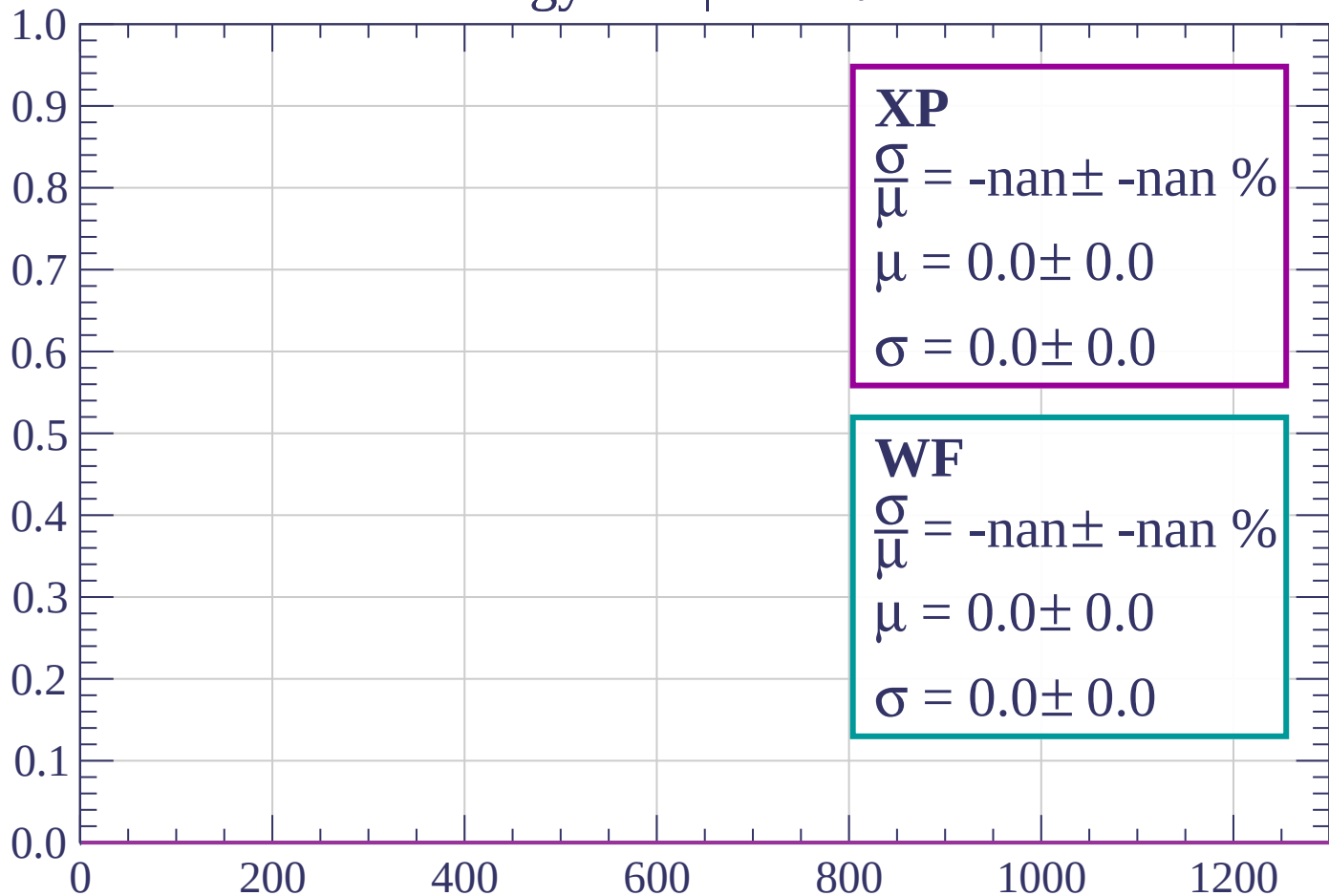
Count



dE/dx (ADC counts/cm)

Energy loss | $60 < \theta < 63$

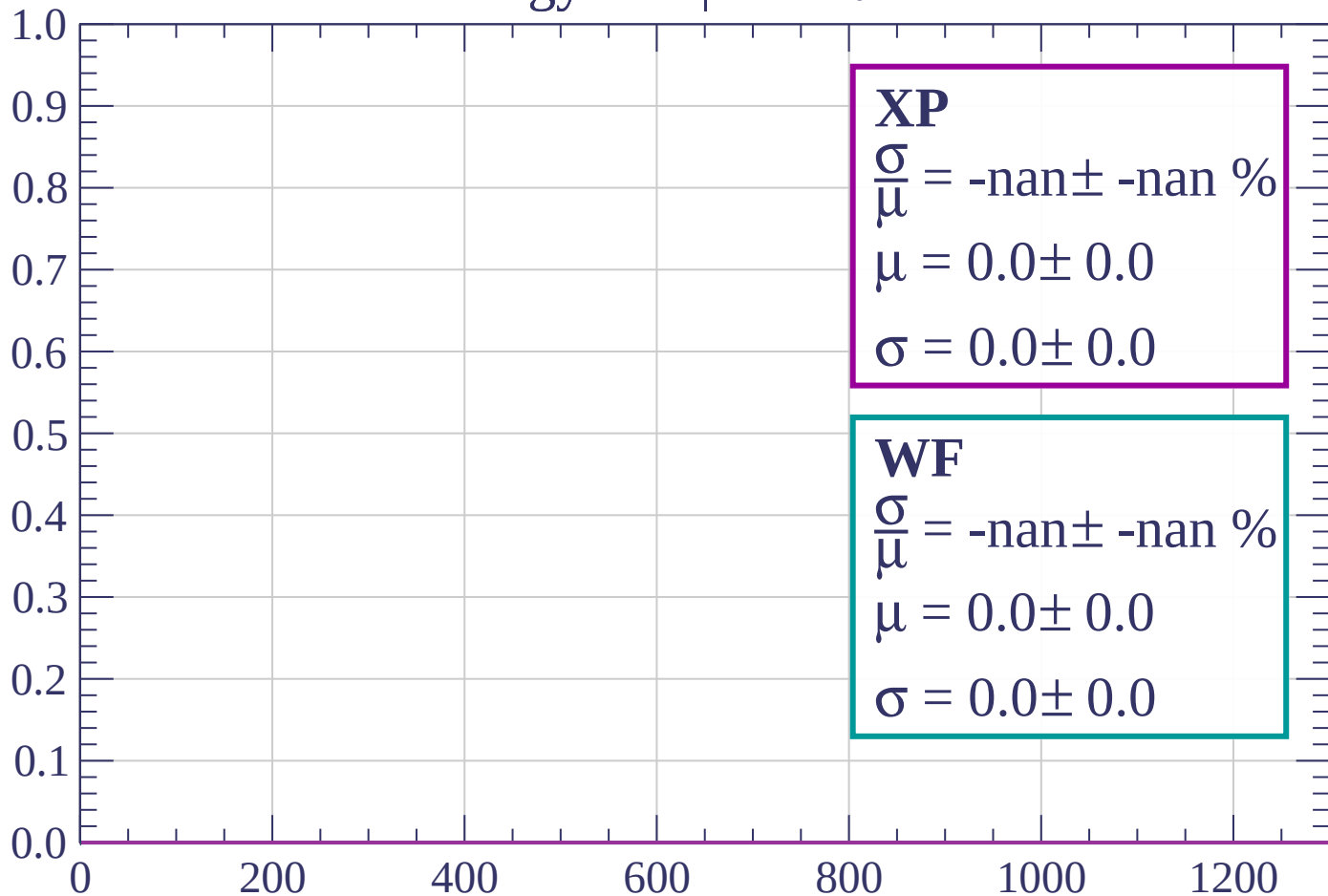
Count



dE/dx (ADC counts/cm)

Energy loss | $63 < \theta < 66$

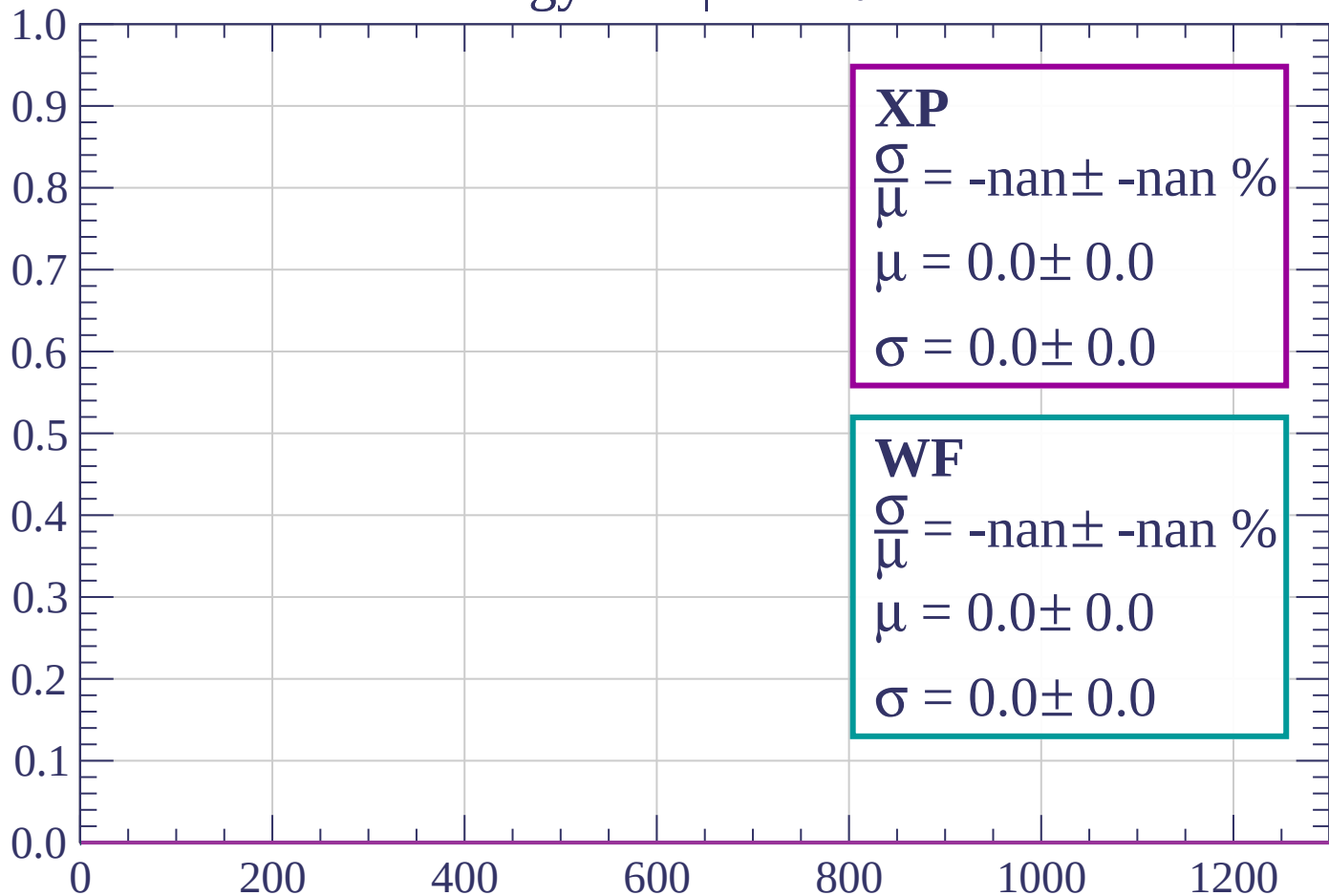
Count



dE/dx (ADC counts/cm)

Energy loss | $66 < \theta < 69$

Count



XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

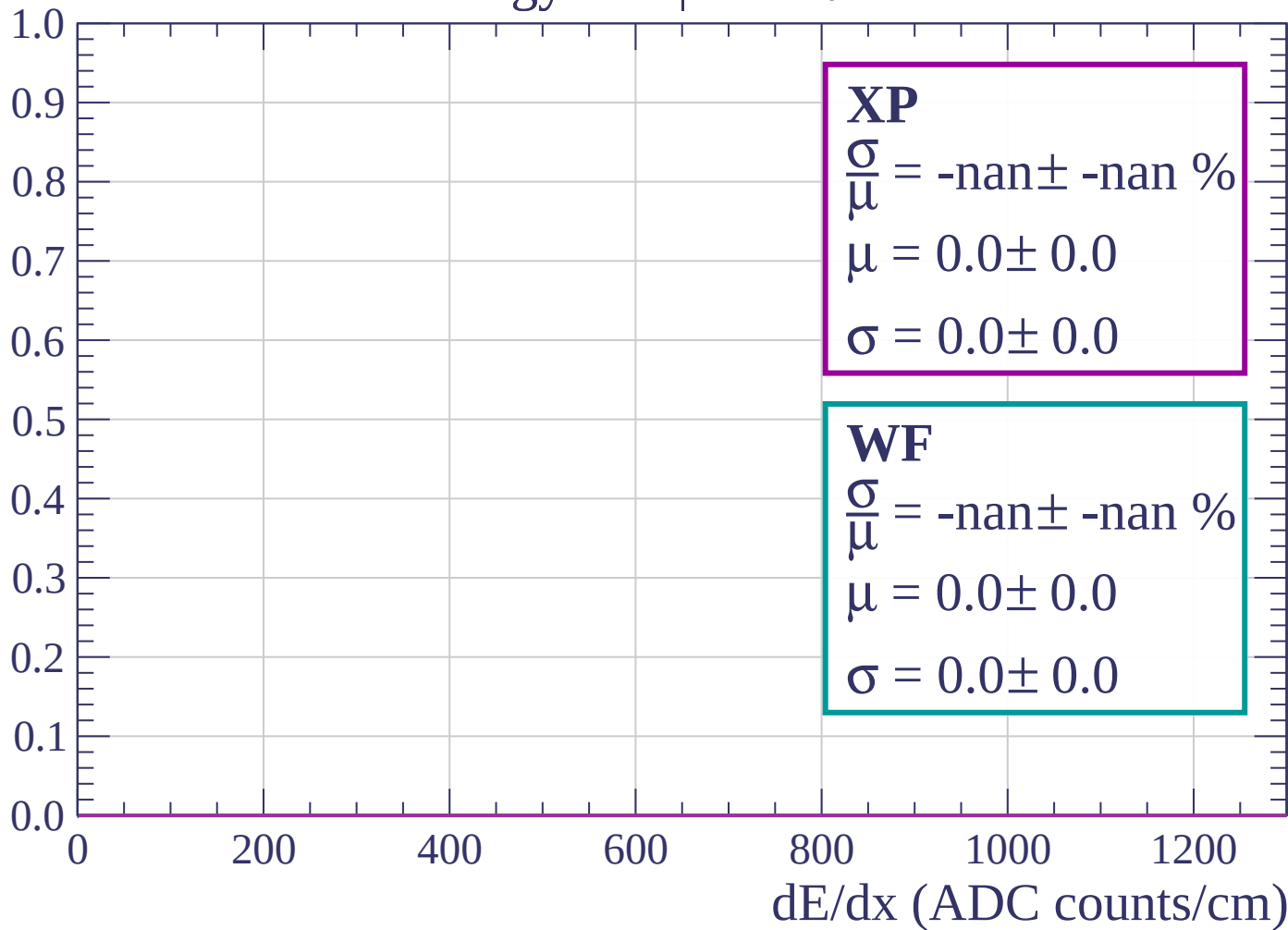
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

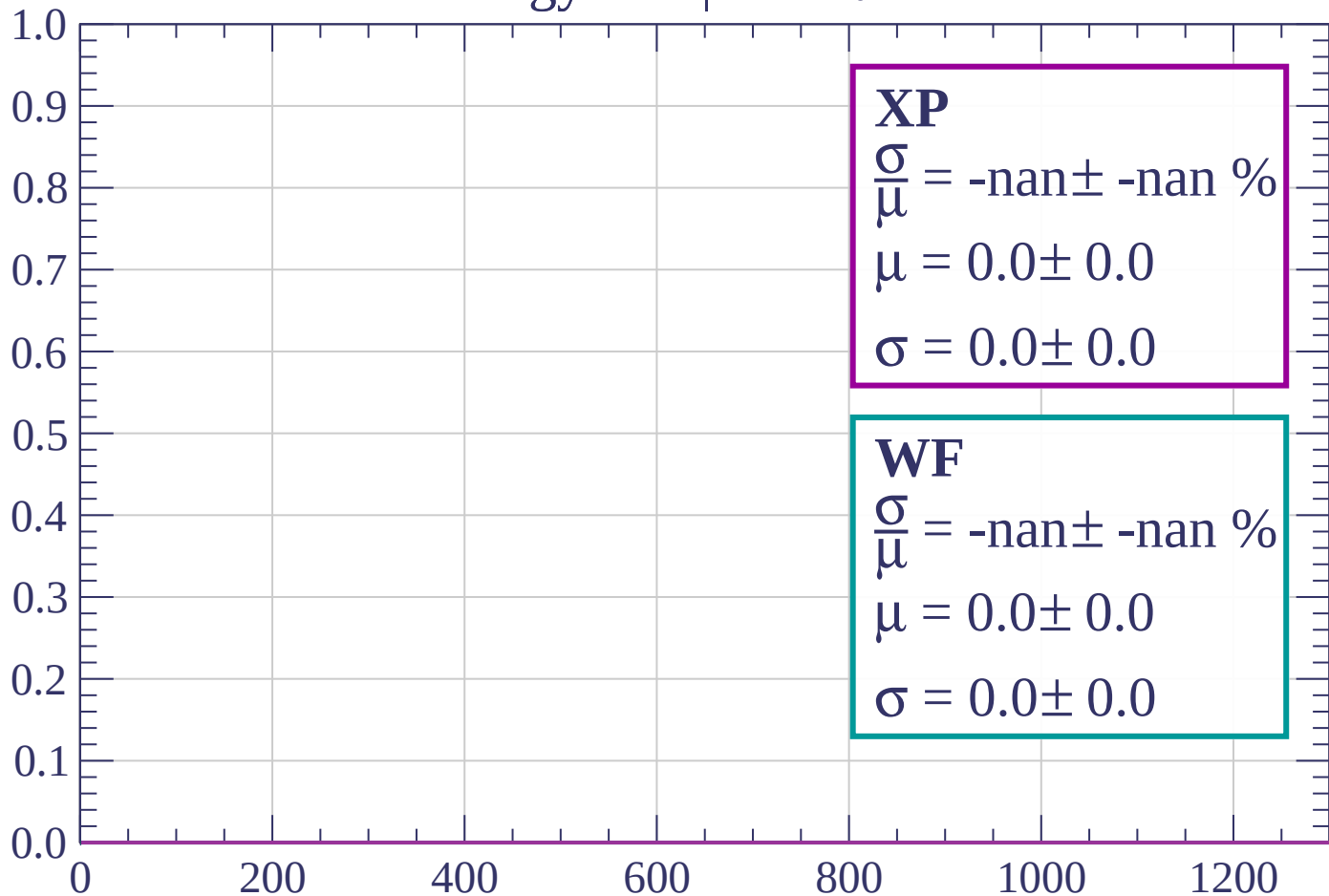
Energy loss | $69 < \theta < 72$

Count



Energy loss | $72 < \theta < 75$

Count



XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

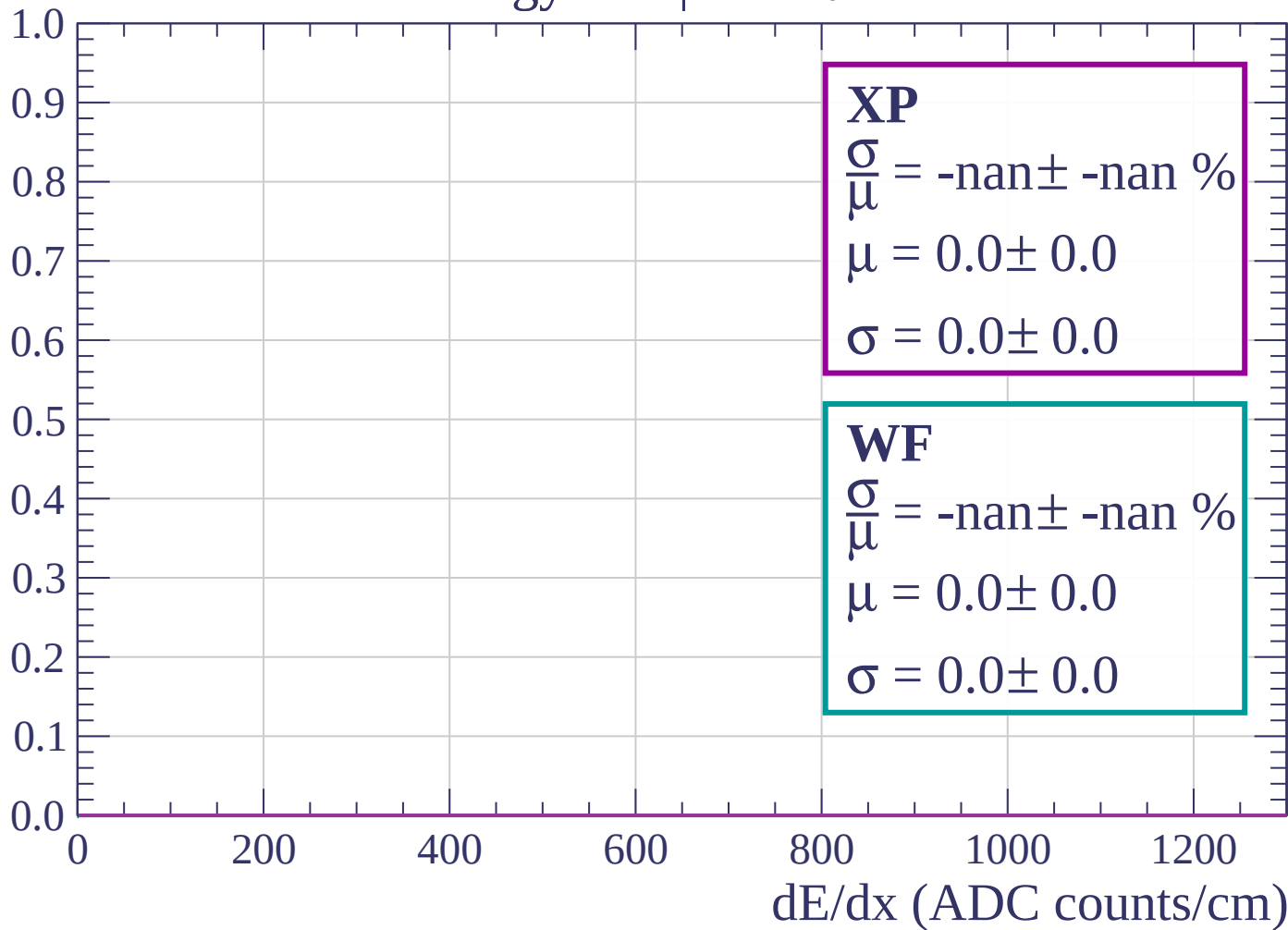
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

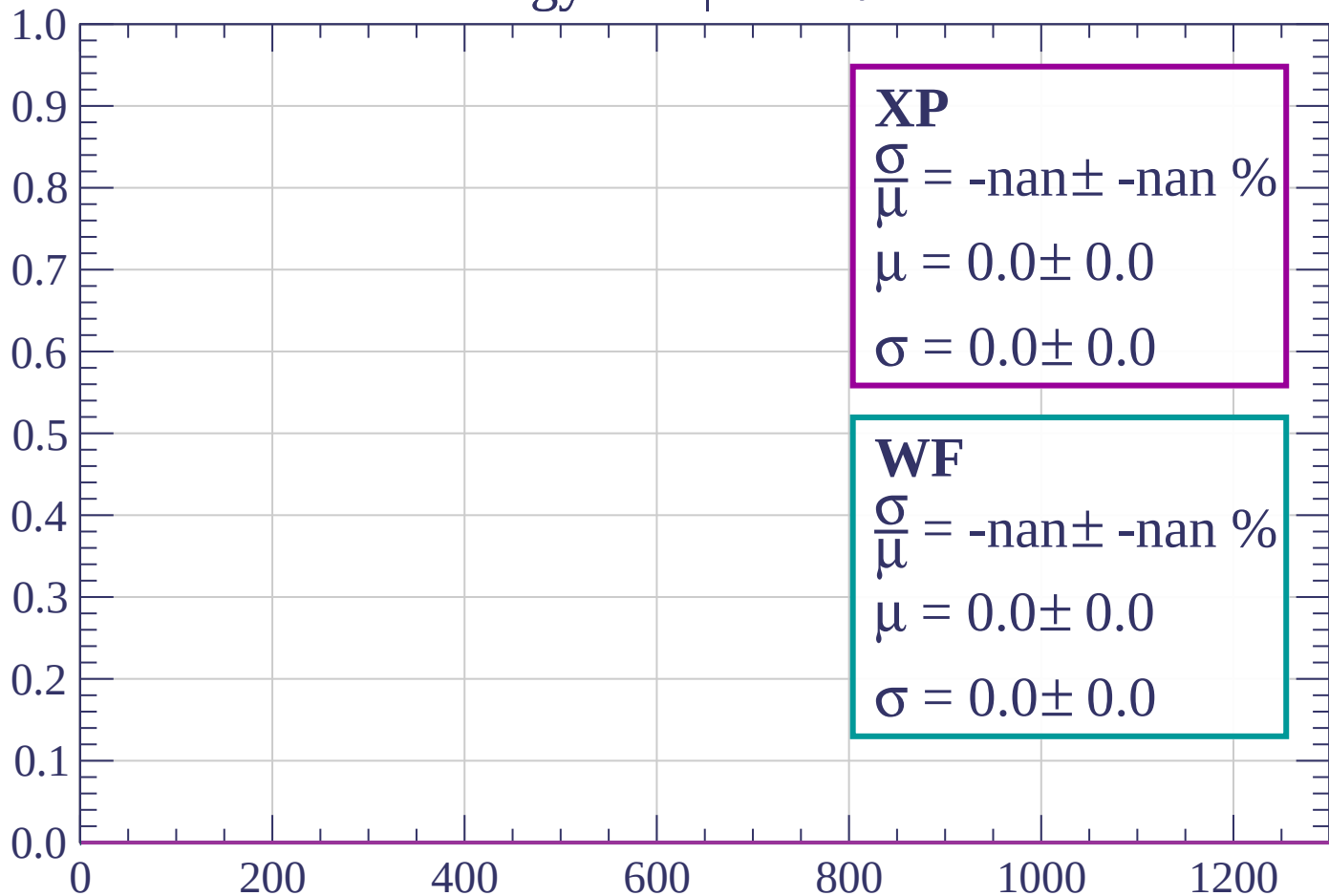
Energy loss | $75 < \theta < 78$

Count



Energy loss | $78 < \theta < 81$

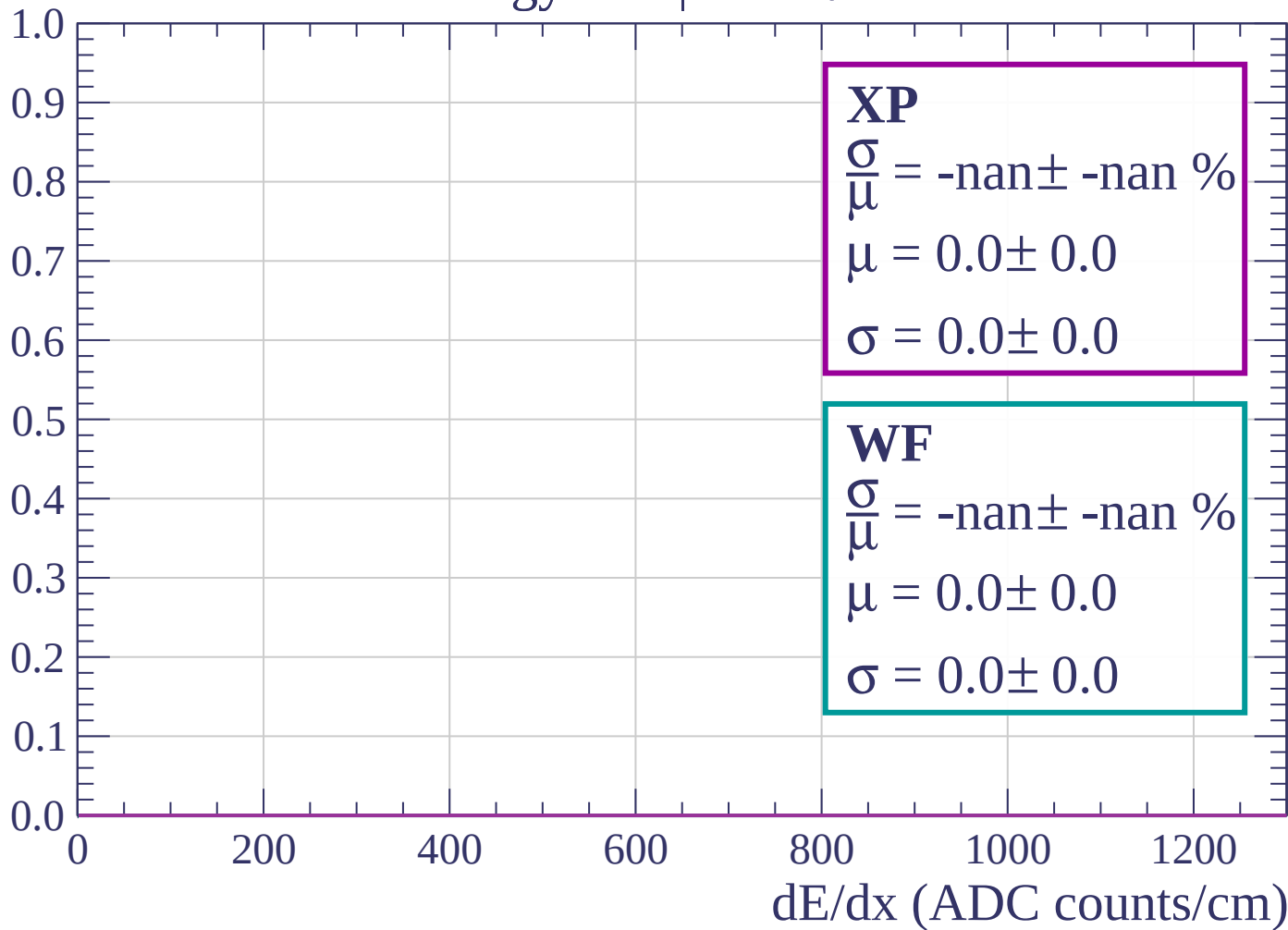
Count



dE/dx (ADC counts/cm)

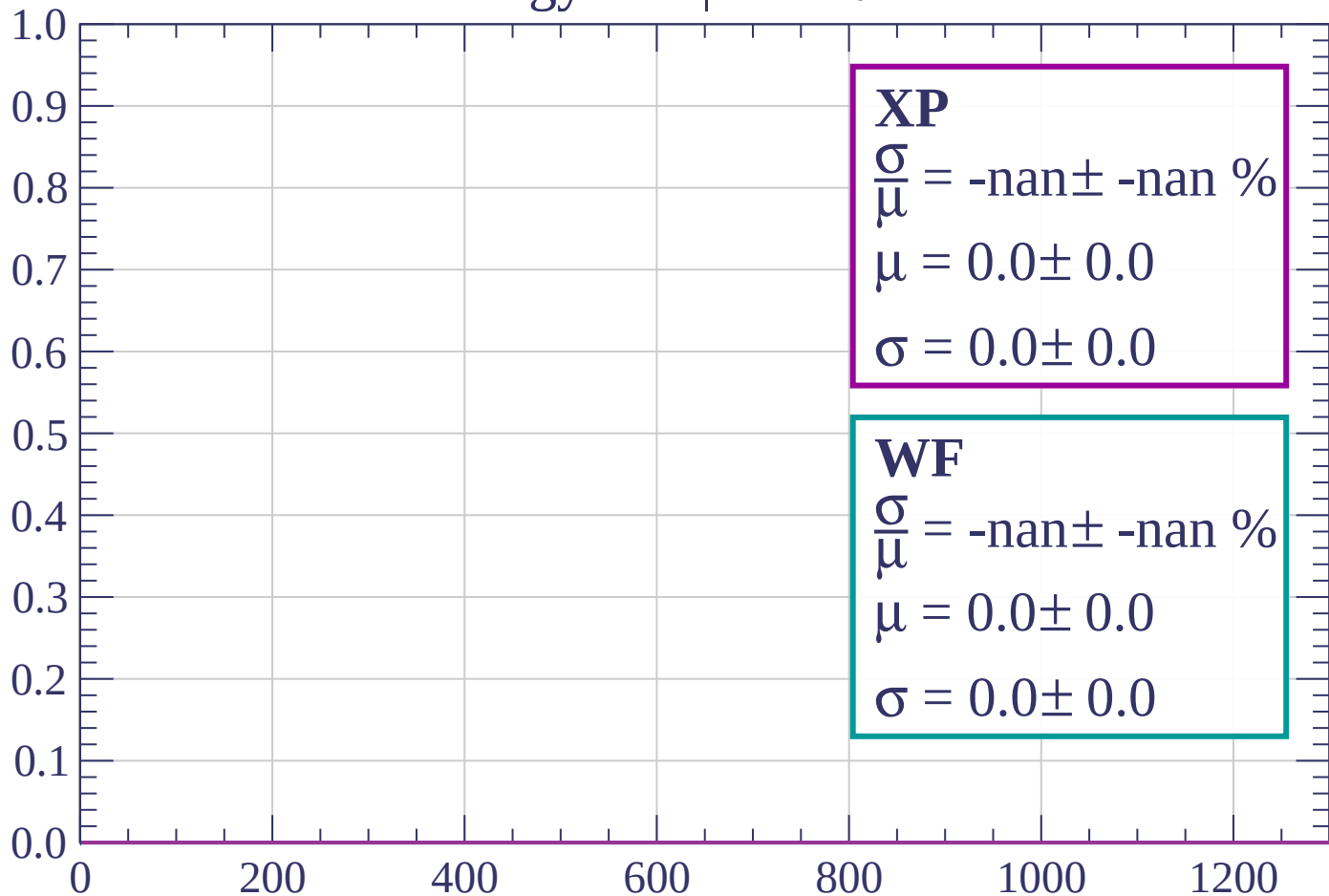
Energy loss | $81 < \theta < 84$

Count



Energy loss | $84 < \theta < 87$

Count



XP

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

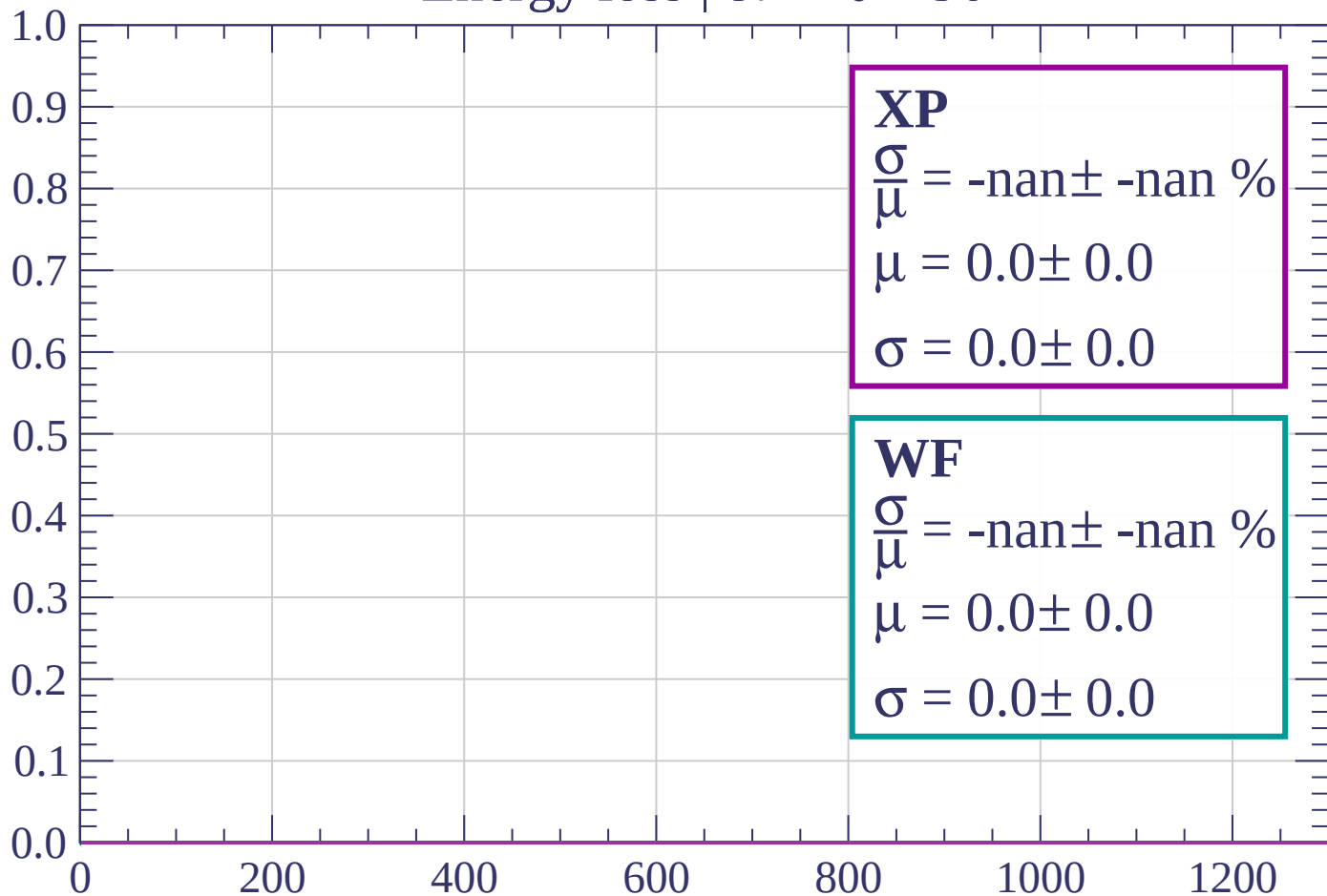
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



XP

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

WF

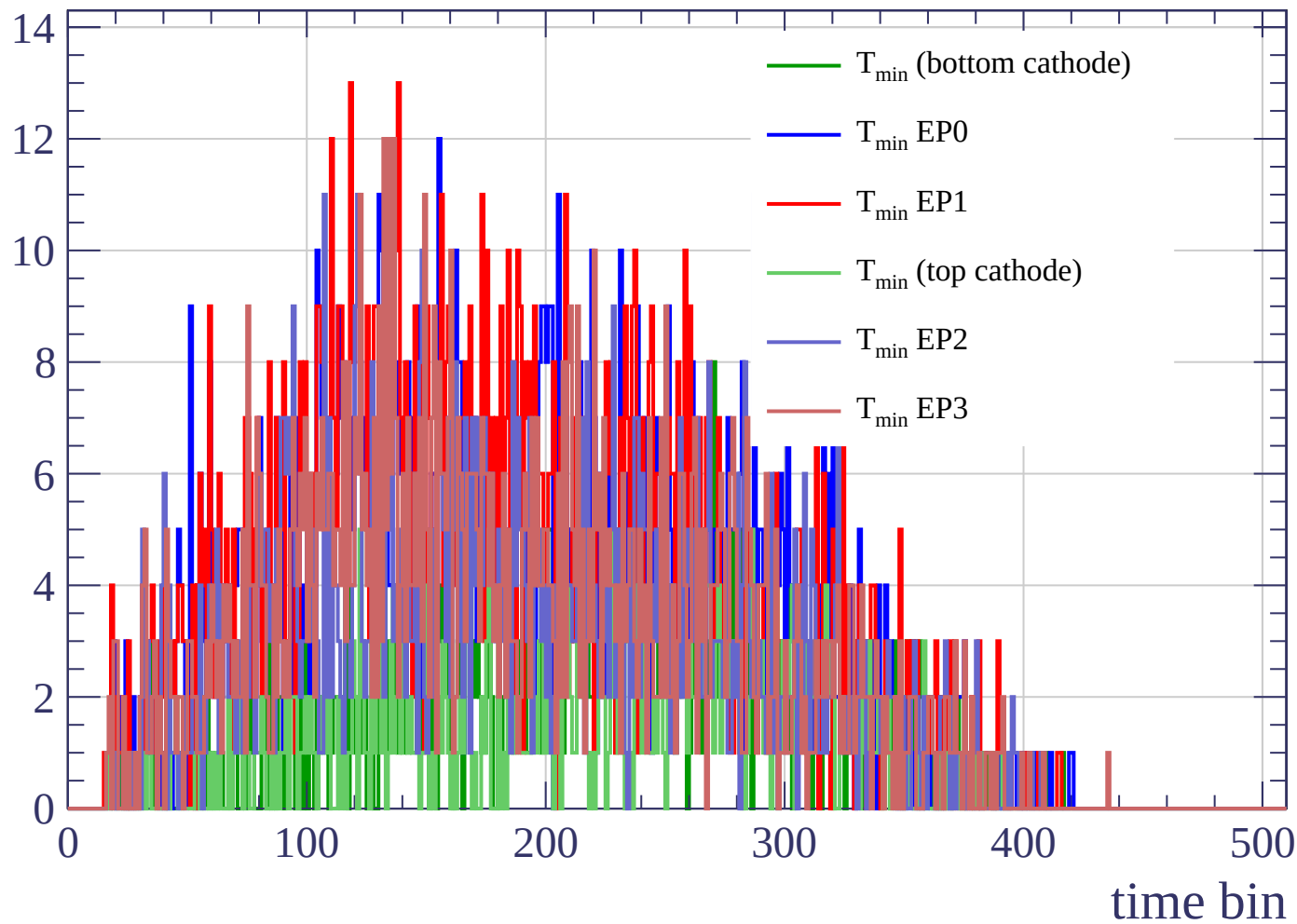
$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx (ADC counts/cm)

Count



Count

